Infinite Factor Structure in Conditional Asset Pricing

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Abstract

The return covariance matrix is widely assumed to have an eigenvalue cliff: a few large eigenvalues, then a bounded bulk, as in the approximate factor structure of Chamberlain and Rothschild (1983). This premise underlies factor tests, principal-component methods, and large-covariance estimation. However, estimating covariances from returns pooled across states means viewing a conditional world unconditionally. We show that, generically, this view removes the cliff: a conditional model with a few factors is, unconditionally, an infinite-factor model in which infinitely many eigenvalues diverge. This holds for the characteristic-managed portfolios that machine-learning methods trade, and for individual stocks whose loadings differ persistently across firms. Yet as long as conditional expected returns admit no asymptotic arbitrage, the maximal Sharpe ratio stays finite. For generic expected returns, infinitely many components are priced, so no finite set of principal components spans expected returns. Reducing returns to a few principal components therefore discards priced risk.

Keywords: approximate factor structure, managed portfolios, principal components, comovement, machine learning

JEL: G12, C38, C58

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1 Introduction

The covariance matrix of asset returns is widely assumed to have an eigenvalue cliff: as the cross-section grows, a few eigenvalues diverge while the rest remain bounded. This spectral separation defines the approximate factor model of Chamberlain and Rothschild (1983). Together with the absence of asymptotic arbitrage, it delivers the finite-factor restrictions on expected returns of the arbitrage pricing theory (APT) of Ross (1976). The same premise underlies much of empirical practice: tests of factor models, principal-component estimators of latent factors (Lettau and Pelger, 2020b; Giglio and Xiu, 2021), estimators of the number of factors, and large-covariance estimation.

We show that the cliff does not survive the conditioning information on which modern empirical asset pricing relies. Firm characteristics enter returns in two ways. They set the weights of large menus of managed portfolios (Kozak, Nagel and Santosh, 2020; Lettau and Pelger, 2020b; Feng, Giglio and Xiu, 2020; Giglio and Xiu, 2021; Gu, Kelly and Xiu, 2021), and they set the factor loadings of individual stocks. Conditional on the current characteristics, a few factors may generate all systematic risk. The unconditional second-moment matrix, however, averages over the movement of characteristics through time, and that averaging can turn a low-dimensional conditional structure into an unconditional one with infinitely many diverging eigenvalues. Empirical work cannot sidestep this averaging: without a model of the conditional structure, a sample covariance matrix pools returns across states and estimates the unconditional one. The dimension of systematic risk in a given state is one thing; the dimension generated across states is another.

Managed portfolios provide the first channel. A single stock can generate infinitely many linearly independent payoffs by scaling its return with different functions of its characteristics. Theorem 1 shows that, for a sufficiently rich menu, the eigenvalue at every fixed rank of the managed-return second-moment matrix grows in proportion to the number of portfolios P. The result imposes no factor structure on stock returns and holds even for a single stock.

Expanding the menu therefore adds new risk directions rather than redundant exposures to a fixed set of factors. Diversification across stocks need not undo this. If characteristics and returns were i.i.d. across stocks, managed returns would converge to constants as the number of stocks N grows. Common variation in characteristics, together with a common return component, can prevent this collapse: a common state shifts the cross-sectional distribution of characteristics, and nonlinear portfolio rules turn these shifts into distinct diversified payoffs. Under these conditions, the divergence survives as N grows.

Individual stocks provide the second channel. We consider a conditional K-factor model in which loadings depend on characteristics, so that systematic risk is exactly K-dimensional at every date (Proposition 2). Observed characteristics need not capture all persistent differences across firms. Following Gagliardini, Ossola and Scaillet (2016), a latent fixed effect enters the loadings, so firms with the same observed characteristics can respond differently to the same common state. When the common state takes infinitely many values, Theorem 3 shows that, generically, the number of unconditional eigenvalues that grow in proportion to N equals the number of distinct fixed-effect values that persist in the limit. With M firm types, there are M such eigenvalues; with infinitely many fixed-effect values, there are infinitely many. A fixed number of conditional factors is thus compatible with infinitely many strong unconditional directions. The conditional model remains valid; what fails is its extrapolation to the unconditional second-moment matrix.

These results change what the absence of arbitrage implies for expected returns. With a cliff, the absence of asymptotic arbitrage yields an approximate finite-factor representation of expected returns. Our setting keeps the no-arbitrage premise but not the cliff. Theorems 5 and 6 show that, for generic expected returns, infinitely many groups of principal components earn risk premia while the maximal Sharpe ratio stays finite; for individual stocks, this requires persistent firm differences and common states that both take infinitely many values. A group collects the components whose eigenvalues grow in the same proportion to the size

of the cross-section. No finite set of principal components therefore spans expected returns exactly. Retaining more groups raises the attainable Sharpe ratio toward that of all groups, but every fixed truncation attains a strictly lower one. The omitted risk also differs from that in the APT. There, the residual pricing error is idiosyncratic. Here, every omitted priced group carries systematic risk, because its eigenvalues keep growing with the cross-section. Truncating the spectrum to a few leading components, as in Lettau and Pelger (2020b); Giglio and Xiu (2021); Fan, Guo and Zheng (2022), therefore discards priced systematic risk. Nor are the omitted components weak factors: their eigenvalues are small relative to the leading ones but still grow in proportion to the number of assets.

These results offer a unified reading of evidence that appears puzzling under the cliff assumption. Leading latent factors of individual stocks earn negligible premia, while portfolios hedged against them earn Sharpe ratios above one (Lopez-Lira and Roussanov, 2023). Premia appear on risk that an approximate factor model classifies as diversifiable (Dello-Preite, Uppal, Zaffaroni and Zviadadze, 2025), and stochastic discount factors estimated from large menus load on components far beyond the leading few (Didisheim, Ke, Kelly and Malamud, 2024). With a cliff, large premia on such small sources of risk imply either near-arbitrage (Kozak, Nagel and Santosh, 2018) or compensation for idiosyncratic risk. Without a cliff, they imply neither: the components beyond the leading few carry systematic risk, are generically priced, and leave the maximal Sharpe ratio finite. In a finite sample, the same components look like weak factors (Onatski, 2012; Lettau and Pelger, 2020a; Giglio, Xiu and Zhang, 2025), because their eigenvalues are small relative to the leading ones even though they grow with the cross-section. Priced residual risk and weak factors are thus two views of the same structure. Because the framework is general, it identifies which components are priced but not the size of their premia, which requires more specific assumptions about expected returns.

Our results require tracking infinitely many systematic directions at once, as both the

portfolio menu and the stock cross-section grow. For managed portfolios, this must work whatever the return structure of the underlying stocks; for individual stocks, it must turn a conditional factor model with finitely many factors into an unconditional spectrum with infinitely many. Standard tools for approximate factor models cannot do this: they count a fixed number of factors against a bounded remainder. We instead represent each object as an operator. A menu of managed portfolios acts on the space of payoffs it can generate. The cross-section of stocks acts on the space of state-response profiles, which record how firms' factor loadings move with the common state. Eigenvalue growth then becomes a spectral property of the limiting operator, and risk premia become properties of expected returns measured against it. To our knowledge, this operator-theoretic approach has not been used to study factor structure in asset pricing.

The analysis provides a population mechanism consistent with the empirical virtue of complexity documented by Kelly, Malamud and Zhou (2024); Kelly, Kuznetsov, Malamud and Zhang (2026); Didisheim et al. (2024). Expanding a menu of characteristic-managed portfolios can add both risk and expected return, rather than merely improve the representation of a fixed set of factors. This argues for modeling risk premia with many factors (Didisheim et al., 2024) and for regularizing covariance matrices, for example through the ridge prior of Kozak et al. (2020), rather than truncating them at a fixed rank. Truncation discards small-eigenvalue directions, which statistics based on inverse covariance matrices weight most heavily. Our companion papers develop random-matrix corrections for the bias that regularization introduces (Chernov, Kelly, Malamud and Schwab, 2025, 2026).

Our results relate to several strands of the literature. The proliferation of characteristics and candidate factors is documented by Cochrane (2011); McLean and Pontiff (2016); Harvey, Liu and Zhu (2016); Feng et al. (2020); Bryzgalova, Huang and Julliard (2023). The use of conditioning information to construct managed portfolios dates to Hansen and Richard (1987). Characteristic-based portfolio policies (Brandt, Santa-Clara and Valkanov,

2009), semiparametric characteristic factor models (Connor, Hagmann and Linton, 2012), and conditional-loading models (Kelly, Pruitt and Su, 2019) provide leading examples of the portfolio rules and return structures considered here. Instrumented principal components target a conditional stock model, whereas our analysis characterizes the unconditional spectra of managed portfolios and individual stocks. Many machine-learning methods likewise construct characteristic-managed portfolios (Gu et al., 2021; Chen, Pelger and Zhu, 2024; Bryzgalova, Pelger and Zhu, 2025; Cong, Feng, He and He, 2025).

A second line of literature studies large cross-sections through principal-component truncation (Giglio and Xiu, 2021; Lettau and Pelger, 2020b). These methods build on the asymptotic principal-component analysis of approximate factor models developed by Connor and Korajczyk (1986, 1993). A related literature estimates the number of factors directly (Bai and Ng, 2002; Onatski, 2009; Ahn and Horenstein, 2013; Fan et al., 2022). Both approaches rely on a finite-dimensional spectral restriction: beyond a finite number of factors, the remaining eigenvalues must stay bounded. Our results show that this restriction can fail for rich menus of managed portfolios and, generically, for individual stock returns when persistent firm differences and common states both take infinitely many values.

A third line of literature studies the strength of cross-sectional dependence rather than the number of factors. Bailey, Kapetanios and Pesaran (2016) summarize dependence through the growth order of the largest covariance eigenvalue, while Chudik, Pesaran and Tosetti (2011) formalize intermediate forms of dependence between strong factors and idiosyncratic noise. Applying this perspective to factor-model residuals, Terenteva, Ahlgren and Antell (2026) find pervasive comovement after conditioning on standard characteristic factors. Such evidence shows that a finite factor specification is incomplete. The growth order of the largest residual eigenvalue does not, however, determine whether the omitted structure contains finitely or infinitely many directions. Our results characterize the latter possibility: a rich menu of managed portfolios has infinitely many diverging eigenvalues, and so do

unconditional stock returns when persistent firm differences and common states both take infinitely many values.

The pricing results are closely related to the question of whether a finite number of factors spans the stochastic discount factor (Kozak and Nagel, 2024). Our contribution is to distinguish exact spanning by finitely many principal components from approximation of the maximal Sharpe ratio in a rich managed-return space. Davis and Lopez-Lira (2026) obtain a related conclusion from estimation uncertainty in equilibrium, whereas the mechanism studied here operates in population and does not require estimation uncertainty.

The remainder of the paper is organized as follows. Section 2 introduces individual stock returns, managed portfolios, and expanding portfolio menus. Section 3 establishes the divergence of infinitely many eigenvalues for managed portfolios, both at fixed N and with a growing stock cross-section. Section 4 imposes a conditional factor model and characterizes the conditional and unconditional factor structures of individual stocks. Section 5 develops the pricing and econometric implications, including an exactly solvable Gaussian example, the pricing of infinitely many groups of principal components, the absence of weak factors, and covariance matrix estimation. The Appendix contains the proofs.

2 Preliminaries

2.1 Individual stocks

There are N stocks with excess returns $r_{t+1} = (r_{i,t+1})_{i=1}^{N} \in \mathbb{R}^N$. Throughout, N may be fixed or growing. Each stock i has d conditioning characteristics $X_{i,t} \in \mathbb{R}^d$, such as book-to-market, size, and momentum. Stack these characteristics in $X_t = (X_{1,t}, \ldots, X_{N,t}) \in \mathbb{R}^{N \times d}$.[1]

[1] A characteristic common to all stocks, such as a macroeconomic variable, is naturally accommodated as a common coordinate that does not vary with i.

The conditional second and first moments of returns are

$$\Sigma_N(X_t) = E[r_{t+1}r'_{t+1} \mid X_t] \in \mathbb{R}^{N\times N}, \qquad \mu_N(X_t) = E[r_{t+1} \mid X_t]. \tag{1}$$

We assume throughout that $\Sigma_N(X_t)$ is finite and positive definite almost surely. The joint distribution of returns and characteristics is the same at every date. When studying the unconditional stock second-moment matrix $\Sigma_N = E[r_{t+1}r'_{t+1}]$, we require $E[\operatorname{tr}\Sigma_N(X_t)] < \infty$.[2] For managed portfolios, the moment conditions apply to the portfolio returns themselves, as specified in Assumption 1. Second moments and covariances differ by the rank-one term $\mu\mu'$, with $\mu = E[r_{t+1}]$, so their eigenvalues interlace (Lemma A.11). We work with second moments throughout. We write $\lambda_j(\cdot)$ for the j-th largest eigenvalue of a symmetric matrix, counted with multiplicity.

The APT of Ross (1976) starts from a factor structure for returns. Chamberlain and Rothschild (1983) characterize an approximate factor structure through the covariance spectrum: after a fixed number of leading eigenvalues, the rest stay uniformly bounded as N grows. When many stocks carry strong exposures to every nonzero combination of the common factors, those leading eigenvalues diverge. We call the resulting split the eigenvalue cliff: finitely many diverging eigenvalues above a uniformly bounded remainder. Random matrix theory studies a close relative, the spiked covariance model, in which a few population eigenvalues stand apart from a bulk (Johnstone, 2001; Baik, Ben Arous and Péché, 2005). We borrow its term bulk for the bounded remainder, although our leading eigenvalues diverge with the dimension rather than staying fixed.

Empirically, researchers look for the cliff in scree plots and test for it formally (Bai and Ng, 2002; Onatski, 2009, 2010; Ahn and Horenstein, 2013). The formal procedures ask whether a fixed, finite number of eigenvalues separates from the bulk, and Connor and Korajczyk

[2]Second moments also arise in the tradable stochastic discount factor normalized as $m_{t+1} = 1 - r'_{t+1}a$. Remark A.12 shows that this discount factor is proportional to the one built from covariances.

(1993) and Bai and Ng (2002) apply them to individual-stock returns. Others take the cliff as given: Fan, Liao and Mincheva (2013) impose an approximate factor structure to estimate the covariance matrix of individual stocks. The cliff is also imposed on portfolio returns. Lettau and Pelger (2020b) assume a low-dimensional factor structure for 370 decile portfolios sorted on anomaly characteristics. Giglio and Xiu (2021) require the factor eigenvalues of 647 portfolios to grow linearly with their number. Fan et al. (2022) impose the same structure to estimate the number of factors in 100 portfolios double-sorted on size and book-to-market.

2.2 Managed portfolios

A managed portfolio assigns to stock i a weight that depends on that stock's own characteristics, for example tilting toward stocks with high book-to-market ratios, small capitalization, or high momentum. A menu of P such portfolios, $f^{(P)}$, is a collection of weighting functions $f_k^{(P)} : \mathbb{R}^d \to \mathbb{R}$, $k = 1, \ldots, P$.[3] Portfolio k's excess return is the cross-sectional average

$$R_{k,t+1} \;=\; \frac{1}{N}\sum_{i=1}^{N} f_k^{(P)}(X_{i,t})\, r_{i,t+1}\,, \qquad k = 1, \ldots, P\,. \tag{2}$$

In vector form, with $f_k^{(P)}(X_t) = (f_k^{(P)}(X_{i,t}))_{i=1}^N$, the return is $R_{k,t+1} = \frac{1}{N} f_k^{(P)}(X_t)' r_{t+1}$.[4] We collect portfolio returns into a vector $R_{t+1} = (R_{k,t+1})_{k=1}^P \in \mathbb{R}^P$.

A managed portfolio converts a conditional trading strategy into an unconditional return series that can be studied with standard asset-pricing tools. Those tools estimate unconditional moments, because a time series of returns pools observations across states. Our object

[3] The superscript (P) indicates that the weights may depend on the number of portfolios P. The returns below also depend on P; we suppress this index. Expanding a menu may change its existing weights as well as add portfolios.

[4] The $1/N$ is the natural cross-sectional-average normalization. Under the large-N weight and moment bounds imposed below, it prevents portfolio scale from growing mechanically with N. It does not by itself guarantee convergence. As a common rescaling, it leaves rank, Sharpe ratios, and relative spectral shape unchanged; absolute eigenvalue magnitudes do change.

of interest is therefore the unconditional second-moment matrix of these portfolios,

$$\Psi_P \;=\; E[R_{t+1}R_{t+1}'] \;\in\; \mathbb{R}^{P\times P}\,. \tag{3}$$

The spectrum of Ψ_P depends on whether the weights respond to states. Static portfolios, whose weights stay fixed across states, cannot add payoffs: N independent static portfolios generate the same payoffs as the N underlying stocks. With more than N of them, some returns are exact combinations of others, and Ψ_P has zero eigenvalues. In the APT of Ross (1976) and Chamberlain and Rothschild (1983), the cross-section expands by adding assets. Adding static portfolios is no substitute.

Managed portfolios are different. Even a single traded asset can support arbitrarily many independent managed returns when its characteristics take infinitely many values, because the weights can respond differently to each characteristic value.[5] The positive definite conditional second moment assumed above ensures that distinct weighting functions produce distinct returns. Consequently, a growing menu can add new payoffs even when the underlying stock universe is fixed.[6] Assumption 1 controls how the menu behaves as it expands and requires it to retain exposure to these payoffs.

Assumption 1 (Portfolio menu and payoff variety) *At every fixed N, let $R_{k,t+1}$ denote the managed return from menu $f^{(P)}$, as in (2); the menus need not be nested. All limits and suprema in P are taken along one specified increasing sequence of menu sizes. For weighting functions with random labels drawn independently of the panel, the conditions hold almost surely conditional on the label sequence. Every $R_{k,t+1}$ is square integrable.*[7] *There is a constant $C_{\text{menu}} < \infty$, the same for every P and every arbitrary managed portfolio, such*

[5] Remark D.5 gives the single-asset statement, and Lemma D.4 the N-stock version.

[6] If the characteristic values appearing across all stocks instead come from a set of m values, there are at most m independent managed returns, because the same weighting function applies to every stock. Then Ψ_P has at least $P-m$ zero eigenvalues for $P > m$. Greater menu variety cannot create the missing payoffs.

[7] Throughout, $\|Y\|_{L^2} = (E[Y^2])^{1/2}$.

that for every measurable ϕ with square-integrable return $R^{\phi}_{t+1} := N^{-1}\sum_{i=1}^{N}\phi(X_{i,t})r_{i,t+1}$, the combination

$$A_P R^{\phi}_{t+1} := \frac{1}{P}\sum_{k=1}^{P} R_{k,t+1}E[R_{k,t+1}R^{\phi}_{t+1}] \tag{4}$$

satisfies

$$A^{\phi}_{t+1} := \lim_{P\to\infty} A_P R^{\phi}_{t+1} \qquad \textit{exists in } L^2, \tag{5}$$

$$\sup_{P\geq 1}\|A_P R^{\phi}_{t+1}\|_{L^2} \leq C_{\text{menu}}\|R^{\phi}_{t+1}\|_{L^2}. \tag{6}$$

The menu retains every nonzero R^{ϕ}_{t+1} in the limit:

$$A^{\phi}_{t+1} \neq 0 \qquad \textit{whenever} \qquad R^{\phi}_{t+1} \neq 0 \quad \textit{in } L^2. \tag{7}$$

There are arbitrarily many linearly independent square-integrable returns R^{ϕ}_{t+1}. If $N\to\infty$, these conditions hold for each N along the sequence with the same constant C_{menu}, independent of N.

Given an arbitrary managed portfolio return R^{ϕ}_{t+1}, (4) forms a combination of the menu returns in which each $R_{k,t+1}$ receives a weight proportional to its co-moment $E[R_{k,t+1}R^{\phi}_{t+1}]$ with R^{ϕ}_{t+1}. Condition (5) requires this combination to settle to a well-defined limit as the menu expands, and (6) prevents it from becoming arbitrarily large relative to R^{ϕ}_{t+1}. Because the nonzero eigenvalues of A_P coincide with those of Ψ_P/P (Lemma D.3), these conditions control the spectrum of Ψ_P.

Condition (7) requires the menu to retain exposure to every nonzero managed portfolio return R^{ϕ}_{t+1}: the settled combination A^{ϕ}_{t+1} must be nonzero.

Assumption 1 imposes no functional form on the individual weighting functions. When stock second moments are finite, characteristics that take infinitely many values already

supply the required payoff variety (Lemma D.4). The assumption states payoff variety directly, so it also covers stocks with infinite unconditional second moments, provided there are arbitrarily many independent square-integrable managed returns. Appendix C.2 introduces the menus used in empirical asset pricing: random features (Didisheim et al., 2024), fixed bases, and sorts and trees (Fama and French, 1993; Daniel, Grinblatt, Titman and Wermers, 1997; Moritz and Zimmermann, 2016; Bryzgalova et al., 2025; Cong et al., 2025). For sorts and trees we use expected-count versions, whose weights use the expected rather than the realized number of stocks in a cell. Appendix F verifies Assumption 1 for these constructions at a fixed N. Appendix G supplies uniform-in-N menu bounds; the conditions for a growing cross-section are collected in Assumption 2.

3 The portfolio eigenvalue cliff fails

The eigenvalue at every fixed rank of the managed-return second-moment matrix grows proportionally to the number of portfolios P (Theorem 1 below). We write $\Psi_{P,N}$ for Ψ_P when displaying the number of underlying stocks. For fixed N, Assumption 1 suffices: there are arbitrarily large collections of linearly independent, square-integrable managed returns, and the expanding menu retains exposure to them. No factor structure is required for the underlying stock returns.

An additional condition is needed when the stock cross-section grows. To see why, suppose the return-characteristic pairs are independent and identically distributed across stocks. For each fixed weighting f_k, the law of large numbers gives

$$R_{k,t+1} = \frac{1}{N}\sum_{i=1}^{N} f_k(X_{i,t})r_{i,t+1} \xrightarrow[N\to\infty]{\text{a.s.}} a_k, \qquad a_k := E[f_k(X_{1,t})r_{1,t+1}]. \tag{8}$$

Writing $\bar{a}_P = (a_1, \ldots, a_P)'$, Lemma H.1 establishes that, for each fixed P,

$$\Psi_{P,N} \longrightarrow \bar{a}_P \bar{a}_P' \qquad \text{as } N \to \infty.$$

The limiting returns are deterministic. Their second-moment matrix has rank at most one, reflecting mean returns rather than stochastic comovement. Cross-sectional diversification can thus remove all comovement among managed returns, leaving at most one positive eigenvalue.

Common variation in characteristics, combined with a common return component, can prevent this collapse. Assumption 2 states the required conditions informally; Appendix D gives the formal version.

Assumption 2 (Common-shock limit: Informal version of Assumption D.2)

Write

$$R^{\phi}_{N,t+1} = \frac{1}{N} \sum_{i=1}^{N} \phi(X_{i,t}) r_{i,t+1}, \qquad E_t[\,\cdot\,] = E[\,\cdot \mid X_t].$$

When $N \to \infty$, there is a common state W_t with the following properties.

(i) There are arbitrarily large finite collections of fixed weightings whose managed returns have uniformly bounded second moments and for which every linear combination with a nonzero coefficient vector has a positive limiting second moment.

(ii) For every fixed measurable weighting ϕ satisfying $\sup_N E[(R^{\phi}_{N,t+1})^2] < \infty$, there is a function v_ϕ such that

$$E_t\left[(R^{\phi}_{N,t+1})^2\right] \longrightarrow v_\phi(W_t)$$

in L^1.

(iii) For every such ϕ with $E[(R^{\phi}_{N,t+1})^2] > 0$ for all sufficiently large N,

$$\liminf_{N\to\infty} \lim_{P\to\infty} \frac{\frac{1}{P}\sum_{k=1}^{P}\left(E[R_{k,t+1}R^{\phi}_{N,t+1}]\right)^2}{E[(R^{\phi}_{N,t+1})^2]} > 0.$$

Condition (i) requires arbitrarily many linearly independent payoff directions to survive diversification. Condition (ii) requires the common state to summarize the information in characteristics relevant for conditional portfolio second moments in the large-N limit. Condition (iii) prevents the menu's average squared exposure to a nonzero managed return from vanishing relative to that return's second moment. Section 4 constructs a conditional factor model for stocks around this common state.

Let $N \to N_{\max}$, where $N_{\max} < \infty$ denotes a fixed cross-section of $N_{\max}$ stocks and $N_{\max} = \infty$ denotes a growing cross-section. Define the normalized eigenvalue lower limits by

$$\lambda^*_j := \liminf_{N\to N_{\max}} \liminf_{P\to\infty} \frac{\lambda_j(\Psi_{P,N})}{P}, \qquad j \geq 1. \tag{9}$$

For a fixed cross-section, the outer operation is evaluation at $N = N_{\max}$. These quantities measure how much normalized eigenvalue growth survives the expanding menu, without requiring the normalized eigenvalues to converge.

Theorem 1 (The portfolio eigenvalue cliff fails) *Impose Assumption 1. When $N_{\max} = \infty$, also impose Assumption D.2. For every fixed $j \geq 1$,*

$$0 < \lambda^*_j \leq \limsup_{N\to N_{\max}} \limsup_{P\to\infty} \frac{\lambda_j(\Psi_{P,N})}{P} \leq C_{\text{menu}}. \tag{10}$$

The constants λ^*_j are nonincreasing in j.

The theorem bounds the normalized eigenvalues away from zero and infinity at every

fixed rank; it does not, in general, establish their convergence. Additional structure on the construction of the menu yields limits. Appendix F develops three constructions: random features, fixed menus, and partitions (multivariate sorting). Under the corresponding assumptions, Proposition F.7 establishes convergence for fixed N, and Proposition I.3 establishes it for a growing stock cross-section:

$$\lim_{N \to N_{\max}} \lim_{P \to \infty} \frac{\lambda_j(\Psi_{P,N})}{P} = \lambda_j^* \in (0, C_{\text{menu}}]. \tag{11}$$

When $N_{\max} = \infty$, we denote these growth constants by κ_j. Both (10) and (11) concern fixed j and take the limit in P before the limit in N when the stock cross-section grows (Remark H.11).

Theorem 1 rules out an unconditional approximate factor structure in the spectral sense of Chamberlain and Rothschild (1983). Such a structure would require all but finitely many covariance eigenvalues to remain bounded as the menu expands.[8] In contrast, Theorem 1 implies that, however many leading principal components are removed, the largest remaining eigenvalue still grows proportionally to P.

Theorem 1 reflects the behavior of the whole matrix: Ψ_P grows proportionally to P, and Ψ_P/P settles, in the sense of (5), to a limit with infinitely many positive eigenvalues. The eigenvalue at every fixed rank therefore grows proportionally to P.

The matrix Ψ_P/P is the co-moment weighting (4) written in portfolio weights. A vector $a \in \mathbb{R}^P$ of weights on the menu produces the return $\sum_{l=1}^P a_l R_{l,t+1}$. Weighting each menu return by its co-moment with this return gives

$$\frac{1}{P} \sum_{k=1}^{P} R_{k,t+1} E\left[R_{k,t+1} \sum_{l=1}^{P} a_l R_{l,t+1} \right] = \sum_{k=1}^{P} R_{k,t+1} \widetilde{a}_k, \qquad \widetilde{a} = \frac{\Psi_P}{P}\, a. \tag{12}$$

[8] The covariance and second-moment matrices differ by a term of rank at most one formed from mean returns. The conclusion therefore extends from Ψ_P to the covariance matrix (Lemma A.11).

The weighting A_P thus maps the portfolio with weights a into the portfolio with weights $\Psi_P a/P$, and the two share their nonzero eigenvalues (Lemma D.3).

Condition (5) makes this weighting settle as the menu expands: for every managed return R^{ϕ}_{t+1},

$$\lim_{P\to\infty} \frac{1}{P} \sum_{k=1}^{P} R_{k,t+1} E[R_{k,t+1} R^{\phi}_{t+1}] = A^{\phi}_{t+1}. \tag{13}$$

The matrix Ψ_P/P settles with it, and its eigenvalues are bounded below by those of the limiting weighting $R^{\phi}_{t+1} \mapsto A^{\phi}_{t+1}$ (Lemma D.7).

Condition (7) makes these limiting eigenvalues positive. The limiting weighting sends no nonzero managed return to zero, and there are arbitrarily many linearly independent managed returns, so it has infinitely many positive eigenvalues. The eigenvalue of Ψ_P/P at every fixed rank therefore stays bounded away from zero, while (6) bounds it from above.

4 Individual stocks in a conditional factor model

Individual stocks can lose the unconditional cliff as well, but only when firms differ persistently. A symmetric benchmark shows why. Suppose that

$$r_{i,t+1} = B(X_{i,t}) G_{t+1}.$$

Given a common state W_t, let characteristics be independent draws from the same distribution, and let the factor be independent of those draws. Stocks then have identical second moments, and every pair of distinct stocks has the same co-moment: $E[r_{i,t+1} r_{j,t+1}]$ is independent of (i, j). This symmetry gives every zero-cost portfolio a bounded second moment when its weights have unit length. Hence $E[rr']$ has at most one eigenvalue growing like N, with the equal-weighted market portfolio as eigenvector. Whether that one eigenvalue

actually grows depends on the market portfolio. As N grows, the equal-weighted market return converges to $E[B(X_{i,t}) \mid W_t]\, G_{t+1}$, the average loading times the factor. Provided the factor's second moment is nondegenerate, the market eigenvalue therefore grows like N exactly when the average loading is nonzero with positive probability (Remark B.10).

Persistent differences in loadings can break this symmetry. We represent them by stock fixed effects: firm attributes not captured by observed characteristics. Stock-specific coefficients of this kind are common in conditional asset pricing (Gagliardini et al., 2016). When the common state takes infinitely many values, the number of eigenvalues of the unconditional second-moment matrix Σ_N that grow like N generically equals the number of distinct fixed-effect values that persist in the limit.

Assumption 3 (Conditional factor model with fixed effects) *Each stock has a fixed effect $S_i \in \mathcal{T}$. The fixed effects form a nonrandom infinite array $(S_i)_{i\geq 1}$. As more stocks are included, their mix of fixed effects approaches a distribution ν. Let M be the size of its support. The array itself may contain more distinct values than remain in this limiting support.*

Each period has a common state W_t, as in Assumption 2, with the same distribution P_W at every date. Given that state, characteristics are independent across stocks and follow the same distribution for every fixed effect.[9]

Stock returns follow the K-factor model

$$r_{i,t+1} = B(X_{i,t}, S_i)' G_{t+1} + \varepsilon_{i,t+1}, \qquad i = 1, \ldots, N. \tag{14}$$

Conditional on the date-t characteristics and fixed effects $(X_t; S) \equiv (X_{i,t}, S_i)_{i\leq N}$, the factors

[9] This distribution can change with the common state. Assumption B.3 in Appendix B states the precise regularity conditions.

have mean and second moment

$$E[G_{t+1} \mid X_t, S] = \mu_G(X_t; S), \qquad E[G_{t+1}G'_{t+1} \mid X_t, S] = \Omega(X_t; S). \tag{15}$$

The mean vector μ_G and second-moment matrix Ω depend on the cross-section only through the empirical distribution of characteristics and fixed effects. Relabeling the stocks leaves these moments unchanged.[10]

The residuals are conditionally mean zero, independent of the factors, and uncorrelated across stocks:

$$E[\varepsilon_{i,t+1}\varepsilon_{j,t+1} \mid X_t, S] = \delta_{i,j}\sigma_\varepsilon^2(X_{i,t}, S_i), \tag{16}$$

where $\delta_{i,j}$ equals one if $i = j$ and zero otherwise, and, for constants $0 < \underline{\sigma}^2 \leq \bar{\sigma}^2 < \infty$,

$$\underline{\sigma}^2 \leq \sigma_\varepsilon^2(x, s) \leq \bar{\sigma}^2$$

for almost every x and every s.

Persistent firm differences affect loadings through $B(X_{i,t}, S_i)$, so firms with the same observed characteristics may still have different loadings. Given the common state, characteristics follow the same distribution across fixed effects. The fixed effect plays the role for stocks that the portfolio index plays for a menu: $B(\cdot, S_i)$ selects the loading function of stock i, as $f_k^{(P)}$ selects the weighting function of portfolio k. The number of distinct fixed effects therefore plays the role of the number of distinct weightings.

With finitely many fixed-effect values, the stocks form firm types: all stocks of one type share one value. With a single value ($M = 1$), persistent heterogeneity disappears, provided every stock carries that value; otherwise a vanishing share of stocks could still carry other

[10] Their dependence on N is suppressed; Appendix B imposes the limits needed as the stock count grows.

values. With infinitely many values, exposures can differ persistently across arbitrarily many fixed-effect values.

Let

$$B_N(X_t; S) = \big(B(X_{1,t}, S_1), \ldots, B(X_{N,t}, S_N)\big)' \in \mathbb{R}^{N \times K}$$

denote the matrix of factor loadings, and let $\Sigma_N(X_t; S) := E[r_{t+1} r'_{t+1} \mid X_t, S]$. The conditional mean-zero and factor-independence restrictions on residuals in Assumption 3 imply

$$\Sigma_N(X_t; S) = B_N(X_t; S)\Omega(X_t; S)B_N(X_t; S)' + \operatorname{diag}\big(\sigma_\varepsilon^2(X_{i,t}, S_i)\big). \tag{17}$$

The conditional second-moment matrix of individual stocks has a strong-factor structure.

Proposition 2 (Strong K-factor structure of stock returns) *Suppose Assumption 3 and Assumption B.3(i)–(iv) hold. Then the ordered eigenvalues of*

$$\Sigma_N(X_t; S) = E[r_{t+1} r'_{t+1} \mid X_t, S]$$

satisfy:

(i) For $k = 1, \ldots, K$, the limit $\lim_{N\to\infty} \lambda_k\big(\Sigma_N(X_t; S)\big)/N$ exists and is strictly positive almost surely.

(ii) For $k = K+1, \ldots, N$,

$$\underline{\sigma}^2 \le \lambda_k\big(\Sigma_N(X_t; S)\big) \le \bar{\sigma}^2$$

uniformly in N.

Proposition 2 is the conditional counterpart of the approximate factor structure of Chamberlain and Rothschild (1983). Their definition requires only that the first K eigenvalues be unbounded. Proposition 2 delivers the stronger version assumed by Bai and Ng (2002), in which these eigenvalues grow linearly with N and the rest stay bounded. Conditional on the current cross-section, K factors therefore span the systematic component of stock returns. The conditioning matters: the unconditional second moment averages over realizations of the cross-section, and this averaging need not preserve the count K. Because a sample covariance matrix pools returns across realizations of the cross-section, it estimates this unconditional moment rather than the conditional one.

Unconditional averaging can leave stocks with a single growing eigenvalue, with finitely many, or with infinitely many. The responses $b(s, w) = E[B(X_{i,t}, s) \mid W_t = w]$ are the average factor loadings of a firm with fixed effect s when the common state is w; they are the same for every stock i because characteristics share one distribution given the state. Genericity here concerns these response functions, with the laws of the common state and the fixed effects held fixed; Definition B.5 gives its precise meaning. Infinitely many growing eigenvalues require two sources of variation. First, persistent firm differences must retain an infinite range as the number of stocks grows: infinitely many distinct fixed effects are not enough if all but finitely many of them lose their share of the cross-section. Second, the common state must take infinitely many values. For generic responses of firms to the common state, these two conditions also suffice.[11] The next theorem makes this statement precise for the stocks' unconditional second-moment matrix.

Theorem 3 (The unconditional stock cliff fails) *Let Assumptions 3 and B.3 hold. If both the distribution of the common state, P_W, and the limiting distribution of fixed effects,*

[11] The exceptions are knife-edge cases. For example, infinitely many eigenvalues fail to grow if the responses of all firm types are exact combinations of the responses of finitely many of them. An arbitrarily small change in the responses breaks such an exact relation.

ν, have infinite support, then, for generic response configurations,

$$\lim_{N\to\infty} \frac{\lambda_j\big(E[r_{t+1}r'_{t+1}]\big)}{N} > 0 \tag{18}$$

for every fixed j.

If ν has $M < \infty$ support points, at most M eigenvalues satisfy (18). If also P_W has infinite support, exactly M do for generic configurations. If the fixed-effect array takes exactly M distinct values, each with positive limiting share, the $N-M$ eigenvalues on the within-type contrast directions stay bounded and converge to at most M finite levels.

Because the S_i are fixed, the unconditional second moment averages over time while holding firm identities constant. Stocks with the same fixed effect are interchangeable. With finitely many types, the eigenvectors of the growing eigenvalues are combinations of the equal-weighted portfolios within each type. A type average need not itself be a principal component.

Averaging is also what makes the conditional and unconditional factor counts differ. Within a fixed effect, characteristics still generate cross-sectional variation in loadings, supporting the K conditional factors of Proposition 2. Averaging over characteristics removes the stock-specific identities of these exposures and leaves a bounded component within each fixed effect. The responses $b(s, w)$ determine the growing eigenvalues, so the number of strong unconditional stock factors depends on firm heterogeneity and state variation, not on the conditional factor count K.

5 Implications

Sections 3 and 4 show that managed portfolios have no eigenvalue cliff under the stated conditions and that stocks generically lose it when persistent firm differences and common states both take infinitely many values. Four implications follow. First, an exactly solvable

Gaussian model delivers the entire spectrum in closed form. For both random features and partitions, the spectrum descends as a staircase. For random features, the staircase becomes shallower as the weights become more nonlinear. Partitions produce a flatter staircase than random features for every degree of nonlinearity. The remaining three implications do not rely on the Gaussian model. Second, infinitely many groups of principal components generically earn nonzero expected returns while the maximal Sharpe ratio stays finite. Third, the spectrum contains no weak factors: what looks like a weak factor is a strong factor with a small growth constant. Fourth, covariance estimation cannot assume a fixed number of factors with a bounded remainder: every fixed truncation leaves out diverging eigenvalues.

5.1 An exactly solvable case: the shape of the spectrum

Theorem 1 shows that managed portfolios have no eigenvalue cliff, but it does not say what spectrum replaces the cliff. With Gaussian characteristics and a Gaussian-shaped conditional second moment of the factor, the growth constants of all eigenvalues have a closed form: they form a geometric staircase. The staircase looks like a cliff when the menu is close to linear, and it becomes shallower as the menu becomes more nonlinear, as characteristics become more correlated across stocks, or as factor risk becomes less concentrated.

Consider the conditional factor model of Assumption 3 in its simplest form. Each stock has d characteristics driven by a common Gaussian state:

$$X_i = \sqrt{\rho}\, W + \sqrt{1-\rho}\, \eta_i, \qquad W, \eta_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d). \tag{19}$$

The parameter $\rho \in (0,1)$ is the correlation between the same characteristic of two different stocks. It is also the share of each characteristic's variance that comes from the common state W. Returns follow the one-factor model

$$r_i = G + \varepsilon_i, \tag{20}$$

where the residuals satisfy Assumption 3, so their conditional variances are bounded above and away from zero. Conditional on W, the factor G is independent of the characteristic draws.

Following Didisheim et al. (2024), who adapt the random Fourier features of Rahimi and Recht (2007), we first construct managed portfolios from random Fourier features. Each portfolio carries a label $\theta = (\omega, b)$. Labels are drawn independently across portfolios and independently of the return-characteristic panel, with

$$\omega \sim \mathcal{N}(0, \gamma^2 I_d), \qquad b \sim \mathrm{U}[0, 2\pi], \qquad \gamma > 0.$$

The label fixes the weight function:

$$f(x; \theta) = \sqrt{2}\cos(\omega' x + b), \qquad f_k^{(P)} = f(\cdot; \theta_k), \qquad R_k = \frac{1}{N}\sum_{i=1}^{N} f_k^{(P)}(X_i)\, r_i. \tag{21}$$

The bandwidth γ controls how sharply the portfolio weights respond to characteristics and, therefore, how nonlinear they are. These portfolios are a special case of the random-feature menu of Appendix C.2.1.

In parallel with random features, we also study partition menus. Portfolio sorts have a nonparametric partitioning interpretation (Cattaneo, Crump, Farrell and Schaumburg, 2020; Cattaneo, Crump, Feng and Wang, 2026). Divide the characteristic space into P cells of equal probability under the cross-sectional distribution $\mathcal{N}(0, I_d)$: multivariate analogs of double- and triple-sorts used in asset pricing. The portfolio for a cell assigns weight P/N to each stock in that cell and zero to every other stock. The weight uses the expected cell count N/P rather than the realized count, so it stays P/N however many stocks the cell holds, and an empty cell gives a zero portfolio. Refine the cells by successive bisection, so $P = 2^n$: cycle through the d characteristics, splitting every cell at the median, within that cell, of the selected characteristic. Unlike the random-feature menu, the partition menu

has no nonlinearity parameter. The menu grows richer only as the cells are refined. See Appendix C.2.3 for technical details on such partition menus.

A portfolio's return splits into a systematic part, shared across stocks through the factor G, and idiosyncratic noise that diversification removes. In a large cross-section (as $N \to \infty$), only the systematic part survives, and each portfolio return becomes a position in the factor G whose size depends on the common state:

$$\lim_{N\to\infty} R_k \;=\; \bar{f}_k(W)\, G, \qquad \bar{f}_k(W) \;=\; E\big[f_k^{(P)}(X_i) \mid W\big], \tag{22}$$

almost surely. For random features, $\bar{f}_k(W) = \bar{f}(\theta_k; W)$ with $\bar{f}(\theta; W) = E[f(X_i; \theta) \mid W]$, the notation of Appendix K. The limiting spectrum is determined by the second moments of these loadings across states, weighted by the factor's conditional second moment.

The second-moment matrix $\Psi_P = E[RR']$ depends on the factor only through its conditional second moment, which we take to be

$$E[G^2 \mid W] = e^{-a\|W-s\|^2}, \qquad a \geq 0, \qquad s \in \mathbb{R}^d. \tag{23}$$

Here a sets how tightly factor risk concentrates in the state space, and s locates it. When a is close to zero, $E[G^2 \mid W]$ is close to a constant. The second moment peaks at one when $W = s$, and when a is large it decays quickly as W moves away from s.

The conditional factor premium $E[G \mid W]$ may also vary with W, subject only to the automatic bound $E[G \mid W]^2 \leq E[G^2 \mid W]$; it does not affect the second-moment spectrum.

To index the steps of the spectrum, define

$$J_m := \binom{m+d}{d} \quad (m \geq 0), \qquad J_{-1} := 0.$$

Step m comprises the ranks $J_{m-1} < j \leq J_m$. Write c_{feat} and c_{sort} for the leading growth

constants of the random-feature menu and partition menu, respectively. Write $Q_{\text{feat},a}$ and $Q_{\text{sort},a}$ for the ratios of their second growth constants to their first. The scales give the height of the first step; the theorem establishes that the same ratios govern every subsequent step.

Theorem 4 (The spectrum for random features and partitions) *Assume (19) with $\rho \in (0,1)$, returns $r_i = G + \varepsilon_i$ satisfying Assumption 3, and (23). Conditional on W, let G be independent of the characteristics. For either the random-feature menu (21) with $\gamma > 0$, or the equal-probability partition menu with $P = 2^n$ that bisects the characteristic coordinates in turn, the limits κ_j in (11) with $N_{\max} = \infty$ exist, almost surely over the labels for random features, and, for $m = 0, 1, 2, \ldots$ and $J_{m-1} < j \leq J_m$, satisfy*

$$\textit{(a) random features:} \qquad \kappa_j = c_{\text{feat}}\, Q_{\text{feat},a}^m,$$

$$\textit{(b) partitions:} \qquad \kappa_j = c_{\text{sort}}\, Q_{\text{sort},a}^m,$$

where $c_{\text{feat}}, c_{\text{sort}} > 0$ do not vary with j and $Q_{\text{feat},a}, Q_{\text{sort},a} \in (0,1)$.

Consistent with Theorem 1,[12] the eigenvalue of Ψ_P at every fixed rank grows like P: $\lambda_j(\Psi_P) \sim P\kappa_j$. Theorem 4 gives the growth constants κ_j exactly. They form a geometric staircase (Figure 1). Step m holds $\binom{m+d-1}{d-1}$ eigenvalues with a common growth constant: one for $m = 0$, d for $m = 1$, and $d(d+1)/2$ for $m = 2$. The steps grow in size when $d > 1$; for $d = 1$, each contains one eigenvalue. When the number of characteristics d is large, step m holds about $d^m/m!$ eigenvalues.

Each eigenvector $v = (v_k)_{k=1}^P$ of the matrix Ψ_P corresponds to a managed portfolio $\sum_k v_k R_{k,t+1}$ and, through (22), to a portfolio weight function (henceforth, eigenfunction) $\sum_k v_k \bar{f}_k(W)$. In the limit as $P \to \infty$, these eigenfunctions are built from Hermite polynomials. For random features, they are Hermite polynomials multiplied by a bell-shaped Gaussian weight that shrinks them toward zero in extreme states. For equal-probability

[12] See Appendix K for a proof of Theorem 4.

partitions with constant factor risk ($a = 0$), they are the Hermite polynomials themselves (Proposition K.9).

The eigenfunctions line up with the steps of the staircase. For partitions with $a = 0$ the match is exact; for random features, it holds for the Hermite polynomial that multiplies the bell-shaped Gaussian weight. Step $m = 0$ holds the largest eigenvalue, whose eigenfunction is a constant: the corresponding portfolio return is the factor G itself. Step $m = 1$ holds d equal eigenvalues, which correspond to factor-timing strategies $W_\ell G$, $\ell = 1, \ldots, d$, linear in the common state. Step $m = 2$ holds $d(d+1)/2$ equal eigenvalues, which correspond to strategies $W_{\ell_1} W_{\ell_2} G$ quadratic in the state, and so on. Each step thus captures a higher degree of nonlinearity in the managed portfolios, and each falls below the one above it by the same factor, $Q_{\text{feat},a}$ for random features and $Q_{\text{sort},a}$ for partitions. The contribution of each degree of nonlinearity to managed-portfolio risk therefore decays exponentially in m.

Equation (K.26) in Appendix K gives $Q_{\text{feat},a}$ and $Q_{\text{sort},a}$ in closed form. $Q_{\text{feat},a}$ increases with γ by (K.27): the more nonlinear the random features, the more higher-order powers of W contribute to managed-portfolio risk. Both $Q_{\text{feat},a}$ and $Q_{\text{sort},a}$ are monotone increasing in ρ (the importance of the common shock) and vanish when $\rho = 0$: without a common characteristic shock, the diversified spectrum has rank one, leaving only the eigenvalue of the factor G itself. Both decay ratios fall as factor risk becomes more concentrated, that is, as a rises. The location s affects only the scales c_{feat} and c_{sort}.

We also show that

$$Q_{\text{feat},a}(\gamma, \rho) < Q_{\text{sort},a}(\rho)\,, \tag{24}$$

no matter how big γ is (Proposition K.9(i′)). In this sense, partition menus are more nonlinear than any random-feature menu: higher powers of W matter more for their portfolio risk, so multivariate sorts produce a shallower staircase.

Individual stocks keep a single growing eigenvalue. The model has no persistent differ-

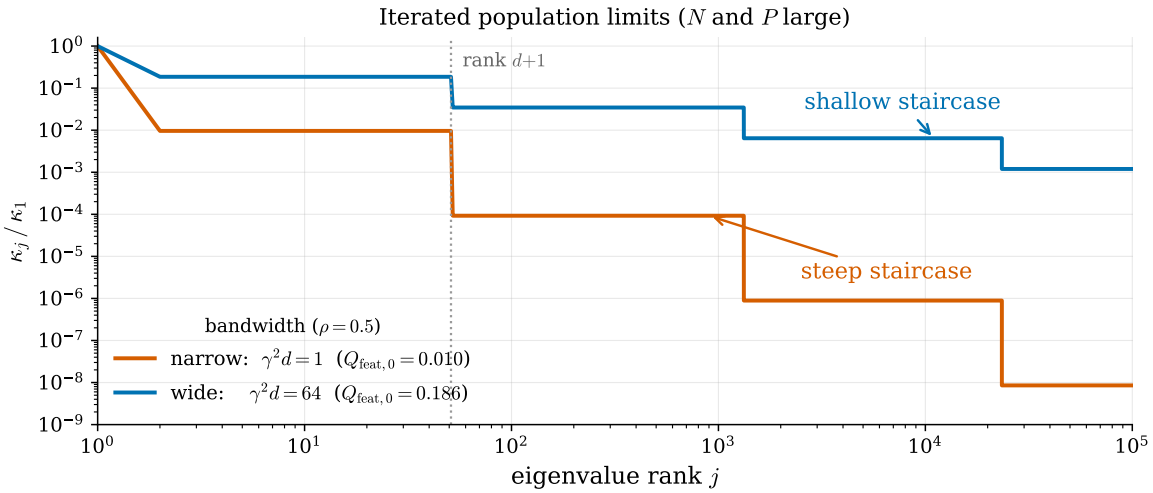


Figure 1: The decay ratio sets the shape of the population spectrum. The figure uses $d = 50$ characteristics, cross-stock characteristic correlation $\rho = 0.5$, constant factor risk ($a = 0$), and bandwidths $\gamma^2 d = 1$ and 64 (so $\gamma^2 = 0.02$ and 1.28). Each curve shows the limiting ratios κ_j/κ_1 from Theorem 4. Step m contains $\binom{m+d-1}{d-1}$ eigenvalues, each with relative size $Q_{\text{feat},0}^m$; the first two steps contain $d + 1 = 51$. The decay ratio $Q_{\text{feat},0}$ is about 0.010 at the narrow bandwidth and 0.186 at the wide one.

ences across stocks, so $M = 1$: conditional on W_t, all stocks are alike. Their unconditional second-moment matrix is $E[G^2]\,\mathbf{1}\mathbf{1}' + \sigma_0^2 I_N$, with $\sigma_0^2 = E[\sigma_\varepsilon^2(X_i)]$, as in the case $M = 1$ of Theorem 3. Exactly one eigenvalue grows with N, and it belongs to the equal-weighted market portfolio. The common state changes the managed portfolios' exposures to the single factor, even when factor risk is constant. Stocks with identical loadings retain one growing eigenvalue; the varying managed exposures generate infinitely many.

At a narrow bandwidth, the spectrum only appears to have a cliff. In the constant-risk case $a = 0$, when γ is small, the staircase in Figure 1 is steep: each step lies about two orders of magnitude below the one before, so the first $d + 1$ eigenvalues look like a cliff. The corresponding principal components are close to the market portfolio

$$\frac{1}{N}\sum_i r_i,$$

and span approximately the same space as the d portfolios that are linear in the characteristics,

$$\frac{1}{N}\sum_i X_{i,\ell} r_i, \qquad \ell = 1, \ldots, d.$$

Here $X_{i,\ell}$ is stock i's ℓ-th characteristic. As $N \to \infty$, these portfolios converge to G and $\sqrt{\rho}\, W_\ell G$, the portfolios described after Theorem 4. Thus, even though stock returns follow a one-factor model, Ψ_P has $d+1$ dominant eigenvalues that may appear to be the only ones growing like P. Yet all other eigenvalues grow like P as well, only with a much smaller growth constant κ_j. In the population, increasing γ raises the relative sizes κ_j/κ_1 of the higher steps and replaces the apparent cliff with a shallow staircase. It also shrinks the common systematic scale c_{feat}.

5.2 Risk premia: infinitely many components are priced

The eigenvalue results concern risk. Which components earn premia is a separate question. Infinitely many components can be priced even when the maximal Sharpe ratio is finite. For generic expected returns and a fixed stock count, every finite selection of leading managed-portfolio components then leaves a Sharpe-ratio shortfall, although a long enough selection makes it arbitrarily small. Measuring that shortfall requires a full benchmark.

Write $\mu_{P,N} = E[R_{t+1}]$ for the mean vector of the P menu returns with N stocks. An investor choosing weights $a \in \mathbb{R}^P$ receives $a'R_{t+1}$, with mean $a'\mu_{P,N}$ and variance $a'\Psi_{P,N}a - (a'\mu_{P,N})^2$. Let S_{full} be the supremum of the Sharpe ratios available from all such choices of weights and menu sizes:

$$\mathrm{S}^2_{\text{full}} := \sup_{N,P} \sup_{\substack{a\in\mathbb{R}^P \\ a'\Psi_{P,N}a>0}} \frac{(a'\mu_{P,N})^2}{a'\Psi_{P,N}a - (a'\mu_{P,N})^2}. \tag{25}$$

Thus the investor may use any portfolio in the menu, with no restriction to leading principal

components. When the stock count is fixed, the supremum holds $N = N_{\max}$ fixed. When it grows, the supremum includes every stock count in the specified sequence.[13]

To study premia, fix the characteristic laws, conditional second moments $\Sigma = (\Sigma_N)_N$, and menu of an economy satisfying Theorem 1. These determine $\Psi_{P,N}$ and its principal components. Vary the conditional means while keeping the conditional covariance positive semidefinite:

$$\mu_N(X_t)\mu_N(X_t)' \preceq \Sigma_N(X_t) \qquad \text{almost surely.} \tag{26}$$

Let $\mathcal{P}(\Sigma)$ be the class of such means whose expected returns on managed portfolios converge when the stock count grows. We also impose technical conditions ensuring that the principal-component groups converge. Definition J.1 in Appendix J states these conditions precisely. The growth constants in (11), denoted by κ_j, then exist and are positive by Theorem 1.

We compare this full benchmark with portfolios that use only leading principal-component groups. The grouping matters when several components have the same growth constant. The eigendecomposition of $\Psi_{P,N}$ is

$$\Psi_{P,N} = \sum_{j=1}^{P} \lambda_j(\Psi_{P,N}) u_j u_j', \qquad \lambda_1(\Psi_{P,N}) \geq \cdots \geq \lambda_P(\Psi_{P,N}) \geq 0,$$

where $u_j' u_{j'} = \mathbf{1}_{\{j=j'\}}$. The component weights u_j depend on P and N. The j-th principal-component portfolio is

$$\mathrm{PC}_{j,t+1} = u_j' R_{t+1}. \tag{27}$$

[13] Menu sizes also range over their specified sequence. A positive numerator with zero variance gives an infinite Sharpe ratio. We take the nonnegative square root, since either sign of the weights is available.

Its second moment and expected return are

$$E[\mathrm{PC}_{j,t+1}^2] = \lambda_j(\Psi_{P,N}), \qquad E[\mathrm{PC}_{j,t+1}] = u_j' \mu_{P,N}. \tag{28}$$

For $\lambda_j(\Psi_{P,N}) > 0$, its uncentered Sharpe ratio is

$$\mathrm{SR}_j = \frac{u_j' \mu_{P,N}}{\sqrt{\lambda_j(\Psi_{P,N})}}.$$

Components with the same limiting growth constant form a group E. Individual eigenvectors within such a group need not be uniquely determined, but the best Sharpe ratio obtainable by combining them is independent of that choice. For a positive growth constant κ_{E}, fix a closed interval $I_{\mathsf{E}} \subset (0, \infty)$ that contains κ_{E} in its interior and no other growth constant. Holding this interval fixed as P and N grow, define

$$\mathsf{E}_{P,N} := \left\{ j \in \{1, \ldots, P\} : \frac{\lambda_j(\Psi_{P,N})}{P} \in I_{\mathsf{E}} \right\}.$$

The set $\mathsf{E}_{P,N}$ collects the components whose normalized eigenvalues approach κ_{E}, even when their finite-menu eigenvalues differ. The group's squared uncentered Sharpe ratio is

$$\mathrm{SR}_{\mathsf{E},P,N}^2 = \sum_{j \in \mathsf{E}_{P,N}} \frac{(u_j' \mu_{P,N})^2}{\lambda_j(\Psi_{P,N})}. \tag{29}$$

Define the limiting squared uncentered Sharpe ratio of a group by

$$\mathrm{SR}_{\mathsf{E},\infty}^2 := \liminf_{N \to N_{\max}} \liminf_{P \to \infty} \mathrm{SR}_{\mathsf{E},P,N}^2.$$

A limiting group is priced if $\mathrm{SR}_{\mathsf{E},\infty}^2 > 0$: some portfolio in the group then has nonzero expected return. Order the groups by decreasing growth constant. For a fixed J, restrict the investor to combinations of the first J groups, and write S_J for the best Sharpe ratio in

the limit. Let $\mathrm{S}_{\mathrm{sys}}$ be the supremum of these values as J increases. Thus S_J uses only the first J groups, while $\mathrm{S}_{\mathrm{sys}}$ allows arbitrarily many of them. To compute these ratios, let x_J be the sum of the first J groups' squared uncentered Sharpe ratios and x_∞ the sum over all groups. Then[14]

$$\begin{aligned} x_J &:= \sum_{\text{first } J \text{ groups } \mathsf{E}} \mathrm{SR}^2_{\mathsf{E},\infty}, \qquad \mathrm{S}^2_J := \frac{x_J}{1-x_J}, \\ x_\infty &:= \sum_{\mathsf{E}} \mathrm{SR}^2_{\mathsf{E},\infty}, \qquad\qquad\quad \mathrm{S}^2_{\mathrm{sys}} := \frac{x_\infty}{1-x_\infty}. \end{aligned} \tag{30}$$

A ratio of one in the uncentered formulas gives an infinite Sharpe ratio. For each fixed J, take the menu and stock limits in the order used above; only then increase J. The full benchmark $\mathrm{S}_{\mathrm{full}}$ in (25) allows every menu portfolio, including components outside any fixed list of leading groups. Whether $\mathrm{S}_{\mathrm{sys}}$ recovers this full benchmark is part of the theorem.

For $\delta \in [0,1)$, define

$$\mathcal{P}_\delta(\Sigma) := \big\{\mu \in \mathcal{P}(\Sigma) : \mu_N \mu_N' \preceq (1-\delta)\Sigma_N \text{ a.s. for every } N\big\}.$$

Thus $\mathcal{P}_0(\Sigma) = \mathcal{P}(\Sigma)$. For $\delta > 0$, the conditional squared Sharpe ratio is bounded by $(1-\delta)/\delta$ in every state. The same bound holds unconditionally, uniformly over portfolios and stock counts. This rules out asymptotic arbitrage in the means-and-variances sense used by the APT.[15]

Genericity concerns expected returns with the second moments and menu held fixed. A zero premium on a given group requires an exact restriction on the means, which an arbitrarily small compatible change can break. Generic expected returns are therefore

[14] For a return with mean m and second moment q, put $u = m^2/q$. Its squared Sharpe ratio is $m^2/(q-m^2) = u/(1-u)$, an increasing function of u.

[15] A nonnegative-payoff arbitrage depends on the full return distribution. For conditionally Gaussian returns with mean μ_N and covariance $\Sigma_N - \mu_N\mu_N'$, every nonzero managed portfolio loses money with positive probability when $\delta > 0$.

typical: the exceptions are confined to such knife-edge restrictions, and Definition J.1 makes this precise for $\mathcal{P}(\Sigma)$ and $\mathcal{P}_\delta(\Sigma)$.

For a growing stock count, the equality conclusion uses two additional conditions: the Sharpe contribution from omitted components must become negligible as we retain more groups, even as the stock count grows, and adding stocks must not lower the full optimum. Definition J.2 in Appendix J gives the precise conditions, including a sufficient restriction on optimal portfolio weights.

Theorem 5 (Infinitely many managed-portfolio components are priced) *Impose the assumptions of Theorem 1 and Definition J.1, for either $N_{\max} < \infty$ or $N_{\max} = \infty$. For every fixed collection of characteristic laws, conditional second-moment family Σ, and menu, a generic $\mu \in \mathcal{P}(\Sigma)$ satisfies*

$$\mathrm{SR}^2_{\mathsf{E},\infty} > 0 \quad \textit{for every group } \mathsf{E} \textit{ with } \kappa_{\mathsf{E}} > 0, \qquad \mathrm{S}^2_{\mathrm{full}} < \infty.$$

There are infinitely many such groups. For every $\delta \in (0,1)$, the same pricing conclusion is generic in $\mathcal{P}_\delta(\Sigma)$, and $\mathrm{S}^2_{\mathrm{full}} \le (1-\delta)/\delta$ throughout that class. On the same generic sets,

$$\mathrm{S}_J < \mathrm{S}_{\mathrm{sys}} \quad \textit{for every finite } J, \qquad \mathrm{S}_J \uparrow \mathrm{S}_{\mathrm{sys}} \le \mathrm{S}_{\mathrm{full}} \quad \textit{as } J \to \infty.$$

If $N_{\max} < \infty$, then $\mathrm{S}_{\mathrm{sys}} = \mathrm{S}_{\mathrm{full}}$. For $N_{\max} = \infty$, any $\mu \in \mathcal{P}(\Sigma)$ with $\mathrm{S}_{\mathrm{full}} < \infty$ satisfying Definition J.2 has $\mathrm{S}_{\mathrm{sys}} = \mathrm{S}_{\mathrm{full}}$.

For generic means and a fixed stock count, no finite number of leading groups attains $\mathrm{S}_{\mathrm{full}}$, but enough groups can get as close to it as desired.

The equality $\mathrm{S}_{\mathrm{sys}} = \mathrm{S}_{\mathrm{full}}$ has a concrete portfolio meaning. Under the theorem's assumptions, any portfolio available from the full menu can be approximated by combining sufficiently many limiting groups: both its expected return and its variance can be matched

arbitrarily closely. That approximation is why allowing J to grow recovers the full Sharpe benchmark.[16]

For individual stocks, we ask the same questions about static portfolios. Hold fixed the unconditional second moments from Theorem 3, and write

$$\Sigma_N = E[r_{t+1} r_{t+1}'], \qquad \bar{\mu}_N = E[r_{t+1}].$$

The superscript r marks quantities built from stock returns. The full benchmark $\mathrm{S}^r_{\text{full}}$ allows any static stock portfolio, with any stock count in the sequence:

$$(\mathrm{S}^r_{\text{full}})^2 := \sup_N \sup_{a \in \mathbb{R}^N \setminus \{0\}} \frac{(a'\bar{\mu}_N)^2}{a'\Sigma_N a - (a'\bar{\mu}_N)^2}. \tag{31}$$

The numerator is the squared expected return of $a'r_{t+1}$ and the denominator is its variance, with the same convention at zero variance as in (25). The fixed-effect array $(S_i)_{i \geq 1}$ also stays fixed. For $\delta \in [0, 1)$, let $\mathcal{P}^r_\delta(\Sigma)$ consist of sequences of unconditional mean vectors $\bar{\mu} = (\bar{\mu}_N)_N$, one for each stock count N, satisfying

$$\bar{\mu}_N \bar{\mu}_N' \preceq (1 - \delta)\Sigma_N \qquad \text{for every } N,$$

for which the expected return on every portfolio that weights stocks by a bounded continuous function of their fixed effects converges as more stocks are included.[17] Set $\mathcal{P}^r(\Sigma) = \mathcal{P}^r_0(\Sigma)$. As for managed portfolios, $\delta > 0$ bounds the squared Sharpe ratio of every static stock portfolio by $(1 - \delta)/\delta$.

Let $\kappa^r_j = \lim_{N \to \infty} \lambda_j(\Sigma_N)/N$ be the stock growth constants. For a positive value κ^r_{E}, fix a closed interval $I_{\mathsf{E}} \subset (0, \infty)$ that contains it in its interior and no other stock growth constant.

[16] The proof of Theorem 5 in Appendix J establishes the group limits in (J.10) and the full-menu approximation in (J.15).

[17] For each fixed bounded continuous function g of the fixed effect, require convergence of $E[N^{-1} \sum_{i \leq N} g(S_i) r_{i,t+1}]$. Definition J.3 in Appendix J.2 states the precise condition and the metric for genericity.

With orthonormal eigenvectors $u_{j,N}$ of Σ_N, define

$$\mathsf{E}_N^r = \{j \in \{1,\dots,N\} : \lambda_j(\Sigma_N)/N \in I_{\mathsf{E}}\},$$
$$\mathrm{SR}_{\mathsf{E},N}^{2,r} = \sum_{j \in \mathsf{E}_N^r} \frac{(u_{j,N}'\bar{\mu}_N)^2}{\lambda_j(\Sigma_N)}, \qquad \mathrm{SR}_{\mathsf{E},\infty}^{2,r} = \liminf_{N\to\infty} \mathrm{SR}_{\mathsf{E},N}^{2,r}.$$

As for managed portfolios, S_J^r is the best limiting Sharpe ratio using only the first J positive groups, ordered by decreasing growth constant, and $\mathrm{S}_{\mathrm{sys}}^r$ is its supremum as more groups are included. Let x_J^r sum their squared uncentered Sharpe ratios and x_{sys}^r sum over all positive groups. If fewer than J groups exist, retain all of them; an empty sum is zero. Thus

$$x_{\mathrm{sys}}^r = \sum_{\mathsf{E}} \mathrm{SR}_{\mathsf{E},\infty}^{2,r}, \qquad (\mathrm{S}_J^r)^2 = \frac{x_J^r}{1-x_J^r}, \qquad (\mathrm{S}_{\mathrm{sys}}^r)^2 = \frac{x_{\mathrm{sys}}^r}{1-x_{\mathrm{sys}}^r}, \tag{32}$$

with the same convention at an uncentered ratio of one. The stock recovery conditions impose the same two requirements as for managed portfolios: a vanishing contribution from omitted components and a full optimum that does not fall as stocks are added. They are stated in Definition J.4 in Appendix J.2. Genericity in $\mathcal{P}^r(\Sigma)$ and $\mathcal{P}_\delta^r(\Sigma)$ has the same meaning as for managed portfolios, with the metric in Definition J.3.

Theorem 6 (Unconditional stock components are priced) *Let $(\Sigma_N)_N$ be the unconditional second-moment family of a stock model satisfying the assumptions of Theorem 3. For every such family and fixed-effect array, a generic $\bar{\mu} \in \mathcal{P}^r(\Sigma)$ satisfies*

$$\mathrm{SR}_{\mathsf{E},\infty}^{2,r} > 0 \quad \textit{for every group } \mathsf{E} \textit{ with } \kappa_{\mathsf{E}}^r > 0, \qquad (\mathrm{S}_{\mathrm{full}}^r)^2 < \infty.$$

If P_W and ν have infinite support, there are infinitely many such groups for generic response configurations. For every $\delta \in (0,1)$, the same pricing conclusion is generic in $\mathcal{P}_\delta^r(\Sigma)$, and

$(\mathrm{S}^r_{\text{full}})^2 \leq (1-\delta)/\delta$ *throughout that class. On the same generic sets,*

$$\mathrm{S}^r_J \uparrow \mathrm{S}^r_{\text{sys}} \leq \mathrm{S}^r_{\text{full}} \quad \textit{as } J \to \infty.$$

If there are infinitely many positive groups, then $\mathrm{S}^r_J < \mathrm{S}^r_{\text{sys}}$ *for every finite* J*. Any* $\bar{\mu} \in \mathcal{P}^r(\Sigma)$ *with* $\mathrm{S}^r_{\text{full}} < \infty$ *satisfying Definition J.4 has* $\mathrm{S}^r_{\text{sys}} = \mathrm{S}^r_{\text{full}}$.

For generic stock means, the number of positive groups matters. If there are only finitely many, using all of them attains $\mathrm{S}^r_{\text{sys}}$. If there are infinitely many, every finite selection omits a priced group whose premium can improve the Sharpe ratio. Even then, including enough groups makes the remaining shortfall relative to $\mathrm{S}^r_{\text{sys}}$ as small as desired. The number of groups is determined by second moments; their pricing depends on expected returns.

When the stock count grows, approaching S_{sys} or $\mathrm{S}^r_{\text{sys}}$ need not recover the full optimum. As more stocks enter, a sequence of portfolios can retain a substantial Sharpe ratio yet involve components farther and farther down the spectrum. Every fixed list of leading groups then misses this contribution in the limit, although an investor allowed to use all components can still earn it. The recovery conditions rule out a persistent contribution of this kind. They also require the full optimum not to fall with the stock count, since S_{full} and $\mathrm{S}^r_{\text{full}}$ include opportunities at smaller stock counts as well.[18] Under these additional conditions, keeping more groups approaches the full Sharpe benchmark in both theorems.

Classical APT combines a low-dimensional covariance structure with the absence of asymptotic arbitrage to obtain a finite-dimensional model of expected returns up to bounded total squared pricing errors (Ross, 1976). Managed portfolios keep the absence of asymptotic arbitrage but lose the low-dimensional covariance structure, and by Theorem 5 their expected returns generically load on infinitely many principal-component groups. Individual stocks

[18] For stocks, a higher optimum at a finite stock count can occur even when every mean vector lies in the leading eigenspace; see (J.29) in Appendix J.2. For a consistently specified stock universe, adding stocks cannot lower the static optimum because the new stocks can receive zero weight. For managed portfolios, the analogous nesting must hold for the spaces of attainable managed returns.

lose that structure too when persistent firm differences and the common state both take infinitely many values, and Theorem 6 then gives the same conclusion. No finite collection of those groups spans expected returns exactly. With a finite maximal Sharpe ratio, a single payoff in the L^2 closure of the return space can still price every return in that space. For finite cross-sections, Filipović and Schneider (2026) characterize conditional representations with traded factors, regression loadings, and unpriced residual risk. Under the law of one price, such representations attain the maximal conditional Sharpe ratio.

This failure of finite spanning is consistent with the out-of-sample evidence of Didisheim et al. (2024): their stochastic discount factor continues to load on factors far beyond the leading few, and truncating it to five principal components cuts its out-of-sample Sharpe ratio from 3.7 to about 1.3. The near-arbitrage argument of Kozak et al. (2018) does not conflict with Theorem 5. It bounds the maximal Sharpe ratio, which the theorem keeps finite, and it delivers a few-factor stochastic discount factor only when a few components dominate the covariance matrix. Without a cliff, the same bound leaves room for infinitely many priced groups.

The pricing of infinitely many groups is also consistent with Lopez-Lira and Roussanov (2023). The first fifty latent factors of individual stock returns capture virtually all of their common time-series variation, yet exposure to these factors earns negligible premia, while a characteristic-based portfolio hedged against them earns a Sharpe ratio above one. Theorems 5 and 6 say that infinitely many groups are priced, not how large each premium is, so they are consistent with this pattern but do not predict its magnitude.

Dello-Preite et al. (2025) also find premia beyond the leading factors, from the opposite covariance premise. They retain an approximate factor structure and let expected returns compensate investors for diversifiable risk. In our setting each fixed principal component retains risk as the menu expands: its second moment grows like P with a positive growth constant. Our results suggest that premia attributed to unsystematic risk in an approximate

factor model may reflect components deep in a spectrum without a cliff. We do not show that the two sets of premia coincide.

Davis and Lopez-Lira (2026) also find that many components are priced, but the source is estimation uncertainty in equilibrium. As the number of assets grows, pricing errors scale with N/T, and the estimated stochastic discount factor loads mainly on estimation error rather than on fundamental risk. Ours is a population result that survives at $T = \infty$, driven by cross-sectional comovement in characteristics. The two mechanisms are complementary: even with infinite data, infinitely many groups of principal components are generically priced, and finite samples add estimation error on top.

5.3 No weak factors

In an approximate factor model, a factor is strong if the corresponding eigenvalue of the return covariance matrix grows proportionally with the number of assets. A weak factor's eigenvalue remains bounded or grows more slowly (Onatski, 2012; Lettau and Pelger, 2020a; Bailey et al., 2016; Giglio et al., 2025).

The classification presumes the cliff: finitely many factors set against a remainder of bounded eigenvalues. The cliff survives in two settings. Conditional on date-t characteristics, stock returns have K eigenvalues that grow like N, and the other $N - K$ stay bounded (Proposition 2). Unconditionally, $N - M$ eigenvalues stay bounded when firms come in M types (Theorem 3). In both settings, the classification applies. It does not apply to managed portfolios: by Theorem 1, the eigenvalue of $\Psi_{P,N}$ at every fixed rank grows like P, and its growth constant stays positive as the number of stocks grows. Nor does it apply to unconditional stock returns when fixed effects and common states both take infinitely many values: the eigenvalue at every fixed rank then grows like N for generic response configurations (Theorem 3). Neither object has a weak factor.

In a finite sample, empirical analysis labels a component weak if its eigenvalue is deemed

small relative to the leading ones. A small eigenvalue need not be bounded: it can grow like P with a small growth constant. For example, in the Gaussian model of Theorem 4, each step of the staircase lies below the previous one by the decay ratio. In Figure 1, with $d = 50$ characteristics and the narrow bandwidth, the decay ratio is about 0.01. The first two steps hold $d + 1 = 51$ components, close to the market portfolio and the d portfolios linear in the characteristics. Every later component is at least two orders of magnitude smaller than these 51, so the spectrum drops sharply after rank 51. A finite-factor reading treats the components after the drop as weak or as noise, yet every one of them grows like P.

These components are also priced: for generic expected returns, every group with a positive growth constant has a positive Sharpe ratio, and infinitely many groups qualify (Theorem 5; Theorem 6 gives the parallel result for stocks). A component that empirical analysis labels weak is therefore a strong component with a small growth constant: its eigenvalue grows like P, and it is priced. Truncating the spectrum after the leading components, as a weak-factor or noise reading of the sample suggests, discards priced risk. The omitted groups lower the attainable Sharpe ratio. Because the maximal Sharpe ratio is finite, the loss relative to the full collection of limiting groups shrinks to zero as more of those groups are retained.

Sections 5.2 and 5.3 thus connect two literatures that are usually read separately. Evidence of premia beyond the leading factors (Lopez-Lira and Roussanov, 2023; Dello-Preite et al., 2025; Didisheim et al., 2024) and evidence of weak factors (Onatski, 2012; Lettau and Pelger, 2020a; Giglio et al., 2025) concern the same components: systematic directions with small growth constants. A finite sample makes them look weak, and a cliff-based model makes their premia look like near-arbitrage or compensation for diversifiable risk. Without a cliff, they are strong, systematic, and generically priced, while the maximal Sharpe ratio stays finite.

5.4 Covariance matrix estimation

A test of an asset-pricing model asks whether its factors attain the maximal Sharpe ratio or whether adding test assets raises it. The GRS test (Gibbons et al., 1989) is the canonical example: its statistic is a scaled estimate of the increase in maximal squared Sharpe ratio. In population, that increase is zero when the factors price the test assets. The usual sample test requires an invertible residual covariance matrix. Related efficiency calculations invert the return covariance or second-moment matrix.

Inverting these matrices raises two separate problems. The first is a sample problem. The sample second-moment matrix becomes ill-conditioned as P approaches the sample length T and singular when $P > T$. This dimension-counting problem exists regardless of the population factor structure and has nothing to do with the absence of a population cliff. When $P > T$, the sample second-moment matrix has at most T positive eigenvalues.[19]

The second is a population problem. Faced with sample ill-conditioning, researchers often assume that the population matrix consists of a few pervasive factor directions and a remainder with bounded eigenvalues. Under the conditions of Theorem 1, managed portfolios do not admit this decomposition (Lemma A.11).

The population problem concerns unconditional second moments of managed returns. Conditional on characteristics, stock returns keep the cliff: K eigenvalues grow like N, and the rest stay bounded (Proposition 2). Instrumented principal components (Kelly et al., 2019) estimate this conditional structure, with stock loadings linear in characteristics. Principal components of unconditional managed returns target a different object, which has no cliff under Theorem 1 whatever the factor structure of stocks. Even when stocks have a single factor with constant loadings, as in (20), nonlinear portfolio weights give managed portfolios state-dependent exposures, and their spectrum has no cliff (Theorem 4).

[19] Appendix L gives two finite-sample benchmarks for eigenvalue dispersion and the absence of gaps. The gap result requires centered returns with independent standardized components; it is not asserted for the paper's dependent managed returns.

Whether a cliff exists matters for covariance estimation, because factor-based estimators presume one. They represent covariance through a few factor exposures and a separately modeled remainder: Fan, Fan and Lv (2008) use observed factors, and Fan et al. (2013) estimate latent factors from leading principal components with their principal orthogonal complement thresholding (POET) estimator. Conditional stock returns satisfy this presumption (Proposition 2); unconditional managed returns do not. For managed portfolios, any fixed principal-component truncation discards eigenvalues that diverge with P. Under Theorem 5, it also generically discards priced groups and lowers the attainable Sharpe ratio relative to S_{sys}. This loss shrinks to zero as more limiting groups are retained, and at a fixed stock count the limiting groups recover S_{full} (Theorem 5). With a growing stock count, the gap described in Section 5.2 can remain.

Other methods restrict cross-sectional dependence. The alpha tests of Fan, Liao and Yao (2015) use sparse residual covariance estimation, and the Sharpe-ratio inference of Caner, Medeiros and Vasconcelos (2023) uses residual precision estimation in factor models. Pesaran and Yamagata (2024) allow weak cross-sectional dependence and correct its effect without estimating an invertible covariance matrix. These restrictions must hold for the managed returns themselves. Even if individual-stock returns satisfy sparse covariance or precision restrictions, characteristic-based portfolio weights need not preserve them (Chernov et al., 2025). Factor truncation and restrictions on cross-sectional dependence both require that, once finitely many factor directions are removed, the residual second-moment matrix have a bounded spectrum. Unconditional managed returns fail this requirement: however many leading components are removed, the largest remaining eigenvalue still grows with the number of portfolios.

How much of the total second moment a truncation captures depends on the menu. The trace of $\Psi_{P,N}$ is the menu's total second moment, summed over portfolios. With the number of stocks fixed, a growing number of leading components captures almost all of this total

for random-feature menus (Appendix E). For refining partitions, any number of components that is small relative to P captures a vanishing share. Neither fact settles estimation, because tests and efficient portfolios use the inverse matrix. The inverse weights each direction by the reciprocal of its eigenvalue. At any finite menu size, the small-eigenvalue directions that a truncation drops can therefore matter for estimated maximal Sharpe ratios and for the residual-covariance inverse in the GRS statistic.

Regularization addresses the inverse without choosing a number of factors. Examples are ridge regularization, including the prior of Kozak et al. (2020), and the linear and nonlinear shrinkage estimators of Ledoit and Wolf (2004, 2017, 2020). None of these estimators presumes a cliff. The optimality results of Ledoit and Wolf (2017, 2020) do assume population eigenvalues in a compact subset of $(0, \infty)$, which managed portfolios violate because their leading eigenvalues grow like P. Regularization makes inversion feasible when P is comparable to or larger than T, but it biases both the estimated covariance matrix and the resulting Sharpe-ratio statistics. Valid inference must correct this bias.

Our companion papers supply those corrections. Chernov et al. (2025) debias the high-dimensional GRS statistic and restore a valid test of portfolio efficiency when P is comparable to T. Chernov et al. (2026) similarly correct the bias that high dimensionality and ridge regularization introduce into estimates of the maximal Sharpe ratio. Those results do not depend on the theorems in this paper. Our results undercut only regularization that relies on a cliff: a fixed factor count with a bounded remainder.

6 Conclusion

The eigenvalue cliff, a few diverging eigenvalues followed by a bounded bulk, is the premise of the approximate factor model and of the empirical methods built on it. Conditioning information undermines this premise. In modern empirical asset pricing, this information typically consists of firm characteristics, which set the weights of managed portfolios and,

in a conditional factor model, the factor loadings of individual stocks. These loadings also reflect persistent differences across firms. Within a period, a few factors may then generate all systematic risk. Across periods, characteristics move, and the weights and loadings move with them. Model-free empirical work relies on the unconditional second-moment matrix, which averages over this movement, and the average is an infinite-factor model. For managed portfolios, this happens whatever the factor structure of stocks. Common variation in characteristics lets it survive diversification. For individual stocks, it happens generically when persistent firm differences and common states both take infinitely many values.

Losing the cliff does not mean losing the absence of arbitrage. As long as conditional expected returns admit no asymptotic arbitrage, the maximal Sharpe ratio stays finite. What fails is the APT's conclusion that finitely many factors price assets. For generic expected returns, infinitely many groups of principal components earn risk premia. No finite set of components therefore spans expected returns. Pricing itself does not fail: a single payoff formed from all returns, possibly as a limit of portfolios, still prices every return. The failure concerns principal components.

For empirical work, the components beyond any fixed cutoff are neither noise nor weak factors. This view reconciles evidence of large premia on small sources of risk with the evidence on weak factors. Their eigenvalues are small relative to the leading ones but still grow with the number of assets, and they are priced. Truncating at a fixed number of components therefore discards priced risk and lowers the attainable Sharpe ratio. Retaining more groups narrows the loss. With a growing stock count, however, even all groups together can fall short of the full maximal Sharpe ratio. Factor-based covariance estimators presume a cliff. Conditional stock returns have one; unconditional managed returns do not. Regularization that does not presume a cliff avoids this problem, provided one corrects for the bias it introduces.

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A Mathematical Tools

The appendices follow logical dependence rather than the order of the main text: the proofs for the Hermite-degree spectrum come after the appendix on risk premia and spanning because they use its matrix–operator dictionary.

This appendix collects the operator facts used throughout: Hilbert–Schmidt and trace-class operators, the min–max principle and Weyl's inequality, the relation between the spectra of AB and BA, and an almost-sure lower bound for empirical random-feature kernels. It closes with the effect of centering on eigenvalues and the normalization of the tradable discount factor.

Definition A.1 (Hilbert–Schmidt integral operator) The integral operator

$$(T_K f)(z) \;=\; \int K(z,\tilde{z})\, f(\tilde{z})\, dP(\tilde{z})$$

on $L^2(P)$ is Hilbert–Schmidt if its kernel is square-integrable, in which case

$$\|T_K\|_{\mathrm{HS}}^2 = \iint K(z,\tilde{z})^2\, dP(z)\, dP(\tilde{z}) = \|K\|_{L^2}^2 < \infty. \tag{A.1}$$

Definition A.2 (Trace-class integral operator) The integral operator T_K on $L^2(P)$ is trace-class if its singular values $\{s_i\}$ are summable, $\|T_K\|_1 = \sum_i s_i < \infty$. For positive self-adjoint T_K the singular values coincide with the eigenvalues $\{\lambda_i\}$, and $\|T_K\|_1 = \sum_i \lambda_i = \operatorname{tr} T_K$.

Lemma A.3 (Tonelli) *Let $(\mathcal{A},\mu)$ and $(\mathcal{B},\nu)$ be σ-finite measure spaces and $h : \mathcal{A}\times\mathcal{B} \to [0,\infty]$ measurable and nonnegative. Then the iterated integrals agree, regardless of finiteness,*

$$\int_{\mathcal{A}}\int_{\mathcal{B}} h(a,b)\,\nu(db)\,\mu(da) = \int_{\mathcal{B}}\int_{\mathcal{A}} h(a,b)\,\mu(da)\,\nu(db). \tag{A.2}$$

In particular, taking μ to be counting measure on $\mathbb{N}$, summation and integration of nonnegative terms may be exchanged: $\sum_k \int_{\mathcal{B}} h(k,b)\,\nu(db) = \int_{\mathcal{B}} \sum_k h(k,b)\,\nu(db)$.

Lemma A.4 (Parseval) *Let $\{e_k\}$ be an orthonormal basis of a Hilbert space $\mathcal{H}$. Then for every $u \in \mathcal{H}$,*

$$\sum_k |\langle e_k, u\rangle|^2 = \|u\|^2. \tag{A.3}$$

Lemma A.5 (Min–max principle) *Let $\mathcal{A}$ be a bounded self-adjoint operator on $\mathcal{H}$, a separable Hilbert space of infinite dimension. Define its Courant–Fischer values, counted from the top of the spectrum,*

$$\lambda_j(\mathcal{A}) \;:=\; \sup_{\substack{V\subseteq\mathcal{H}\\ \dim V = j}} \; \inf_{\substack{\psi\in V\\ \|\psi\|=1}} \; \big\langle \psi,\ \mathcal{A}\,\psi \big\rangle\,, \qquad j = 1, 2, \ldots$$

Then for each j exactly one of the following holds.

(a) $\lambda_j(\mathcal{A}) > \sup \sigma_{\mathrm{ess}}(\mathcal{A})$, and $\lambda_j(\mathcal{A})$ is the j-th largest eigenvalue of $\mathcal{A}$ lying above the essential spectrum, counted with multiplicity.

(b) $\lambda_j(\mathcal{A}) = \sup \sigma_{\mathrm{ess}}(\mathcal{A})$, in which case $\lambda_k(\mathcal{A}) = \sup \sigma_{\mathrm{ess}}(\mathcal{A})$ for every $k \geq j$, and $\mathcal{A}$ has at most $j-1$ eigenvalues above $\sup \sigma_{\mathrm{ess}}(\mathcal{A})$, counted with multiplicity.

In particular,

$$\lambda_j(\mathcal{A}) \;\geq\; \sup \sigma_{\mathrm{ess}}(\mathcal{A}) \quad \text{for every } j, \qquad \text{and} \qquad \lambda_j(\mathcal{A}) \;\downarrow\; \sup \sigma_{\mathrm{ess}}(\mathcal{A}) \ \text{ as } j \to \infty. \tag{A.4}$$

For a noncompact operator, $\lambda_j(\mathcal{A})$ always denotes this Courant–Fischer value, which need not be an isolated eigenvalue.

Proof of Lemma A.5. This is the min–max principle of Reed and Simon (1978, Theorem XIII.1). That theorem is stated for an operator B bounded below, with the min–max values $\mu_j(B) = \sup_{\phi_1,\dots,\phi_{j-1}} \inf\{\langle\psi, B\psi\rangle : \|\psi\| = 1,\ \psi \perp \phi_1,\dots,\phi_{j-1}\}$ counted from the bottom of the spectrum, and it gives the dichotomy above with $\inf\sigma_{\mathrm{ess}}(B)$ in place of $\sup\sigma_{\mathrm{ess}}(\mathcal{A})$. Apply it to $B = -\mathcal{A}$, which is bounded below because $\mathcal{A}$ is bounded. Since $\sigma_{\mathrm{ess}}(-\mathcal{A}) = -\sigma_{\mathrm{ess}}(\mathcal{A})$, we have $\inf\sigma_{\mathrm{ess}}(-\mathcal{A}) = -\sup\sigma_{\mathrm{ess}}(\mathcal{A})$. The spectral theorem identifies the displayed max–min value with $\sup\{c : \dim\operatorname{ran}\mathbf{1}_{(c,\infty)}(\mathcal{A}) \geq j\}$: a spectral subspace above c supplies the lower bound, while every j-dimensional subspace intersects its orthogonal complement if that spectral subspace has dimension less than j. This gives $\mu_j(-\mathcal{A}) = -\lambda_j(\mathcal{A})$ and the stated dichotomy. If case (a) holds at every rank, the decreasing eigenvalues have finite multiplicity and their limit is a spectral accumulation point, hence belongs to the essential spectrum. If case (b) occurs, the sequence is already constant thereafter. Both cases give (A.4). □

Lemma A.6 (Compressions along increasing subspaces) *Let $\mathcal{A}$ be a bounded self-adjoint operator on a separable Hilbert space $\mathcal{H}$. Let $\mathcal{V}_1 \subseteq \mathcal{V}_2 \subseteq \cdots$ be closed subspaces of $\mathcal{H}$ with orthogonal projections Π_P, let $\mathcal{V}_\infty$ be the closure of $\bigcup_P \mathcal{V}_P$, and let Π_∞ be its orthogonal projection, so that $\Pi_P \to \Pi_\infty$ strongly. For a closed subspace $\mathcal{V}$ with projection Π, write $\lambda_j(\Pi\mathcal{A}\Pi|_{\mathcal{V}})$ for the Courant–Fischer value of Lemma A.5 with the variational subspaces restricted to $\mathcal{V}$. Then, for every fixed $j \leq \dim\mathcal{V}_\infty$,*

$$\lambda_j\big(\Pi_P\mathcal{A}\Pi_P|_{\mathcal{V}_P}\big) \;\longrightarrow\; \lambda_j\big(\Pi_\infty\mathcal{A}\Pi_\infty|_{\mathcal{V}_\infty}\big).$$

If $\mathcal{V}_\infty = \mathcal{H}$, the limit is $\lambda_j(\mathcal{A})$.

Proof of Lemma A.6. Every j-dimensional $V \subseteq \mathcal{V}_P$ lies in $\mathcal{V}_\infty$, so restricting the variational subspaces gives the upper bound $\lambda_j(\Pi_P \mathcal{A} \Pi_P|_{\mathcal{V}_P}) \le \lambda_j(\Pi_\infty \mathcal{A} \Pi_\infty|_{\mathcal{V}_\infty})$. For the lower bound, fix $\varepsilon > 0$ and a j-dimensional $V \subseteq \mathcal{V}_\infty$ whose minimum Rayleigh quotient is within ε of $\lambda_j(\Pi_\infty \mathcal{A} \Pi_\infty|_{\mathcal{V}_\infty})$. Strong convergence is uniform on the unit sphere of V, so for P large the projection $\Pi_P V \subseteq \mathcal{V}_P$ is j-dimensional and its Rayleigh quotients converge to those of V. The argument uses the values computed within $\mathcal{V}_P$, which for finite-dimensional $\mathcal{V}_P$ are the eigenvalues of a finite matrix; adding a zero orthogonal complement could change them when they are negative. □

Lemma A.7 (Weyl's inequality) *Let A and E be self-adjoint bounded operators on a Hilbert space $\mathcal{H}$ (for example, compact self-adjoint operators and real symmetric matrices), and let $\lambda_1(\cdot) \ge \lambda_2(\cdot) \ge \cdots$ denote the Courant–Fischer values of Lemma A.5 in decreasing order (for matrices, see Bhatia, 1997, Corollary III.1.2). For every admissible index k,*

$$\lambda_k(A) + \lambda_{\inf}(E) \;\le\; \lambda_k(A+E) \;\le\; \lambda_k(A) + \lambda_{\sup}(E), \tag{A.5}$$

where $\lambda_{\sup}(E) = \sup_x \frac{\langle x, Ex\rangle}{\|x\|^2}$ and $\lambda_{\inf}(E) = \inf_x \frac{\langle x, Ex\rangle}{\|x\|^2}$. Consequently:

1. *(Perturbation bound) for every k,*

$$\big|\lambda_k(A) - \lambda_k(A+E)\big| \;\le\; \|E\|_{\mathrm{op}}; \tag{A.6}$$

2. *(Positive perturbation) if $E \succeq 0$, then for every k,*

$$\lambda_k(A) \;\le\; \lambda_k(A+E) \;\le\; \lambda_k(A) + \|E\|_{\mathrm{op}}. \tag{A.7}$$

Proof of Lemma A.7. By the Courant–Fischer definition, for any bounded self-adjoint T,

$$\lambda_k(T) = \sup_{\substack{V \subseteq \mathcal{H} \\ \dim V = k}} \inf_{\substack{x \in V \\ \|x\|=1}} \langle x, Tx \rangle. \tag{A.8}$$

For any unit vector x, $\langle x, (A+E)x\rangle = \langle x, Ax\rangle + \langle x, Ex\rangle$ with $\lambda_{\inf}(E) \le \langle x, Ex\rangle \le \lambda_{\sup}(E)$. Substituting the upper bound into the min–max formula, the constant $\lambda_{\sup}(E)$ passes through both the inner inf and outer sup, giving $\lambda_k(A+E) \le \lambda_k(A) + \lambda_{\sup}(E)$; the lower bound is symmetric. This is (A.5). Since $\|E\|_{\mathrm{op}} = \max(|\lambda_{\sup}(E)|, |\lambda_{\inf}(E)|)$, applying (A.5) to $A+E$ and to $(A+E)+(-E) = A$ yields $|\lambda_k(A) - \lambda_k(A+E)| \le \|E\|_{\mathrm{op}}$, which is 1. If $E \succeq 0$ then $\lambda_{\inf}(E) \ge 0$ and $\lambda_{\sup}(E) = \|E\|_{\mathrm{op}}$, and (A.5) reduces to 2. □

Lemma A.8 *Let $A : \mathcal{H}_2 \to \mathcal{H}_1$ and $B : \mathcal{H}_1 \to \mathcal{H}_2$ be bounded linear operators between Hilbert spaces (in particular, $A = M$ an $m \times n$ matrix and $B = M'$ its transpose). Then AB (on $\mathcal{H}_1$) and BA (on $\mathcal{H}_2$) have the same nonzero spectrum. Their generalized eigenspaces at each nonzero eigenvalue have equal dimension. An isolated nonzero eigenvalue of finite algebraic multiplicity for either product has the same algebraic and geometric multiplicities for the other. If $B = A^*$, the restrictions of AA^* and A^*A to their kernel complements are unitarily equivalent; their ordered variational values coincide, with zeros appended in finite dimension.*

Proof of Lemma A.8. Equality of the nonzero spectra follows from the resolvent identity

$$(I - zBA)^{-1} = I + zB(I - zAB)^{-1}A$$

whenever the inverse on the right exists, and the same identity with A, B interchanged. For multiplicity, fix a nonzero eigenvalue λ and $n \ge 1$. The intertwining identity $B(AB - \lambda I)^n = (BA - \lambda I)^n B$ maps $\ker(AB - \lambda I)^n$ into $\ker(BA - \lambda I)^n$. It is injective there: if $Bv = 0$, then $(AB - \lambda I)^n v = (-\lambda)^n v$, so membership in the kernel forces $v = 0$. Interchanging A

and B gives the reverse dimension inequality. Thus the generalized eigenspaces have the same dimension for every n. If λ is isolated, the same intertwining identity holds for the resolvents and hence for their Riesz projections. The map B is an isomorphism between the corresponding Riesz subspaces: its inverse is $A(BA)^{-1}$ restricted to the latter subspace, where BA is invertible because $\lambda \neq 0$. This transfers finite algebraic multiplicity and proves equality; $n = 1$ gives geometric multiplicity. Finally, polar decomposition $A = U|A|$ gives $AA^* = U(A^*A)U^*$, with U unitary between the kernel complements. Positivity and zero padding give equality of all ordered variational values. □

Lemma A.9 *For a measure μ_θ on Θ and real measurable features, let*

$$K(z, \tilde{z}) = \int_\Theta \phi_\theta(z)\, \phi_\theta(\tilde{z})\, \mu_\theta(d\theta), \qquad \int_\Theta \phi_\theta(z)^2\, \mu_\theta(d\theta) < \infty$$

on a full-measure set $\mathcal{Z}_0$. Then K is symmetric and positive semidefinite on $\mathcal{Z}_0 \times \mathcal{Z}_0$.

Proof of Lemma A.9. On $\mathcal{Z}_0 \times \mathcal{Z}_0$, Cauchy–Schwarz bounds the absolute feature integral by $\sqrt{K(z,z)K(\tilde{z},\tilde{z})} < \infty$. Symmetry follows from

$$K(z, \tilde{z}) = \int_\Theta \phi_\theta(z)\phi_\theta(\tilde{z})\, \mu_\theta(d\theta) = K(\tilde{z}, z).$$

For positive semidefiniteness, take any $z_1, \ldots, z_n$ and scalars $c_1, \ldots, c_n$. Then

$$\sum_{i,j} c_i c_j\, K(z_i, z_j) = \int_\Theta \Big(\sum_i c_i\, \phi_\theta(z_i) \Big)^2 \mu_\theta(d\theta) \geq 0. \tag{A.9}$$

□

Lemma A.10 *Let $\mathbb{P}$ be a probability measure on a measurable space $\mathcal{Z}$, let $\theta_1, \theta_2, \ldots \overset{iid}{\sim} \mu_\theta$,*

and define the empirical kernel

$$K_P(z,\tilde{z}) = \frac{1}{P}\sum_{k=1}^{P} \phi_{\theta_k}(z)\,\phi_{\theta_k}(\tilde{z}), \tag{A.10}$$

the sample analogue of $K(z,\tilde{z}) = \int_\Theta \phi_\theta(z)\phi_\theta(\tilde{z})\,\mu_\theta(d\theta)$. *Suppose that (i)* T_K *is Hilbert–Schmidt; (ii)* $\phi_\theta \in L^2(\mathbb{P})$ *for* μ_θ*-almost every* θ*; and (iii) the diagonal* $\int_\Theta \phi_\theta(z)^2\,\mu_\theta(d\theta)$ *is finite for* $\mathbb{P}$*-almost every* z*, so that the mixture* $K(z,\tilde{z})$ *converges absolutely for a.e.* $(z,\tilde{z})$. *Let* $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$ *be the eigenvalues of* T_K *and* $e_1, e_2, \ldots$ *orthonormal eigenfunctions. Then, with* $\lambda_j(\cdot)$ *the Courant–Fischer values of Lemma A.5, for every fixed* j*,*

$$\liminf_{P\to\infty} \lambda_j(T_{K_P}) \;\geq\; \lambda_j(T_K) = \lambda_j \qquad \textit{almost surely.} \tag{A.11}$$

Proof of Lemma A.10. Append zero spectral values if the rank is finite. The lower bound is immediate when $\lambda_j = 0$, so it suffices to treat a fixed j with $\lambda_j > 0$.

Fix j and let $V = \operatorname{span}\{e_1, \ldots, e_j\}$. The compression of the positive self-adjoint operator T_{K_P} to V is the $j \times j$ matrix $M_P = [\langle e_i, T_{K_P} e_m\rangle]_{i,m\leq j}$ with entries

$$\langle e_i, T_{K_P} e_m\rangle = \frac{1}{P}\sum_{k=1}^{P} \langle e_i, \phi_{\theta_k}\rangle\,\langle e_m, \phi_{\theta_k}\rangle. \tag{A.12}$$

Since $\phi_\theta \in L^2(\mathbb{P})$ for μ_θ-almost every θ, the inner products $\langle e_i, \phi_\theta\rangle$ are defined μ_θ-almost everywhere. We first establish

$$\int_\Theta \langle e_i, \phi_\theta\rangle\langle e_m, \phi_\theta\rangle\,\mu_\theta(d\theta) \;=\; \langle e_i, T_K e_m\rangle \;=\; \lambda_i\,\delta_{im}\,, \qquad i,m \leq j. \tag{A.13}$$

The Hilbert–Schmidt hypothesis alone does not directly license interchanging $\int \mu_\theta(d\theta)$ with the inner product. A trace condition would justify that step, but is not assumed here. We

obtain (A.13) by cutting on the diagonal instead of on Θ. Set

$$D(z) \;:=\; \int_\Theta \phi_\theta(z)^2\,\mu_\theta(d\theta)\,, \qquad \mathcal{S}_b \;:=\; \{z : D(z) \le b\}\,, \tag{A.14}$$

so $D < \infty$ a.e. by hypothesis and $\mathcal{S}_b \uparrow \mathcal{Z}$ up to a null set. Write $\mathcal{G}$ for the map $f \mapsto (\theta \mapsto \langle f, \phi_\theta\rangle)$, a priori defined only where the inner products make sense.

The forms agree on functions supported in $\mathcal{S}_b$. Let f vanish off $\mathcal{S}_b$. By Cauchy–Schwarz, $\int_\Theta |\phi_\theta(z)\phi_\theta(u)|\,\mu_\theta(d\theta) \le \sqrt{D(z)D(u)}$, so

$$\iint |f(z)f(u)| \int_\Theta |\phi_\theta(z)\phi_\theta(u)|\,\mu_\theta(d\theta)\,d\mathbb{P}(z)\,d\mathbb{P}(u) \;\le\; b\,\|f\|^2_{L^1(\mathbb{P})} \;\le\; b\,\|f\|^2_{L^2(\mathbb{P})} \;<\; \infty.$$

The integrand is absolutely integrable, so Fubini applies and $\langle f, T_K f\rangle = \int_\Theta \langle f, \phi_\theta\rangle^2\,\mu_\theta(d\theta) = \|\mathcal{G}f\|^2_{L^2(\mu_\theta)}$.

$\mathcal{G}$ is bounded on all of $L^2(\mathbb{P})$. For general f, put $f_b = f\mathbf{1}_{\mathcal{S}_b}$, so $f_b \to f$ in $L^2(\mathbb{P})$ and hence $\langle f_b, \phi_\theta\rangle \to \langle f, \phi_\theta\rangle$ for μ_θ-almost every θ. Fatou's lemma and the previous paragraph give

$$\int_\Theta \langle f, \phi_\theta\rangle^2\,\mu_\theta(d\theta) \;\le\; \liminf_b \int_\Theta \langle f_b, \phi_\theta\rangle^2\,\mu_\theta(d\theta) \;=\; \liminf_b\, \langle f_b, T_K f_b\rangle \;\le\; \|T_K\|\,\|f\|^2_{L^2(\mathbb{P})},$$

so $\mathcal{G}$ is everywhere defined with $\|\mathcal{G}\|^2 \le \|T_K\|$.

Conclusion. The two quadratic forms $f \mapsto \langle f, T_K f\rangle$ and $f \mapsto \|\mathcal{G}f\|^2$ are bounded and agree on $\bigcup_b L^2(\mathcal{S}_b)$, which is dense in $L^2(\mathbb{P})$; hence they agree everywhere, that is $T_K = \mathcal{G}^*\mathcal{G}$. Polarizing, $\langle e_i, T_K e_m\rangle = \langle \mathcal{G}e_i, \mathcal{G}e_m\rangle = \int_\Theta \langle e_i, \phi_\theta\rangle\langle e_m, \phi_\theta\rangle\,\mu_\theta(d\theta)$, which with $T_K e_m = \lambda_m e_m$ is (A.13).

Each summand of M_P is therefore integrable: by Cauchy–Schwarz,

$$E\big|\langle e_i, \phi_\theta\rangle\langle e_m, \phi_\theta\rangle\big| \le (E\langle e_i, \phi_\theta\rangle^2)^{1/2}(E\langle e_m, \phi_\theta\rangle^2)^{1/2} = (\lambda_i\lambda_m)^{1/2} < \infty,$$

using (A.13) on the diagonal. The strong law of large numbers, applied to each of the finitely many entries, gives almost surely

$$(M_P)_{im} \xrightarrow[P\to\infty]{} E\big[\langle e_i, \phi_\theta\rangle\langle e_m, \phi_\theta\rangle\big] = \langle e_i, T_K e_m\rangle = \lambda_i\, \delta_{im}, \tag{A.15}$$

by (A.13), so $M_P \to \mathrm{diag}(\lambda_1, \ldots, \lambda_j)$ and hence $\lambda_j(M_P) \to \lambda_j$ a.s. By Lemma A.5, the j-th largest eigenvalue of a positive operator dominates that of any j-dimensional compression, so $\lambda_j(T_{K_P}) \geq \lambda_j(M_P)$. Therefore

$$\liminf_{P\to\infty} \lambda_j(T_{K_P}) \geq \lim_{P\to\infty} \lambda_j(M_P) = \lambda_j$$

almost surely. □

Lemma A.11 (Centering and pervasive factor loadings) *Let $Y \in \mathbb{R}^n$ have finite second moments, $\Psi_Y = E[YY']$, $\mu = E[Y]$, and $V = \mathrm{Var}(Y)$. Then, for $1 \leq j < n$,*

$$\lambda_{j+1}(\Psi_Y) \leq \lambda_j(V) \leq \lambda_j(\Psi_Y). \tag{A.16}$$

For a sequence with dimension $n \to \infty$, if $\liminf_n \lambda_j(\Psi_{Y,n})/n > 0$ for every fixed j, then

$$\liminf_{n\to\infty} \frac{\lambda_j(V_n)}{n} \geq \liminf_{n\to\infty} \frac{\lambda_{j+1}(\Psi_{Y,n})}{n} > 0. \tag{A.17}$$

The growth constants need not agree before and after centering. For a covariance decomposition $V_N = B_N B_N' + E_N$, with B_N having K columns and $0 \preceq E_N \preceq C I_N$, one has $\lambda_{K+1}(V_N) \leq C$. If in addition $\lambda_K(B_N' B_N) \to \infty$, exactly K eigenvalues diverge.

Proof of Lemma A.11. The identity $\Psi_Y = V + \mu\mu'$ and rank-one interlacing give (A.16). Dividing by the cross-sectional size transfers positive fixed-index lower limits from Ψ_Y to V. The rank of $B_N B_N'$ is at most K, so Weyl's inequality bounds $\lambda_{K+1}(V_N)$ by C; its

K potentially nonzero eigenvalues are those of $B_N'B_N$. Since $V_N \succeq B_N B_N'$, the additional loading condition makes all K diverge. Merely writing a K-factor representation does not give this condition: $B_N = (1, 0, \ldots, 0)'$ and $E_N = I_N$ produce eigenvalues $2, 1, \ldots, 1$. □

Remark A.12 (Normalization of the tradable discount factor) Given X_t and excess returns r_{t+1}, write $\mu = E[r_{t+1} \mid X_t]$ and $\Sigma = E[r_{t+1}r_{t+1}' \mid X_t] \succ 0$. Within the normalized affine family $m_{t+1} = 1 - r_{t+1}'a_{\text{SDF}}$, the unique coefficient satisfying $E[m_{t+1}r_{t+1} \mid X_t] = 0$ is $a_{\text{SDF}} = \Sigma^{-1}\mu$. It also uniquely minimizes $E[(1 - r_{t+1}'a_{\text{SDF}})^2 \mid X_t]$ over that family. If $V = \Sigma - \mu\mu' \succ 0$ and $\text{SR}^2_{\text{cond}} = \mu' V^{-1}\mu$, then

$$1 - r_{t+1}'\Sigma^{-1}\mu = \frac{1 - (r_{t+1} - \mu)'V^{-1}\mu}{1 + \text{SR}^2_{\text{cond}}}.$$

Thus the second-moment and covariance formulas give proportional tradable pricing kernels under two normalizations: unit intercept and unit conditional mean. Orthogonality to excess returns alone does not fix their scale or establish pointwise positivity. The unconditional formulas follow by dropping the conditioning.

B Stock Assumptions and Unconditional Factor Counts

B.1 Common states and fixed-effect-dependent laws

Assumption 3 uses a common conditional characteristic law across stocks. We first allow this law to depend on fixed effects, then explain the exchangeability representation behind the common-state formulation. The following assumption replaces only the common-law clause of Assumption 3; its nonrandom fixed-effect array, conditional factor model, conditional moments, and residual conditions remain in force. The stock results require regularity of the fixed-effect array and its responses. The characteristic-law extension itself imposes only the conditional independence stated here.

Assumption B.1 (Fixed-effect characteristic laws) Each period, conditional on the common state W_t and the nonrandom fixed effects, characteristics are independent across stocks, with

$$X_{i,t} \sim p(\cdot \mid S_i, W_t). \tag{B.1}$$

Here $(s, w) \mapsto p(\cdot \mid s, w)$ is a Borel probability kernel on the characteristic space. The baseline in Assumption 3 is the special case $p(\cdot \mid s, w) = p_0(\cdot \mid w)$ for every s, where p_0 denotes the common conditional characteristic law.

In this extension a fixed effect can change both the loading $B(x, s)$ and the conditional distribution of characteristics. With finitely many fixed effects, the model is a collection of conditionally independent fixed-effect classes, each with its own conditional characteristic law. With infinitely many fixed effects, the formulation also allows every stock to have a different fixed effect; no repeated observations of an individual fixed effect are required.

Theorem B.2 (de Finetti–Hewitt–Savage representation) *Let $(X_i)_{i\geq 1}$ be an infinite exchangeable sequence of random variables taking values in a Polish space $\mathcal{S}$ (in our application $\mathcal{S} = \mathbb{R}^d$); that is, for every n the law of $(X_1, \ldots, X_n)$ is invariant under all permutations of the indices. Then there exists a random probability measure μ on $\mathcal{S}$, the directing random measure, such that, conditionally on μ, the X_i are i.i.d. with common law μ. Taking $W = \mu$ gives a probability kernel $w \mapsto p(\cdot \mid w)$ such that, conditionally on W, the X_i are i.i.d. with law $p(\cdot \mid W)$. Writing P_W for the law of $W = \mu$, the marginal law of each X_i is*

$$p(dx) = \int p(dx \mid w)\, P_W(dw).$$

The sequence is i.i.d. if and only if its directing measure μ is almost surely nonrandom. When $W = \mu$, this is equivalent to a degenerate law for W.

For $\mathcal{S} = \{0, 1\}$ this is de Finetti's original theorem (de Finetti, 1937); the extension to Polish spaces is due to Hewitt and Savage (1955). In the baseline of Assumption 3, the theorem applies to the entire characteristic sequence at each date. For finitely many types, the corresponding partially exchangeable representation (Kallenberg, 2005) requires invariance of the joint law under separate permutations within each infinite fixed-effect class. It represents the classes as independent conditional on their collection of directing measures, which can be included in a single common state W_t. Under Assumption B.1, conditional independence is imposed directly. With infinitely many fixed effects, and particularly when fixed effects occur only once, within-class exchangeability alone does not yield that assumption.

For $M < \infty$ types with positive limiting shares π_ℓ, the conditional strong law within each fixed-effect class and convergence of the shares imply that the pooled empirical characteristic law converges almost surely to $\sum_{\ell=1}^{M} \pi_\ell p(\cdot \mid s_\ell, W_t)$. More generally define the limiting-population conditional and marginal laws by

$$p(\cdot \mid w) := \int_{\mathcal{T}} p(\cdot \mid s, w)\, \nu(ds), \qquad p(\cdot) := \int p(\cdot \mid w)\, P_W(dw).$$

B.2 Stock regularity and conditional strong-factor structure

Assumption B.3 (Stock-level non-degeneracy and regularity) (i) Almost surely,

$$\begin{aligned} \frac{1}{N} B_N(X_t; S)' B_N(X_t; S) = \frac{1}{N} \sum_{i \le N} B(X_{i,t}, S_i) B(X_{i,t}, S_i)' \\ \longrightarrow \Sigma_B(W_t) \succ 0, \end{aligned} \tag{B.2}$$

where

$$\Sigma_B(w) := \int_{\mathcal{T}} \int B(x, s) B(x, s)'\, p(dx \mid s, w)\, \nu(ds). \tag{B.3}$$

(ii) The conditional factor second moment has an almost-sure nonsingular limit,

$$\Omega(X_t; S) \longrightarrow \Omega(W_t) \succ 0.$$

This limit is imposed directly for general fixed-effect arrays. At each fixed N, the conditional factor mean and second moment are continuous functionals of the marked empirical law, with its weak topology.

(iii) The residual variance bounds of Assumption 3 hold.

(iv) The stock-level primitives are uniformly bounded:

$$\sup_{x,s} \|B(x,s)\| < \infty, \qquad \sup_{N} \sup_{X,S} \|\Omega(X;S)\| < \infty. \tag{B.4}$$

(v) For the unconditional stock result, the fixed-effect space $\mathcal{T}$ and common-state space W are Polish, with Borel laws ν and P_W. The deterministic array satisfies

$$\nu_N := N^{-1} \sum_{i=1}^{N} \delta_{S_i} \Rightarrow \nu. \tag{B.5}$$

Its conditional mean loading

$$b(s,w) := \int B(x,s)\, p(dx \mid s,w)$$

has a bounded jointly continuous version in (s,w), with the equality holding for every s and P_W-almost every w.

Throughout the appendices, the nonrandom array S is a suppressed parameter: $\Sigma_N(X_t)$, $\mu_G(X_t)$ and $\Omega(X_t)$ stand for $\Sigma_N(X_t; S)$, $\mu_G(X_t; S)$ and $\Omega(X_t; S)$. Because S is nonrandom, $\sigma(X_t, S_{1:N}) = \sigma(X_t)$, so conditioning on X_t alone gives the same moments.

For the limiting population, sample two fixed effects independently from ν, then sample their characteristics independently conditional on these fixed effects and W. Their characteristic pair law is

$$P_2(dz, du) = \int p(dz \mid w)\, p(du \mid w)\, P_W(dw), \qquad p(\cdot \mid w) = \int p(\cdot \mid s, w)\, \nu(ds). \tag{B.6}$$

The random fixed effects in this construction describe population sampling; the actual array (S_i) remains nonrandom. With finitely many types and convergent shares, (B.6) is also the limit of the law of a uniformly sampled ordered pair of distinct stocks. At finite N, sampling without replacement generally gives a different pair law. In the baseline, the formula holds for any two distinct stocks. If the pooled directing law $p(\cdot \mid W)$ is nonrandom, then $P_2 = p \otimes p$. This last statement concerns the pooled population; in the extension it does not assert that stocks with different fixed effects have identical characteristic laws.

Sufficient conditions for Assumption B.3(i). With finitely many types and convergent shares, conditional independence and the bounded loadings yield (B.2) by the strong law within each type. For a general regular fixed-effect array, a sufficient additional condition is that $s \mapsto \int B(x,s)B(x,s)'p(dx \mid s, w)$ be bounded and continuous for P_W-almost every w. Conditional independence and boundedness make the centered empirical average converge almost surely to zero, while (B.5) gives convergence of its conditional mean. Without this condition or another suitable law of large numbers, weak convergence of the fixed-effect array alone implies neither (B.2) nor convergence of the characteristic empirical law to the mixture $p(\cdot \mid w)$; the limit therefore remains the explicit requirement in Assumption B.3(i). Its nonsingularity is a requirement on $\Sigma_B(w)$ for P_W-almost every w: in almost every state, no nonzero $a \in \mathbb{R}^K$ has $a'B(x,s) = 0$ for $\nu(ds)\, p(dx \mid s, w)$-almost every (s, x). Averaging over states makes $\iint B(x,s)B(x,s)'\, p(dx \mid s)\, \nu(ds)$ nonsingular as well. The converse fails, which is why the conditional form is the one imposed. Here $p(dx \mid s) = \int p(dx \mid s, w)\, P_W(dw)$.

Remark B.4 (The stock second moment is uniformly elliptic at fixed N) The factor model of Assumption 3 and the bounds in Assumption B.3(iii)–(iv) control the stock second moment at each fixed cross-section. Write P_N for the law of the N-stock characteristic panel X. We display S in the factor moment $\Omega(X;S)$ here. Set

$$\bar{q} \;:=\; \sup_{x,s}\; \sup_{N}\; \operatorname*{ess\,sup}_{X}\; B(x,s)'\,\Omega(X;S)\,B(x,s) \;<\; \infty, \tag{B.7}$$

finite by (B.4) of Assumption B.3(iv). The supremum runs over the finite-N panel X rather than over a limiting state, because it is $\Omega(X;S)$ that must be dominated at every N; the limit $\Omega(w)$ would not do it. With (B.7) the decomposition (17) gives, at every N and for P_N-almost every X,

$$\underline{\sigma}^2\, I_N \;\preceq\; \Sigma_N(X) \;\preceq\; \bar{\Sigma}_N\, I_N\,, \qquad \bar{\Sigma}_N \;:=\; N\bar{q} + \bar{\sigma}^2\,. \tag{B.8}$$

The lower bound follows because $B_N\Omega(X;S)B_N'$ is positive semidefinite. For the upper bound,

$$\|B_N\Omega(X;S)B_N'\| \le \operatorname{tr}\big(B_N\Omega(X;S)B_N'\big) = \sum_{i\le N} B(x_i,S_i)'\Omega(X;S)B(x_i,S_i) \le N\bar{q}.$$

In particular $\Sigma_N(X)^{1/2}$ and $\Sigma_N(X)^{-1/2}$ are essentially bounded at each fixed N.

Proof of Proposition 2. By (17), $\Sigma_N(X_t;S) = B_N\,\Omega(X_t;S)\,B_N' + D_N$ with $B_N = B_N(X_t;S) \in \mathbb{R}^{N\times K}$ containing the loadings with fixed effects (its i-th row is $B(X_{i,t},S_i)'$), and $D_N = \operatorname{diag}(\sigma_\varepsilon^2(X_{i,t},S_i))$, so $\|D_N\| \le \bar{\sigma}^2$. By Lemma A.8, the nonzero eigenvalues of the $N\times N$ matrix $B_N\Omega(X_t;S)B_N'$ coincide with those of the $K\times K$ matrix $\Omega(X_t;S)^{1/2}(B_N'B_N)\Omega(X_t;S)^{1/2}$. Define

$$\Omega_B(w) := \Omega(w)^{1/2}\Sigma_B(w)\Omega(w)^{1/2}, \tag{B.9}$$

which is positive definite for P_W-almost every w by Assumption B.3(i)–(ii). That assumption and continuity of the matrix square root give $\Omega(X_t;S)^{1/2}(B_N'B_N/N)\Omega(X_t;S)^{1/2} \to \Omega_B(W_t)$ almost surely. Hence the at most K nonzero eigenvalues of $B_N\Omega(X_t;S)B_N'$, each rescaled by $\frac{1}{N}$, converge to the eigenvalues of $\Omega_B(W_t)$, while its remaining $N-K$ eigenvalues are zero. Weyl's inequality (A.7) from Lemma A.7 gives

$$\lambda_k(B_N\Omega(X_t;S)B_N') \le \lambda_k(\Sigma_N(X_t;S)) \le \lambda_k(B_N\Omega(X_t;S)B_N') + \|D_N\|.$$

Hence, for $k \le K$, $\lambda_k(\Sigma_N(X_t;S))/N \to \lambda_k(\Omega_B(W_t)) > 0$ (the $\|D_N\|/N \to 0$ perturbation is negligible), which proves Proposition 2(i). For $k > K$, $\lambda_k(\Sigma_N(X_t;S)) \le \lambda_k(B_N\Omega(X_t;S)B_N') + \|D_N\| \le \bar{\sigma}^2$, while $\lambda_k(\Sigma_N(X_t;S)) \ge \underline{\sigma}^2$ since $\Sigma_N(X_t;S) \succeq D_N \succeq \underline{\sigma}^2 I$ by Assumption B.3(iii). □

B.3 Response configurations and genericity

Definition B.5 (Response configurations and genericity) Fix $\mathcal{T}, \mathsf{W}, \nu, P_W$ as in Assumption B.3(v), $K \ge 1$, a measurable factor second moment $\Omega(w) \succ 0$ for P_W-almost every w with $C_\Omega := \operatorname{ess\,sup}_w \|\Omega(w)\| < \infty$, and a constant $L > 0$. The space of response configurations is

$$\mathfrak{C}_L := \{b \in C_b(\mathcal{T} \times \mathsf{W}; \mathbb{R}^K) : \|b\|_\infty \le L\}, \qquad d(b,\widetilde{b}) := \|b - \widetilde{b}\|_\infty. \tag{B.10}$$

A stock model has configuration b when, for every $s \in \mathcal{T}$, $\int B(x,s)\, p(dx \mid s,w) = b(s,w)$ for P_W-almost every w. The requirement applies to every sampled fixed effect, including those with zero ν-mass. Thus the laws of fixed effects and states, and the factor second moments, are held fixed; the conditional mean responses vary. The space $\mathfrak{C}_L$ is a closed subset of a Banach space and hence a complete metric space. A property is generic here if it holds on a residual subset, meaning that its exceptions form a countable union of nowhere dense sets.

The count is governed by a single object. For a fixed effect s and state w, let $\bar{B}_s(w) = b(s,w) = \int B(x,s)\, p(dx \mid s,w)$ be the expected loading of a stock with fixed effect s in state w. Let

$$\varphi_s \;:=\; \Omega(\cdot)^{1/2}\, \bar{B}_s(\cdot) \;\in\; \mathcal{H}_W := L^2(P_W; \mathbb{R}^K) \tag{B.11}$$

be its state-response profile: the vector-valued function of the state that records how a fixed-effect-s stock's exposure moves as the state moves. The count is set by the spread of these profiles across fixed effects, carried by the fixed-effect overlap operator on $\mathcal{H}_W$,

$$\mathcal{O} \;:=\; E_{S\sim\nu}\big[\varphi_S \otimes \varphi_S\big] \;=\; \int \varphi_s \otimes \varphi_s \, \nu(ds), \tag{B.12}$$

the uncentered second-moment operator of the profiles under the fixed-effect law. The bound (B.7) passes to $\Omega(W)$ by Assumption B.3(ii); conditional Jensen then gives

$$b(s,w)'\Omega(w)b(s,w) \le E[B(X,s)'\Omega(w)B(X,s) \mid s,w] \le \bar{q}. \tag{B.13}$$

Thus $\mathcal{O}$ is positive semidefinite and trace-class, with $\operatorname{tr}\mathcal{O} = E_S\|\varphi_S\|^2_{\mathcal{H}_W} \le \bar{q}$. Its rank

$$K^* \;:=\; \operatorname{rank}(\mathcal{O}) \;=\; \dim \overline{\operatorname{span}}\{\varphi_s : s \in \operatorname{supp}\nu\} \tag{B.14}$$

is the number of independent state-responses the fixed-effect law carries. On M points, $\nu = \sum_\ell \pi_\ell \delta_{s_\ell}$, the operator is $\mathcal{O} = \sum_\ell \pi_\ell\, \varphi_{s_\ell} \otimes \varphi_{s_\ell}$, of rank at most M, with nonzero eigenvalues those of the $M \times M$ fixed-effect overlap matrix $\big[\sqrt{\pi_\ell \pi_{\ell'}}\, \langle \varphi_{s_\ell}, \varphi_{s_{\ell'}} \rangle_{\mathcal{H}_W}\big]$. The span formula uses continuity of $s \mapsto \varphi_s$; without continuity it is the ν-essential closed span.

It is convenient to read $\mathcal{O}$ through the map it factors across. Let

$$\mathcal{J}_\nu :\; L^2(\nu) \;\to\; \mathcal{H}_W\,, \qquad \mathcal{J}_\nu g \;=\; \int g(s)\, \varphi_s \, \nu(ds)\,, \tag{B.15}$$

so that $\mathcal{J}_\nu^* h = \langle \varphi_\cdot, h \rangle$ and $\mathcal{O} = \mathcal{J}_\nu \mathcal{J}_\nu^*$. Since $\int \|\varphi_s\|^2 \nu(ds) = \operatorname{tr} \mathcal{O} < \infty$, the map $\mathcal{J}_\nu$ is Hilbert–Schmidt, and because $\ker \mathcal{O} = \ker \mathcal{J}_\nu^*$,

$$K^* \;=\; \operatorname{rank} \mathcal{O} \;=\; \operatorname{rank} \mathcal{J}_\nu. \tag{B.16}$$

For each configuration $b \in \mathfrak{C}_L$ write $\varphi_{b,s} = \Omega^{1/2} b(s, \cdot)$, $\mathcal{J}_{\nu,b} g = \int g(s) \varphi_{b,s} \, \nu(ds)$, and $\mathcal{O}_b = \mathcal{J}_{\nu,b} \mathcal{J}_{\nu,b}^*$. Dominated convergence gives norm-continuity of $s \mapsto \varphi_{b,s} \in \mathcal{H}_W$, with the bound

$$\sup_s \|\varphi_{b,s}\|_{\mathcal{H}_W}^2 \le C_\Omega L^2.$$

Proposition B.6 (Generic unconditional stock factor counts) *Suppose P_W has infinite support and* $\operatorname{supp} \nu$ *has cardinality $M \in \{1, 2, \dots, \infty\}$. For every positive integer $\ell \le M$, the set*

$$\mathfrak{U}_\ell := \{ b \in \mathfrak{C}_L : \lambda_\ell(\mathcal{O}_b) > 0 \}$$

is relatively open and dense in $\mathfrak{C}_L$. Consequently,

$$\{ b \in \mathfrak{C}_L : \operatorname{rank} \mathcal{O}_b = M \} = \bigcap_{1 \le \ell \le M} \mathfrak{U}_\ell \tag{B.17}$$

is a dense G_δ and is residual. For finite M, it is the open dense set $\mathfrak{U}_M$. For $M = \infty$, denote this set by $\mathfrak{C}_\infty$; every fixed-index eigenvalue of $\mathcal{O}_b$ is positive on $\mathfrak{C}_\infty$.

Proof of Proposition B.6. Openness. The rank-one operator identity and Cauchy–Schwarz give

$$\|\mathcal{O}_b - \mathcal{O}_{\widetilde{b}}\|_{\mathrm{op}} \le C_\Omega (\|b\|_\infty + \|\widetilde{b}\|_\infty) \|b - \widetilde{b}\|_\infty. \tag{B.18}$$

Weyl's inequality therefore makes each eigenvalue continuous, so $\mathfrak{U}_\ell$ is relatively open.

Density at every finite rank. Choose ℓ distinct points in $\operatorname{supp}\nu$ and ℓ distinct points in $\operatorname{supp} P_W$. Disjoint metric neighborhoods of these points admit bounded continuous functions $u_1, \dots, u_\ell$ on $\mathcal{T}$ and $v_1, \dots, v_\ell$ on W, respectively, with disjoint supports within each family, and with each function positive on a neighborhood of its chosen point. Thus

$$a_i := \int u_i^2 \, d\nu > 0, \qquad q_i := \int v_i(w)^2 e_1' \Omega(w) e_1 \, P_W(dw) > 0,$$

where $e_1 \in \mathbb{R}^K$ is the first coordinate vector. Positivity of q_i uses $\Omega(w) \succ 0$ almost surely. Define the bounded continuous perturbation

$$D(s,w) := \sum_{i=1}^{\ell} u_i(s) v_i(w) e_1.$$

Given $b \in \mathfrak{C}_L$ and a prescribed neighborhood of b, first replace b by $b_0 = (1-\delta)b$ with $\delta > 0$ sufficiently small. Then $\|b_0\|_\infty < L$ and $b_0 + tD$ remains in that neighborhood and in $\mathfrak{C}_L$ for all sufficiently small $|t|$. For $h_j = \Omega^{1/2} v_j e_1 \in \mathcal{H}_W$, form the $\ell \times \ell$ matrix

$$A_{ji}(t) := \langle \mathcal{J}_{\nu, b_0 + tD} u_i, h_j \rangle_{\mathcal{H}_W} = A_{ji}(0) + t \, \delta_{ij} a_i q_i.$$

Its determinant is a polynomial of degree ℓ with nonzero leading coefficient $\prod_{i=1}^{\ell} a_i q_i$. It has only finitely many roots, so arbitrarily small t give an invertible matrix. For these t, $\operatorname{rank} \mathcal{J}_{\nu, b_0+tD} \geq \ell$, equivalently $\lambda_\ell(\mathcal{O}_{b_0+tD}) > 0$. This proves density of $\mathfrak{U}_\ell$.

Simultaneous genericity. Baire's theorem on the complete space $\mathfrak{C}_L$ makes the countable intersection in (B.17) dense. Its complement is a countable union of relatively closed sets with empty interior, proving residuality. If M is finite, $\dim L^2(\nu) = M$ bounds the rank by M, so that intersection equals $\mathfrak{U}_M$. If $M = \infty$, positive eigenvalues at every finite index are equivalent to infinite rank. □

Lemma B.7 (Common-law realization) *Every configuration in (B.10) is realized by a bounded stock model satisfying Assumptions 3 and B.3. The characteristic law can be held fixed across all configurations.*

Proof of Lemma B.7. Since W is Polish, choose a Borel injection $\iota : \mathsf{W} \to [0,1]$ with Borel inverse on its image. Take independent vectors $\eta_i \in \{-1,1\}^K$ with independent symmetric coordinates, independent of W, and put

$$A = \Omega(W), \qquad X_i = (\iota(W), \eta_i, A), \qquad B(X_i, s) = b(s, W) + \eta_i.$$

Choose a version of Ω taking values in the bounded set $\{A \succeq 0 : \|A\| \leq C_\Omega\}$ everywhere. The characteristic domain is $\iota(\mathsf{W}) \times \{-1,1\}^K \times \{A \succeq 0 : \|A\| \leq C_\Omega\}$, a Borel subset of a finite-dimensional Euclidean space after vectorizing the matrix coordinate. On this characteristic domain define $B((z, e, A), s) = b(s, \iota^{-1}(z)) + e$, and extend it boundedly and measurably outside the domain if necessary. Its norm is at most $L + \sqrt{K}$. Conditional on W, the X_i are i.i.d., and their law is the same for all fixed effects and all configurations b. Only the loading function varies with the configuration. Take $G = A^{1/2}\zeta$, where ζ is independent, mean zero, and has covariance I_K, and add independent bounded residuals of a fixed positive variance. Since the matrix coordinate A is observed in every X_i, the conditional factor mean is zero and its second moment is exactly A. The latter is the empirical average of the matrix coordinate, a bounded continuous functional of the empirical characteristic law. Moreover, $E[B(X_i, s) \mid W] = b(s, W)$ for every s.

For each fixed w, the map $s \mapsto b(s,w)b(s,w)'$ is bounded and continuous. Writing $Y_i = B(X_i, S_i) = b(S_i, W) + \eta_i$, equation (B.5) and the strong law for independent bounded variables, applied conditionally on W, give

$$\frac{1}{N}\sum_{i=1}^{N} Y_i Y_i' \longrightarrow I_K + \int b(s,W)b(s,W)'\,\nu(ds) \succ 0 \qquad \text{almost surely.}$$

Thus Assumption B.3(i) also holds, even at configurations with zero conditional mean responses. This construction shows that the configuration space contains realizable stock models throughout, including all the perturbations used in the density proof.

□

By Lemma B.7, every configuration in (B.10), including each perturbed configuration in the proof of Proposition B.6, is realized by an admissible stock model. The genericity statement is relative to the specified response space and uniform topology. It does not assert genericity in every restricted parametric family of characteristic laws and loadings. For example, if all $b(s, w)$ are constant in w and Ω is constant, the rank is at most K. More generally, if the state law has only $J < \infty$ support points, then $\operatorname{rank} \mathcal{O}_b \leq KJ$. Infinite state variation is therefore necessary for the infinite-rank conclusion.

Lemma B.8 (Averages over a regular array of fixed effects) *Let F be a bounded continuous map from the space of fixed effects $\mathcal{T}$ into a separable Hilbert space, and let the array satisfy (B.5), as required by Assumption B.3(v). Then $\frac{1}{N}\sum_{i\leq N} F(S_i) \to \int F\, d\nu$ in norm.*

Proof of Lemma B.8. Write $\nu_N = \frac{1}{N}\sum_{i\leq N} \delta_{S_i}$ and $C_F = \sup_s \|F(s)\|$. Weak convergence on the Polish space implies uniform tightness of $\{\nu_N, \nu\}$ (Dudley, 2002, Chapter 11). Given $\varepsilon > 0$, choose a compact set $C \subset \mathcal{T}$ to which all these laws assign mass at least $1 - \varepsilon$. Since $F(C)$ is compact, there is a finite-dimensional subspace with orthogonal projection P such that $\sup_{s\in C} \|(I-P)F(s)\| < \varepsilon$. Every coordinate of PF is bounded and continuous, so weak convergence gives $\int PF\, d\nu_N \to \int PF\, d\nu$ in this finite-dimensional space. The remaining terms satisfy

$$\limsup_N \left\| \int F\, d\nu_N - \int F\, d\nu \right\| \leq 2\varepsilon + 2C_F\varepsilon.$$

Letting $\varepsilon \downarrow 0$ proves norm convergence. □

We apply this with $F(s) = \varphi_s \otimes \varphi_s$, continuous by Assumption B.3(v) and dominated convergence, and bounded in Hilbert–Schmidt norm by (B.13). When the fixed effects are an i.i.d. sample the same conclusion is Mourier's strong law for integrable Hilbert-space-valued variables; with the array fixed, regularity is what replaces it.

Proposition B.9 (Fixed effects set the count) *Let Assumptions 3 and B.3 hold, allowing the common characteristic law in Assumption 3 to be replaced by Assumption B.1. Let $\mathcal{O}$ be the fixed-effect overlap operator (B.12). Then:*

(i) ***The growing eigenvalues.*** *For every fixed j,*

$$\frac{\lambda_j(\Sigma_N)}{N} \longrightarrow \lambda_j(\mathcal{O}),$$

and exactly $K^* = \operatorname{rank}(\mathcal{O})$ of these limits are positive. If $K^* < \infty$, every fixed-index eigenvalue beyond K^* is $o(N)$. When P_W has infinite support, Proposition B.6 gives $K^* = \infty$ generically for infinite-support ν, and $K^* = M$ generically when ν has $M < \infty$ support points.

(ii) ***The rest, with finitely many values.*** *When the array takes exactly M values, each with positive limiting share, write $d_\ell^{(N)}$ for the common unconditional second moment of a stock with fixed effect ℓ and $c_{\ell\ell}^{(N)}$ for the co-moment of two distinct stocks with that fixed effect. For each ℓ, Σ_N acts on the contrast subspace of fixed-effect class ℓ, of dimension $N_\ell - 1$, as multiplication by $d_\ell^{(N)} - c_{\ell\ell}^{(N)}$, and this eigenvalue converges to $d_\ell^\infty - c_{\ell\ell}^\infty$. These account for $N - M$ bounded eigenvalues. Of the remaining M eigenvalues, exactly K^* have positive limits after division by N.*

Proof of Proposition B.9. Part (i). Write $B_i = B(X_{i,t}, S_i)$, $\Omega_N = \Omega(X_t; S)$, and $L_B = \sup_{x,s} \|B(x,s)\|$. Let $\Gamma_N = [\langle \varphi_{S_i}, \varphi_{S_j} \rangle_{\mathcal{H}_W}]$ be the Gram matrix of the sampled profiles. For

$i \neq j$, conditional independence gives

$$E[B_i'\Omega(W_t)B_j] = E_W[\bar{B}_{S_i}(W)'\Omega(W)\bar{B}_{S_j}(W)] = \langle \varphi_{S_i}, \varphi_{S_j} \rangle_{\mathcal{H}_W}.$$

Hence each off-diagonal entry of $\Sigma_N - \Gamma_N$ is bounded in absolute value by $L_B^2 E\|\Omega_N - \Omega(W_t)\|$. Its diagonal entries are bounded by $2\bar{q} + \bar{\sigma}^2 =: c_0$, using (B.4) and (B.7). The operator norm is bounded by the maximal absolute row sum, so

$$\left\| \frac{\Sigma_N - \Gamma_N}{N} \right\|_{\text{op}} \leq L_B^2 E\|\Omega_N - \Omega(W_t)\| + \frac{c_0}{N} \longrightarrow 0. \tag{B.19}$$

The expectation tends to zero by Assumption B.3(ii), (iv) and bounded convergence. This argument applies even when every sampled fixed effect is distinct. By Lemma A.8 the nonzero eigenvalues of Γ_N are those of $\sum_{i \leq N} \varphi_{S_i} \otimes \varphi_{S_i}$ on $\mathcal{H}_W$. By Lemma B.8 applied to $F(s) = \varphi_s \otimes \varphi_s$,

$$\frac{1}{N} \sum_{i \leq N} \varphi_{S_i} \otimes \varphi_{S_i} \;\longrightarrow\; E_{S \sim \nu}[\varphi_S \otimes \varphi_S] \;=\; \mathcal{O}$$

in Hilbert–Schmidt norm, so $\lambda_j(\Gamma_N)/N \to \lambda_j(\mathcal{O})$ for every fixed j (Weyl, Lemma A.7). Combining this with (B.19) gives $\lambda_j(\Sigma_N)/N \to \lambda_j(\mathcal{O})$, and exactly $K^* = \text{rank}(\mathcal{O})$ of the limits are positive. Proposition B.6 supplies the generic counts.

Part (ii). Suppose the array takes exactly M values, and let $\mathcal{G}_1, \ldots, \mathcal{G}_M$ collect the stocks by fixed effect, $N_\ell = |\mathcal{G}_\ell|$, $N_\ell/N \to \pi_\ell > 0$. For sufficiently large N, every $N_\ell \geq 2$. Invariance under relabeling within each class makes the diagonal second moments $d_\ell^{(N)}$ constant within a class and the off-diagonal co-moments $c_{\ell\ell'}^{(N)}$ constant for each pair of classes, including distinct stocks in the same class. With $C_N = [c_{\ell\ell'}^{(N)}]$ and U_N the matrix of class indicators, we have the exact decomposition

$$\Sigma_N = G_N + \Delta_N, \qquad G_N = U_N C_N U_N', \qquad \Delta_N = \text{diag}(d_\ell^{(N)} - c_{\ell\ell}^{(N)} : i \in \mathcal{G}_\ell). \tag{B.20}$$

The diagonal matrix Δ_N is constant within each class. Let $V := \mathrm{span}\{\mathbf{1}_{\mathcal{G}_1}, \ldots, \mathbf{1}_{\mathcal{G}_M}\}$, of dimension M. The range of $U_N C_N U_N'$ lies in V and it annihilates $V^\perp$, while Δ_N preserves V and $V^\perp$ since $\mathrm{diag}(\mathbf{1}_{\mathcal{G}_\ell})\mathbf{1}_{\mathcal{G}_{\ell'}} = \delta_{\ell\ell'}\mathbf{1}_{\mathcal{G}_\ell}$. So on the part of $V^\perp$ carried by fixed effect ℓ, dimension $N_\ell - 1$, Σ_N acts as the scalar $d_\ell^{(N)} - c_{\ell\ell}^{(N)}$, contributing $N_\ell - 1$ eigenvalues and accounting for $\sum_\ell (N_\ell - 1) = N - M$ of them. Each converges, $d_\ell^{(N)} - c_{\ell\ell}^{(N)} \to d_\ell^\infty - c_{\ell\ell}^\infty$ by bounded convergence as in Part (i), now applied to diagonal as well as off-diagonal moments. These $N - M$ eigenvalues therefore settle onto at most M finite levels. In the orthonormal basis $\mathbf{1}_{\mathcal{G}_\ell}/\sqrt{N_\ell}$ of V, the restriction of Σ_N/N converges to $[\sqrt{\pi_\ell \pi_{\ell'}} c_{\ell\ell'}^\infty]$, whose nonzero eigenvalues are those of $\mathcal{O}$. Exactly K^* eigenvalues of this block have positive limits. □

Proof of Theorem 3. Proposition B.9(i) identifies the normalized stock-eigenvalue limits with those of the overlap operator. Proposition B.6 gives its generic rank, and Proposition B.9(ii) gives the bounded within-type eigenvalues, including the exchangeable case $M = 1$. These conclusions apply under the common-law setting of Assumption 3 and under its extension in Assumption B.1. □

Remark B.10 (The exchangeable case) For the case $M = 1$ with no sampled fixed effects outside $\mathrm{supp}\,\nu$, suppress the common fixed effect in B and σ_ε^2. Then (B.20) is the equicorrelation identity $\Sigma_N = (d_N - c_N)I_N + c_N \mathbf{1}\mathbf{1}'$. It gives the market eigenvalue $d_N + (N-1)c_N$, with eigenvector $\mathbf{1}$, and Proposition B.9(ii) gives the $N - 1$ equal eigenvalues $d_N - c_N$, with

$$\begin{aligned} c_\infty &= E_W[\bar{B}(W)'\Omega(W)\bar{B}(W)], \\ d_\infty - c_\infty &= E_W[\mathrm{tr}(\Omega(W)\mathrm{Cov}_{x|W}(B))] + \int \sigma_\varepsilon^2 \, dp. \end{aligned}$$

The first quantity, c_∞, is positive exactly when $\bar{B}(W)$ is not almost surely zero, which holds on the open dense set in Proposition B.6. The unconditional average loading may

nevertheless vanish by cancellation across states. When $c_\infty = 0$, even the market eigenvalue is $o(N)$; when $c_\infty > 0$, it is the largest eigenvalue for all sufficiently large N.

C Standard Menu Constructions and Conditions

This appendix restates the menu conditions of Assumption 1 and describes the standard menu constructions: random features, a fixed basis, and partitions, each with the conditions it needs. Later appendices verify Assumption 1 for each construction.

C.1 Formal menu conditions

Definition C.1 (Formal menu conditions) For reference, we restate Assumption 1. At every fixed N, let $R_{k,t+1}$ denote the managed return from menu $f^{(P)}$, as in (2); the menus need not be nested. All limits and suprema in P are taken along one specified increasing sequence of menu sizes. For weighting functions with random labels drawn independently of the panel, the conditions hold almost surely conditional on the label sequence. Every $R_{k,t+1}$ is square integrable. There is a constant $C_{\text{menu}} < \infty$, the same for every P and every arbitrary managed portfolio, such that for every measurable ϕ with square-integrable return $R^{\phi}_{t+1} := N^{-1}\sum_{i=1}^{N} \phi(X_{i,t}) r_{i,t+1}$, the combination

$$A_P R^{\phi}_{t+1} := \frac{1}{P}\sum_{k=1}^{P} R_{k,t+1} E[R_{k,t+1} R^{\phi}_{t+1}] \tag{C.1}$$

satisfies

$$A^{\phi}_{t+1} := \lim_{P\to\infty} A_P R^{\phi}_{t+1} \qquad \text{exists in } L^2, \tag{C.2}$$

$$\sup_{P\geq 1} \|A_P R^{\phi}_{t+1}\|_{L^2} \leq C_{\text{menu}} \|R^{\phi}_{t+1}\|_{L^2}. \tag{C.3}$$

The menu retains every nonzero R^{ϕ}_{t+1} in the limit:

$$A^{\phi}_{t+1} \neq 0 \qquad \text{whenever} \qquad R^{\phi}_{t+1} \neq 0 \quad \text{in } L^2. \tag{C.4}$$

There are arbitrarily many linearly independent square-integrable returns R^{ϕ}_{t+1}. If $N \to \infty$, these conditions hold for each N along the sequence with the same constant C_{menu}, independent of N.

C.2 Types of weight menus

Empirical asset pricing builds managed portfolios in two main ways: from parameterized weight functions, or from partitions of the characteristic space. We use three such menus by name: random-feature menus and fixed-basis menus, both built from parameterized weight functions, and partition menus. The three subsections below state their conditions. Throughout this subsection, $p = N^{-1}\sum_{i\leq N} \text{law}(X_{i,t})$ is the pooled characteristic law; μ_θ denotes a distribution of portfolio labels.

Parameterized weight functions. The first approach specifies a family of weight functions $f : \mathbb{R}^d \times \Theta \to \mathbb{R}$. A menu of P portfolios is then indexed by labels $\theta_1, \ldots, \theta_P \in \Theta$, with $f^{(P)}_k = f(\cdot; \theta_k)$. The labels may be drawn randomly from a distribution $\mu_\theta(d\theta)$ on Θ, as in Didisheim et al. (2024); fixed in advance, as in Chen, Roussanov and Wang (2022); or estimated from the data, as in Kelly et al. (2026). The random-feature and fixed-basis conditions here concern labels chosen independently of returns; menus whose labels are estimated from the same returns fall outside them.

The first, the random-feature menu, consists of random Fourier features,

$$f_k(x) = \sqrt{2}\cos(\omega_k' x + b_k), \qquad \theta_k = (\omega_k, b_k),$$

whose weight functions do not depend on P at all; only the number of draws does. The second, the fixed-basis menu, is

$$f_k^{(P)}(x) = P^{1/2} q_k H_k(x),$$

where $\{H_k\}$ is an orthonormal basis of $L^2(p)$ and the coefficients q_k are bounded. Examples include Hermite polynomials for a Gaussian p, or a complete spline system with refining knots, orthonormalized in $L^2(p)$.

Sorts, partitions, and trees. The second approach partitions the characteristic space into cells $C_1, \ldots, C_P$. Portfolio k holds equally weighted stocks whose characteristics fall in C_k:

$$f_k^{(P)}(x) = P\,\mathbf{1}\{x \in C_k\}.$$

For $d = 1$, characteristic-quantile cells give an expected-count version of a univariate sort. For $d > 1$, they give double, triple, and more general multivariate sorts.[20]

Cattaneo et al. (2020) develop estimation and inference for characteristic-sorted portfolios, including the choice of the number of portfolios. Cattaneo et al. (2026) study beta-sorted portfolios, where the sorting variables are factor exposures estimated in a first step.

Partition menus also encompass tree-based procedures. A decision tree recursively splits $\mathbb{R}^d$ into cells, and its leaves define the portfolios. Adding a split refines the menu by dividing one existing cell into two. Thus tree leaves form partition menus. The partition conditions below cover trees whose splits are fixed in advance and whose levels satisfy the

[20]The weight $P\mathbf{1}\{x \in C_k\}$ normalizes by expected cell size, $p(C_k) \approx 1/P$, rather than by the realized number of stocks in the cell. The usual empirical sort, $n_{k,t}^{-1} \sum_{i \in C_k} r_{i,t+1}$, is not exactly of the managed form in (2), since $n_{k,t}$ depends on the entire cross-section rather than on stock i's own characteristics. The two do not coincide exactly in expectation because the realized reciprocal cell count is nonlinear. They can become asymptotically equivalent in a joint limit in which expected cell counts diverge and concentrate uniformly. That is not the fixed-N, growing-P limit studied here: expected cell counts then tend to zero. The expected-count version belongs to (2) and remains well defined even when a cell is empty.

mass conditions. Moritz and Zimmermann (2016) construct conditional portfolio sorts using trees; Bryzgalova et al. (2025) use tree leaves to form cross-sections of test assets; and Cong et al. (2025) grow trees to expand the efficient frontier.

A growing menu must settle and remain asymptotically complete. Conditions (C.2)–(C.3) of Definition C.1 make its co-moment operators converge on each managed return, with one bound valid for every arbitrary managed portfolio. Asymptotic completeness, condition (C.4), ensures that the limit detects every nonzero arbitrary managed portfolio. Random features achieve it through the label distribution, a fixed basis exhausts $L^2(p)$, and generating partitions resolve every measurable characteristic weighting.

C.2.1 Random-feature conditions

Let the weight functions belong to a jointly measurable family $f(\cdot;\theta)$, and draw an infinite sequence of labels independently from $\mu_\theta(d\theta)$ and independently of the panel, so that

$$f_k = f(\cdot;\theta_k), \qquad R_\theta := \frac{1}{N} f(X_t;\theta)' r_{t+1}.$$

Suppose that the corresponding managed returns have finite second moment on average over the labels:

$$\int_\Theta E[R_\theta^2]\,\mu_\theta(d\theta) < \infty. \tag{C.5}$$

Suppose also that the returns

$$R_{\theta_1}, \ldots, R_{\theta_P}$$

are almost surely linearly independent for every finite P. This forces the managed-payoff space to be infinite dimensional, and hence the characteristic support to be infinite; it supplies

that part of Assumption 1. Finally, require that no nonzero arbitrary managed portfolio is orthogonal to the feature family: for every nonzero square-integrable

$$R^{\phi}_{t+1} = \frac{1}{N}\sum_{i=1}^{N} \phi(X_{i,t}) r_{i,t+1},$$

the set

$$\left\{\theta \in \Theta : E[R_\theta R^{\phi}_{t+1}] \neq 0\right\}$$

has strictly positive μ_θ-measure.[21]

C.2.2 Fixed-basis conditions

Let $\{H_k\}_{k\geq 1}$ be an infinite orthonormal basis of $L^2(p)$, and let $\{q_k\}_{k\geq 1}$ be nonzero coefficients with $q_{\max} := \sup_k |q_k| < \infty$. Suppose that

$$\underline{\sigma}^2 I_N \preceq \Sigma_N(X_t) \preceq \bar{\Sigma}_N I_N \qquad \text{almost surely,}$$

where $\underline{\sigma}^2 > 0$ and $\bar{\Sigma}_N < \infty$, although $\bar{\Sigma}_N$ may grow with N. Use the menu

$$f^{(P)}_k(x) = P^{1/2} q_k H_k(x), \qquad k = 1, \ldots, P,$$

with this common basis and these coefficients at every P.

[21] Linear independence of every finite menu does not by itself guarantee the final richness condition. For example, let $d = N = 1$, $f(x;\theta) = \cos(\theta x)$, $\theta \sim \mathrm{U}[0,1]$, let X be standard normal, and let r be independent of X with $E[r^2] = 1$. Every finite collection of features is almost surely linearly independent, yet every odd weighting function is orthogonal to the entire feature family. The random phase in our Fourier-feature menu rules out this particular failure by introducing both even and odd components.

C.2.3 Partition conditions

Fix an increasing sequence of integers $P_n \to \infty$. For each $P = P_n$, partition the characteristic space into cells $C_1^{(P)}, \ldots, C_P^{(P)}$. Let

$$\mathcal{F}_P = \sigma\big(C_1^{(P)}, \ldots, C_P^{(P)}\big).$$

Suppose that the partitions are nested, so that $\mathcal{F}_{P_n} \subseteq \mathcal{F}_{P_{n+1}}$, and generate the characteristic sigma-field:

$$\sigma\left(\bigcup_{n\geq 1} \mathcal{F}_{P_n}\right) = \mathcal{B}(\mathbb{R}^d) \qquad \text{modulo } p\text{-null sets.} \tag{C.6}$$

All limits and bounds indexed by P in this part are along the specified sizes P_n. Portfolio k holds the stocks whose characteristics fall in cell $C_k^{(P)}$:

$$f_k^{(P)}(x) = P\,\mathbf{1}\{x \in C_k^{(P)}\}, \qquad R_{k,t+1}^{(P)} = \frac{P}{N} \sum_{i\,:\,X_{i,t} \in C_k^{(P)}} r_{i,t+1}, \qquad k = 1, \ldots, P. \tag{C.7}$$

Write $C^{(P)}(x)$ for the cell containing x. The cells need not have equal probability, but their masses must be uniformly comparable to $1/P$: for fixed constants $0 < c_- \leq c_+ < \infty$,

$$\frac{c_-}{P} \leq p\Big(C_k^{(P)}\Big) \leq \frac{c_+}{P} \qquad \text{for all } k, P. \tag{C.8}$$

In addition, the refinement density must settle uniformly: for a measurable g with $c_- \leq g \leq c_+$,

$$g_P(x) := P\,p\big(C^{(P)}(x)\big), \qquad \|g_P - g\|_{L^\infty(p)} \longrightarrow 0. \tag{C.9}$$

Equal-mass dyadic partitions, $P_n = 2^n$, satisfy it with $g_{P_n} \equiv g \equiv 1$.[22] At the fixed stock count under consideration, require that the conditional second moment be uniformly elliptic,

$$\Sigma_N(X_t) \succeq \underline{\sigma}^2 I_N \qquad \text{almost surely, for some } \underline{\sigma}^2 > 0, \tag{C.10}$$

and essentially bounded above at that fixed N,

$$\Sigma_N(X_t) \preceq \bar{\Sigma}_N I_N \qquad \text{almost surely, for some } \bar{\Sigma}_N < \infty. \tag{C.11}$$

The upper bound makes every cell return square integrable. The cell-mass ceiling forces p to be nonatomic: an atom of mass $a > 0$ would belong to a cell of mass at least a, contradicting $p(C_k^{(P_n)}) \leq c_+/P_n$. Settling and asymptotic completeness use the two-sided stock-moment bounds, the mass conditions (C.8)–(C.9), and the generating condition (C.6).

D Managed-Payoff Operators and the General Lower Bound

This appendix defines the operators and settled forms that a growing menu generates, proves the lower bound in Theorem 1, and states a formal common-shock condition that is sufficient for Assumption 2.

Write $Z = (r_{t+1}, X_t)$, let $\mathbb{P}$ be its law, and let $\tilde{Z}$ be an independent copy. Define

$$\mathcal{D} \;:=\; \left\{ \Phi_\psi := \tfrac{1}{N} \textstyle\sum_{i \leq N} \psi(X_{i,t})\, r_{i,t+1} \;\in L^2(\mathbb{P}) \;:\; \psi \text{ measurable} \right\} \tag{D.1}$$

for the space of arbitrary managed portfolios of (C.2). Thus $\Phi_\psi = R^\psi$ in the notation of the main text. Only the payoff is required to be square integrable; uniform ellipticity, when

[22] Exact equal mass and nesting need not be compatible at every integer P. Comparable masses alone do not imply uniform convergence of the refinement density: splitting the cells one at a time between successive dyadic levels gives a nested, generating, comparable-mass sequence for which g_P oscillates.

imposed for a construction, additionally implies $\psi \in L^2(p)$, where $p = \frac{1}{N}\sum_i \mathrm{law}(X_{i,t})$ is the average cross-sectional law of a characteristic. This is a linear space, and every R_k lies in it. Write A_P for the operator the menu generates on $L^2(\mathbb{P})$,

$$A_P\Phi \;:=\; \frac{1}{P}\sum_{k=1}^{P} R_k\,\langle R_k, \Phi\rangle\,, \qquad \langle R_k, \Phi\rangle := E_{\tilde{Z}}\big[R_k(\tilde{Z})\,\Phi(\tilde{Z})\big], \tag{D.2}$$

and say the menu **settles weakly** if, for all $\Phi, \Phi' \in \mathcal{D}$,

$$q_P(\Phi, \Phi') \;:=\; \langle A_P\Phi, \Phi'\rangle \;=\; \frac{1}{P}\sum_{k=1}^{P}\langle R_k, \Phi\rangle\,\langle R_k, \Phi'\rangle \;\longrightarrow\; q(\Phi, \Phi') \tag{D.3}$$

for a settled bilinear form q on $\mathcal{D}$. Order that form by its max–min values,

$$\alpha_j \;:=\; \sup_{V\subseteq\mathcal{D},\ \dim V = j}\ \min_{\Phi\in V,\ \|\Phi\|=1} q(\Phi,\Phi) \;\geq\; 0. \tag{D.4}$$

We set $\alpha_j = 0$ if $j > \dim\mathcal{D}$.

The numbers α_j are the Courant–Fischer values of the settled form. They provide lower bounds for the constants λ_j^* of Theorem 1; equality requires additional spectral convergence, and is not inferred from strong settling alone.

Throughout, we index the settled limit by the arbitrary managed portfolio rather than by the weighting behind it, writing A^{Φ_ψ} for the A^ψ of (C.2); in this notation (C.2) reads $A_P\Phi \to A^\Phi$ in L^2 for every $\Phi \in \mathcal{D}$.

Remark D.1 (Single-stock and panel weightings) Condition (C.2) checks the menu against portfolios built the way the menu is, through a weighting ψ of a single stock's characteristics. Widening it to an arbitrary $\eta : \mathbb{R}^{N\times d} \to \mathbb{R}^N$, with portfolio return $\frac{1}{N}r'_{t+1}\eta(X_t)$, can make (C.4) fail even with two stocks. Suppose for this paragraph that the characteristics are identically distributed, so that no per-stock reweighting intervenes. Then $\langle R_k, \frac{1}{N}r'\eta(X)\rangle = \frac{1}{N^2}\sum_i E[f_k(X_{i,t})\,u_i(X_t)]$ with $u = \Sigma_N(X_t)\eta(X_t)$, so A^η depends on

η only through the one-coordinate marginal $z \mapsto \sum_i E[u_i(X_t) \mid X_{i,t} = z]$. With $N = 2$, $\Sigma_2 = I_2$ and exchangeable characteristics, the constant panel weighting $\eta = (1, -1)'$ yields a nonzero payoff orthogonal to every menu return: its second moment is $1/2$, whereas $E[f_k(X_1) - f_k(X_2)] = 0$. Thus its settled limit A^η vanishes for every menu. The restriction to ψ matches the construction in (2): the menu's own weights are functions of one stock's characteristics, so the arbitrary managed portfolios are drawn from the same class as the menu returns.

Restricting attention to $\mathcal{D}$ costs nothing and is what makes the condition reachable from Assumption 1. It costs nothing because every R_k lies in $\mathcal{D}$, so A_P maps $\mathcal{D}$ into itself and annihilates $\mathcal{D}^\perp$; its nonzero eigenvalues are therefore those of its compression to $\mathcal{D}$, and enlarging the supremum in (D.4) beyond $\mathcal{D}$ could only add subspaces on which q vanishes. It is what makes the condition reachable because (C.2) is a statement about arbitrary managed portfolios and says nothing outside $\mathcal{D}$. Remark D.1 explains why $\mathcal{D}$ is built from weightings of a single stock's characteristics. The lower-bound argument works on finite-dimensional subspaces of $\mathcal{D}$, so it needs no uniform operator bound: on each such subspace, bilinearity upgrades pointwise settling to uniform convergence. The bound in (C.3) is imposed on A_P and passes to its strong limit, with the same constant C_{menu} uniformly over arbitrary managed portfolios and P, and also over N in the growing-stock case. It bounds the size of spectral mass, but does not prevent a bounded amount from moving through directions that change with P.

The empirical co-moment kernel and its stronger convergence condition are

$$K_P(Z, \tilde{Z}) := \frac{1}{P} \sum_{k=1}^{P} R_k^{(P)}(Z) R_k^{(P)}(\tilde{Z}), \tag{D.5}$$

and

$$K_P \longrightarrow \mathcal{C} \qquad \text{in } L^2(\mathbb{P} \otimes \mathbb{P}). \tag{D.6}$$

Here $\mathcal{C}$ is the limiting co-moment kernel. Its integral operator on $L^2(\mathbb{P})$ is $T_{\mathcal{C}}\Phi(Z) := E_{\tilde{Z}}[\mathcal{C}(Z,\tilde{Z})\Phi(\tilde{Z})]$. This condition concerns a kernel; it is stronger than strong convergence of its integral operators on individual payoffs.

Assumption D.2 (Formal common-shock regularity) Let $\mathcal{V}$ be a separable, infinite-dimensional Hilbert space of measurable characteristic weightings, continuously embedded in $L^2(p)$ for the fixed reference characteristic law p. It contains every menu weighting and every fixed measurable weighting ϕ with $\sup_N E[(R^{\phi}_{N,t+1})^2] < \infty$, and each of its elements satisfies this moment bound; $\mathcal{V} = L^2(p)$ is the leading case. Changing the norm changes both the unit ball used for uniform convergence and the domain available for completeness. Require completeness of the limiting menu on characteristic weightings. For a random-feature menu this means that, for $h \in L^2(p)$,

$$\int h(x) f(x;\theta)\, p(dx) = 0 \quad \text{for } \mu_\theta\text{-almost every } \theta \quad \Longrightarrow \quad h = 0 \quad \text{in } L^2(p).$$

For a fixed basis use the complete basis and nonzero coefficients of Appendix C.2.2. For a partition menu impose the generating condition (C.6), the mass bounds (C.8), and uniform mass settling (C.9). These are restrictions on the portfolio menu; no conditional stock-factor model is imposed here. For each weighting f, write $f_N(X_t) = (f(X_{i,t}))_{i=1}^N$. There is a measurable family $\{\mathcal{Q}_w : w \in \mathsf{W}\}$ of bounded, self-adjoint, positive semidefinite operators on $\mathcal{V}$, satisfying $\operatorname{ess\,sup}_w \|\mathcal{Q}_w\|_{\mathrm{op}} < \infty$, such that

$$\sup_{\|f\|_{\mathcal{V}} \le 1} E\left| \frac{1}{N^2} f_N(X_t)' \Sigma_N(X_t) f_N(X_t) - \langle f, \mathcal{Q}_{W_t} f \rangle_{\mathcal{V}} \right| \xrightarrow[N\to\infty]{} 0. \tag{D.7}$$

Define the positive operator Q_∞ by $\langle f, Q_\infty h\rangle_{\mathcal{V}} = E\langle f, \mathcal{Q}_W h\rangle_{\mathcal{V}}$ and require

$$\operatorname{rank}(Q_\infty) = \infty. \tag{D.8}$$

Finally, write q_N for the settled menu form (D.3) in the N-stock economy and $\Phi_{N,\phi} = R^{\phi}_{N,t+1}$. For a single weighting ϕ,

$$q_N(\Phi_{N,\phi}, \Phi_{N,\phi}) = \lim_{P\to\infty} \frac{1}{P} \sum_{k=1}^{P} \big(E[R_{k,t+1} R^{\phi}_{N,t+1}]\big)^2.$$

For every fixed measurable ϕ with $\sup_N E[(R^{\phi}_{N,t+1})^2] < \infty$ and $E[(R^{\phi}_{N,t+1})^2] > 0$ for all sufficiently large N, require

$$\liminf_{N\to\infty} \frac{q_N(\Phi_{N,\phi}, \Phi_{N,\phi})}{E[(R^{\phi}_{N,t+1})^2]} > 0. \tag{D.9}$$

Condition (D.8) is equivalent to $\dim \operatorname{ess}\, \overline{\operatorname{span}}_w \operatorname{range}(\mathcal{Q}_w) = \infty$, the essential closed span being the smallest closed subspace containing $\operatorname{range}(\mathcal{Q}_w)$ for P_W-almost every w, which is unaffected by the versions chosen on a null set. Indeed, positivity gives $\ker Q_\infty = \{f : \mathcal{Q}_W f = 0 \text{ almost surely}\}$, and taking orthogonal complements gives the equivalence.

The common-state rank condition describes the limiting economy. For fixed complete menu geometries it also determines the rank of the limiting portfolio spectrum: the large-stock limits for random-feature and partition menus identify the number of positive growth constants with this rank. It is separate from topological genericity: an infinite-rank condition does not itself establish genericity on an unspecified space of models.

Assumption D.2 is sufficient for Assumption 2: the positive operator Q_∞ has arbitrarily large finite-dimensional subspaces on which its quadratic form is positive definite, and (D.7) identifies that form with the limiting return second moments. The uniformity over the full unit ball in (D.7) is stronger than the pointwise convergence needed in Assumption 2. Infinite support of W_t alone does not imply (D.8): infinitely many states may generate the same conditional law and hence the same $\mathcal{Q}_w$.

Lemma D.3 (The settling chain) (D.6) $\Rightarrow$ (C.2) $\Rightarrow$ (D.3), *with* $A^\Phi = T_{\mathcal{C}}\Phi$ *and* $q(\Phi, \Phi') = \langle A^\Phi, \Phi' \rangle$*; and* $\lambda_j(\Psi_P)/P = \lambda_j(A_P)$ *for every* j*.*

Proof of Lemma D.3. For the first implication, A_P is the integral operator with kernel $K_P(Z, \tilde{Z}) = \frac{1}{P}\sum_k R_k(Z)R_k(\tilde{Z})$ of (D.5), so for any Φ, $\|A_P\Phi - T_{\mathcal{C}}\Phi\|_{L^2} \le \|K_P - \mathcal{C}\|_{L^2(\mathbb{P}\otimes\mathbb{P})}\|\Phi\|_{L^2}$, which (D.6) sends to zero; the limit is $A^\Phi = T_{\mathcal{C}}\Phi$. For the second, the inner product is continuous, so $A_P\Phi \to A^\Phi$ in L^2 gives $q_P(\Phi, \Phi') = \langle A_P\Phi, \Phi' \rangle \to \langle A^\Phi, \Phi' \rangle$ for every Φ'. The identity $\lambda_j(\Psi_P)/P = \lambda_j(A_P)$ is Lemma A.8 applied to the Gram matrix $\frac{1}{P}\Psi_P = \left[\frac{1}{P}\langle R_k, R_l \rangle\right]$ and its associated finite-rank operator A_P. □

Weak settling constrains only the scalar co-moments $\langle A_P\Phi, \Phi' \rangle$; it does not by itself give a norm limit for $A_P\Phi$.

Lemma D.4 (Rich characteristics make $\mathcal{D}$ infinite-dimensional) *If* $\Sigma_N(X_t)$ *is positive definite almost surely,* $E[\operatorname{tr}\Sigma_N(X_t)] < \infty$*, and the pooled characteristic law* p *has infinite support (equivalently, the fixed-*N *panel* X_t *has infinite support), then* $\dim \mathcal{D} = \infty$*.*

Proof of Lemma D.4. Let $\mathcal{S}_b \subset L^2(p)$ be the vector space of simple functions. The moment assumption makes $J_N\psi := \Phi_\psi$ square integrable for every $\psi \in \mathcal{S}_b$, because

$$E[\Phi_\psi^2] \le N^{-2}E[\|\psi_N(X_t)\|^2 \operatorname{tr}\Sigma_N(X_t)] \le N^{-1}\|\psi\|_\infty^2 E[\operatorname{tr}\Sigma_N(X_t)] < \infty.$$

Thus $J_N\mathcal{S}_b \subset \mathcal{D}$ by (D.1). If $J_N\psi = 0$ in L^2, then the law of iterated expectations and (1) give

$$0 = E[\Phi_\psi^2] = \frac{1}{N^2}E[\psi_N(X_t)'\Sigma_N(X_t)\psi_N(X_t)]\,.$$

The integrand is nonnegative, and $\Sigma_N(X_t)$ is positive definite almost surely by the nondegeneracy imposed after (1). Hence $\psi(X_{i,t}) = 0$ almost surely for every i. Since $p =$

$N^{-1}\sum_i \mathrm{law}(X_{i,t})$, this is equivalent to $\psi = 0$ in $L^2(p)$. Thus J_N is injective on $\mathcal{S}_b$. Infinite support of p supplies arbitrarily many disjoint positive-mass sets, whose indicators are linearly independent; hence $\dim \mathcal{S}_b = \infty$ and therefore $\dim \mathcal{D} = \infty$. We neither define nor need J_N on all of $L^2(p)$: that stronger assertion would require an additional operator-domain bound. Notice that asymptotic completeness is not used in this dimension argument; it is a separate condition on the settled form. □

Remark D.5 (The managed-payoff space of one stock) For $N = 1$, define the measure $\nu_r(dx) = E[r_{t+1}^2 \mid X_t = x]P_X(dx)$. It is σ-finite. The map $f \mapsto f(X_t)r_{t+1}$ satisfies

$$E[(f(X_t)r_{t+1})^2] = \int f(x)^2 \, \nu_r(dx).$$

It is therefore an isometry from $L^2(\nu_r)$ onto the managed-payoff space, which has infinite dimension exactly when ν_r has infinite support. Under the standing positive conditional second moment, ν_r and P_X have the same null sets, so infinite support of characteristics suffices. Lemma D.4 gives the corresponding result for multiple stocks.

The infinite-support hypothesis of Lemma D.4 is not a restriction on the menu and cannot be dropped: if the characteristics take finitely many values then $L^2(p)$ is finite-dimensional, $\mathcal{D}$ is too, and $\alpha_j = 0$ for every j beyond $\dim \mathcal{D}$, whatever the menu. The retained-exposure condition (C.4) does not itself exclude this degeneracy, so Assumption 1 explicitly requires arbitrarily many independent square-integrable managed returns. Lemma D.4 supplies a sufficient condition for this requirement; Theorem 1 imposes payoff variety directly and does not require $E[\mathrm{tr}\,\Sigma_N(X_t)] < \infty$.

Lemma D.6 (Completeness and positive lower bounds) *Under* (D.3), $\alpha_j > 0$ *for every* j *exactly when the settled form* q *has infinite rank. If* $\dim \mathcal{D} = \infty$, *strong settling* (C.2) *and asymptotic completeness* (C.4) *imply this infinite-rank condition.*

Proof of Lemma D.6. Each q_P is symmetric and positive semidefinite, so q is too, being its pointwise limit.

The first assertion uses only (D.3); the second needs the representation of q that (C.2) supplies, since (C.4) is a statement about A^Φ.

Asymptotic completeness is positive definiteness of q. Cauchy–Schwarz for a positive semidefinite form gives $q(\Phi,\Phi')^2 \le q(\Phi,\Phi)\,q(\Phi',\Phi')$, so $q(\Phi,\Phi)=0$ forces $\langle A^\Phi,\Phi'\rangle = q(\Phi,\Phi')=0$ for every $\Phi'\in\mathcal{D}$. Each $A_P\Phi$ is a finite combination of the R_k and so lies in $\mathcal{D}$; hence A^Φ lies in the closure of $\mathcal{D}$, and by continuity the display extends from $\mathcal{D}$ to that closure, giving $\|A^\Phi\|^2=0$. Conversely $A^\Phi=0$ gives $q(\Phi,\Phi)=0$. So (C.4) says exactly that $q(\Phi,\Phi)>0$ for every nonzero $\Phi\in\mathcal{D}$.

Infinite rank is what positivity needs. Write $\ker q = \{\Phi\in\mathcal{D} : q(\Phi,\Phi)=0\}$, a subspace by the Cauchy–Schwarz step, and $\operatorname{rank} q := \dim(\mathcal{D}/\ker q)$. If $\operatorname{rank} q \ge j$, choose $\Phi_1,\dots,\Phi_j\in\mathcal{D}$ independent modulo $\ker q$ and set $V=\operatorname{span}\{\Phi_1,\dots,\Phi_j\}$, so that $V\cap\ker q=\{0\}$. On V the map $\Phi\mapsto q(\Phi,\Phi)$ is a quadratic form on a finite-dimensional space, hence continuous, and strictly positive off the origin; its minimum over the compact unit sphere of V is attained and strictly positive, so $\alpha_j \ge \min_{\Phi\in V,\|\Phi\|=1} q(\Phi,\Phi)>0$ by (D.4). Conversely, if $\operatorname{rank} q = m<\infty$ then every V of dimension $j>m$ meets $\ker q$ nontrivially, the minimum in (D.4) is zero for every such V, and $\alpha_j=0$. So $\alpha_j>0$ for every j exactly when $\operatorname{rank} q=\infty$.

The dimension hypothesis. Completeness makes $\ker q=\{0\}$, so $\operatorname{rank} q=\dim\mathcal{D}=\infty$ under the stated hypothesis. Lemma D.4 supplies infinite dimension from infinite characteristic support. □

Lemma D.7 (Weak settling gives the lower bound) *Under (D.3), for every fixed j,*

$$\liminf_{P\to\infty}\ \frac{\lambda_j(\Psi_P)}{P}\ \ge\ \alpha_j. \tag{D.10}$$

Proof of Lemma D.7. By Lemma D.3, $\lambda_j(\Psi_P)/P=\lambda_j(A_P)$, and $\langle A_P\Phi,\Phi\rangle = q_P(\Phi,\Phi)\ge 0$. The claim is immediate if $\alpha_j=0$. Otherwise take $0<a<\alpha_j$. By (D.4) there is a j-

dimensional $V \subseteq \mathcal{D}$ with $\min_{\Phi \in V,\, \|\Phi\|=1} q(\Phi, \Phi) > a$. The Courant–Fischer min–max principle gives

$$\frac{\lambda_j(\Psi_P)}{P} = \lambda_j(A_P) \;\geq\; \min_{\Phi \in V,\; \|\Phi\|=1} q_P(\Phi, \Phi).$$

On the finite-dimensional V, convergence of the finitely many entries $q_P(e_m, e_n)$, $m, n \leq j$, in a basis of V gives, by bilinearity, uniform convergence of q_P to q on the unit sphere of V, so the right-hand side tends to a limit $> a$. Hence $\liminf_P \lambda_j(\Psi_P)/P \geq a$, and letting $a \uparrow \alpha_j$ gives (D.10). □

Lemma D.8 (Uniform exposure on fixed finite spans) *Under Assumptions 1 and 2, for every j there exist a j-dimensional space V_j of fixed weightings with uniformly bounded return second moments, $c_j > 0$, and $N_j < \infty$ such that*

$$0 < c_j E[(R^{\phi}_{N,t+1})^2] \leq q_N(\Phi_{N,\phi}, \Phi_{N,\phi}), \qquad \phi \in V_j \setminus \{0\}, \quad N \geq N_j. \tag{D.11}$$

Proof of Lemma D.8. Let $\mathcal{F}$ be the vector space of fixed measurable weightings with

$$\sup_N E[(R^{\phi}_{N,t+1})^2] < \infty.$$

For $\phi, \psi \in \mathcal{F}$, put $\mathfrak{G}_N(\phi, \psi) = E[R^{\phi}_{N,t+1} R^{\psi}_{N,t+1}]$. Condition (ii) of Assumption 2 implies $\mathfrak{G}_N(\phi, \phi) \to \mathfrak{G}_\infty(\phi, \phi) := E[v_\phi(W)]$. Polarization gives a positive bilinear limit $\mathfrak{G}_\infty$ on $\mathcal{F}$. Condition (i) supplies a j-dimensional $V_j \subset \mathcal{F}$ on which $\mathfrak{G}_\infty$ is positive definite. On this finite-dimensional space, convergence of the matrix entries gives, for all sufficiently large N,

$$\tfrac{1}{2}\mathfrak{G}_\infty(\phi, \phi) \leq \mathfrak{G}_N(\phi, \phi) \leq 2\mathfrak{G}_\infty(\phi, \phi), \qquad \phi \in V_j.$$

Write $b_N(\phi, \psi) = q_N(\Phi_{N,\phi}, \Phi_{N,\psi})$. The uniform bound in Assumption 1 gives $0 \leq b_N(\phi, \phi) \leq C_{\text{menu}} \mathfrak{G}_N(\phi, \phi)$. Consequently, the quadratic forms b_N are equicontinuous on the compact

sphere $\{\phi \in V_j : \mathfrak{G}_\infty(\phi,\phi) = 1\}$: by Cauchy–Schwarz for a positive form,

$$|b_N(\phi,\phi) - b_N(\psi,\psi)| \le 2C_{\text{menu}}\|\phi-\psi\|_{\mathfrak{G}_\infty}(\|\phi\|_{\mathfrak{G}_\infty} + \|\psi\|_{\mathfrak{G}_\infty}).$$

Suppose their minima on this sphere were not bounded away from zero. There would then be $N_\ell \to \infty$ and sphere points $\phi_\ell \to \phi_*$ such that $b_{N_\ell}(\phi_\ell,\phi_\ell) \to 0$. Equicontinuity would imply $b_{N_\ell}(\phi_*,\phi_*) \to 0$. But $\mathfrak{G}_N(\phi_*,\phi_*) \to 1$, and Condition (iii) of Assumption 2 gives $\liminf_N b_N(\phi_*,\phi_*)/\mathfrak{G}_N(\phi_*,\phi_*) > 0$, a contradiction. Thus $b_N(\phi,\phi) \ge \delta_j \mathfrak{G}_\infty(\phi,\phi)$ eventually for some $\delta_j > 0$. Taking $c_j = \delta_j/2$ proves (D.11), including its strict positivity. □

The bound (D.11) is therefore a consequence of the stated main-text assumptions. The economy supplies nondegenerate limiting returns; pointwise retained exposure and the uniform menu bound then prevent an increasingly weak combination within any fixed finite collection. Weightings with uniformly bounded second moments form a linear class, so no separate admissibility condition on their combinations is needed.

Proof of Theorem 1, the bound (10). Write $\Psi_{P,N}$ for Ψ_P in the N-stock economy. Assumption 1 gives (C.2), which by Lemma D.3 implies (D.3) with the settled form $q(\Phi,\Phi') = \langle A^\Phi, \Phi'\rangle$ on $\mathcal{D}$. At a given N, denote its max–min values by $\alpha_{N,j}$. Lemma D.7 gives $\liminf_P \lambda_j(\Psi_P)/P \ge \alpha_{N,j}$ for every fixed j. The payoff variety in Assumption 1 gives $\dim \mathcal{D} = \infty$. Together with asymptotic completeness (C.4), Lemma D.6 therefore gives $\alpha_{N,j} > 0$ for every j. Moreover, for every unit $\Phi \in \mathcal{D}$,

$$q(\Phi,\Phi) = \langle A^\Phi, \Phi\rangle \le \|A^\Phi\|_{L^2}\|\Phi\|_{L^2} \le C_{\text{menu}},$$

so (D.4) gives $\alpha_{N,j} \le C_{\text{menu}}$ uniformly in N. For the finite-menu upper bound, every nonzero eigenvector of A_P lies in $\text{range}(A_P) \subseteq \mathcal{D}$. Hence the finite-$P$ bound in (C.3) gives $\|A_P\|_{\text{op}} \le$

C_{menu} on its nonzero spectral subspace. By Lemma D.3,

$$\frac{\lambda_j(\Psi_P)}{P} = \lambda_j(A_P) \le C_{\text{menu}} \qquad \text{for every } j, P, N.$$

If $N_{\max} < \infty$, the main-text constant $\lambda_j^* := \liminf_{P\to\infty} \lambda_j(\Psi_{P,N_{\max}})/P$ therefore lies in $[\alpha_{N_{\max},j}, C_{\text{menu}}]$. If $N_{\max} = \infty$, Assumption D.2 implies Assumption 2. Fix j and use the space V_j supplied by Lemma D.8. Its image $V_{N,j} := \{\Phi_{N,f} : f \in V_j\}$ has dimension j for $N \ge N_j$, and (D.11) gives

$$\alpha_{N,j} \ge \min_{\Phi\in V_{N,j},\ \|\Phi\|_{L^2}=1} q_N(\Phi,\Phi) \ge c_j.$$

Consequently,

$$0 < c_j \le \lambda_j^* := \liminf_{N\to\infty} \liminf_{P\to\infty} \frac{\lambda_j(\Psi_{P,N})}{P} \le C_{\text{menu}}.$$

The sequence λ_j^* is nonincreasing because every finite-menu spectrum is ordered. This proves (10). Lemma D.8 derives the bound from the economy's limiting payoff variety and the menu's pointwise exposure. □

The growth of each eigenvalue at a fixed position follows from retained exposure to fixed arbitrary managed portfolios. For a fixed nonzero R_{t+1}^{ϕ}, the settling and retained-exposure conditions give

$$\frac{1}{P}\sum_{k=1}^{P} R_{k,t+1} E[R_{k,t+1} R_{t+1}^{\phi}] \longrightarrow A^{\phi} \quad \text{in } L^2, \qquad A^{\phi} \neq 0. \tag{D.12}$$

The connection to Ψ_P can be seen by taking a combination of the current menu returns:

$$\sum_{k=1}^{P} R_{k,t+1} E\Big[R_{k,t+1} \sum_{l=1}^{P} a_l R_{l,t+1}\Big] = \sum_{k=1}^{P} R_{k,t+1} \tilde{a}_k, \qquad \tilde{a} = \Psi_P a. \tag{D.13}$$

The menu's response to this combination is recorded by the second-moment matrix. To obtain a bound for the eigenvalue at position j, the proof starts from j independent arbitrary managed portfolios held fixed as P grows. The menu retains every combination of those returns, which gives a positive lower bound at that position. Infinite characteristic variety lets us repeat the argument for any fixed j. This does not say that all P eigenvalues are large at once; the size of the bound can decline with j. Section 5.3 develops that distinction.

The bound (D.10) is one-sided. For an orthonormal sequence e_P, take $A_P = e_P\langle e_P, \cdot\rangle$. Then $\lambda_1(A_P) = 1$ for every P, whereas $A_P\Phi \to 0$ for every fixed Φ. Thus even strong settling alone need not preserve spectral values. Kernel convergence (D.6) rules this out: it implies convergence in Hilbert–Schmidt norm and hence in operator norm. Lemma A.7 then gives $\lambda_j(A_P) \to \lambda_j(T_{\mathcal{C}})$ for every fixed j.

E Portfolio Kernels and Spectral Limits

This appendix represents a random-feature menu by its portfolio kernel. It shows that the lower bound of Theorem 1 survives at fixed N under a Hilbert–Schmidt condition weaker than the integrability condition (C.5), and it reduces the kernel spectrum to an eigenvalue problem in characteristics.

E.1 The portfolio kernel and the integrability conditions

For labels $\theta \sim \mu_\theta$, define the matrix kernel $k(X, \tilde{X}) := \int f(X;\theta) f(\tilde{X};\theta)'\, \mu_\theta(d\theta)$. Because the weight on stock i depends only on its own characteristics, $f(X;\theta) = (f(x_i;\theta))_{i=1}^N$, the entries of this $N \times N$ matrix are values of a single scalar kernel k_0 of two stocks' characteristics,

$$\begin{aligned} k_0(z,u) &= \int f(z;\theta)\, f(u;\theta)\, \mu_\theta(d\theta)\,, \\ \big[k(X,\tilde{X})\big]_{ij} &= \int f(x_i;\theta)\, f(\tilde{x}_j;\theta)\, \mu_\theta(d\theta) \;=\; k_0(x_i,\tilde{x}_j)\,, \end{aligned} \tag{E.1}$$

that is, $k(X, \tilde{X}) = \big(k_0(x_i, \tilde{x}_j)\big)_{i,j=1}^{N}$.

Write $Z = (r, X)$ for a generic copy of (r_{t+1}, X_t) and define the scalar managed-return map

$$g(Z;\theta) \;=\; \tfrac{1}{N}\, f(X;\theta)' r\,, \tag{E.2}$$

so that $R_{k,t+1} = g(Z_{t+1};\theta_k)$. Write $g_\theta := g(\cdot;\theta)$ when viewing a managed return as an element of $L^2(\mathbb{P})$. The associated positive semidefinite scalar kernel (the "portfolio kernel" of Kelly et al., 2026) and its integral operator on L^2 of the law of Z are

$$K(Z, \tilde{Z}) \;=\; \int g(Z;\theta)\, g(\tilde{Z};\theta)\, \mu_\theta(d\theta) \;=\; \tfrac{1}{N^2}\, r' k(X, \tilde{X}) \tilde{r}\,, \tag{E.3}$$

and

$$(T_K \phi)(Z) \;=\; E\Big[K(Z, \tilde{Z})\, \phi(\tilde{Z})\Big]\,, \tag{E.4}$$

respectively. Here $\tilde{Z}$ is an independent copy of Z.

We impose the diagonal condition when invoking mixture representations:

$$k_0(z, z) = \int f(z;\theta)^2\, \mu_\theta(d\theta) < \infty \quad p\text{-almost everywhere.} \tag{E.5}$$

It makes $K(Z, Z) = N^{-2} r' k(X, X) r$ finite almost surely. The integrability condition (C.5) for random-feature menus requires the managed portfolios to have finite second moment on average over the label. By Tonelli's theorem and conditioning on X, it is equivalent to the trace condition

$$E[\mathrm{tr}(k(X, X)\Sigma_N(X))] \;=\; \int E\big[(f(X;\theta)' r)^2\big]\, \mu_\theta(d\theta) \;<\; \infty. \tag{E.6}$$

We replace it by the following weaker requirement.

Assumption E.1 (Hilbert–Schmidt portfolio kernel) Let $\tilde{X}$ be an independent copy of X. We have

$$\operatorname{tr} E\Big[k(X,\tilde{X})\,\Sigma_N(\tilde{X})\,k(\tilde{X},X)\,\Sigma_N(X)\Big] \;<\; \infty\,. \tag{E.7}$$

Furthermore,

$$E[(f(X;\theta)'r)^2] \;<\; \infty \tag{E.8}$$

for $\mu_\theta(d\theta)$-almost every θ.

Let $Z, \tilde{Z}$ be independent copies of (r, X). Using $E[r\,r' \mid X] = \Sigma_N(X)$ and the cyclicity of the trace,

$$\begin{aligned} E\Big[K(Z,\tilde{Z})^2\Big] &= \frac{1}{N^4}\,E\Big[r'k(X,\tilde{X})\tilde{r}\,\tilde{r}'k(\tilde{X},X)r\Big] \\ &= \frac{1}{N^4}\,E\Big[r'k(X,\tilde{X})\,\Sigma_N(\tilde{X})\,k(\tilde{X},X)\,r\Big] \\ &= \frac{1}{N^4}\,\operatorname{tr} E\Big[k(X,\tilde{X})\,\Sigma_N(\tilde{X})\,k(\tilde{X},X)\,rr'\Big] \\ &= \frac{1}{N^4}\,\operatorname{tr} E\Big[k(X,\tilde{X})\,\Sigma_N(\tilde{X})\,k(\tilde{X},X)\,\Sigma_N(X)\Big]\,, \end{aligned} \tag{E.9}$$

where N^{-4} is the square of the N^{-2} normalization in the portfolio kernel (E.3).

Lemma E.2 *Let K be a measurable symmetric kernel, finite and positive semidefinite on $\mathcal{Z}_0 \times \mathcal{Z}_0$ for a $\mathbb{P}$-full set $\mathcal{Z}_0$. If $E[K(Z,\tilde{Z})^2] < \infty$ for independent $Z, \tilde{Z} \sim \mathbb{P}$, then the integral operator T_K on $L^2(\mathbb{P})$ is Hilbert–Schmidt, compact, self-adjoint, and positive.*

For the portfolio kernel $K(Z,\tilde{Z}) = N^{-2}r'k(X,\tilde{X})\tilde{r}$, take $p = N^{-1}\sum_i \operatorname{law}(X_i)$. The lemma applies whenever k_0 is finite and positive semidefinite on $\mathcal{X}_0 \times \mathcal{X}_0$ for a p-full set $\mathcal{X}_0$,

and K is square integrable. Indeed, every stock characteristic lies in $\mathcal{X}_0$ almost surely, and

$$\sum_{u,v=1}^{n} a_u a_v K(Z_u, Z_v) = \frac{1}{N^2} \sum_{u,v=1}^{n} \sum_{i,j=1}^{N} a_u r_{u,i} a_v r_{v,j} k_0(x_{u,i}, x_{v,j}) \geq 0.$$

Symmetry follows from $k(\tilde{X}, X) = k(X, \tilde{X})'$.

Proof of Lemma E.2. Square integrability gives $\|T_K\|_{\mathrm{HS}} = \|K\|_{L^2} < \infty$, and symmetry makes T_K self-adjoint. To prove positivity, set $\mathcal{S}_b = \{Z \in \mathcal{Z}_0 : K(Z, Z) \leq b\}$ and let $f \in L^2(\mathbb{P})$ vanish off $\mathcal{S}_b$. For independent copies $Z_1, \ldots, Z_n$, positive semidefiniteness gives

$$0 \leq E\left[\frac{1}{n^2} \sum_{u,v=1}^{n} f(Z_u) f(Z_v) K(Z_u, Z_v)\right] = \left(1 - \frac{1}{n}\right) \langle f, T_K f \rangle + \frac{1}{n} E[f(Z)^2 K(Z, Z)].$$

The diagonal terms are bounded by bf^2, and the off-diagonal terms are integrable by Cauchy–Schwarz using $K \in L^2(\mathbb{P} \otimes \mathbb{P})$. Letting $n \to \infty$ proves $\langle f, T_K f \rangle \geq 0$. The union $\bigcup_b L^2(\mathcal{S}_b)$ is dense because the diagonal is finite almost surely. Boundedness extends positivity to all of $L^2(\mathbb{P})$. □

Lemma E.3 *Under the mixture representation $K(z, \tilde{z}) = \int_\Theta g_\theta(z)\, g_\theta(\tilde{z})\, \mu_\theta(d\theta)$ with the normalized managed return $g_\theta(z) = \frac{1}{N} f(X; \theta)' r$ of (E.2), if $E[K(Z, Z)] < \infty$ then T_K is trace-class, with*

$$\begin{aligned} \operatorname{tr} T_K = \sum_j \lambda_j = E[K(Z, Z)] &= \frac{1}{N^2} E\big[\operatorname{tr}\big(k(X, X)\, \Sigma_N(X)\big)\big] \\ &= \frac{1}{N^2} \int_\Theta E\big[(f(X; \theta)' r)^2\big]\, \mu_\theta(d\theta) < \infty. \end{aligned} \tag{E.10}$$

The portfolio normalization makes the trace of T_K equal to N^{-2} times the quantity in (E.6); their finiteness is equivalent.

Proof of Lemma E.3. The feature representation gives $|K(z, \tilde{z})|^2 \leq K(z, z) K(\tilde{z}, \tilde{z})$, so $E[K(Z, \tilde{Z})^2] \leq (E[K(Z, Z)])^2 < \infty$. Thus T_K is bounded and Hilbert–Schmidt from the

outset. For every $f \in L^2(\mathbb{P})$, Cauchy–Schwarz gives

$$\int_\Theta \left(\int |f(z) g_\theta(z)| \, d\mathbb{P}(z) \right)^2 \mu_\theta(d\theta) \leq \|f\|_2^2 E[K(Z,Z)] < \infty.$$

Fubini therefore yields $\langle f, T_K f \rangle = \int \langle f, g_\theta \rangle^2 \mu_\theta(d\theta) \geq 0$. For any orthonormal basis (e_j), Tonelli and Parseval (Lemmas A.3 and A.4) give

$$\sum_j \langle e_j, T_K e_j \rangle = \int_\Theta \sum_j \langle e_j, g_\theta \rangle^2 \mu_\theta(d\theta) = \int_\Theta \|g_\theta\|_2^2 \mu_\theta(d\theta) = E[K(Z,Z)].$$

The finite diagonal sum of the positive operator proves trace class and the trace identity. The remaining equalities follow by conditioning rr' on X and using $g_\theta(Z) = N^{-1} f(X;\theta)' r$. □

Lemma E.4 *Write $M(\theta,\theta) := E[g(Z;\theta)^2] \in [0,\infty]$ and $\Theta_\infty := \{\theta : M(\theta,\theta) < \infty\}$. For $(\theta,\tilde\theta) \in \Theta_\infty^2$, define $M(\theta,\tilde\theta) := E[g(Z;\theta) g(Z;\tilde\theta)]$. Under (E.7) and (E.5),*

$$\int_{\Theta_\infty} \int_{\Theta_\infty} M(\theta,\tilde\theta)^2 \, \mu_\theta(d\theta) \, \mu_\theta(d\tilde\theta) \;\leq\; E\big[K(Z,\tilde Z)^2\big] \;<\; \infty\,, \tag{E.11}$$

so M is square integrable on $\Theta_\infty \times \Theta_\infty$. Suppose in addition that θ ranges over $\mathbb{R}^L$, that $\theta \mapsto g(Z;\theta)$ is continuous for $\mathbb{P}$-almost every Z, and that the off-diagonal products are bounded in L^1 locally uniformly: for every bounded open $U \subseteq \mathbb{R}^L$,

$$C_U \;:=\; \sup\big\{\, E\big|g(Z;\theta)\, g(Z;\tilde\theta)\big| \;:\; \theta,\tilde\theta \in U,\; \theta \neq \tilde\theta \,\big\} \;<\; \infty\,. \tag{E.12}$$

Then $M(\theta,\theta) \leq C_U < \infty$ for every $\theta \in U$; that is, (E.8) holds for every θ.

Proof of Lemma E.4. We truncate the label set and compare operators. Put $\Theta_n := \{\theta : M(\theta,\theta) \leq n\}$, so $\Theta_n \uparrow \Theta_\infty$, and let $K_n(Z,\tilde Z) := \int_{\Theta_n} g_\theta(Z) g_\theta(\tilde Z) \, \mu_\theta(d\theta)$.

Step 1: the identity on the truncation. For $\theta, \tilde{\theta} \in \Theta_n$, Cauchy–Schwarz gives $E|g_\theta g_{\tilde{\theta}}| \le \sqrt{M(\theta,\theta)M(\tilde{\theta},\tilde{\theta})} \le n$, so the integrand is absolutely integrable over $\Theta_n^2 \times \mathcal{Z}^2$ and Fubini applies:

$$\int_{\Theta_n}\int_{\Theta_n} M^2 \, \mu_\theta(d\theta)\, \mu_\theta(d\tilde{\theta}) \;=\; E\big[K_n(Z,\tilde{Z})^2\big] \;\le\; n^2 \;<\; \infty, \tag{E.13}$$

using independence of Z and $\tilde{Z}$ to factor the inner expectation into $M(\theta,\tilde{\theta})^2$. In particular $K_n \in L^2(\mathbb{P}\otimes\mathbb{P})$, so T_{K_n} is Hilbert–Schmidt, and so is $T_K - T_{K_n} = T_{K-K_n}$.

Step 2: both pieces are positive. Let $\mathcal{S}_m = \{K(Z,Z) \le m\}$. For ϕ vanishing off $\mathcal{S}_m$, the bound $\int_\Theta |g_\theta(z) g_\theta(u)| \mu_\theta(d\theta) \le \sqrt{K(z,z)K(u,u)} \le m$ licenses Fubini, so $\langle \phi, T_{K_n}\phi\rangle = \int_{\Theta_n} \langle \phi, g_\theta\rangle^2 \mu_\theta(d\theta) \ge 0$ and likewise $\langle \phi, T_{K-K_n}\phi\rangle = \int_{\Theta\setminus\Theta_n} \langle \phi, g_\theta\rangle^2 \mu_\theta(d\theta) \ge 0$. Condition (E.5) makes the quadratic form $K(Z,Z) = N^{-2} r' k(X,X) r$ finite almost surely. Thus the sets $\mathcal{S}_m$ exhaust $\mathcal{Z}$, $\bigcup_m L^2(\mathcal{S}_m)$ is dense in $L^2(\mathbb{P})$, and both forms are bounded; hence $0 \preceq T_{K_n} \preceq T_K$.

Step 3: ordering controls the Hilbert–Schmidt norm. If $0 \preceq A \preceq B$ are Hilbert–Schmidt, then $\operatorname{tr}\big(A(B-A)\big) \ge 0$. To see this, note first that a product of two Hilbert–Schmidt operators is trace class, so the trace is defined and absolutely convergent. Writing $C := B - A \succeq 0$, which is Hilbert–Schmidt hence compact, diagonalize $C = \sum_k \mu_k \, e_k \otimes e_k$ with $\mu_k \ge 0$ and $\{e_k\}$ orthonormal; then $\operatorname{tr}(AC) = \sum_k \langle e_k, ACe_k\rangle = \sum_k \mu_k \langle e_k, Ae_k\rangle \ge 0$ since $A \succeq 0$. Consequently $\operatorname{tr} A^2 \le \operatorname{tr}(AB) \le (\operatorname{tr} A^2)^{1/2} (\operatorname{tr} B^2)^{1/2}$ by Cauchy–Schwarz for the Hilbert–Schmidt inner product, whence $\operatorname{tr} A^2 \le \operatorname{tr} B^2$. Applied to $T_{K_n} \preceq T_K$, this is $E[K_n^2] \le E[K^2]$.

Step 4: pass to the limit. Combining with (E.13), $\int_{\Theta_n}\int_{\Theta_n} M^2 \, \mu_\theta(d\theta)\, \mu_\theta(d\tilde{\theta}) \le E[K^2]$ for every n, and monotone convergence as $\Theta_n \uparrow \Theta_\infty$ gives (E.11). A square-integrable function is finite almost everywhere, which is the first claim.

What this argument does not reach is the part of the label space outside Θ_∞, where $M(\theta,\theta) = \infty$ and Cauchy–Schwarz gives no control of the off-diagonal entries. Extending (E.11) to all of Θ^2 would need the interchange of E with $\int \mu_\theta(d\theta)\mu_\theta(d\tilde{\theta})$, and a sufficient

condition for that interchange at fixed N is $\iint (E|g_\theta g_{\tilde\theta}|)^2 \, \mu_\theta(d\theta) \, \mu_\theta(d\tilde\theta) < \infty$. The trace condition (C.5) would be sufficient by Cauchy–Schwarz, but is not assumed here. The second claim supplies a sufficient condition for (E.8), namely $\mu_\theta(\Theta_\infty) = 1$. By (E.9) the right-hand side of (E.11) equals

$$N^{-4} \operatorname{tr} E[k(X, \tilde X)\Sigma_N(\tilde X)k(\tilde X, X)\Sigma_N(X)],$$

finite under (E.7). Once the second claim establishes $\mu_\theta(\Theta_\infty) = 1$, it follows that $M \in L^2(\mu_\theta \otimes \mu_\theta)$, and in particular $M(\theta, \tilde\theta)$ is finite for $(\mu_\theta \otimes \mu_\theta)$-almost every $(\theta, \tilde\theta)$.

For the second claim, fix a bounded open $U \subseteq \mathbb{R}^L$ and $\theta_0 \in U$, and take $\theta_n \in U$ with $\theta_n \to \theta_0$ and $\theta_n \neq \theta_0$ for every n. By the assumed continuity in θ, for $\mathbb{P}$-almost every Z

$$\big|g(Z;\theta_0)\, g(Z;\theta_n)\big| \xrightarrow[n\to\infty]{} g(Z;\theta_0)^2 \geq 0\,,$$

so Fatou's lemma and (E.12) give

$$M(\theta_0,\theta_0) = E\big[g(Z;\theta_0)^2\big] \leq \liminf_{n\to\infty} E\big|g(Z;\theta_0)\, g(Z;\theta_n)\big| \leq C_U < \infty\,,$$

which is (E.8) at θ_0. □

Lemma E.4 controls the off-diagonal label kernel on Θ_∞^2 up to a $(\mu_\theta \otimes \mu_\theta)$-null set. The square-integrability of the portfolio kernel does not imply that Θ_∞ has full measure. The following construction verifies the failure within the managed-return model itself. Let $N = 1$, $r = 1$, and let X take distinct values $x_{k,a}$, $k \geq 1$, $a \in \{0, 1\}$, with probabilities $p_{k,a} = 2^{-k-1}$. For $\theta \sim \mathrm{U}[0, 1]$, set

$$f(x_{k,0};\theta) = \frac{\cos(2\pi k\theta)}{\sqrt{kp_{k,0}}}, \qquad f(x_{k,1};\theta) = \frac{\sin(2\pi k\theta)}{\sqrt{kp_{k,1}}}.$$

The feature functions are continuous in the label, and orthogonality of the trigonometric

functions gives the portfolio kernel

$$K(x_{k,a}, x_{l,b}) = \frac{\delta_{kl}\delta_{ab}}{2kp_{k,a}}, \qquad \|T_K\|_{\mathrm{HS}}^2 = 2\sum_{k\geq 1}\frac{1}{(2k)^2} < \infty. \tag{E.14}$$

Its diagonal is finite at every characteristic value. Nevertheless,

$$E[f(X;\theta)^2] = \sum_{k\geq 1}\frac{\cos^2(2\pi k\theta) + \sin^2(2\pi k\theta)}{k} = \infty \qquad \text{for every } \theta.$$

Thus (E.7) and diagonal finiteness do not imply (E.8), even for pointwise continuous label features. The local bound (E.12) necessarily fails, by the Fatou argument in Lemma E.4.

The next theorem weakens the random-feature trace condition to Assumption E.1. Its conclusion is a lower bound rather than an exact limit.

Theorem E.5 (The eigenvalue cliff fails at finite N: extension) *Let N be fixed, let θ_k be i.i.d. with law μ_θ and independent of the panel, and set $f_k = f(\cdot;\theta_k)$. Suppose Assumption E.1 holds, $K(Z,Z) < \infty$ almost surely, and $\operatorname{rank} T_K = \infty$. Then the positive eigenvalues of T_K $\tau_1 \geq \tau_2 \geq \cdots > 0$ satisfy $\sum_j \tau_j^2 < \infty$ and, almost surely,*

$$\liminf_{P\to\infty}\frac{\lambda_j(\Psi_P)}{P} \geq \tau_j > 0 \qquad \textit{for every fixed } j. \tag{E.15}$$

Every fixed-index eigenvalue therefore diverges, so a fixed factor count cannot leave a uniformly bounded residual spectrum.

Proof of Theorem E.5. Recall the scalar managed-return map (E.2), the portfolio kernel (E.3), and the associated integral operator T_K in (E.4). Write $\tau_j = \lambda_j(T_K)$ for the eigenvalues of T_K; these are the constants in the statement. They are lower bounds here; when the trace condition holds, they equal the exact limits $\lim_{P\to\infty}\lambda_j(\Psi_P)/P$, the constants λ_j^* of Theorem 1 at $N_{\max} = N$.

Step 1: T_K is Hilbert–Schmidt. Equation (E.9) and Assumption E.1 give $K \in L^2(\mathbb{P}\otimes\mathbb{P})$. The mixture representation (E.3), the almost-sure finiteness of $K(Z,Z)$, and Lemma A.9 make K finite and positive semidefinite on a $\mathbb{P}$-full set. Lemma E.2 therefore makes T_K Hilbert–Schmidt, self-adjoint, and positive, with

$$\sum_j (\tau_j)^2 \;=\; \operatorname{tr} T_K^2 \;=\; \frac{1}{N^4}\operatorname{tr} E\Big[k(X,\tilde{X})\,\Sigma_N(\tilde{X})\,k(\tilde{X},X)\,\Sigma_N(X)\Big] \;<\; \infty\,.$$

Finally, the explicit infinite-rank hypothesis and positivity of T_K imply that infinitely many τ_j are strictly positive. When the settled menu form is $\langle\cdot, T_K\cdot\rangle$, Lemma D.6 supplies infinite rank from completeness on an infinite-dimensional managed-payoff space.

Step 2: finite-P spectrum. For the i.i.d. draw $\theta_1,\dots,\theta_P$, consider the Monte Carlo kernel

$$K_P(Z,\tilde{Z}) \;=\; \frac{1}{P}\sum_{k=1}^{P} g(Z;\theta_k)\,g(\tilde{Z};\theta_k)\,. \tag{E.16}$$

The operator T_{K_P} is finite rank, and its nonzero eigenvalues coincide with those of the $P\times P$ Gram matrix with entries $\frac{1}{P}E[g(Z;\theta_k)g(Z;\theta_l)] = \frac{1}{P}E[R_kR_l] = \frac{1}{P}(\Psi_P)_{kl}$. Thus the nonzero eigenvalues of T_{K_P} equal those of $\frac{1}{P}\Psi_P$:

$$\lambda_j(T_{K_P}) \;=\; \frac{\lambda_j(\Psi_P)}{P}\,. \tag{E.17}$$

Step 3: lower bound on the spectrum. All three hypotheses of Lemma A.10 hold: T_K is Hilbert–Schmidt by Step 1; $g(\cdot;\theta)\in L^2(\mathbb{P})$ for μ_θ-almost every θ by (E.8) of Assumption E.1; and the diagonal is finite almost surely by the theorem's explicit hypothesis. Condition (E.5) is one sufficient condition for this last requirement, since then $K(Z,Z) = N^{-2}r'k(X,X)r$ is a finite quadratic form. The lemma applies: comparing T_{K_P} with the compression of its action to the top-j eigenspace of T_K and invoking the strong law of large numbers entrywise,

for every fixed j

$$\liminf_{P\to\infty} \frac{\lambda_j(\Psi_P)}{P} \;=\; \liminf_{P\to\infty} \lambda_j(T_{K_P}) \;\geq\; \lambda_j(T_K) \;=\; \tau_j\,, \tag{E.18}$$

which is (E.15). Since infinitely many $\tau_j > 0$, infinitely many eigenvalues of Ψ_P diverge at least at the linear rate P. □

E.2 From portfolios to characteristics

We restrict T_K to the economically relevant invariant subspace of functions linear in returns, $\Phi(Z) = r'\eta(X)$ with $\eta : \mathbb{R}^{N\times d} \to \mathbb{R}^N$, and integrate the returns out. The restriction discards nothing: every eigenfunction of T_K with a nonzero eigenvalue is linear in r, as the proof of the proposition below shows. Write $\Sigma_N(X) = E[r_{t+1}r'_{t+1} \mid X_t = X]$. For this subsection assume the two-sided bounds $\underline{\sigma}^2 I_N \preceq \Sigma_N(X) \preceq \bar{\Sigma}_N I_N$, which hold by (B.8) in the factor model, and define the symmetrized matrix kernel

$$\tilde{k}(X,\tilde{X}) \;=\; \Sigma_N(X)^{1/2}\, k(X,\tilde{X})\, \Sigma_N(\tilde{X})^{1/2}\,. \tag{E.19}$$

Let $(\tilde{r},\tilde{X})$ be an independent copy of (r,X). Using $E[\tilde{r}\,\tilde{r}' \mid \tilde{X}] = \Sigma_N(\tilde{X})$, the eigenvalue equation $T_K\Phi = \lambda\Phi$ reduces, with the factor $\frac{1}{N^2}$ inherited from the portfolio kernel (E.3), to

$$\tfrac{1}{N^2}\int k(X,\tilde{X})\,\Sigma_N(\tilde{X})\,\eta(\tilde{X})\,P_N(d\tilde{X}) \;=\; \lambda\,\eta(X)\,. \tag{E.20}$$

Equivalently, in terms of the symmetrized weighting $\tilde{\phi}(X) = \Sigma_N(X)^{1/2}\eta(X)$, the eigenvalue equation (E.20) becomes the symmetric problem

$$\tfrac{1}{N^2}\int \tilde{k}(X,\tilde{X})\,\tilde{\phi}(\tilde{X})\,P_N(d\tilde{X}) \;=\; \lambda\,\tilde{\phi}(X)\,, \tag{E.21}$$

generated by the symmetrized kernel $\tilde{k}$ of (E.19); this is the self-adjoint form of the same eigenfunction equation.

Proposition E.6 *At fixed N, suppose Assumption E.1 holds and $\underline{\sigma}^2 I_N \preceq \Sigma_N(X) \preceq \bar{\Sigma}_N I_N$ for P_N-almost every X, with $0 < \underline{\sigma}^2 \leq \bar{\Sigma}_N < \infty$. The nonzero eigenvalues of the portfolio kernel operator (E.4) (restricted to the invariant subspace of functions linear in r) coincide with the nonzero eigenvalues of the symmetric matrix kernel operator*

$$(\tilde{\mathcal{T}}\tilde{\phi})(X) \;=\; \tfrac{1}{N^2}\int \tilde{k}(X,\tilde{X})\,\tilde{\phi}(\tilde{X})\,P_N(d\tilde{X}) \tag{E.22}$$

acting on $L^2(P_N;\mathbb{R}^N)$, the space of square-integrable functions $\tilde{\phi}:\mathbb{R}^{N\times d}\to\mathbb{R}^N$ with

$$\int \|\tilde{\phi}(X)\|^2 P_N(dX) < \infty.$$

The map $\eta \mapsto \Sigma_N(X)^{1/2}\eta$ is a similarity transform: it preserves eigenvalues while making the operator self-adjoint, so the spectrum can be read off a problem in the N characteristics alone.

Proof of Proposition E.6. The map $L\eta = r'\eta(X)$ satisfies

$$\underline{\sigma}^2\|\eta\|^2_{L^2(P_N)} \leq \|L\eta\|^2_{L^2(\mathbb{P})} \leq \bar{\Sigma}_N\|\eta\|^2_{L^2(P_N)}.$$

Its range is therefore closed. The kernel representation implies that T_K annihilates its orthogonal complement, so self-adjointness puts the range of T_K inside this closed linear-return space. In particular, every eigenfunction with a nonzero eigenvalue has the form $r'\eta(X)$ with $\eta \in L^2(P_N;\mathbb{R}^N)$. Equation (E.9) makes T_K and $\tilde{\mathcal{T}}$ Hilbert–Schmidt under the stated condition.

Let $\lambda \neq 0$ be an eigenvalue of T_K with eigenfunction $\Phi(Z) = r'\eta(X)$, so that η solves

(E.20). Multiplying (E.20) on the left by $\Sigma_N(X)^{1/2}$ gives

$$\tfrac{1}{N^2}\int \Sigma_N(X)^{1/2}k(X,\tilde{X})\,\Sigma_N(\tilde{X})^{1/2}\left[\Sigma_N(\tilde{X})^{1/2}\eta(\tilde{X})\right]P_N(d\tilde{X}) \;=\; \lambda\,\Sigma_N(X)^{1/2}\eta(X)\,. \quad \text{(E.23)}$$

With $\tilde{\phi}(X) = \Sigma_N(X)^{1/2}\eta(X)$ and $\tilde{k}$ as in (E.19), this is exactly $\tilde{\mathcal{T}}\tilde{\phi} = \lambda\tilde{\phi}$. Conversely, if $\tilde{\mathcal{T}}\tilde{\phi} = \lambda\tilde{\phi}$ with $\lambda \neq 0$, then $\eta = \Sigma_N(X)^{-1/2}\tilde{\phi}$ solves (E.20). Here almost-sure invertibility of $\Sigma_N(X)$ would not suffice: for $\eta \mapsto \Sigma_N(X)^{1/2}\eta$ to map $L^2(P_N;\mathbb{R}^N)$ onto itself, with an inverse that also maps it into itself, both $\Sigma_N(X)^{1/2}$ and $\Sigma_N(X)^{-1/2}$ must be essentially bounded. This is exactly the uniform ellipticity assumed in the proposition: for P_N-a.e. X,

$$\underline{\sigma}\,\|\eta(X)\| \;\le\; \|\Sigma_N(X)^{1/2}\eta(X)\| \;\le\; \bar{\Sigma}_N^{1/2}\,\|\eta(X)\|\,,$$

so $\eta \in L^2(P_N;\mathbb{R}^N)$ if and only if $\Sigma_N(X)^{1/2}\eta \in L^2(P_N;\mathbb{R}^N)$. The map $\eta \mapsto \Sigma_N(X)^{1/2}\eta$ is therefore an eigenvalue-preserving bijection between the nonzero-eigenvalue eigenspaces of the two operators. Finally, $\tilde{\mathcal{T}}$ is self-adjoint on $L^2(P_N;\mathbb{R}^N)$ because $\tilde{k}(X,\tilde{X})' = \tilde{k}(\tilde{X},X)$. □

F Finite-Cross-Section Menus

Throughout this section, N is fixed. The construction conditions are stated in Appendices C.2.1, C.2.2, and C.2.3. Whenever an exact fixed-index limit exists, we write

$$\lim_{P\to\infty}\frac{\lambda_j(\Psi_P)}{P} = \lambda_j^*(N), \qquad j \text{ fixed.} \quad \text{(F.1)}$$

For partition menus the limit is along the specified sizes P_n. These constants agree with the fixed-N definition in Theorem 1.

F.1 Settling menus and random features

This subsection proves the exact limit (F.1) for settling menus, whose empirical kernel (D.5) converges as in (D.6). Paragraph (i) below derives that convergence for random labels, and paragraph (ii) imposes it for deterministic ones. The managed-payoff space is (D.1).

The constants $\lambda_j^*(N)$ have a simple interpretation. Reweight the limiting menu into combinations with zero pairwise second moments and order them by second moment: $\lambda_j^*(N)$ is the second moment of the j-th such combination, the j-th principal-component portfolio of the menu in the limit. Formally, $\lambda_j^*(N) = \lambda_j(T_{\mathcal{C}})$ is the j-th eigenvalue of the integral operator $T_{\mathcal{C}}$ on $L^2(\mathbb{P})$ with kernel $\mathcal{C}$, the limit in (D.6). Equation (F.1) then says that the j-th eigenvalue of Ψ_P is, for large P, about $P\lambda_j^*(N)$. That infinitely many $\lambda_j^*(N)$ are positive is the statement that the menu keeps finding new return directions with positive second moments however far down the list one looks.

Kernel convergence makes the sequence $\lambda_1^*(N) \geq \lambda_2^*(N) \geq \cdots \geq 0$ square-summable, so the constants decline to zero. The decay bounds their size rather than their number. Infinitely many of them are strictly positive exactly when $T_{\mathcal{C}}$ has infinite rank, which completeness (C.4) on an infinite-dimensional $\mathcal{D}$ supplies (Lemma D.6).

(i) Random labels. Draw the labels $\theta_1, \ldots, \theta_P$ independently from μ_θ and set $f_k = f(\cdot; \theta_k)$. The trace condition (C.5), with $R_\theta = N^{-1} f(X_t; \theta)' r_{t+1}$, then delivers (D.6) through a law of large numbers, with limit $\mathcal{C} = K$. The positive-μ_θ-measure completeness condition in Appendix C.2.1 verifies (C.4). Without (C.5), this route to Assumption 1 and the exact limit is unavailable, although Assumption 1 need not fail. Under the weaker Hilbert–Schmidt condition, Theorem E.5 still gives the lower bound.

(ii) Deterministic labels. Kernel convergence (D.6) may instead be imposed directly. Under $\Sigma_N(X) \preceq \bar{\Sigma}_N I_N$, for a limit $\mathcal{W} \in L^2(p \otimes p)$, let

$$\mathcal{W}_P(u,v) = \frac{1}{P}\sum_{k \le P} f_k^{(P)}(u) f_k^{(P)}(v), \qquad \mathcal{C}(Z,\tilde{Z}) = N^{-2} r' [\mathcal{W}(X_i, \tilde{X}_{i'})]_{i,i'} \tilde{r}.$$

Conditioning as in (E.9) gives

$$\|K_P - \mathcal{C}\|_{L^2(\mathbb{P}\otimes\mathbb{P})} \le \frac{\bar{\Sigma}_N}{N} \|\mathcal{W}_P - \mathcal{W}\|_{L^2(p\otimes p)}. \tag{F.2}$$

If also $\Sigma_N(X) \succeq \underline{\sigma}^2 I_N$, the reverse bound holds with $\underline{\sigma}^2/N$. Thus raw weight-kernel convergence is sufficient under the upper bound and equivalent to kernel convergence under both bounds. Conversely, applying the lower bound to $K_P - K_{P'}$ makes $\mathcal{W}_P$ Cauchy in $L^2(p \otimes p)$. The upper bound then identifies the kernel limit with the displayed $\mathcal{C}$.

Lemma F.1 (Limits of settling menus) *At fixed N, suppose (D.6) holds. For every fixed j,*

$$\lambda_j(\Psi_P)/P \to \lambda_j(T_{\mathcal{C}}).$$

Conditions (C.2)–(C.3) hold with

$$C_{\text{menu}} = \sup_P \|K_P\|_{L^2} < \infty.$$

If $\dim \mathcal{D} = \infty$ and (C.4) holds, every limit is positive. Under the random-feature conditions of Appendix C.2.1, these conclusions and Assumption 1 hold almost surely, with $\mathcal{C} = K$ and $\sum_j \lambda_j(T_K) < \infty$. For i.i.d. labels satisfying (C.5),

$$\sup_{P \ge 1} \frac{\operatorname{tr} \Psi_P}{P} < \infty \qquad \textit{almost surely.} \tag{F.3}$$

Proof of Lemma F.1. The proof establishes the limit (F.1). The argument is at fixed N and uses no structure on $\Sigma_N(X_t)$ or $\mu(X_t)$; the menu may depend on P and may be deterministic or random.

Step 1: the general argument under (D.6). Write $g_k^{(P)}(Z) = \frac{1}{N} f_k^{(P)}(X)'r$ for the managed return generated by the weight $f_k^{(P)}$, as in (E.2). Since $\Psi_{P,jk} = E[g_j^{(P)} g_k^{(P)}] = \langle g_j^{(P)}, g_k^{(P)} \rangle_{L^2(\mathbb{P})}$, Ψ_P is their Gram matrix. By Lemma A.8 its nonzero eigenvalues coincide with those of the finite-rank operator $\mathcal{S}_P = \sum_{k=1}^{P} g_k^{(P)} \otimes g_k^{(P)}$ on $L^2(\mathbb{P})$, and $\frac{1}{P}\mathcal{S}_P$ is exactly the integral operator T_{K_P} with kernel K_P in (D.5), the menu's empirical counterpart of the portfolio kernel (E.3). Hypothesis (D.6) is precisely that $K_P \to \mathcal{C}$ in $L^2(\mathbb{P} \otimes \mathbb{P})$; the limit's operator $T_{\mathcal{C}}$ is then Hilbert–Schmidt, and its eigenvalues $\lambda_1^*(N) \geq \lambda_2^*(N) \geq \cdots \geq 0$ are square-summable, $\sum_j (\lambda_j^*(N))^2 = \|\mathcal{C}\|^2_{L^2(\mathbb{P}\otimes\mathbb{P})} < \infty$. Since the Hilbert–Schmidt norm dominates the operator norm,

$$\left\|T_{K_P} - T_{\mathcal{C}}\right\|_{\text{op}} \;\leq\; \left\|T_{K_P} - T_{\mathcal{C}}\right\|_{\text{HS}} \;=\; \left\|K_P - \mathcal{C}\right\|_{L^2(\mathbb{P}\otimes\mathbb{P})} \;\longrightarrow\; 0\,,$$

we obtain settling on every $\Phi \in L^2(\mathbb{P})$ and, simultaneously for all such Φ,

$$C_{\text{menu}} := \sup_P \|K_P\|_{L^2(\mathbb{P}\otimes\mathbb{P})} < \infty, \qquad \|T_{K_P}\Phi\|_{L^2},\ \|T_{\mathcal{C}}\Phi\|_{L^2} \leq C_{\text{menu}}\|\Phi\|_{L^2}.$$

Thus (C.2)–(C.3) hold with one constant independent of the arbitrary managed portfolio. At fixed N it may depend on N and, for random labels, on the realized infinite label sequence. Weyl's inequality (Lemma A.7, part 1) gives, for every fixed j, $\left|\lambda_j(T_{K_P}) - \lambda_j(T_{\mathcal{C}})\right| \leq \|T_{K_P} - T_{\mathcal{C}}\|_{\text{op}} \to 0$. As $\lambda_j(T_{K_P}) = \lambda_j(\mathcal{S}_P)/P = \lambda_j(\Psi_P)/P$ and $\lambda_j(T_{\mathcal{C}}) = \lambda_j^*(N)$, this is (F.1). Finally, infinitely many $\lambda_j^*(N)$ are positive if and only if $T_{\mathcal{C}}$ has infinite rank. The settled form here is $\langle \cdot, T_{\mathcal{C}} \cdot \rangle$, so Lemma D.6 supplies this rank condition from completeness (C.4) on an infinite-dimensional $\mathcal{D}$. Under these hypotheses every $\lambda_j^*(N)$ is strictly positive and every

eigenvalue of Ψ_P indexed by a fixed j grows at the rate P. Nothing here required the menu to be drawn at random or to be the same for each P.

Step 2: random labels. Let $\theta_1, \dots, \theta_P$ be i.i.d. draws from μ_θ and $f_k = f(\cdot; \theta_k)$, so that K_P of (D.5) is the Monte-Carlo kernel of the portfolio kernel K in (E.3). We claim that condition (C.5) alone implies (D.6), with $\mathcal{C} = K$. The convergence required is in $L^2(\mathbb{P} \otimes \mathbb{P})$, so the relevant law of large numbers is its vector-valued counterpart rather than the scalar one: the strong law for i.i.d. random elements of a separable Banach space, due to Mourier (1953). In its modern form (Ledoux and Talagrand, 1991, Corollary 7.10), if $\xi, \xi_1, \xi_2, \dots$ are i.i.d. random elements of a separable Banach space $\mathbb{B}$, then $\frac{1}{n}\sum_{k \le n} \xi_k \to E\,\xi$ almost surely in the norm of $\mathbb{B}$ if and only if ξ is Bochner integrable, that is, if and only if $E\|\xi\|_{\mathbb{B}} < \infty$ (Diestel and Uhl, 1977, Theorem II.2.2).

We apply this with $\mathbb{B} = L^2(\mathbb{P} \otimes \mathbb{P})$, which is a separable Hilbert space, and $\xi_\theta := g(\cdot; \theta) \otimes g(\cdot; \theta)$. Its norm is $\|\xi_\theta\|_{L^2(\mathbb{P}\otimes\mathbb{P})} = \|g(\cdot;\theta)\|^2_{L^2(\mathbb{P})} = E[g(Z;\theta)^2]$, so the Bochner-integrability criterion reads

$$\int_\Theta \|\xi_\theta\|_{L^2(\mathbb{P}\otimes\mathbb{P})}\, \mu_\theta(d\theta) \;=\; \int_\Theta E\big[g(Z;\theta)^2\big]\, \mu_\theta(d\theta) \;=\; E[K(Z,Z)] \;=\; \operatorname{tr} T_K \;<\; \infty,$$

which is (C.5), equivalently N^{-2} times (E.6); the trace identity follows from Lemma E.3. Mourier's strong law therefore applies and gives $K_P = \frac{1}{P}\sum_{k \le P} \xi_{\theta_k} \to K$ almost surely in $L^2(\mathbb{P}\otimes\mathbb{P})$, which is (D.6) with $\mathcal{C} = K$. Step 1 then yields (F.1) with $\lambda_j^*(N) = \lambda_j(T_K)$, almost surely. Moreover T_K is trace-class by Lemma E.3, so in this case the constants are summable rather than only square-summable:

$$\sum_j \lambda_j^*(N) \;=\; \operatorname{tr} T_K \;=\; E[K(Z,Z)] \;<\; \infty, \tag{F.4}$$

and every sampled return lies in L^2 almost surely, since the nonnegative integral (C.5) is finite. On the same probability-one event, the uniform bound and settling in Step 1 hold for

all arbitrary managed portfolios. For any nonzero R^{ϕ}_{t+1}, Fubini and Cauchy–Schwarz give

$$\langle R^{\phi}_{t+1}, T_K R^{\phi}_{t+1} \rangle = \int_{\Theta} E[R_\theta R^{\phi}_{t+1}]^2 \, \mu_\theta(d\theta) > 0 :$$

the integral is finite by (C.5), and strictly positive by the completeness hypothesis in Appendix C.2.1. Hence $T_K R^{\phi}_{t+1} \neq 0$, proving (C.4). Almost-sure linear independence for every finite menu implies infinitely many independent managed returns; in particular p cannot have finite support. Thus the managed-return space is infinite-dimensional, T_K has infinite rank, and every $\lambda^*_j(N)$ is positive.

Under the Hilbert–Schmidt condition (E.7) alone, $\operatorname{tr} T_K$ may be infinite, the random element ξ_θ is then not Bochner integrable, and the hypothesis of Ledoux and Talagrand (1991, Corollary 7.10) fails, which makes this exact-limit argument unavailable; Theorem E.5 therefore states the lower bound that its separate proof establishes.

The trace identity also gives (F.3). The diagonal of Ψ_P is $(\Psi_P)_{kk} = E[g(Z;\theta_k)^2]$, and $\theta \mapsto E[g(Z;\theta)^2]$ has finite mean $\int E[g(Z;\theta)^2]\,\mu_\theta(d\theta) = E[K(Z,Z)] < \infty$, so by the strong law of large numbers

$$\sum_{j\geq 1} \frac{\lambda_j(\Psi_P)}{P} \;=\; \operatorname{tr}\Big(\tfrac{1}{P}\Psi_P\Big) \;=\; \frac{1}{P}\sum_{k=1}^{P} E\big[g(Z;\theta_k)^2\big] \;\xrightarrow[P\to\infty]{\text{a.s.}}\; E[K(Z,Z)] \;=\; \sum_{j\geq 1} \lambda^*_j(N)\,. \quad \text{(F.5)}$$

Every term in this convergent sequence is finite almost surely, proving (F.3); thus $\operatorname{tr}\Psi_P = O(P)$ at each fixed N.

Step 3: deterministic labels. Here (D.6) is imposed directly, so Step 1 applies verbatim. □

F.2 A fixed basis

The fixed-basis menu is $f^{(P)}_k(x) = P^{1/2} q_k H_k(x)$ of Appendix C.2.2, with $\{H_k\}$ an orthonormal basis of $L^2(p)$ and $\{q_k\}$ bounded. Two things separate it from random features. The menu

of size P consists of the first P basis functions, not a fresh sample, so nothing is drawn at random and there is no law of large numbers to appeal to. When $\inf_k |q_k| > 0$, (D.6) fails, so the route of Appendix F.1 is closed in that case. What replaces the law of large numbers is monotonicity. Enlarging the menu adds one positive rank-one term, so every eigenvalue can only rise, and the limit exists because the whole family is uniformly bounded.

Use the pooled law $p = N^{-1}\sum_i p_i$, where $p_i = \text{law}(X_{i,t})$; identical distributions across stocks are unnecessary. The conditional second moment obeys

$$\underline{\sigma}^2 I_N \preceq \Sigma_N(X_t) \preceq \bar{\Sigma}_N I_N \qquad \text{almost surely,} \tag{F.6}$$

for constants $0 < \underline{\sigma}^2 \leq \bar{\Sigma}_N < \infty$, the ceiling free to grow with N, so a strong factor in the stocks is permitted. Condition (F.6) coincides with the sort's (C.10)–(C.11) and is a condition on the stocks' second moment alone: no factor structure is used anywhere in this subsection. Write

$$Y_k := \frac{1}{N}\sum_{i=1}^{N} H_k(X_{i,t})\, r_{i,t+1}\,, \qquad \mathsf{M}_{kl} := E[Y_k Y_l] = \frac{1}{N^2} E\big[H_k(X_t)'\Sigma_N(X_t)H_l(X_t)\big] \tag{F.7}$$

for the basis portfolios and their second moments. Write $D_q = \text{diag}(q_1, q_2, \dots)$ and $D_{q,P} = \text{diag}(q_1, \dots, q_P)$. Define the bounded synthesis map $J_N : L^2(p) \to L^2(\mathbb{P})$ by $J_N\psi = \Phi_\psi = N^{-1}\sum_i \psi(X_{i,t})r_{i,t+1}$, and write $\psi_N = (\psi(X_{i,t}))_{i\leq N}$. Since $E\|\psi_N\|^2 = N\|\psi\|^2_{L^2(p)}$, (F.6) gives

$$\frac{\underline{\sigma}^2}{N}\|\psi\|^2_{L^2(p)} \leq \|J_N\psi\|^2_{L^2} \leq \frac{\bar{\Sigma}_N}{N}\|\psi\|^2_{L^2(p)}. \tag{F.8}$$

The lower bound also shows that any measurable weighting generating a square-integrable return belongs to $L^2(p)$, so this domain includes every arbitrary managed portfolio in

Assumption 1. For completeness, the adjoint representation of the co-moments is

$$\begin{aligned} V_\psi(z) &:= N^2(J_N^* J_N \psi)(z) = \sum_{i=1}^{N} \frac{dp_i}{dp}(z) E[(\Sigma_N(X_t)\psi_N)_i \mid X_{i,t} = z], \\ \langle Y_k, \Phi_\psi \rangle &= \frac{1}{N^2} \langle H_k, V_\psi \rangle_{L^2(p)}. \end{aligned} \tag{F.9}$$

The densities $dp_i/dp \leq N$ account for possibly different characteristic laws. In particular the implication

$$V_\psi = 0 \quad \Longrightarrow \quad \|\Phi_\psi\|_{L^2}^2 = \frac{1}{N^2}\langle \psi, V_\psi \rangle = 0 \tag{F.10}$$

is a consequence, rather than an additional assumption. By the lower bound in (F.8), it also implies $\psi = 0$ in $L^2(p)$.

Proposition F.2 (The fixed-basis menu) *Let p be the pooled characteristic law, let $\{H_k\}$ be an infinite orthonormal basis of $L^2(p)$, let $\{q_k\}$ be bounded and nonzero, and let (F.6) hold. Then at every P, $\frac{1}{P}\Psi_P = D_{q,P}\,[\mathsf{M}_{kl}]_{k,l\leq P}\,D_{q,P}$. The ratios $\lambda_j(\Psi_P)/P$ are nondecreasing in P, and*

$$\lim_{P\to\infty} \frac{\lambda_j(\Psi_P)}{P} \;=\; \lambda_j^*(N) \;=\; \lambda_j\big(D_q \mathsf{M} D_q\big) \;\leq\; \frac{q_{\max}^2\,\bar{\Sigma}_N}{N} \;<\; \infty, \qquad q_{\max} := \sup_k |q_k|. \tag{F.11}$$

Here $\lambda_j(D_q\mathsf{M}D_q)$ is a min–max value, which need not be an attained eigenvalue. Assumption 1 holds at every fixed N, with $C_{\text{menu}} = q_{\max}^2\bar{\Sigma}_N/N$, and every $\lambda_j^*(N) > 0$. If moreover $\mathsf{M} = I$, the constants are the upper order statistics of the sequence (q_k^2), with the j-th upper order statistic defined as $\inf_{|F|=j-1}\sup_{k\notin F} q_k^2$ for finite $F \subset \mathbb{N}$. If $\inf_k |q_k| > 0$, then (D.6) fails.

Proof of Proposition F.2. Step 1: square integrability and the exact factorization. By (F.8), $\|Y_k\|_{L^2}^2 \leq \bar{\Sigma}_N/N$, so every finite-menu return is square integrable. The

infinite orthonormal basis and the injectivity of J_N in (F.8) give $\dim \mathcal{D} = \infty$. By (2),

$$R^{(P)}_{k,t+1} \;=\; \tfrac{1}{N}\sum_i f^{(P)}_k(X_{i,t})\, r_{i,t+1} \;=\; P^{1/2} q_k\, Y_k\,, \qquad \text{so} \qquad \Psi_{P,kl} \;=\; E[R_k R_l] \;=\; P\, q_k q_l\, \mathsf{M}_{kl}\,,$$

which is the stated identity. The menu depends on P only through the common factor $P^{1/2}$, which the normalization Ψ_P/P removes.

Step 2: from Gram matrix to operator. $\frac{1}{P}\Psi_P$ is the Gram matrix of $q_1 Y_1, \ldots, q_P Y_P$ in $L^2(\mathbb{P})$, so by Lemma A.8 its nonzero eigenvalues are those of

$$T_P \;:=\; \sum_{k\le P} q_k^2\, Y_k \otimes Y_k \qquad \text{on } L^2(\mathbb{P}), \qquad \text{whence} \qquad \frac{\lambda_j(\Psi_P)}{P} = \lambda_j(T_P).$$

Step 3: monotonicity and a uniform bound. $T_{P+1} - T_P = q_{P+1}^2\, Y_{P+1} \otimes Y_{P+1} \succeq 0$, so Lemma A.7, part 2 gives $\lambda_j(T_{P+1}) \ge \lambda_j(T_P)$ for every j: the ratios are nondecreasing in P. For the bound, Parseval and $Y_k = J_N H_k$ give, for every $\Phi \in L^2(\mathbb{P})$,

$$\langle \Phi, T_P \Phi\rangle \le q_{\max}^2 \sum_{k\ge 1} \langle H_k, J_N^*\Phi\rangle^2_{L^2(p)} = q_{\max}^2 \|J_N^*\Phi\|^2_{L^2(p)} \le \frac{q_{\max}^2 \bar{\Sigma}_N}{N} \|\Phi\|^2_{L^2}.$$

Hence $\sup_P \|T_P\| \le C_{\text{menu}} = q_{\max}^2 \bar{\Sigma}_N / N$, uniformly over arbitrary managed portfolios. This operator-norm bound uses only the upper bound in (F.6). It stays finite even when $\inf_k |q_k| > 0$, where the Hilbert–Schmidt constant of a settling menu diverges (Step 7).

Step 4: the limit, and settling. By Steps 2–3 the increasing sequence T_P has a bounded positive limit $T_\infty = \sum_{k\ge 1} q_k^2 Y_k \otimes Y_k$, defined as a quadratic form by $\langle \Phi, T_\infty \Phi\rangle = \sum_k q_k^2 \langle Y_k, \Phi\rangle^2$, which the Step-3 bound makes a bounded form and hence a bounded positive operator; the L^2 convergence $T_P\Phi \to T_\infty \Phi$ is proved directly in this step. To identify its spectral values, we prove $\lambda_j(T_P) \to \lambda_j(T_\infty)$ by two inequalities. For the upper one, $T_P \preceq T_\infty$ gives $\lambda_j(T_P) \le \lambda_j(T_\infty)$ by Lemma A.7, part 2. For the lower one, fix $\delta > 0$ and a j-dimensional V with $\min_{\Phi\in V, \|\Phi\|=1} \langle \Phi, T_\infty \Phi\rangle \ge \lambda_j(T_\infty) - \delta$. No restriction on V is needed:

by Step 3 the series $\sum_k q_k^2 \langle Y_k, \Phi\rangle^2$ converges for every $\Phi \in L^2(\mathbb{P})$, so its tail beyond P tends to zero pointwise. On the finite-dimensional V the forms $\langle \cdot, T_P \cdot\rangle$ therefore converge to $\langle \cdot, T_\infty \cdot\rangle$ uniformly on the unit sphere, by bilinearity from the finitely many entries, exactly as in the proof of Lemma D.7. Hence $\liminf_P \lambda_j(T_P) \geq \lambda_j(T_\infty) - \delta$. Finally the uniform bound of Step 3 makes the synthesis map $Ce_k = q_k Y_k$ bounded on ℓ^2. Since $T_\infty = CC^*$ and $D_q \mathsf{M} D_q = C^*C$, their ordered variational values coincide by the polar-decomposition part of Lemma A.8, proving (F.11). For settling, note that the left side of (C.2) is exactly $T_P \Phi_\psi$:

$$\frac{1}{P}\sum_{k\leq P} R_k \, E[R_k \Phi_\psi] \;=\; \frac{1}{P}\sum_{k\leq P} P\, q_k^2\, Y_k \, \langle Y_k, \Phi_\psi\rangle \;=\; T_P \Phi_\psi.$$

For a positive operator S, $\|S\Phi\|^2 \leq \|S\| \, \langle \Phi, S\Phi\rangle$; applied to $S = T_\infty - T_P \succeq 0$ this gives $\|T_\infty \Phi - T_P \Phi\|^2 \leq \|T_\infty\| \sum_{k>P} q_k^2 \langle Y_k, \Phi\rangle^2 \to 0$, the tail of a convergent series. So (C.2) holds with $A^{\Phi_\psi} = T_\infty \Phi_\psi$.

Step 5: completeness and positivity. By the Cauchy–Schwarz step in the proof of Lemma D.6, $A^{\Phi_\psi} = 0$ if and only if $\langle \Phi_\psi, T_\infty \Phi_\psi\rangle = \sum_k q_k^2 \langle Y_k, \Phi_\psi\rangle^2 = 0$, that is, if and only if $\langle Y_k, \Phi_\psi\rangle = 0$ for every k with $q_k \neq 0$. When every $q_k \neq 0$ this reads $\langle H_k, V_\psi\rangle = 0$ for every k, hence $V_\psi = 0$ because $\{H_k\}$ is a basis. So (F.10) proves (C.4) without a separate completeness assumption. For positive constants, identify M with $U^* J_N^* J_N U$ on ℓ^2, where $Ue_k = H_k$. Equation (F.8) gives $\mathsf{M} \succeq (\underline{\sigma}^2/N) I$, hence the min–max principle on the first j coordinates yields

$$\lambda_j^*(N) = \lambda_j(D_q \mathsf{M} D_q) \geq \frac{\underline{\sigma}^2}{N} \min_{1\leq k\leq j} q_k^2 > 0.$$

Step 6: the constants, and when they are q_j^2. If $\mathsf{M} = I$ (the basis portfolios uncorrelated with unit second moment, $\frac{1}{N^2} E[H_k(X_t)' \Sigma_N(X_t) H_l(X_t)] = \delta_{kl}$), then $D_q \mathsf{M} D_q = D_q^2$ is diagonal

and its min–max values are

$$\lambda_j^*(N) = \inf_{\substack{F\subset\mathbb{N}\\|F|=j-1}} \sup_{k\notin F} q_k^2.$$

When a decreasing rearrangement exists, this is the j-th entry of that rearrangement. In general the supremum need not be attained: for $q_k = 1 - 1/(k+1)$, every $\lambda_j^*(N) = 1$, although no q_k^2 equals one. The hypothesis $\mathsf{M} = I$ is a genuine restriction: at $N = 1$ it reads $E[H_k(X)H_l(X)\Sigma_1(X)] = \delta_{kl}$, which orthonormality of $\{H_k\}$ in $L^2(p)$ delivers only when $\Sigma_1 \equiv 1$ almost everywhere.

Step 7: (D.6) fails. By (F.6), $\|Y_k\|^2 = \mathsf{M}_{kk} \geq \frac{\sigma^2}{N^2} E\|H_k(X_t)\|^2 = \underline{\sigma}^2/N$, so with $\underline{q} := \inf_k |q_k| > 0$,

$$\|K_P\|^2_{L^2(\mathbb{P}\otimes\mathbb{P})} = \|T_P\|^2_{\mathrm{HS}} = \sum_{k,l\leq P} q_k^2 q_l^2 \langle Y_k, Y_l\rangle^2 \;\geq\; \sum_{k\leq P} q_k^4 \|Y_k\|^4 \;\geq\; P\,\frac{\underline{q}^4\underline{\sigma}^4}{N^2} \;\longrightarrow\; \infty,$$

so K_P cannot converge in $L^2(\mathbb{P}\otimes\mathbb{P})$ and the fixed basis is not an application of Appendix F.1. □

For random features, the menu is an i.i.d. sample and the constants are the eigenvalues of a compact limit reached by a law of large numbers. Here the menu is a fixed list, the constants are reached from below by a monotone sequence, and the limiting operator need not be compact, which is why the $\lambda_j^*(N)$ of (F.11) carry no guarantee of decay beyond what $\{q_k\}$ and M impose.

F.3 Partition menus

Sorted portfolios, the workhorse of empirical asset pricing, are the leading partition menu. They are also the sharpest illustration of what this paper is about, because a sort refines:

the number of portfolios is the number of cells, so enlarging the menu means cutting the characteristic space finer, and there is no separate dial to turn.

Lemma F.3 (Partition floor) *Under the partition conditions of Appendix C.2.3, every eigenvalue of $P^{-1}\Psi_P$ is at least $c_-\underline{\sigma}^2/N > 0$.*

Proof of Lemma F.3. We have

$$R_{k,t+1} \;=\; \frac{P}{N}\sum_{i=1}^{N} \mathbf{1}\{X_{i,t} \in C_k^{(P)}\} r_{i,t+1} \tag{F.12}$$

so that, for any portfolio,

$$\begin{aligned} \sum_k a_k R_{k,t+1} \;&=\; \frac{P}{N}\sum_k a_k \sum_{i=1}^{N} \mathbf{1}\{X_{i,t} \in C_k^{(P)}\} r_{i,t+1} \\ &= \frac{P}{N}\sum_{i=1}^{N} r_{i,t+1}\Big(\sum_k a_k \mathbf{1}\{X_{i,t} \in C_k^{(P)}\}\Big) \;=\; \frac{P}{N}\sum_{i=1}^{N} r_{i,t+1} a_{k(i)}\,, \end{aligned} \tag{F.13}$$

where $k(i)$ is the unique cell of the partition to which stock i belongs at time t. Thus, with $\tilde{a}_t = (a_{k(i)})_{i=1}^N$, we get

$$\begin{aligned} a' P^{-1}\Psi_P a \;&=\; \frac{1}{P}\,E\left[\left(\sum_k a_k R_{k,t+1}\right)^2\right] \\ &= \frac{P}{N^2}\,E\left[\Big(\sum_{i=1}^{N} a_{k(i)}\, r_{i,t+1}\Big)^2\right] \\ &= \frac{P}{N^2} E[\tilde{a}_t' \Sigma_N(X_t)\tilde{a}_t] \underbrace{\geq}_{\text{(C.10)}} \frac{P}{N^2}\,\underline{\sigma}^2 E[\|\tilde{a}_t\|^2] \;=\; \frac{P}{N^2}\underline{\sigma}^2\, E\Big[\sum_{i=1}^{N} a_{k(i)}^2\Big] \\ &= \frac{P}{N^2}\,\underline{\sigma}^2 \sum_{k=1}^{P} a_k^2 \sum_{i=1}^{N} E\big[\mathbf{1}\{X_{i,t} \in C_k^{(P)}\}\big] \;=\; \frac{P}{N^2}\,\underline{\sigma}^2 \sum_{k=1}^{P} a_k^2\, N\, p\big(C_k^{(P)}\big) \\ &\underbrace{\geq}_{p(C_k^{(P)}) \geq c_-/P} \frac{P}{N^2}\,\underline{\sigma}^2\,\frac{N c_-}{P}\,\|a\|^2 \;=\; \frac{c_-\,\underline{\sigma}^2}{N}\,\|a\|^2\,. \end{aligned} \tag{F.14}$$

As a is arbitrary, every eigenvalue of $P^{-1}\Psi_P$ is at least $c_-\underline{\sigma}^2/N > 0$: the disjoint cells give the partition menu a uniform floor, and no eigenvalue is small. □

Why partitions need a separate argument. The bounds (C.10)–(C.11) and (C.8) put each cell return's second moment between positive multiples of P/N at fixed N. The floor in Lemma F.3 implies

$$\|K_P\|^2_{L^2(\mathbb{P}\otimes\mathbb{P})} = \mathrm{tr}((\Psi_P/P)^2) \geq P{c_-}^2\underline{\sigma}^4/N^2 \longrightarrow \infty.$$

Thus kernel convergence (D.6) fails, even though the operator can settle strongly. Every eigenvalue diverges, a stronger conclusion than the fixed-rank statement for random features. The fixed basis shares this floor when $\inf_k |q_k| > 0$ (Proposition F.2). Partitions have no separate coefficient sequence: refining their cells also increases the menu size.

Lemma F.4 (Partition settling and spectral limits) *Suppose the partition conditions in Appendix C.2.3 hold. At fixed N, Assumption 1 holds with $C_{\text{menu}} = c_+\bar{\Sigma}_N/N$. Let $J_N h = N^{-1}\sum_i h(X_i)r_i$, $Q_N = J_N^* J_N$, and let $M_{\sqrt{g}}$ denote multiplication by $\sqrt{g}$ on $L^2(p)$. Along the specified partition sizes,*

$$\frac{\lambda_j(\Psi_P)}{P} \longrightarrow \lambda_j(M_{\sqrt{g}}Q_N M_{\sqrt{g}}) \geq \frac{c_-\underline{\sigma}^2}{N} \qquad \textit{for every fixed } j,$$

where λ_j on the right is a Courant–Fischer value.

Proof of Lemma F.4. Let $\psi_N = (\psi(X_i))_{i=1}^N$. Fix an arbitrary managed portfolio $\Phi_\psi = N^{-1}\psi_N' r \in L^2$. The lower ellipticity bound implies $\psi \in L^2(p)$. Define the finite signed measure

$$\eta_\psi(A) := \sum_{i\leq N} E[\mathbf{1}_A(X_i)(\Sigma_N(X)\psi_N)_i]\,.$$

It is absolutely continuous with respect to the pooled law $p = N^{-1}\sum_i \mathrm{law}(X_i)$; write $V_\psi = d\eta_\psi/dp$. The upper bound in (C.11), conditional Jensen, and $d\,\mathrm{law}(X_i)/dp \le N$ give $V_\psi \in L^2(p)$. Directly from the definitions,

$$\begin{aligned} E[R_k\Phi_\psi] &= \frac{P}{N^2}\int_{C_k^{(P)}} V_\psi\, dp, \\ A_P\Phi_\psi &= \frac{1}{N^2}\Phi_{h_P}, \\ h_P &= P\int_{C^{(P)}(\,\cdot\,)} V_\psi\, dp = g_P E[V_\psi \mid \mathcal{F}_P]. \end{aligned}$$

Full generation, martingale convergence, and (C.9) imply $h_P \to gV_\psi$ in $L^2(p)$. The map $h \mapsto \Phi_h$ is continuous from $L^2(p)$ to $L^2(\mathbb{P})$ because

$$E[\Phi_h^2] \le \frac{\bar\Sigma_N}{N^2}E\|h_N\|^2 = \frac{\bar\Sigma_N}{N}\|h\|^2_{L^2(p)}.$$

Thus $A_P\Phi_\psi \to A^{\Phi_\psi} := N^{-2}\Phi_{gV_\psi}$ in L^2. The operator bound in (C.3) is uniform in P: for $a \in \mathbb{R}^P$, the upper mass bound and (C.11) give

$$a'\frac{\Psi_P}{P}a \le \frac{P\bar\Sigma_N}{N^2}\sum_{i,k} a_k^2 \Pr(X_i \in C_k^{(P)}) \le \frac{c_+\bar\Sigma_N}{N}\|a\|^2,$$

and the nonzero spectrum of A_P is that of Ψ_P/P. Finally, if $A^{\Phi_\psi} = 0$, positive definiteness and $g \ge c_-$ imply $V_\psi = 0$ in $L^2(p)$, whereas

$$\int \psi V_\psi\, dp = E[\psi_N'\Sigma_N(X)\psi_N] = N^2E[\Phi_\psi^2].$$

Hence $\Phi_\psi = 0$, proving asymptotic completeness. Each cell return is square integrable by the upper stock-moment bound; the upper cell-mass bound for arbitrarily large P implies that p has no atoms. Lemma D.4 then gives $\dim \mathcal{D} = \infty$. This verifies Assumption 1 under

the partition conditions of Appendix C.2.3. Comparable masses without (C.9) do not give this conclusion.

The exact fixed-index limit for this class also needs no pair-density condition. Using J_N and Q_N from the statement, let $\mathcal{V}_P = L^2(\mathcal{F}_P)$ and $\Pi_P = E[\cdot \mid \mathcal{F}_P]$ be the cell-constant subspace and its orthogonal projection. In the orthonormal cell basis $\mathbf{1}_{C_k}/\sqrt{p(C_k)}$, one has

$$\frac{\Psi_P}{P} = \Pi_P M_{\sqrt{g_P}} Q_N M_{\sqrt{g_P}} \Pi_P\big|_{\mathcal{V}_P}. \tag{F.15}$$

Here $\|Q_N\| \le \bar{\Sigma}_N/N$ and $\Pi_P \to I$ strongly by the generating condition (C.6). Uniform mass settling permits replacement of g_P by g in operator norm. By Lemma A.6, the Courant–Fischer values of the resulting increasing compressions converge to those of $M_{\sqrt{g}} Q_N M_{\sqrt{g}}$. This proves (F.1) along the stipulated refining sizes.

□

The structure of the limiting spectrum. We now state what partitions yield beyond (F.1). Fix N, and let p be the pooled characteristic law of Appendix C.2: the average cross-sectional distribution of characteristics, and the distribution a sort is formed against in practice. Nothing is assumed about how the stocks differ from one another; in particular, they need not be exchangeable. The mass ceiling (C.8) makes the pooled characteristic law nonatomic (Appendix C.2.3): no single value has positive probability under p. This permits equal-mass cells along the dyadic subsequence. An atom of mass $a > 0$ would instead prevent its containing cell from having mass $1/P < a$. Nonatomic marginals alone do not exclude identical realized characteristics across stocks; the pair-law condition imposed below controls that possibility.

Two distinct objects drive the spectrum, and both grow at rate P, so growth orders cannot tell them apart. Averaging across the cross-section, let ς^2 on the one hand, and the

pair law $P_{2,N}$ together with c on the other, be defined by

$$\begin{aligned}\int_A \varsigma^2\, dp &= \frac{1}{N}\sum_{i\le N} E\big[\mathbf{1}_A(X_{i,t})\,[\Sigma_N(X_t)]_{ii}\big],\\ \int_{A\times B} c\, dP_{2,N} &= \frac{1}{N(N-1)}\sum_{i\neq j} E\big[\mathbf{1}_A(X_{i,t})\,\mathbf{1}_B(X_{j,t})\,[\Sigma_N(X_t)]_{ij}\big],\end{aligned}$$

for Borel sets A, B, with $P_{2,N} := \frac{1}{N(N-1)}\sum_{i\neq j}\mathrm{law}(X_{i,t}, X_{j,t})$ the average law of the characteristics of two distinct stocks at stock count N. The limiting pooled pair law remains P_2 as in (B.6). Thus $\varsigma^2(x)$ is the conditional second moment of a single stock with characteristic x, and $c(x,\tilde{x})$ is the conditional co-moment of two different stocks with characteristics x and $\tilde{x}$; both are averaged over the cross-section, and each reduces to the obvious conditional expectation when the stocks happen to be exchangeable. Let ϱ measure how far the characteristics of two stocks are from being drawn independently of one another: $\varrho \equiv 1$ when they are independent and identically distributed, and ϱ records their cross-sectional dependence otherwise. Formally $\varrho = dP_{2,N}/d(p\otimes p)$ is the density of $P_{2,N}$ against $p\otimes p$, when that density exists. For the whole-spectrum description, we additionally assume $P_{2,N} \ll p\otimes p$ and $\iint (c\varrho)^2\, dp\, dp < \infty$. These are additional comovement conditions, not consequences of Appendix C.2.3. Its upper stock-moment bound does imply $\operatorname{ess\,sup}\varsigma^2 < \infty$. For $N = 1$, omit the pair-law formulas and set the comovement kernel to zero; the own-variance formulas remain valid.

For both Hilbert spaces used here, set $\tilde{c} = (N-1)c\varrho/N$. On $L^2(p)$ or $L^2(\mathfrak{m})$, $T_{\tilde{c}}$ denotes integration against that kernel using the measure of the stated space. The limiting cell-uniform law is $d\mathfrak{m} = dp/g$; it is a probability law because $d\mathfrak{m}_P = dp/g_P$ gives every current cell mass $1/P$ and $g_P \to g$ uniformly with a positive lower bound. Multiplication by a function h is denoted M_h.

Proposition F.5 (Sorted portfolios: the whole spectrum) *Suppose the partition con-*

ditions in Appendix C.2.3 hold, with

$$P_{2,N} \ll p \otimes p, \qquad c\varrho \in L^2(p \otimes p).$$

Set $d\mathfrak{m}_P = dp/g_P$, $d\mathfrak{m} = dp/g$, *and* $a_+ = \operatorname{ess\,sup}(g\varsigma^2/N)$. *Then* $\mathfrak{m}_P \to \mathfrak{m}$ *in total variation,* $\mathfrak{m}$ *is a probability law, and*

$$\frac{Pc_-\underline{\sigma}^2}{N} I_P \preceq \Psi_P \preceq \frac{Pc_+\bar{\Sigma}_N}{N} I_P. \tag{F.16}$$

Moreover:

(i) $P^{-1}\sum_{j=1}^{P} \delta_{\lambda_j(\Psi_P)/P}$ *converges weakly to the law of* $g(X)\varsigma^2(X)/N$ *under* $X \sim \mathfrak{m}$.

(ii) For every fixed j, $\lambda_j(\Psi_P)/P \to \lambda_j^*(N) = \lambda_j(\mathcal{A}_g)$, *where*

$$\mathcal{A}_g = M_{g\varsigma^2/N} + M_g T_{\tilde{c}} M_g \quad \text{on } L^2(\mathfrak{m}).$$

These Courant–Fischer values satisfy $\lambda_j^*(N) \geq a_+$ *and* $\lambda_j^*(N) \downarrow a_+$; *each value above* a_+ *is a discrete eigenvalue, counted with multiplicity.*

(iii) Kernel convergence (D.6) fails, although (C.2)–(C.3) hold.

The law $\mathfrak{m}$ is the limiting distribution of a uniformly selected current cell; it need not assign equal probability to the cells of an earlier partition. The bulk in item (i) is the cell-weighted own-second-moment distribution. Its support is bounded below by $c_-\underline{\sigma}^2/N$. For equal-mass cells it is the law of $\varsigma^2(X)/N$ under p. Item (ii) describes the fixed-index top of the spectrum. Its excess over a_+ records comovement above the upper bulk edge; for every $\delta > 0$, only finitely many values exceed $a_+ + \delta$. There may be infinitely many positive excesses converging to zero, or none. In the latter case the top of the spectrum is flat at

a_+, although the bulk need not be a point mass. Compact comovement can also produce isolated eigenvalues below the lower bulk edge, which item (ii) does not describe.

Proof of Proposition F.5. The eigenvalue floor uses lower ellipticity and the lower cell-mass bound. The whole-spectrum description additionally uses the stated comovement, refinement, and generating conditions; no factor structure is used. Settling and the exact fixed-index limit are established in Lemma F.4.

Write $w_k(X) \in \mathbb{R}^N$ for the weight vector of cell k, with i-th entry $w_{k,i}(X) = \frac{P}{N}\mathbf{1}\{x_i \in C_k\}$, so that $R_{k,t+1} = w_k(X)' r_{t+1}$ and

$$\Psi_{P,kl} \;=\; E\big[w_k(X)'\,\Sigma_N(X)\,w_l(X)\big] \;=\; \frac{P^2}{N^2}\sum_{i,j\le N} E\big[\mathbf{1}\{x_i \in C_k\}\,\mathbf{1}\{x_j \in C_l\}\,[\Sigma_N(X)]_{ij}\big]. \quad \text{(F.17)}$$

Step 1: the bounds (F.16). Lemma F.3 gives the lower bound; the operator bound in Lemma F.4 gives the upper bound.

Step 2: the compression identity. Split (F.17) into its diagonal ($i = j$) and off-diagonal ($i \neq j$) parts, $\Psi_P = D_P + E_P$. A diagonal term survives only when $k = l$, the cells being disjoint, so D_P is the diagonal matrix

$$D_{P,kk} \;=\; \frac{P^2}{N^2}\sum_{i\le N} E\big[\mathbf{1}_{C_k}(X_{i,t})\,[\Sigma_N(X)]_{ii}\big] \;=\; \frac{P^2}{N}\int_{C_k}\varsigma^2\,dp \;=\; \frac{P}{N}\cdot P\int_{C_k}\varsigma^2\,dp\,,$$

directly by the defining relation for ς^2 above, which is a cross-sectional average over i. The off-diagonal part involves two distinct stocks, so it must be evaluated under their joint law $P_{2,N}$ rather than under the product of the marginals: the characteristics need not be independent across stocks, and in general are not. Writing $\varrho = dP_{2,N}/d(p\otimes p)$,

$$E_{P,kl} \;=\; \frac{P^2}{N^2}\,N(N-1)\int_{C_k}\int_{C_l} c\;dP_{2,N} \;=\; P^2\int_{C_k}\int_{C_l}\tilde{c}\;dp\,dp\,, \qquad \tilde{c} \;:=\; \frac{N-1}{N}\,c\,\varrho\,,$$

the second equality being the change of measure from $P_{2,N}$ to $p\otimes p$, whose density is ϱ.

First take equal-mass cells. We claim $\frac{1}{P}E_P$ is not an approximation to the comovement operator but is exactly that operator, compressed to the sort. Let $T_{\tilde{c}}$ be the integral operator on $L^2(p)$ with kernel $\tilde{c}$, and put

$$e_k \;:=\; \sqrt{P}\,\mathbf{1}_{C_k}\,, \qquad k = 1, \dots, P.$$

Because the cells have equal p-mass, $\|e_k\|^2_{L^2(p)} = P\,p(C_k) = 1$, and because they are disjoint the e_k are orthogonal: $\{e_1, \dots, e_P\}$ is an orthonormal system in $L^2(p)$. Its span $\mathcal{V}_P$ is the space of functions constant on each cell. Compute

$$\big\langle e_k,\; T_{\tilde{c}}\, e_l \big\rangle_{L^2(p)} \;=\; P \int_{C_k}\!\!\int_{C_l} \tilde{c}\; dp\, dp \;=\; \frac{1}{P}\, E_{P,kl}\,. \tag{F.18}$$

So $\frac{1}{P}E_P$ is precisely the matrix, in the orthonormal basis $\{e_k\}$, of the compression $\Pi_P\, T_{\tilde{c}}\, \Pi_P$, where Π_P is the orthogonal projection of $L^2(p)$ onto $\mathcal{V}_P$. Equation (F.18) is an exact identity.

The diagonal block is the same kind of object. Let $M_{\varsigma^2/N}$ denote multiplication by ς^2/N on $L^2(p)$. Since the cells are disjoint, $\langle e_k,\, M_{\varsigma^2/N}\, e_l\rangle = \delta_{kl}\,\frac{P}{N}\int_{C_k}\varsigma^2\,dp = \frac{1}{P}D_{P,kl}$, so $\frac{1}{P}D_P$ is the compression of $M_{\varsigma^2/N}$. Adding the two blocks,

$$\frac{1}{P}\,\Psi_P \;=\; \Pi_P\,\mathcal{A}_{\text{sort}}\,\Pi_P\,, \qquad \mathcal{A}_{\text{sort}} \;:=\; \underbrace{T_{\tilde{c}}}_{\text{compact}} \;+\; \underbrace{M_{\varsigma^2/N}}_{\text{multiplication}}\,, \tag{F.19}$$

again exactly. The definitions give $Q_N = J_N^* J_N = \mathcal{A}_{\text{sort}}$ on $L^2(p)$, connecting this formula to Lemma F.4. The whole of $\frac{1}{P}\Psi_P$ is therefore the compression of one bounded self-adjoint operator $\mathcal{A}_{\text{sort}}$ on $L^2(p)$, and everything the proposition asserts is a statement about $\sigma(\mathcal{A}_{\text{sort}})$.

For general masses the same computation in the orthonormal basis $\hat{e}_k := \mathbf{1}_{C_k}/\sqrt{p(C_k)}$ replaces (F.19) by the exact diagonal congruence

$$\frac{1}{P}\,\Psi_P \;=\; D^{1/2}\,\big(\Pi_P\,\mathcal{A}_{\text{sort}}\,\Pi_P\big)\,D^{1/2}\,, \qquad D := \operatorname{diag}\big(P\,p(C_k^{(P)})\big)\,. \tag{F.20}$$

Since $D^{1/2}$ acts on $\mathcal{V}_P$ as multiplication by $\sqrt{g_P}$, $g_P = P\,p(C^{(P)}(\cdot))$, and g_P is $\mathcal{F}_P$-measurable, (F.20) is a compression of the single operator $M_{\sqrt{g_P}}\mathcal{A}_{\mathrm{sort}}M_{\sqrt{g_P}}$ rather than a two-sided decoration of a compression. The limiting object under the unitary change from $L^2(p)$ to $L^2(\mathfrak{m})$ is $\mathcal{A}_g = M_{g\varsigma^2/N} + M_g T_{\tilde{c}} M_g$ on the cell-uniform measure $L^2(\mathfrak{m})$ of items (i)–(ii). The bulk is then the spectral distribution of the multiplication $M_{g\varsigma^2/N}$ (the law of $g\varsigma^2/N$ under $\mathfrak{m}$), and the top the discrete eigenvalues the compact $M_g T_{\tilde{c}} M_g$ lifts above it. Equal mass is $D = I$, $\mathcal{A}_g = \mathcal{A}_{\mathrm{sort}}$.

Two standard facts finish Step 2. First, $\iint \tilde{c}^2\,dp\,dp < \infty$ makes $T_{\tilde{c}}$ Hilbert–Schmidt, hence compact. It need not be positive semidefinite: $\tilde{c}$ is built from an off-diagonal covariance, and stocks may comove negatively, so $T_{\tilde{c}}$ can have eigenvalues of either sign.

Second, $\Pi_P \to \Pi_\infty$ strongly, where $\Pi_\infty := E[\,\cdot \mid \mathcal{F}_\infty]$ is the orthogonal projection of $L^2(p)$ onto $L^2(\mathcal{F}_\infty)$, $\mathcal{F}_\infty := \sigma\big(\bigcup_P \mathcal{F}_P\big)$. Because the cells are nested, the projection onto $\mathcal{V}_P$ is the conditional expectation $\Pi_P = E[\,\cdot \mid \mathcal{F}_P]$, $\mathcal{F}_P := \sigma(C_1, \ldots, C_P)$: for $\psi \in L^2(p)$ and $x \in C_k$, $(\Pi_P\psi)(x) = \frac{1}{p(C_k)}\int_{C_k} \psi\,dp = E[\psi \mid \mathcal{F}_P](x)$. The $\mathcal{F}_P$ increase, so martingale convergence in $L^2(p)$ gives $\Pi_P\psi \to \Pi_\infty\psi$ for every ψ. This projection convergence holds for any refining partition. Under the generating condition (C.6), $\mathcal{F}_\infty = \mathcal{B}$ mod p, so $\Pi_\infty = I$; in general a sort resolves only the comovement its own σ-field can see, the compression $\Pi_\infty T_{\tilde{c}} \Pi_\infty$. For the compact $T_{\tilde{c}}$, strong convergence of the projections upgrades to operator-norm convergence of the compressions,

$$\big\|\Pi_P\, T_{\tilde{c}}\, \Pi_P \;-\; \Pi_\infty\, T_{\tilde{c}}\, \Pi_\infty\big\|_{\mathrm{op}} \;\longrightarrow\; 0\,.$$

For each fixed j, Lemma A.6 shows that the eigenvalue $\lambda_j(E_P/P)$ converges to the Courant–Fischer value of $\Pi_\infty T_{\tilde{c}} \Pi_\infty$ restricted to $L^2(\mathcal{F}_\infty)$. Compactness makes these limiting Courant–Fischer values tend to zero. The operator-norm convergence also gives convergence of the singular values, which controls eigenvalues of either sign in the bulk comparison. We write

$T_{\tilde{c}}$ for the off-diagonal operator. Note that N is fixed here, so the factor $\frac{N-1}{N}$ does not disappear: $\lambda_j(T_{\tilde{c}}) = \frac{N-1}{N}\lambda_j(T_{c_\varrho})$ exactly, and it is $T_{\tilde{c}}$ rather than T_{c_ϱ} that enters $\mathcal{A}_{\mathrm{sort}}$.

Step 3: the fixed-j limits, item (ii). The cells are $\mathcal{F}_\infty$-measurable, so every $\mathcal{V}_P \subseteq L^2(\mathcal{F}_\infty)$ and $\bigcup_P \mathcal{V}_P$ is dense in $L^2(\mathcal{F}_\infty)$ (this is $\Pi_P \to \Pi_\infty$). By (F.19) and Courant–Fischer,

$$\lambda_j\big(\tfrac{1}{P}\Psi_P\big) \;=\; \max_{V\subseteq\mathcal{V}_P,\ \dim V = j}\ \min_{0\neq\psi\in V} \frac{\langle\psi, \mathcal{A}_{\mathrm{sort}}\psi\rangle}{\|\psi\|^2}\,,$$

a maximum over j-dimensional subspaces of $\mathcal{V}_P$. Since $\mathcal{V}_P \uparrow L^2(\mathcal{F}_\infty)$, Lemma A.6 shows that this converges to the j-th Courant–Fischer value of $\mathcal{A}_{\mathrm{sort}}$ on $L^2(\mathcal{F}_\infty)$. Hence, for every fixed j,

$$\frac{\lambda_j(\Psi_P)}{P} \;\longrightarrow\; \lambda_j^*(N) \;:=\; \lambda_j\big(\Pi_\infty\,\mathcal{A}_{\mathrm{sort}}\,\Pi_\infty\big).$$

This uses only that the sort refines; it does not require the cuts to separate points. Under (C.6), $\Pi_\infty = I$ and $\lambda_j^*(N) = \lambda_j(\mathcal{A}_{\mathrm{sort}})$. Either way $\lambda_j^*(N) \geq c_-\underline{\sigma}^2/N > 0$ by the floor (F.16). Thus the direct fixed-index spectral conclusion extends beyond the generating menus for which Lemma F.4 verifies Assumption 1. The remainder of this proof assumes (C.6), so that $\Pi_\infty = I$ and $\lambda_j^*(N) = \lambda_j(\mathcal{A}_{\mathrm{sort}})$, and establishes items (i) and (ii) of Proposition F.5.

Step 4: general masses. The display above is the equal-mass reduction (F.19); at general masses the exact statement is the congruence (F.20), and the fixed-j limit is reached from it in two moves. Because $g_P = P\,p(C^{(P)}(\cdot))$ is constant on cells, hence $\mathcal{F}_P$-measurable, multiplication by it commutes with the conditional expectation Π_P, and $D^{1/2}$ acting on $\mathcal{V}_P$ is exactly $M_{\sqrt{g_P}}$. So (F.20) reads

$$\frac{1}{P}\,\Psi_P \;=\; \Pi_P\,\mathcal{B}_P\,\Pi_P \quad \text{on } \mathcal{V}_P, \qquad \mathcal{B}_P \;:=\; M_{\sqrt{g_P}}\,\mathcal{A}_{\mathrm{sort}}\,M_{\sqrt{g_P}}\,, \tag{F.21}$$

and, writing $\mathcal{B} := M_{\sqrt{g}}\,\mathcal{A}_{\mathrm{sort}}\,M_{\sqrt{g}}$,

$$\big\|\mathcal{B}_P - \mathcal{B}\big\|_{\mathrm{op}} \;\le\; 2\,{c_+}^{1/2}\,\big\|\mathcal{A}_{\mathrm{sort}}\big\|_{\mathrm{op}}\,\big\|\sqrt{g_P} - \sqrt{g}\big\|_{\infty} \;\longrightarrow\; 0$$

by (C.9), since $\|M_h\|_{\mathrm{op}} = \|h\|_\infty$ and $|\sqrt{a} - \sqrt{b}| \le |a-b|/(2\sqrt{c_-})$ on $[c_-, c_+]$. First move: Weyl's inequality (Lemma A.7, part 1) replaces $\mathcal{B}_P$ by $\mathcal{B}$ at a cost $\|\mathcal{B}_P - \mathcal{B}\|_{\mathrm{op}} \to 0$, so it is enough to find $\lim_P \lambda_j(\Pi_P \mathcal{B} \Pi_P)$. Second move: $\mathcal{B}$ is a fixed bounded self-adjoint operator, so Lemma A.6 applies to it and gives $\lambda_j(\Pi_P \mathcal{B} \Pi_P) \to \lambda_j(\Pi_\infty \mathcal{B} \Pi_\infty)$. Under (C.6) this is $\lambda_j(\mathcal{B})$, and the unitary $U\psi = g^{-1/2}\psi$ from $L^2(\mathfrak{m})$ to $L^2(p)$ carries $\mathcal{A}_g$ of item (ii) to $\mathcal{B}$,

$$U\,\mathcal{A}_g\,U^{-1} \;=\; M_{g\varsigma^2/N} + M_{\sqrt{g}} T_{\tilde{c}} M_{\sqrt{g}} \;=\; \mathcal{B}\,,$$

so $\lambda_j^*(N) = \lambda_j(\mathcal{A}_g)$ as item (ii) states. Equal mass is $g \equiv 1$, $\mathcal{B} = \mathcal{A}_{\mathrm{sort}}$, and the two displays coincide. Uniform convergence in (C.9) licenses the first move; boundedness of the refinement densities alone does not give the needed operator-norm convergence.

It remains to locate the $\lambda_j^*(N)$. Because $M_{\sqrt{g}} T_{\tilde{c}} M_{\sqrt{g}}$ is compact, Weyl's theorem on the invariance of the essential spectrum under a compact perturbation (Reed and Simon, 1978, Theorem XIII.14) gives

$$\sigma_{\mathrm{ess}}\big(\mathcal{B}\big) \;=\; \sigma_{\mathrm{ess}}\big(M_{g\varsigma^2/N}\big) \;=\; \operatorname{ess\,ran}\big(g\varsigma^2/N\big) \;\subseteq\; \big[\operatorname{ess\,inf}(g\varsigma^2)/N,\; \operatorname{ess\,sup}(g\varsigma^2)/N\big],$$

which is bounded away from zero, consistently with the floor (F.16). Two consequences.

First, apply the min–max principle, Lemma A.5, to the bounded self-adjoint operator $\mathcal{B}$. Its part (A.4) gives

$$\lambda_j^*(N) = \lambda_j(\mathcal{B}) \ge \sup \sigma_{\mathrm{ess}}(\mathcal{B}) = \operatorname{ess\,sup}(g\varsigma^2)/N \qquad \text{for every } j,$$
$$\lambda_j^*(N) \downarrow \operatorname{ess\,sup}(g\varsigma^2)/N.$$

The $\lambda_j^*(N)$ therefore never descend into the bulk: they stall at its upper edge. Lemma A.5 also says which case obtains at each j: either $\lambda_j^*(N)$ is a genuine eigenvalue strictly above the bulk, or $\lambda_j^*(N)$ has already reached the edge and $\lambda_{j'}^*(N) = \operatorname{ess\,sup}(g\varsigma^2)/N$ for all $j' \geq j$.

Second, only finitely many exceed that edge by any fixed margin. If $C_g = M_{\sqrt{g}} T_{\tilde{c}} M_{\sqrt{g}}$ and $s_j(C_g)$ are its singular values, the min–max inequality gives $\lambda_j(\mathcal{B}) \leq \operatorname{ess\,sup}(g\varsigma^2/N) + s_j(C_g)$. Thus, for any $\delta > 0$, the number of j with $\lambda_j^*(N) > \operatorname{ess\,sup}(g\varsigma^2)/N + \delta$ is at most $\#\{j : s_j(C_g) > \delta\}$, which is finite by compactness. Those finitely many values are generated by the reweighted comovement C_g, and they are item (ii) of the proposition. Negative comovement can instead generate isolated values below the lower bulk edge; these are outside the fixed-index top of the spectrum. The bulk itself, item (i), is a statement about the empirical distribution of all P eigenvalues, whose indices grow with P, which is a different object from the fixed-j limits $\lambda_j^*(N)$.

Step 5: the bulk, item (i). Fix a truncation rank r and split $E_P = E_P^{(r)} + R_r$, where $E_P^{(r)}$ collects the r eigenvalues of E_P largest in modulus. (Truncating the top r would not do: E_P is symmetric but not positive semidefinite, so its negative tail must be caught as well.) Then $\|R_r\|_{\mathrm{op}} = |\lambda|_{(r+1)}(E_P) = P\left(|\lambda|_{(r+1)}(T_{\tilde{c}}) + o(1)\right)$, where $|\lambda|_{(1)} \geq |\lambda|_{(2)} \geq \cdots$ are the eigenvalues ordered by modulus, and these tend to zero by compactness. Two standard facts about empirical spectral distributions are used, and we state them. The rank inequality: for symmetric $P \times P$ matrices A, B, $\|F^A - F^B\|_\infty \leq \operatorname{rank}(A-B)/P$ (Bai and Silverstein, 2010, Theorem A.43). And a Lévy-distance form of Weyl's inequality: since $\lambda_j(A) - \lambda_j(B)$ is bounded by $\|A-B\|_{\mathrm{op}}$ for every j (Lemma A.7, part 1), the two distribution functions are within $\|A-B\|_{\mathrm{op}}$ of one another in the Lévy metric. With these, the rank inequality gives $\left\|F^{\Psi_P/P} - F^{(D_P+R_r)/P}\right\|_\infty \leq r/P \to 0$, while Weyl's inequality (Lemma A.7) bounds the Lévy distance between the spectral distributions of $(D_P + R_r)/P$ and D_P/P by $\|R_r\|_{\mathrm{op}}/P = |\lambda|_{(r+1)}(T_{\tilde{c}}) + o(1)$. Letting $P \to \infty$ and then $r \to \infty$, and using $|\lambda|_{(r+1)}(T_{\tilde{c}}) \to 0$, the spectral distributions of Ψ_P/P and of D_P/P have the same weak limit. It remains to

identify the limit law of D_P/P, and the conditional expectation of Step 2 does it exactly. The k-th diagonal entry is $\frac{1}{N}\,P\int_{C_k}\varsigma^2\,dp = \frac{1}{N}\,(\Pi_P\varsigma^2)(x)$ for any $x \in C_k$; that is, the P values on the diagonal of D_P/P are precisely the values taken by the function $\frac{1}{N}\,\Pi_P\varsigma^2 = \frac{1}{N}\,E[\varsigma^2 \mid \mathcal{F}_P]$. Since each cell carries p-mass exactly $1/P$, counting those values with equal weight $1/P$ is the same as integrating against p, so the empirical spectral distribution of D_P/P is the law of $\frac{1}{N}\,E[\varsigma^2 \mid \mathcal{F}_P]$ under p. By (C.6) and martingale convergence (Billingsley, 1995, Section 35), $E[\varsigma^2 \mid \mathcal{F}_P] \to \varsigma^2$ in $L^2(p)$, hence in distribution. Therefore the empirical distribution of $\lambda_j(\Psi_P)/P$ converges weakly to the law of $\varsigma^2(X)/N$ with $X \sim p$, as claimed.

For general comparable masses, the same last identification is made under the cell-uniform law $\mathfrak{m}_P$, defined by $d\mathfrak{m}_P/dp = 1/g_P$. Indeed the k-th diagonal entry of D_P/P is

$$\frac{g_P(x)}{N}\,E[\varsigma^2 \mid \mathcal{F}_P](x), \qquad x \in C_k^{(P)},$$

and giving the P cell values equal weight is exactly integration under $\mathfrak{m}_P$. By (C.9), both $g_P \to g$ and $1/g_P \to 1/g$ uniformly, since inversion is Lipschitz on $[c_-, c_+]$. Also $E[\varsigma^2 \mid \mathcal{F}_P] \to \varsigma^2$ in $L^2(p)$. Boundedness of g_P and ς^2 therefore gives, for every bounded Lipschitz F,

$$\int F\left(\frac{g_P E[\varsigma^2 \mid \mathcal{F}_P]}{N}\right)\frac{dp}{g_P} \longrightarrow \int F\left(\frac{g\varsigma^2}{N}\right)\frac{dp}{g}.$$

Thus the diagonal empirical law converges to the law of $g(X)\varsigma^2(X)/N$ under $X \sim \mathfrak{m}$, $d\mathfrak{m} = dp/g$. The finite-rank/compact-tail comparison above is unaffected by the bounded diagonal congruence in (F.20) (equivalently, apply it to $M_{\sqrt{g_P}}T_{\tilde{c}}M_{\sqrt{g_P}}$, whose singular-value tails are uniformly bounded by $c_+ s_j(T_{\tilde{c}})$). Hence the full empirical spectral distribution has the same limit, which proves item (i) at general masses under the stated uniform mass convergence.

Step 6: failure of kernel convergence. The floor (F.16) gives $\Psi_{P,kk} \geq Pc_-\underline{\sigma}^2/N$ for

every cell. Therefore

$$\|K_P\|^2_{L^2(\mathbb{P}\otimes\mathbb{P})} = \frac{1}{P^2}\sum_{k,l\le P}\Psi^2_{P,kl} \ge \frac{1}{P^2}\sum_{k\le P}\Psi^2_{P,kk} \ge \frac{P{c_-}^2\underline{\sigma}^4}{N^2} \longrightarrow \infty.$$

Thus (D.6) fails. The upper bound in Lemma F.4 and the floor give $\operatorname{tr}\Psi_P \asymp P^2/N$ at fixed N. This construction satisfies strong settling but admits no square-integrable limiting co-moment kernel. Consequently, the first implication of Lemma D.3 does not reverse. □

Remark F.6 (Why a sort cannot settle to a compact kernel) The two summands of $\mathcal{A}_{\text{sort}}$ are of different type, and that is exactly why a refining sort falls outside the settling framework (D.6) of Appendix F.1. The comovement $T_{\tilde{c}}$ is compact, its eigenvalues decaying, the kind of operator a random-feature menu settles to. The own-variance block is the opposite. It is the multiplication operator $M_{\varsigma^2/N}$, whose essential spectrum $\operatorname{ess\,ran}(\varsigma^2/N)$ is bounded away from zero, and it inherits this from the cell structure: the sort's own-overlap kernel

$$k_{\text{sort}}(x,\tilde{x}) \;=\; \sum_k \frac{\mathbf{1}\{x\in C_k\}\,\mathbf{1}\{\tilde{x}\in C_k\}}{p(C_k)}\,, \qquad (\Pi_P\phi)(x) \;=\; \int k_{\text{sort}}(x,\tilde{x})\,\phi(\tilde{x})\,p(d\tilde{x})\,,$$

is the reproducing kernel of the projection Π_P (weight $1/p(C_k)$ on each cell). Unlike a decaying compact kernel, a projection fixes the constant function, $\int k_{\text{sort}}(x,\tilde{x})\,p(d\tilde{x}) = \sum_k \mathbf{1}\{x\in C_k\} = 1$ for every x and every P, and as the cells refine $\Pi_P \to I$, the identity, whose spectrum is flat: every eigenvalue one, nothing decaying, no cliff. For $(p\otimes p)$-almost every pair of distinct points $x \ne \tilde{x}$, the mass ceiling makes the nested cells eventually separate them: $p(C^{(P)}(x)) \le c_+/P \to 0$. Thus $k_{\text{sort}}(x,\tilde{x}) \to 0$, while the diagonal $1/p(C_k)$ blows up as the cells shrink, so the mass escapes to the diagonal, exactly where an L^2 kernel cannot follow it. Hence the empirical co-moment kernel of a refining sort does not converge to a Hilbert–Schmidt $\mathcal{C}$ as (D.6) demands: its own-variance part converges instead to $M_{\varsigma^2/N}$, which lifts every eigenvalue by the floor (F.16) rather than leaving a cliff. The row sums

equal to one identify Π_P as a projection converging to the identity; they are not a near miss of the settling condition.

Two stocks, in closed form. Proposition F.5 holds at any fixed N, and taking $N = 2$ makes every object in it explicit. Nothing in the construction depends on N being large: the cells partition the characteristic space rather than the cross-section, so $P \to \infty$ still means cutting that space ever more finely, and the characteristics are still continuously distributed. Most cells are empty in any given period, which is harmless, because Ψ_P is an expectation over the characteristics rather than a realized matrix.

The economy. One factor, unit betas, homoskedastic residuals: $r_{i,t+1} = G_{t+1} + \varepsilon_{i,t+1}$, where G is independent of the characteristics, $E[G] = 0$, $E[G^2] = 1$, and the residuals are independent of G and the characteristics, mutually uncorrelated, mean zero, and have variance $\sigma^2 > 0$. Then

$$\Sigma_2(X_t) \;=\; \begin{pmatrix} 1+\sigma^2 & 1 \\ 1 & 1+\sigma^2 \end{pmatrix}, \qquad \underline{\sigma}^2 \;=\; \lambda_{\min}\big(\Sigma_2\big) \;=\; \sigma^2. \tag{F.22}$$

This is the whole of Σ, and the two objects are read straight off it: the diagonal gives the own variance $\varsigma^2(x) = 1 + \sigma^2$, and the off-diagonal gives the comovement $c(x, \tilde{x}) = 1$.

The characteristics. Let the two stocks' characteristics have the symmetric density on $[0,1]^2$

$$\begin{aligned} \varrho(x,\tilde{x}) &= 1 + 2\sum_{k\geq 1} \theta^k \cos(k\pi x)\cos(k\pi\tilde{x}) \\ &= \frac{1}{2}\Big[\mathcal{K}_\theta\big(\pi(x-\tilde{x})\big) + \mathcal{K}_\theta\big(\pi(x+\tilde{x})\big)\Big], \qquad 0 \leq \theta < 1, \end{aligned} \tag{F.23}$$

where $\mathcal{K}_\theta(u) = \frac{1-\theta^2}{1-2\theta\cos u+\theta^2}$, which is manifestly nonnegative; the second form therefore shows that ϱ is a genuine density, $\varrho \geq 0$. Since $\int_0^1 \cos(k\pi\tilde{x})\, d\tilde{x} = 0$, both marginals are uniform, so $p = \mathrm{Unif}[0,1]$ and $p \otimes p$ has density one. Thus the pair law $P_{2,N}$ at $N = 2$ has dependence

density ϱ relative to $p \otimes p$, and

$$c\varrho = \varrho,$$

which holds because $c \equiv 1$. The single parameter θ is the strength of cross-sectional dependence: $\theta = 0$ makes the characteristics independent, and $\theta \uparrow 1$ makes them move together.

The spectrum, in closed form. The first line of (F.23) diagonalizes the comovement operator: the constant and the functions $\sqrt{2}\cos(k\pi x)$, $k \geq 1$, are orthonormal in $L^2(p)$ and diagonalize $T_{c\varrho}$ with eigenvalues $1, \theta, \theta^2, \ldots$, one for the level and one for each mode of dependence. With $N = 2$, $T_{\tilde{c}} = \frac{1}{2}T_{c\varrho}$. Carrying these eigenvalues through Proposition F.5 gives

$$\frac{\lambda_j(\Psi_P)}{P} \longrightarrow \lambda_j^*(N) = \frac{1 + \sigma^2 + \theta^{j-1}}{2}, \qquad j = 1, 2, 3, \ldots \tag{F.24}$$

Everything the proposition asserts can now be seen at once. The bulk is the single point $\varsigma^2/N = \frac{1+\sigma^2}{2}$, because ς^2 is constant here. For $0 < \theta < 1$, every $\lambda_j^*(N)$ lies strictly above it and $\lambda_j^*(N) \downarrow \frac{1+\sigma^2}{2}$ at the geometric rate θ. Thus, for $0 < \theta < 1$, there are infinitely many comovement directions above the bulk, and they accumulate at its edge: no gap separates them from it, and no rank marks where comovement ends. Every eigenvalue diverges, and the floor (F.16) reads $\Psi_P \succeq \frac{P}{2}\sigma^2 I_P$, consistent with $\lambda_j^*(N) \geq \frac{1+\sigma^2}{2} > \frac{\sigma^2}{2}$, with the first inequality strict for every j when $\theta > 0$.

What produced them. The economy has two assets and one factor, and every stock has the same beta. Yet the sort reports infinitely many diverging comovement directions, and (F.24) says exactly where they came from: the excess $\lambda_j^*(N) - \frac{1+\sigma^2}{2} = \theta^{j-1}/2$ is half the strength of the j-th mode of cross-sectional dependence in the characteristics. Set $\theta = 0$ (draw the two stocks' characteristics independently) and the spectrum collapses to $\lambda_1^*(N) =$

$\frac{2+\sigma^2}{2}$ and $\lambda_j^*(N) = \frac{1+\sigma^2}{2}$ for $j \geq 2$: one comovement direction, the market, which is the right answer for a one-factor economy. Every direction beyond the first is generated by cross-sectional dependence in the characteristics.

The bulk can hide a factor entirely. Item (ii) of Proposition F.5 asserts only that $\lambda_j^*(N) \geq a_+ = \operatorname{ess\,sup}(g\varsigma^2/N)$. Nothing need lie above it, and the failure is not pathological.

One way is negative comovement. If $T_{\tilde{c}} \preceq 0$, so that stocks comove negatively in every direction, then $C_g \preceq 0$ and $\mathcal{B} \preceq M_{g\varsigma^2/N}$ in the notation of Step 4 of the proof of Proposition F.5. Hence, for equal-mass cells,

$$\lambda_j^*(N) = \operatorname{ess\,sup}(\varsigma^2/N) \qquad \text{for every fixed } j,$$

although the bulk may remain nonconstant. Negative comovement may also create a lower outlier. For example, take $N = 2$, independent uniform characteristics, and

$$\Sigma_2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

For equal-mass cells, $\Psi_P/P = I_P - (2P)^{-1}\mathbf{1}\mathbf{1}'$ has $P-1$ eigenvalues equal to one and one eigenvalue equal to $1/2$. Thus every fixed-index top limit is one, while a separated eigenvalue persists below the bulk.

The second way is more troubling, because the comovement is genuine and positive and still does not show. In the example below, the factor loads only on low-variance stocks. Its contribution is too small to lift that part of the spectrum above the higher variance elsewhere in the cross-section. Take $N = 2$, characteristics independent and uniform, and

$$\beta(x) = 0.3 \cdot \mathbf{1}\{x < \tfrac{1}{2}\}, \qquad \sigma_\varepsilon^2(x) = 0.10 \cdot \mathbf{1}\{x < \tfrac{1}{2}\} + 3.00 \cdot \mathbf{1}\{x \geq \tfrac{1}{2}\},$$

so that $\Sigma_2(X)$ is the one-factor matrix $\beta\beta' + \mathrm{diag}(\sigma_\varepsilon^2)$. The factor is real: stocks with characteristic below $\frac{1}{2}$ load on it. Yet $\operatorname{ess\,sup} \varsigma^2/N = 1.5$, while the whole comovement contributes only $\frac{1}{2}\int \beta^2 dp = 0.0225$, concentrated exactly on the stocks whose own variance is 0.19. The result is $\lambda_j^*(N) = 1.5$ for every j: nothing at all above the bulk. Move the same factor onto the high-variance stocks instead and it reappears: the bulk edge rises to $(3.00 + 0.09)/2 = 1.545$, and $\lambda_1^*(N) = 1.5675$ sits above it. The factor's comovement contribution is unchanged; only its location has moved.

The two failures point in opposite directions, and together they undermine any factor count read off a sorted spectrum. A sort can manufacture diverging directions that are not factors, as (F.24) shows; and it can conceal factors that are. Whether a true factor is visible above the bulk turns on whether it happens to live among stocks whose idiosyncratic variance is large enough to lift it clear, rather than on how strong it is.

The lesson is that the cliff is a property of the menu rather than of the economy. A menu confined to a fixed, finite set of weighting schemes spans only finitely many directions, and so manufactures a finite factor model however rich the underlying cross-section. A narrow positive feature bandwidth can instead produce an apparent finite-factor pattern, as in Section 5.1, while retaining infinitely many positive limiting growth constants. A refining sort goes to the other extreme and manufactures no cliff at all. Both are artifacts of the weighting scheme. The refining sort also shows that (D.6) has content: the sort violates it, and violating it removes the cliff rather than restoring it.

Without the generating condition (C.6), Step 3 of the proof of Proposition F.5 still gives the fixed-index limits. A sort resolves only the comovement its own cuts can see: a sort that refines size alone can separate almost every sampled pair when size has a nonatomic distribution, but it does not recover functions of the other characteristics. By contrast, with a positive continuous density on $\mathbb{R}^d$, cycling through coordinates and bisecting each cell at the conditional median produces a generating dyadic construction.

So the sort can produce a spectrum that looks exactly like an approximate factor structure: a handful of large eigenvalues, then a bulk. That appearance is the trap. An approximate factor structure in the spectral sense of Chamberlain and Rothschild (1983) requires the eigenvalues outside the handful to stay bounded, and here they do not: by (F.16) every one of the P eigenvalues in the bulk diverges at the very same rate P as the handful above it. The bulk is idiosyncratic risk levered up by the sort until it grows exactly as fast as genuine comovement does, rather than idiosyncratic risk that washes out. A researcher who finds a gap in this spectrum and concludes "k factors, and bounded idiosyncratic noise below" has read the gap correctly and the noise catastrophically wrongly.

The floor is a consequence of own-stock second moments and the menu normalization; it does not by itself identify priced risk. Conditional on a cell holding one name, its return is a P/N multiple of that stock's return, so its conditional second moment is $(P/N)^2[\Sigma_N(X)]_{ii}$. For an equal-mass cell, averaging the own-stock contribution over occupancy gives the diagonal term $(P/N)E[\varsigma^2 \mid C]$. Disjointness makes these own-stock contributions diagonal across cells; full cell returns can still comove through different stocks.

A refining sort violates the boundedness half of an approximate factor structure outright, but for a mundane reason: each cell's own second moment grows like P/N. The sort spectrum illustrates the absence of a cliff without isolating the comovement mechanism behind Theorem 1. Proposition F.5 does not say that the sort reveals the infinitely many factors of a random-feature menu: the diverging bulk is the cross-section's own idiosyncratic variance, levered up, rather than new priced risk. Genuine comovement may be there (item (ii) permits infinitely many directions to carry it), but the sort provides no way to tell which eigenvalues carry it and which are levered noise, and, as the hidden-factor example shows, it may reveal none of it at all. That is precisely why the eigenvalue cliff cannot be used to certify a factor model, and why the granularity of a menu must be judged against the size of the cross-section rather than trusted on its own.

F.4 Standard-menu summary

Proposition F.7 (Standard menus satisfy Assumption 1 at fixed N) *At each fixed N, the constructions in Appendices C.2.1, C.2.2, and C.2.3 satisfy Assumption 1 under their stated conditions. For random features the conclusion holds almost surely for the infinite independent label sequence; for partitions all limits are along the specified refining sizes P_n. Each construction also has an exact fixed-index limit (F.1), and all these limits are positive. The constant C_{menu} may depend on N and, for random features, on the realized infinite label sequence.*

The fixed-N conditions alone give neither uniformity in N nor the growing-stock bound in Lemma D.8.

Proof of Proposition F.7. For random features, Lemma F.1 proves settling, completeness, the operator bound, and exact convergence. For a fixed basis, these conclusions are Proposition F.2. For partitions, Lemma F.4 gives Assumption 1, the exact limit, and its positivity. □

For the fixed basis, a sufficient bound uniform in N is $\sup_N(\sup_k |q_k|)^2\bar{\Sigma}_N/N < \infty$. For random features, an integrable envelope $\sup_N E[R^2_{\theta,N}] \le \bar{e}(\theta)$ with $\int \bar{e}\, d\mu_\theta < \infty$ gives a pathwise bound for all P, N by the scalar strong law. These are additional bounds on the growing-stock sequence. For a random-feature menu the settled form is compact, its α_j decline to zero, and asymptotic completeness keeps them positive. For a refining sort the settled form is not compact at all (Remark F.6), and the constants are held up from below by leverage rather than by richness: $\alpha_j \ge c_-\underline{\sigma}^2/N$ for every j. That floor is Lemma F.3.

G Regularity for Growing Managed Menus

Throughout, X and $\tilde{X}$ denote independent copies of the characteristic panel X_t, and P_N is the law of X_t. Capital letters denote the random panel and its entries (X_t, $X_{i,t}$); lowercase x_i

(or x) denotes a realized value of stock i's characteristics, the argument of $B(\cdot,\cdot)$ and $f(\cdot;\theta)$, and we drop the period subscript t where no confusion arises. For the portfolio operators, $\Sigma_N(X)$ always denotes the conditional second moment $E[r_{t+1}r_{t+1}' \mid X_t]$. Any additional stock-level variables that are random are integrated out by this conditional expectation. Portfolio weights depend on neither the integrated-out stock-level variables nor the array S: they are functions of characteristics alone. The size subscript is suppressed where the N-dependence of the spectrum is not at issue.

The factor-model verification uses the following effective conditional moments, defined for a stock drawn from the limiting population: conditional on $W = w$, let (S, X) have law $\nu(ds)p(dx \mid s, w)$. These random fixed effects describe population sampling, while the original fixed-effect array remains nonrandom. Write $p_s(dx) := p(dx \mid s) = \int p(dx \mid s, w)P_W(dw)$. Then

$$\begin{aligned} \bar{\beta}_w(u) &:= E\big[B(u,S) \mid X = u,\, W = w\big], \\ \bar{b}_w(u) &:= E\big[B(u,S)'\Omega(w)B(u,S) \mid X = u,\, W = w\big], \\ \bar{s}(u) &:= E\big[\sigma_\varepsilon^2(u,S) \mid X = u\big]. \end{aligned} \tag{G.1}$$

The first is the effective loading conditional on observable characteristics, the second the effective own systematic second moment, and the third the effective idiosyncratic variance. They enter only as functions of u and w; none creates an additional coordinate in the managed-portfolio space.

Assumption G.1 (Managed-portfolio regularity) We add the following to the conditional factor model of Assumption 3 and to clauses (ii)–(iv) of Assumption B.3, allowing the common characteristic-law clause of Assumption 3 to be replaced by Assumption B.1. Thus the appendix results invoking this assumption cover both the baseline common law and fixed-effect-dependent characteristic laws. Clauses (i) and (v) of Assumption B.3 are

not imposed here: the non-redundant loading limit and the stock-response regularity used in Theorem 3 are separate from the managed-portfolio model space.

(a) **Idiosyncratic variance:** there are constants $0 < \underline{\sigma}^2 \leq \bar{\sigma}^2 < \infty$ with $\underline{\sigma}^2 \leq \sigma_\varepsilon^2(x, s) \leq \bar{\sigma}^2$ for every fixed effect s and p_s-almost every x. These are the residual bounds already imposed in Assumption 3 and recalled in Assumption B.3(iii).

(b) **Uniformly elliptic factor moments:** let W denote the common shock of the cross-section (Assumption 3, or its extension B.1; see also Theorem B.2), P_W its law, and require $p(\cdot \mid w) \ll p$ for P_W-almost every w, and set $\mathsf{d}_w = dp(\cdot \mid w)/dp$. We require that the factor second moment be uniformly elliptic in the state (there are constants $0 < \underline{\omega} \leq \bar{\omega} < \infty$ with $\underline{\omega}\, I_K \preceq \Omega(w) \preceq \bar{\omega}\, I_K$ for P_W-almost every w) and that the conditional densities be uniformly square integrable over the state,

$$\operatorname*{ess\,sup}_w \|\mathsf{d}_w\|_{L^2(p)} \;<\; \infty\,. \tag{G.2}$$

The ellipticity half strengthens Assumption B.3(ii) from almost-sure nonsingularity of the limiting Ω to a uniform bound on $\Omega(w)^{-1}$.

(c) **Nonatomic characteristics:** there are a fixed effect s_0 of positive limiting share $\pi_{s_0} > 0$ and a vector $a_0 \in \mathbb{R}^K$ such that $p_{s_0}(a_0' B(\cdot, s_0) \neq 0) > 0$ and the restriction of p_{s_0} to this set is nonatomic. In particular $L^2(p)$ is infinite-dimensional. This richness is an explicit condition on characteristic variation.

(d) **Complete set of managed portfolios:** no weighting of the characteristics is invisible to the whole menu. Formally, the menu satisfies the completeness clause of Assumption D.2, stated there for random-feature menus (Appendix C.2.1). This is a condition on the menu rather than on the economy. For a partition menu, the counterpart is the generating condition of Appendix C.2.3: $\mathcal{F}_P \uparrow \mathcal{F}_\infty$ and $L^2(\mathcal{F}_\infty) = L^2(p)$, up to null sets.

The cell-constant subspaces $\mathcal{V}_P$ then have dense union in $L^2(p)$. The comparable-mass bounds $c_- \le g_P \le c_+$ and the uniform limit (C.9) make the limiting multiplication by $\sqrt{g}$ boundedly invertible, so it removes no direction. Equal cell masses are not required.

(e) **Finite single-stock kernel:** we impose the diagonal-finiteness condition (E.5) for the single-stock kernel of (E.1). Condition (E.8) bounds the second moment of a managed return at almost every label; (E.5) instead integrates the squared feature over labels at a fixed characteristic. Neither condition alone implies the other. It makes $k_0(z,u)$ absolutely convergent for $p \otimes p$-almost every (z,u) (Cauchy–Schwarz), hence the portfolio kernel $K(Z,Z) = N^{-2} r' k(X,X) r$ finite almost surely, which is the diagonal-finiteness hypothesis of Lemma A.10.

Assumption G.2 (Large-N managed-menu regularity) For a random-feature menu, require $\sup_N \int E[R_{\theta,N}^2]\, \mu_\theta(d\theta) < \infty$. When the same random label sequence is used across stock counts, also require an integrable envelope:

$$\sup_{N \ge 1} E[(R_{N,t+1}^{f(\cdot;\theta)})^2] \le \bar{e}(\theta), \qquad \int \bar{e}(\theta)\, \mu_\theta(d\theta) < \infty. \tag{G.3}$$

For the fixed-basis and partition constructions in this assumption, require that one fixed law p equal the pooled characteristic marginal at every N (as under a common characteristic law). Fix the basis and bounded coefficients $\{q_k\}$ across N. For a partition menu, use one fixed pooled law p, refining partition sequence and limiting cell-mass density g across N. Require a uniform bound on the own second moments $\sup_N \|\varsigma_{(N)}^2\|_\infty < \infty$ and convergence of the dependence-adjusted co-moments $c_{(N)}\varrho_{(N)}$ in $L^2(p \otimes p)$ to a square-integrable limit. Here $\varsigma_{(N)}^2$, $c_{(N)}$, and $\varrho_{(N)}$ are the N-stock versions of the own moment, pair co-moment, and pair density defined in Appendix F.3. In particular, the finite-N pair law is required to have this density.

Remark G.3 (Consequences of the large-N menu bounds) For a random-feature menu with one label sequence across stock counts, the envelope (G.3) and the strong law give $\sup_{P,N} P^{-1} \sum_{k \le P} E[(R_{N,t+1}^{f(\cdot;\theta_k)})^2] < \infty$ almost surely, which supplies a single bound C_{menu} in (C.3) for the realized menus. A bound on the label integral separately at each N does not alone give this pathwise uniformity. For a fixed-basis menu, the stock bounds (B.8) imply

$$\sup_{N \ge 1} \frac{\bar{\Sigma}_N}{N} < \infty. \tag{G.4}$$

Indeed, the characteristic-to-payoff map $J_N f = N^{-1} \sum_i f(X_{i,t}) r_{i,t+1}$ has $\|J_N\|^2 \le \bar{\Sigma}_N / N$, so the menu bound is at most $(\sup_k |q_k|)^2 \sup_N \bar{\Sigma}_N / N$.

Remark G.4 (Uniform and integrated density bounds) Condition (G.2) is a uniform-in-state $L^2(p)$ bound. It is weaker than $\operatorname{ess\,sup}_w \|\mathsf{d}_w\|_{L^\infty(p)} < \infty$, but stronger than the integrated condition $E_W \|\mathsf{d}_W\|_{L^2(p)}^2 < \infty$. These three conditions must not be conflated. In the Gaussian model of Section 5.1, $p = \mathcal{N}(0, I_d)$ and $p(\cdot \mid w) = \mathcal{N}(\sqrt{\rho}\, w, (1-\rho) I_d)$, so a direct Gaussian integral gives

$$\|\mathsf{d}_w\|_{L^2(p)}^2 \;=\; \int \frac{p(u \mid w)^2}{p(u)}\, du \;=\; (1-\rho^2)^{-d/2}\, e^{\rho \|w\|^2/(1+\rho)},$$

and, averaging over $W \sim \mathcal{N}(0, I_d)$ with $E[e^{c\|W\|^2}] = (1-2c)^{-d/2}$ at $c = \rho/(1+\rho) < \frac{1}{2}$,

$$E_W\left[\|\mathsf{d}_W\|_{L^2(p)}^2\right] \;=\; (1-\rho^2)^{-d/2} \left(\tfrac{1-\rho}{1+\rho}\right)^{-d/2} \;=\; (1-\rho)^{-d} \;<\; \infty \qquad (\rho \in (0,1)), \tag{G.5}$$

diverging only as $\rho \to 1$. Nevertheless, $\|\mathsf{d}_w\|_{L^2(p)}^2 = (1-\rho^2)^{-d/2} e^{\rho\|w\|^2/(1+\rho)}$ is unbounded in w, so the Gaussian model does not satisfy (G.2). Its Gaussian calculations therefore require separate direct hypotheses.

H Large-Stock Limits and Genericity

The first lemma formalizes the i.i.d. illustration of Section 3: in an i.i.d. cross-section, every normalized eigenvalue $\lambda_j(\Psi_{P,N})/P$ beyond the first vanishes as the cross-section grows.

Lemma H.1 (Collapse in an i.i.d. cross-section) *Let the pairs $(X_{i,t}, r_{i,t+1})_{i\le N}$ be i.i.d. across i, let $R_{k,t+1} = N^{-1}\sum_{i\le N} f_k(X_{i,t})\, r_{i,t+1}$, and suppose $E[f_k(X_{1,t})^2 r_{1,t+1}^2] < \infty$ for every $k \le P$. The weights may depend on P but are held fixed as N varies. Write $a_k = E[f_k(X_{1,t})\, r_{1,t+1}]$ and $b_{k_1k_2} = E[f_{k_1}(X_{1,t}) f_{k_2}(X_{1,t})\, r_{1,t+1}^2]$. Then, at every N,*

$$E[R_{k_1,t+1}R_{k_2,t+1}] \;=\; a_{k_1}a_{k_2} \;+\; \tfrac{1}{N}\big(b_{k_1k_2} - a_{k_1}a_{k_2}\big), \tag{H.1}$$

the case $k_1 = k_2$ included. With $\bar a_P = (a_k)_{k\le P}$, $\Psi_{P,N} \to \bar a_P \bar a_P'$ as $N \to \infty$ at fixed P. That limit has rank at most one: if $\bar a_P \neq 0$, its only nonzero eigenvalue is $\|\bar a_P\|^2$; if $\bar a_P = 0$, it is zero. Consequently $\lim_{P\to\infty}\lim_{N\to\infty} \lambda_j(\Psi_{P,N})/P = 0$ for every $j \ge 2$. For the order of Theorem 1 ($P \to \infty$ first), put $\Psi_{P,1} = (b_{kl})_{k,l\le P}$ and suppose

$$\sup_{P\ge 1} \frac{\|\Psi_{P,1} - \bar a_P \bar a_P'\|_{\mathrm{op}}}{P} \le C_{\mathrm{iid}} < \infty. \tag{H.2}$$

Then, for every $P \ge j \ge 2$,

$$0 \le \frac{\lambda_j(\Psi_{P,N})}{P} \le \frac{C_{\mathrm{iid}}}{N}, \qquad \lim_{N\to\infty}\limsup_{P\to\infty} \frac{\lambda_j(\Psi_{P,N})}{P} = 0. \tag{H.3}$$

Thus whenever the inner exact limits exist, they collapse in the order of Theorem 1 as well. If $\bar a_P = 0$ for every P, the same conclusions hold also for $j = 1$.

For an explicit example, take $E[r_{i,t+1} \mid X_{i,t}] = 0$, $E[r_{i,t+1}^2 \mid X_{i,t}] = 1$, and the fixed-basis menu $f_k^{(P)} = \sqrt{P}\, q_k H_k$ with $0 < |q_k| \le q_{\max}$. Then $\Psi_{P,1} = P\,\mathrm{diag}(q_1^2, \dots, q_P^2)$ and $\bar a_P = 0$, so

the exact fixed-N constants obey

$$0 < \lambda_j^*(N) := \lim_{P\to\infty} \frac{\lambda_j(\Psi_{P,N})}{P} = \frac{1}{N}\lambda_j(D_q^2) \le \frac{q_{\max}^2}{N} \quad \textit{for every fixed } j. \tag{H.4}$$

Here the infinite diagonal operator is interpreted through its Courant–Fischer values.

Proof of Lemma H.1. Expand the product and split the double sum by whether the two stock indices agree:

$$\begin{aligned} E[R_{k_1,t+1}R_{k_2,t+1}] \;&= \tfrac{1}{N^2}\sum_{i\neq j} E\big[f_{k_1}(X_{i,t})r_{i,t+1}\big]\, E\big[f_{k_2}(X_{j,t})r_{j,t+1}\big] \\ &\quad + \tfrac{1}{N^2}\sum_{i} E\big[f_{k_1}(X_{i,t})f_{k_2}(X_{i,t})r_{i,t+1}^2\big], \end{aligned}$$

the off-diagonal terms factoring because distinct stocks are independent and the diagonal ones being equal across i because the stocks are identically distributed. There are $N(N-1)$ terms of the first kind and N of the second, which is (H.1). In matrix form, $\Psi_{P,N} = \bar{a}_P\bar{a}_P' + N^{-1}(\Psi_{P,1} - \bar{a}_P\bar{a}_P')$; the matrix in parentheses is a covariance and is positive semidefinite. This proves the fixed-P limit and its rank assertion. Since $\bar{a}_P\bar{a}_P'$ has rank at most one, the min–max principle gives $\lambda_j(\Psi_{P,N}) \le N^{-1}\|\Psi_{P,1} - \bar{a}_P\bar{a}_P'\|$ for $j \ge 2$. Condition (H.2) proves (H.3). In the centered fixed-basis example, conditional moments and orthonormality give the displayed diagonal $\Psi_{P,1}$ exactly. Consequently $P^{-1}\Psi_{P,N} = N^{-1}\operatorname{diag}(q_1^2,\dots,q_P^2)$. Its fixed-$j$ variational limits are those of $N^{-1}D_q^2$, proving (H.4); they are positive because at least j coefficients are nonzero, and are bounded by $q_{\max}^2/N$. □

H.1 Scalar operators in the factor model

For finitely many fixed effects $s_1,\dots,s_M$ with positive limiting shares π_ℓ, write $\pi_{N,\ell} = N_\ell/N$. Let $\hat{p}_{N,\ell}$ be the empirical characteristic law within type ℓ. The finite-N conditional moment maps may depend on N; that dependence is suppressed in $\mu_G(X_t;S)$ and $\Omega(X_t;S)$.

Under Assumption B.1, the fixed-effect classes are independent conditional on W_t and the characteristics within type s_ℓ are i.i.d. with law $p_\ell(\cdot \mid W_t) := p(\cdot \mid s_\ell, W_t)$. The conditional strong law therefore gives $\hat{p}_{N,\ell} \Rightarrow p_\ell(\cdot \mid W_t)$ almost surely within every type. Since M is finite and $\pi_{N,\ell} \to \pi_\ell$, the pooled empirical law converges to

$$\hat{p}_N \xrightarrow[N\to\infty]{\text{a.s.}} p(\cdot \mid W_t) := \sum_{\ell=1}^{M} \pi_\ell \, p_\ell(\cdot \mid W_t), \qquad p = \int p(\cdot \mid w) \, P_W(dw) = \sum_{\ell} \pi_\ell \, p_\ell. \tag{H.5}$$

We impose the actual almost-sure limits of the conditional moments given the common state,

$$\mu_G(X_t) \xrightarrow[N\to\infty]{\text{a.s.}} \mu_G(W_t)\,, \qquad \Omega(X_t) \xrightarrow[N\to\infty]{\text{a.s.}} \Omega(W_t)\,, \tag{H.6}$$

measurable with respect to W_t. The second limit is imposed in Assumption B.3(ii). The first is an additional condition, needed only for the premium arguments. Bounded factor second moments also bound the factor means and supply uniform integrability.

Let

$$c_w(z,u) = \bar{\beta}_w(z)' \, \Omega(w) \, \bar{\beta}_w(u)\,. \tag{H.7}$$

Here $c_w(z,u)$ is the systematic co-moment between two distinct stocks with characteristics z and u when the common state is w, the factor-driven part of their return second moment, and $\bar{s}(z)$ of (G.1) is the idiosyncratic variance of a stock with characteristics z. The product form is not an assumption: given the state, the two stocks' characteristics are independent across stocks and their posterior fixed-effect probabilities factorize in the limit, so the conditional expectation of $B(z,S_1)'\Omega(w)B(u,S_2)$ splits into $\bar{\beta}_w(z)'\Omega(w)\bar{\beta}_w(u)$. On the diagonal no such splitting occurs, and it is $\bar{b}_w(z)$ of (G.1), not $\bar{\beta}_w(z)'\Omega(w)\bar{\beta}_w(z)$, that appears.

A managed portfolio $\Pi_N = \frac{1}{N}\sum_{i\le N} \psi(X_{i,t})\, r_{i,t+1}$ uses the same weighting function across

fixed effects. Its limiting systematic second moment can be expressed using the pooled law $p(\cdot \mid W) = \int_{\mathcal{T}} p(\cdot \mid s, W)\, \nu(ds)$ and the posterior loading in (G.1). The pooled law alone does not determine this second moment: the association between characteristics and fixed-effect-dependent loadings also matters.

Bayes' rule gives the following identity for the loading already defined in (G.1):

$$\bar{\beta}_w(u) \;=\; \int_{\mathcal{T}} B(u,s)\, \frac{dp(\cdot \mid s, w)}{dp(\cdot \mid w)}(u)\, \nu(ds)\,. \tag{H.8}$$

This projection calculates systematic co-moments using observable characteristics. Portfolio weights remain functions of the characteristics alone.

In the factor model of Assumption 3, $\Omega(w)$ in (H.7) is the conditional factor second moment of (15) at the common shock $W_t = w$. The own systematic second moment is uniformly bounded by the finite-N constant of (B.7): $\sup_{z,w} c_w(z,z) \le \bar{q}$. Indeed, Jensen's inequality gives $c_w(z,z) \le \bar{b}_w(z)$ in (G.1), and $\bar{b}_w(z) \le \bar{q}$ because $\Omega(w)$ is an almost-sure limit of the matrices $\Omega(X)$ over which (B.7) takes its essential supremum. Let $\mathcal{K}$ be the scalar kernel operator on $L^2(p)$ generated by the single-stock kernel k_0 of (E.1):

$$(\mathcal{K}\phi)(z) = \int k_0(z,u)\phi(u)\, p(du). \tag{H.9}$$

Assumption H.2 (Hilbert–Schmidt portfolio kernel) The operator $\mathcal{K}$ of (H.9) has finite Hilbert–Schmidt norm,

$$\|\mathcal{K}\|_{\mathrm{HS}}^2 \;=\; \iint k_0(z,u)^2\, p(dz)\, p(du) \;<\; \infty\,, \tag{H.10}$$

and each feature is square integrable,

$$\|f(\cdot;\theta)\|_{L^2(p)} \;<\; \infty \qquad \text{for } \mu_\theta\text{-almost every } \theta. \tag{H.11}$$

The feature condition is separate from the Hilbert–Schmidt bound: a square-integrable kernel need not have an integrable diagonal. In particular, it need not satisfy

$$\int \|f(\cdot;\theta)\|^2_{L^2(p)}\,\mu_\theta(d\theta) < \infty.$$

Nevertheless the diagonal-cut argument in the proof of Lemma A.10, applied on $L^2(p)$ using (E.5) and (H.11), gives

$$\langle h, \mathcal{K}h\rangle = \int |\langle h, f(\cdot;\theta)\rangle|^2 \mu_\theta(d\theta) \le \|\mathcal{K}\|\,\|h\|_2^2. \tag{H.12}$$

Completeness in Assumption G.1(d) therefore makes $\mathcal{K}$ injective, without a trace assumption. For a bounded measurable g, write M_g for multiplication by g. Define the comovement operator $\mathcal{H}$ by

$$\langle \phi, \mathcal{H}\psi\rangle_{L^2(p)} = E_W[(\Gamma^\Omega_W\phi)'(\Gamma^\Omega_W\psi)], \qquad \Gamma^\Omega_W\psi := \Omega(W)^{1/2}\int \bar\beta_W(u)\psi(u)\,p(du \mid W). \tag{H.13}$$

The bound on the systematic own second moments and conditional Jensen give

$$\|\Gamma^\Omega_w\psi\|^2 \le \bar q\left(\int |\psi(u)|\,p(du \mid w)\right)^2 \le \bar q\int |\psi(u)|^2\,p(du \mid w).$$

Averaging over W proves $0 \preceq \mathcal{H} \preceq \bar q I$. Equivalently, $(\mathcal{H}\psi)(z) = E[c_W(z, X_2)\psi(X_2) \mid X_1 = z]$ under the limiting pooled pair-and-state law. The posterior loading in this formula retains the association between characteristics and fixed effects, although the portfolio weighting has only a characteristic argument.

For the finite cross-section, let $\bar p_N = \sum_\ell (N_\ell/N)p_\ell$ and define

$$\mathsf{d}_N(z) := \frac{d\bar p_N}{dp}(z) = \sum_{\ell=1}^M \frac{N_\ell}{N}\frac{dp_\ell}{dp}(z). \tag{H.14}$$

Since $dp_\ell/dp \le 1/\pi_\ell$ and $N_\ell/N \to \pi_\ell$, the functions d_N are uniformly bounded and $\|\mathsf{d}_N - 1\|_\infty \to 0$. With (X_1, X_2) a uniformly sampled ordered pair of distinct stocks, put

$$\begin{aligned}(\mathcal{H}^{(N)}\psi)(z) &= E[B(z,S_1)'\Omega(X)B(X_2,S_2)\psi(X_2) \mid X_1 = z],\\ \varsigma^2_{(N)}(z) &= E[B(z,S_1)'\Omega(X)B(z,S_1) + \sigma^2_\varepsilon(z,S_1) \mid X_1 = z].\end{aligned} \tag{H.15}$$

For $N = 1$, omit the off-diagonal operator. The multiplier d_N transfers the finite pooled marginal $\bar{p}_N$ to the fixed reference law p; thus the conditional pair and own-moment formulas represent quadratic forms on that one fixed space.

The compact, positive operator $\mathcal{A} := \mathcal{K}^{1/2}\mathcal{H}\mathcal{K}^{1/2}$ has the same nonzero spectrum as $\mathcal{K}\mathcal{H}$ by Lemma A.8. Define

$$\kappa_j := \lambda_j(\mathcal{A}) = \lambda_j(\mathcal{K}\mathcal{H}). \tag{H.16}$$

Here zeros are appended if the rank is finite. Since $0 \preceq \mathcal{A} \preceq \bar{q}\mathcal{K}$, Lemma A.7 gives $\sum_j \kappa_j^2 \le \bar{q}^2\|\mathcal{K}\|^2_{\mathrm{HS}} < \infty$. Injectivity and dense range of $\mathcal{K}^{1/2}$ imply $\operatorname{rank}(\mathcal{A}) = \operatorname{rank}(\mathcal{H})$: the two operators have the same finite rank if either is finite-rank, and otherwise both have infinite rank.

The range of $\mathcal{H}$ also has a direct statewise description. For every ψ, $\langle \psi, \mathcal{H}\psi\rangle = E\|\Gamma^\Omega_W\psi\|^2$ vanishes exactly when $\Gamma^\Omega_W\psi = 0$ almost surely. Taking orthogonal complements shows that $\overline{\operatorname{range}(\mathcal{H})}$ is the essential closed span of the representers $(\Gamma^\Omega_w)^* a$, $a \in \mathbb{R}^K$: the smallest closed subspace containing these vectors for every a and almost every w. These representers are $u \mapsto a'\Omega(w)^{1/2}\bar{\beta}_w(u)\mathsf{d}_w(u)$. Nonsingularity of $\Omega(w)$ implies that their essential closed span equals that of the loading-weighted densities $a'\bar{\beta}_w\mathsf{d}_w$.

H.2 The large-cross-section limit

Let $T_{K,N}$ denote the portfolio-kernel operator T_K in the N-stock economy. For each fixed N, define its spectral constants by

$$\kappa_{N,j} := \lambda_j(T_{K,N}). \tag{H.17}$$

Under the trace condition these equal $\lambda_j^*(N)$, the exact limits $\lim_{P\to\infty} \lambda_j(\Psi_{P,N})/P$; under the weaker Hilbert–Schmidt hypotheses they give the lower bounds in Theorem E.5.

Large-N regularity for a random-feature menu. The first requirement of Assumption G.2 is the large-N analogue of condition (C.5) of Appendix C.2.1:

$$\sup_{N\geq 1} \frac{1}{N^2} \int E\big[(f(X_t;\theta)' r_{t+1})^2\big]\, \mu_\theta(d\theta) \;<\; \infty\,. \tag{H.18}$$

Lemma H.3 (The large-N operator limit) *Suppose there are finitely many fixed effects with positive limiting shares and impose Assumptions 3, B.3(ii)–(iv), G.1, and H.2. For the kernel spectral constants $\kappa_{N,j}$ of (H.17), there is a nonincreasing sequence $(\kappa_j)_{j\geq 1}$ of nonnegative numbers with $\sum_j \kappa_j^2 < \infty$ such that, for every fixed j,*

$$\kappa_{N,j} \;\xrightarrow[N\to\infty]{}\; \kappa_j\,. \tag{H.19}$$

The number of positive limits is $\operatorname{rank}(\mathcal{H})$.

Proof of Lemma H.3. The argument uses the fixed pooled space $L^2(p)$. Assumption H.2 makes $\mathcal{K}$ Hilbert–Schmidt, and (H.12) gives its injectivity.

For $\psi \in L^2(p)$, put

$$J_N\psi := \frac{1}{N}\sum_{i=1}^{N} \psi(X_i) r_i$$

and define the positive self-adjoint operator Q_N on $L^2(p)$ by

$$\langle \phi, Q_N \psi \rangle_{L^2(p)} := E[(J_N \phi)(J_N \psi)]. \tag{H.20}$$

This definition is valid although the i-th coordinate has marginal $p_{\ell(i)}$: since $p = \sum_\ell \pi_\ell p_\ell$, one has $dp_\ell/dp \le 1/\pi_\ell$, and the finitely many shares N_ℓ/N are uniformly bounded. Assumption B.3(iv) bounds systematic moments by $\bar{q}$, and Assumption 3 bounds residual variances. Together they give a bound on (H.20) that is uniform in N. In particular, Q_N is a bounded operator on the single fixed space $L^2(p)$.

By (E.3) and Fubini's theorem, $T_{K,N} = J_N \mathcal{K} J_N^*$. For the interchange, conditional Cauchy–Schwarz, the bounded stock second moments, and $dp_\ell/dp \le 1/\pi_\ell$ give, for $g \in L^2(\mathbb{P})$,

$$a_g(x) := \frac{1}{N} \sum_i \frac{dp_{\ell(i)}}{dp}(x) E[|r_i g(Z)| \mid X_i = x] \in L^2(p).$$

For any $g, h \in L^2(\mathbb{P})$, the absolute bilinear integral is bounded by

$$\iint |k_0(x,u)| a_g(x) a_h(u)\, p(dx) p(du) \le \|k_0\|_2 \|a_g\|_2 \|a_h\|_2 < \infty.$$

Fubini then identifies the signed counterparts of a_g and a_h as $J_N^* g$ and $J_N^* h$, proving the operator identity. Since $Q_N = J_N^* J_N$, apply Lemma A.8 to $A = J_N \mathcal{K}^{1/2}$. The fixed-$N$ constants $\kappa_{N,j} = \lambda_j(T_{K,N})$ are the ordered spectral values of

$$\mathcal{A}_N := \mathcal{K}^{1/2} Q_N \mathcal{K}^{1/2}. \tag{H.21}$$

This gives the required reduction without assigning a separate weighting to each fixed effect. At fixed N and under two-sided bounds on $\Sigma_N(X)$, Proposition E.6 reduces the same spectrum to an eigenvalue problem in the N stocks' characteristics.

We next identify the limit of Q_N. Expanding (H.20) into diagonal and off-diagonal stock

pairs gives

$$Q_N = \frac{N-1}{N}\widehat{\mathcal{H}}_N + \frac{1}{N}\widehat{V}_N, \tag{H.22}$$

where $\widehat{V}_N = M_{\mathsf{d}_N \varsigma^2_{(N)}}$ and the first term is the off-diagonal quadratic form. If $\mathcal{H}^{(N)}$ denotes the conditional ordered-pair operator in (H.15), then

$$\widehat{\mathcal{H}}_N = M_{\mathsf{d}_N}\mathcal{H}^{(N)}, \qquad \mathsf{d}_N = \frac{d\bar{p}_N}{dp} = \sum_\ell \frac{N_\ell}{N}\frac{dp_\ell}{dp}. \tag{H.23}$$

Indeed, $\mathcal{H}^{(N)}$ conditions on the first characteristic, whose finite-N marginal is $\bar{p}_N = \mathsf{d}_N p$; transferring its quadratic form to $L^2(p)$ contributes the multiplier d_N. Equation (H.22) is therefore exact for every N and every fixed-effect composition, and (H.23) supplies precisely the normalizer recorded in (H.14).

The two terms are uniformly bounded. The Cauchy–Schwarz inequality for the positive form induced by $\Omega(X)$, followed by conditional Jensen, gives

$$\sup_{N\geq N_0} \|\mathcal{H}^{(N)}\|_{\mathrm{op}} \leq C\bar{q}, \qquad \sup_N \|\widehat{\mathcal{H}}_N\|_{\mathrm{op}} \leq C\bar{q}, \qquad \sup_N \|\widehat{V}_N\|_{\mathrm{op}} \leq C(\bar{q}+\bar{\sigma}^2), \tag{H.24}$$

for a finite constant C depending only on the positive limiting shares. Here $dp_\ell/dp \leq 1/\pi_\ell$ and $N_\ell/N \to \pi_\ell$ imply $\|\mathsf{d}_N - 1\|_\infty \to 0$. Choose N_0 so that $1/2 \leq \mathsf{d}_N \leq 3/2$ for $N \geq N_0$; this lower bound permits division by d_N when bounding the unweighted conditional operator $\mathcal{H}^{(N)}$. For each fixed $\psi \in L^2(p)$, the convergence of the fixed-effect shares, the conditional law of large numbers within every fixed effect, and (H.6) give

$$\widehat{\mathcal{H}}_N\psi \longrightarrow \mathcal{H}\psi \quad \text{in } L^2(p). \tag{H.25}$$

To see the L^2 conclusion, condition separately on the finitely many ordered pairs of fixed effects. For each pair, boundedness of $B'\Omega B$ and conditional Jensen dominate the difference

by a fixed $L^2(p_\ell)$ function proportional to $E[|\psi(X_2)| \mid X_1 = \cdot]$. The density bounds $dp_\ell/dp \leq 1/\pi_\ell$ transfer these finitely many bounds to $L^2(p)$. Conditional dominated convergence then applies to each pair, and $N_\ell/N \to \pi_\ell$ completes the sum. Thus, with $D_N := Q_N - \mathcal{H}$, (H.22) implies

$$D_N \longrightarrow 0 \quad \text{strongly}, \qquad \sup_N \|D_N\|_{\text{op}} < \infty. \tag{H.26}$$

Set $\mathcal{A} = \mathcal{K}^{1/2}\mathcal{H}\mathcal{K}^{1/2}$. Since $\mathcal{K}^{1/2}$ is compact, the standard compactness lemma (a uniformly bounded, strongly convergent sequence composed on the right with a compact operator converges in norm) applied to (H.26) yields $\|D_N\mathcal{K}^{1/2}\|_{\text{op}} \to 0$. Hence

$$\|\mathcal{A}_N - \mathcal{A}\|_{\text{op}} \leq \|\mathcal{K}^{1/2}\|_{\text{op}} \, \|D_N\mathcal{K}^{1/2}\|_{\text{op}} \longrightarrow 0. \tag{H.27}$$

Weyl's inequality now gives, for every fixed j,

$$\kappa_{N,j} = \lambda_j(\mathcal{A}_N) \longrightarrow \lambda_j(\mathcal{A}) =: \kappa_j.$$

This proves (H.19).

The rank and square-summability claims follow from the argument after (H.16), applied to this same limiting operator. □

Proposition H.4 (The large-N limit for a random-feature menu) *Suppose that the fixed-effect array takes finitely many distinct values, each with a positive limiting share. The limits are iterated: for each fixed N, let $P \to \infty$ first, which yields the per-portfolio constants $\kappa_{N,j}$ of (H.17), with the exact limit (F.1) proved in Appendix F.1; then let $N \to \infty$. Under Assumptions 1, 3, B.3(ii)–(iv), H.2, and G.1, and condition (H.18), there is an infinite, nonincreasing sequence $\kappa_1 \geq \kappa_2 \geq \cdots \geq 0$ with $\sum_j \kappa_j < \infty$, given by (H.16), such that, for*

every fixed j,

$$\kappa_{N,j} \xrightarrow[N\to\infty]{} \kappa_j\,, \tag{H.28}$$

and the number of positive κ_j equals $\operatorname{rank}(\mathcal{H})$. *Consequently* $\lim_{N\to\infty}\lim_{P\to\infty}\lambda_j(\Psi_{P,N})/P = \kappa_j$ *for every fixed j, almost surely over the random labels.*

Proof of Proposition H.4. The reduction in the proof of Lemma H.3 gives, under the stated hypotheses, the convergence $\kappa_{N,j} \to \kappa_j = \lambda_j(\mathcal{K}\mathcal{H})$ of (H.28) and the rank count, and it does so without using the summability of the κ_j. Two further points upgrade the conclusion to the present statement. First, for each fixed N the $P \to \infty$ step is the exact limit (F.1) proved in Appendix F.1 under the trace condition, so the iterated limit $\lim_{N\to\infty}\lim_{P\to\infty}\lambda_j(\Psi_P)/P = \kappa_j$ is a genuine limit. Second, the large-N trace condition (H.18) gives $\sup_N \sum_j \kappa_{N,j} < \infty$ (recall $\sum_j \kappa_{N,j} = \operatorname{tr} T_K = N^{-2}\int E[(f(X_t;\theta)'r)^2]\mu_\theta(d\theta)$); since $\kappa_{N,j} \to \kappa_j \geq 0$, Fatou's lemma for sums gives

$$\sum_j \kappa_j \;\leq\; \liminf_{N\to\infty} \sum_j \kappa_{N,j} \;<\; \infty\,,$$

so $\mathcal{A} = \mathcal{K}^{1/2}\mathcal{H}\mathcal{K}^{1/2}$ is trace-class and $\sum_j \kappa_j < \infty$, in place of the square-summability $\sum_j \kappa_j^2 < \infty$ of Lemma H.3. □

H.3 The large-cross-section limit for a partition menu

Proposition F.5 is a fixed-N statement, and its constants $\lambda_j^*(N)$ depend on the cross-section: they are the $\lambda_j^*(N) = \lambda_j(\mathcal{A}_{\text{sort}})$ of (F.19), built from the two objects the sort cannot tell apart: the levered own variance ς^2/N and the dependence-adjusted comovement $\tilde{c} = \frac{N-1}{N}c\varrho$ between distinct stocks. This subsection lets $N \to \infty$. The point is that the two objects part company in the limit: the levered own variance is of order $1/N$ and washes out, and what is left is exactly the comovement.

For a partition menu, Assumption G.2 asks that the own variance and the comovement settle as the cross-section grows. The assumption represents all cross-sections on one fixed pooled probability space (X, p) and holds the refining partitions (hence their limiting mass density g) fixed as N varies. On that common space, it requires

$$\sup_N \sup \varsigma_{(N)}^2 \;<\; \infty\,, \qquad \iint \left(c_{(N)}\varrho_{(N)} - c_\infty \varrho_\infty\right)^2 dp\, dp \;\xrightarrow[N\to\infty]{}\; 0\,, \tag{H.29}$$

for a limiting dependence-adjusted comovement $c_\infty\varrho_\infty$ with $\iint (c_\infty\varrho_\infty)^2\, dp\, dp < \infty$, where the subscript records the cross-section size. The common-p representation and the displayed L^2 convergence are triangular-array restrictions; they are not consequences of pointwise de Finetti convergence alone. Under Assumption 3, the following conditions suffice for them. The average marginal p, the pair density ϱ, and the partitions do not depend on N. With several fixed effects, an N-free pair density restricts the fixed-effect composition, because finite-N pair laws generally differ from their limit. Define the conditional factor moment and its limit by

$$\begin{aligned} c_{(N)}(z,u) &= E[B(z,S_1)'\Omega(X_t)B(u,S_2) \mid X_1 = z, X_2 = u], \\ c_\infty(z,u) &= E[B(z,S_1)'\Omega(W_t)B(u,S_2) \mid X_1 = z, X_2 = u]. \end{aligned}$$

The limit c_∞ equals the posterior-mean form $E_W[\bar\beta_W(z)'\Omega(W)\bar\beta_W(u) \mid X_1 = z, X_2 = u]$. Because B is bounded, (H.6) and dominated convergence give $c_{(N)} \to c_\infty$ in the displayed weighted L^2 norm, which is the second half of (H.29). The first requirement is the uniform variance bound of Assumption G.1(a) together with $\bar q < \infty$.

Proposition H.5 (The large-N limit for a partition menu) *Let the partition conditions of Appendix C.2.3, Assumption G.2 for a partition menu, and (H.29) hold. At every $N \geq 2$, require $P_{2,N} \ll p \otimes p$ and $c_{(N)}\varrho_{(N)} \in L^2(p \otimes p)$, where $P_{2,N}$ is the ordered-pair law. Write $T_{c_\infty\varrho_\infty}$ for the limiting comovement, the integral operator on $L^2(p)$ with kernel $c_\infty\varrho_\infty$.*

Then, for every fixed j,

$$\lim_{N\to\infty}\lim_{P\to\infty}\frac{\lambda_j(\Psi_P)}{P} \;=\; \lim_{N\to\infty}\lambda_j^*(N) \;=\; \kappa_j \;:=\; \lambda_j\big(M_{\sqrt{g}}\,T_{c_\infty\varrho_\infty}\,M_{\sqrt{g}}\big)\,,$$

with $\sum_j \kappa_j^2 < \infty$. Under Assumption 3, $T_{c_\infty\varrho_\infty} = \mathcal{H}$, the comovement operator (H.13), so the constants are $\kappa_j = \lambda_j(M_g\mathcal{H}) \geq 0$. Because $g \in [c_-, c_+]$ makes $M_{\sqrt{g}}$ boundedly invertible, the count of positive κ_j is $\operatorname{rank}(\mathcal{H})$ *whatever the cell masses, exactly as for a random-feature menu; for cells of equal mass, where $g \equiv 1$, the weighting drops out and $\kappa_j = \lambda_j(\mathcal{H})$. Without (C.6) the same holds with $\mathcal{H}$ replaced by its compression $E[\,\cdot \mid \mathcal{F}_\infty]\,\mathcal{H}\,E[\,\cdot \mid \mathcal{F}_\infty]$ to the σ-field the sort's cuts generate.*

Proof of Proposition H.5. The inner limit is Step 4 of the proof of Proposition F.5: $\lambda_j(\Psi_P)/P \to \lambda_j^*(N) = \lambda_j(\mathcal{A}_g)$ with $\mathcal{A}_g = M_{g\varsigma^2/N} + M_g T_{\tilde{c}} M_g$ on $L^2(\mathfrak{m})$, at each fixed N. Transport this to $L^2(p)$ by the unitary $U\psi = g^{-1/2}\psi$, under which $U\mathcal{A}_gU^{-1} = M_{g\varsigma^2/N} + M_{\sqrt{g}}T_{\tilde{c}}M_{\sqrt{g}}$; the spectrum is unchanged. For the outer limit, bound the two summands separately. Multiplication by $g\varsigma^2/N$ has operator norm at most $c_+ \sup \varsigma_{(N)}^2/N$, which tends to zero by (C.8) and the first half of (H.29). And $\tilde{c} = \frac{N-1}{N}c_{(N)}\varrho_{(N)}$, so

$$\begin{aligned}\big\|T_{\tilde{c}} - T_{c_\infty\varrho_\infty}\big\|_{\mathrm{op}} \;&\leq\; \big\|T_{\tilde{c}} - T_{c_\infty\varrho_\infty}\big\|_{\mathrm{HS}} \\ &\leq\; \tfrac{N-1}{N}\big\|c_{(N)}\varrho_{(N)} - c_\infty\varrho_\infty\big\|_{L^2(p\otimes p)} \;+\; \tfrac{1}{N}\big\|c_\infty\varrho_\infty\big\|_{L^2(p\otimes p)} \;\longrightarrow\; 0\end{aligned}$$

by the second half of (H.29). Sandwiching between the bounded $M_{\sqrt{g}}$ preserves this,

$$\big\|M_{\sqrt{g}}T_{\tilde{c}}M_{\sqrt{g}} - M_{\sqrt{g}}T_{c_\infty\varrho_\infty}M_{\sqrt{g}}\big\|_{\mathrm{op}} \;\leq\; c_+\,\big\|T_{\tilde{c}} - T_{c_\infty\varrho_\infty}\big\|_{\mathrm{op}} \;\longrightarrow\; 0.$$

So the whole of $U\mathcal{A}_gU^{-1}$ converges in operator norm to $M_{\sqrt{g}}T_{c_\infty\varrho_\infty}M_{\sqrt{g}}$, and Weyl's inequality (Lemma A.7) gives $\lambda_j^*(N) \to \lambda_j(M_{\sqrt{g}}T_{c_\infty\varrho_\infty}M_{\sqrt{g}})$ for every fixed j. Square-summability

follows from the Hilbert–Schmidt property of $T_{c_\infty \varrho_\infty}$, the bound $\iint (c_\infty \varrho_\infty)^2 \, dp \, dp < \infty$, together with $\|M_{\sqrt{g}}\|^2 \le c_+$.

Under Assumptions 3 and B.3(ii),(iv), bounded convergence and the second-moment limit in (H.6) imply that the finite signed measures $c_{(N)}\varrho_{(N)}\, d(p \otimes p)$ converge setwise to the factor-model co-moment measure $c_\infty \, dP_2$. The assumed L^2 limit must therefore be the density of that same measure. This identifies $T_{c_\infty \varrho_\infty} = \mathcal{H}$ in (H.13). Indeed, disintegration against the pair law $P_2 = \varrho_\infty \cdot (p \otimes p)$ of (B.6) gives

$$\begin{aligned}(\mathcal{H}\psi)(z) &= E[c_W(z, X_2)\psi(X_2) \mid X_1 = z] \\ &= \int c_\infty(z,u)\varrho_\infty(z,u)\psi(u)\, p(du).\end{aligned}$$

Here $c_w(z,u) = \bar{\beta}_w(z)'\Omega(w)\bar{\beta}_w(u)$, as in (H.7). Then $\mathcal{H} = E_W[(\Gamma_W^\Omega)^*\Gamma_W^\Omega] \succeq 0$, hence its congruence by $M_{\sqrt{g}}$ is positive. Since $c_- \le g \le c_+$, this multiplier is boundedly invertible and

$$\ker(M_{\sqrt{g}}\mathcal{H}M_{\sqrt{g}}) = M_{\sqrt{g}}^{-1} \ker \mathcal{H}.$$

The number of positive κ_j therefore equals $\operatorname{rank}(\mathcal{H})$, independently of g. Without (C.6), Step 4 delivers $\lambda_j^*(N) = \lambda_j\big(E[\cdot \mid \mathcal{F}_\infty]\, \mathcal{A}_g\, E[\cdot \mid \mathcal{F}_\infty]\big)$, with the projections taken in the appropriate Hilbert space. Since g is $\mathcal{F}_\infty$-measurable, the projection in $L^2(p)$ commutes with $M_{\sqrt{g}}$. Transport from $L^2(\mathfrak{m})$ therefore gives the stated compression, and compression by a fixed projection preserves the norm convergence. □

Two remarks. First, the levered noise that dominates the fixed-N spectrum (the floor (F.16) and the bulk of Proposition F.5(i), both of order ς^2/N) disappears here. It is a fixed-N artifact of the sort, and the large-N limit strips it away, leaving only comovement. The artifact still matters at any finite N: comovement is visible only where it lifts an eigenvalue above the levered-noise edge a_+, and Proposition F.5(ii) guarantees only $\lambda_j^*(N) \ge a_+$.

Second, the limiting constants are the same object in both menu cases, up to the weighting the menu applies: $\kappa_j = \lambda_j(\mathcal{K}\mathcal{H})$ for a random-feature menu (Proposition H.4) and $\kappa_j = \lambda_j(M_g\mathcal{H})$ for a partition menu, the refinement density g of (C.8) playing for the sort the role $\mathcal{K}$ plays for a random-feature menu; for cells of equal mass $g \equiv 1$ and the sort's weighting drops out. Since $\mathcal{K}^{1/2}$ is injective and $M_{\sqrt{g}}$ is boundedly invertible, the two have infinitely many positive constants under exactly the same condition, $\operatorname{rank}(\mathcal{H}) = \infty$. Genericity, however, is proved below only for random-feature menus, in the regular configuration spaces of the next subsection.

H.4 Genericity of the managed-portfolio factor count

This subsection shows that infinite rank of $\mathcal{H}$, and hence infinitely many positive κ_j, is generic in each regular configuration space. It defines these spaces, shows that each is a Baire space of admissible models, and constructs infinite-rank configurations arbitrarily close to every configuration.

Definition H.6 (Model configurations) Fix the portfolio weights f, the type shares $\pi_\ell > 0$, and the marginal laws p_ℓ, with $p = \sum_\ell \pi_\ell p_\ell$. An admissible model is a tuple $\vartheta = (B, P_W, \{p_\ell(\cdot \mid w)\}_{\ell,w}, \Omega)$ satisfying Assumption 3, with its common characteristic-law clause replaced by Assumption B.1, parts (ii)–(iv) of Assumption B.3, and Assumptions G.2, H.2, and G.1. In particular, the type-specific conditional densities $\mathsf{d}_{\ell,w} = dp_\ell(\cdot \mid w)/dp_\ell$ satisfy

$$\int \mathsf{d}_{\ell,w}(x)\, P_W(dw) = 1 \quad p_\ell\text{-a.e.}, \qquad \ell \le M. \tag{H.30}$$

The family is specified type by type; its pooled conditional law alone does not determine the posterior loadings in (G.1). We do not impose Assumption B.3(i), which concerns the stock factor count rather than the managed-payoff operator.

For genericity, fix a loading function B for which at least one admissible model with the fixed menu, shares, and marginals exists. This reference-model requirement is part of the meaning of admissible B. Let C_b be the Banach space of bounded continuous real functions on $\mathbb{R}^d$, with the supremum norm. On $[0,1]$ with Lebesgue measure du, set

$$\mathcal{E} = \prod_{\ell=1}^{M} L^\infty([0,1]; C_b) \times L^\infty([0,1]; \mathrm{Sym}_K).$$

Here Sym_K is the space of real symmetric $K \times K$ matrices. The first factors consist of strongly measurable, essentially bounded C_b-valued maps. A regular configuration $\vartheta = (\mathsf{d}_1, \dots, \mathsf{d}_M, \widetilde{\Omega})$ belongs to $\mathfrak{M}_B \subset \mathcal{E}$ if

$$\begin{aligned} &\int \mathsf{d}_\ell(u,x)\, p_\ell(dx) = 1 \quad \text{for a.e. } u, \qquad \int_0^1 \mathsf{d}_\ell(u,x)\, du = 1 \quad p_\ell\text{-a.e.}, \quad \ell \le M, \\ &\operatorname*{ess\,inf}_u \inf_x \min_{\ell \le M} \mathsf{d}_\ell(u,x) > 0, \qquad \operatorname*{ess\,inf}_u \lambda_{\min}(\widetilde{\Omega}(u)) > 0. \end{aligned} \tag{H.31}$$

Its primitive metric is

$$d_{\mathrm{reg}}(\vartheta, \vartheta') = \max_{\ell \le M} \operatorname*{ess\,sup}_u \|\mathsf{d}_\ell(u,\cdot) - \mathsf{d}'_\ell(u,\cdot)\|_\infty + \operatorname*{ess\,sup}_u \|\widetilde{\Omega}(u) - \widetilde{\Omega}'(u)\|_{\mathrm{op}}. \tag{H.32}$$

Draw U uniformly on $[0,1]$, and draw characteristics independently within and across types conditional on U, with densities $\mathsf{d}_\ell(U,\cdot)$ relative to p_ℓ. The canonical state and its factor second moment are

$$W = \big(\mathsf{d}_\ell(U,\cdot)p_\ell : \ell \le M\big), \qquad \Omega(W) = E[\widetilde{\Omega}(U) \mid W].$$

A property is **generic in $\mathfrak{M}_B$** if its complement is meager in d_{reg}, that is, a countable union of nowhere dense sets.

For either an admissible model or a regular configuration, let $\mathcal{H}_\vartheta$ be the operator in

(H.13) and put

$$\mathcal{A}_\vartheta = \mathcal{K}^{1/2}\mathcal{H}_\vartheta\mathcal{K}^{1/2}, \qquad d_{\mathrm{sp}}(\vartheta,\vartheta') = \|\mathcal{A}_\vartheta - \mathcal{A}_{\vartheta'}\|_{\mathrm{HS}}.$$

The spectral distance d_{sp} is used only for approximation; it is not the topology defining genericity. The eigenvalues of $\mathcal{A}_\vartheta$ are the constants κ_j in (H.16). By Lemma H.3, completeness of the menu gives $\operatorname{rank}\mathcal{A}_\vartheta = \operatorname{rank}\mathcal{H}_\vartheta$.

The regular configurations retain continuous positive conditional densities and uniform moment bounds. They specify models directly; they are not identified through their comovement operators. Their primitive topology is stronger than convergence of the portfolio spectra. The next lemma shows that every regular configuration has an admissible realization and that $\mathfrak{M}_B$ is a Baire space. The lemma after it shows that regular configurations approximate all admissible spectra.

Lemma H.7 (A Baire space of admissible configurations) *For every fixed admissible B, the nonempty space $\mathfrak{M}_B$ is Baire. Every configuration has an admissible realization, and the map $\vartheta \mapsto \mathcal{A}_\vartheta$ is continuous from d_{reg} to the Hilbert–Schmidt norm.*

Proof of Lemma H.7. Each Bochner L^∞ space in $\mathcal{E}$ is Banach; separability of C_b is not required for completeness. Indeed, a Cauchy sequence has a subsequence whose successive essential-supremum distances are summable. Outside one null set this subsequence converges uniformly in its Banach range, and its limit is strongly measurable and essentially bounded. The full sequence then converges in the same norm. The two normalization maps in (H.31) are continuous linear maps into $L^\infty([0,1])$ and $L^2(p_\ell)$, respectively. Their prescribed level sets therefore form a closed affine subspace of $\mathcal{E}$. The two strict uniform lower bounds define an open subset of that subspace: a perturbation smaller than half either lower bound preserves it. This subset is nonempty, since $\mathsf{d}_\ell \equiv 1$ and $\widetilde{\Omega} \equiv I_K$ belong to it. An open subset of a complete metric space is a Baire space, which proves the first assertion.

To realize a configuration, draw $G \mid U \sim \mathcal{N}(0, \widetilde{\Omega}(U))$ independently of the characteristics conditional on U, and use independent centered Gaussian residuals with the prescribed bounded positive variances. The density constraints give exactly the fixed type marginals. For each fixed N, the posterior factor second moment is

$$\Omega_N(X) = \frac{\displaystyle\int_0^1 \widetilde{\Omega}(u) \prod_{i \le N} \mathsf{d}_{\ell(i)}(u, X_i)\, du}{\displaystyle\int_0^1 \prod_{i \le N} \mathsf{d}_{\ell(i)}(u, X_i)\, du}.$$

Both numerator and denominator are continuous in the type empirical laws: for a.e. u, the log densities are bounded continuous functions, and their uniform bounds permit dominated convergence of the likelihood products. The denominator is bounded away from zero at each fixed N. The within-type strong laws identify the vector of directing laws W from the infinite characteristic sample. Conditional on W, that sample contains no additional information about U. Because the stocks form one infinite array, the sigma-fields $\sigma(X_1, \dots, X_N)$ increase. Bounded martingale convergence yields $\Omega_N(X) \to E[\widetilde{\Omega}(U) \mid W] = \Omega(W)$ almost surely. Conditioning preserves both ellipticity bounds; the factor means are zero. The uniform density bounds imply (G.2).

The fixed menu has $\int \|f(\cdot;\theta)\|^2_{L^2(p)} \mu_\theta(d\theta) < \infty$: this follows from the trace condition of any admissible reference model and the residual lower bound, at a stock count containing every type. Bounded stock conditional second moments, Jensen's inequality, and $p_\ell \le p/\pi_\ell$ give a constant C independent of N with

$$E[(R^{\phi}_{N,t+1})^2] \le C\|\phi\|^2_{L^2(p)}.$$

Hence $\bar{e}(\theta) = C\|f(\cdot;\theta)\|_2^2$ supplies (G.3) and the trace bound. The remaining kernel and

menu requirements depend only on the fixed menu and marginal laws. All conditions defining admissibility are therefore satisfied.

For continuity, write $r_\ell = dp_\ell/dp$ and define

$$(b_u a)(x) = \sum_\ell \pi_\ell r_\ell(x) \mathsf{d}_\ell(u,x) B(x,s_\ell)' a, \qquad \mathcal{H}_\vartheta = \int_0^1 b_u \widetilde{\Omega}(u) b_u^* \, du.$$

The identity remains valid after conditioning the factor moment on W, since b_u depends only on the directing laws. On any bounded primitive neighborhood, $\|b_u\|_{\mathrm{HS}}$ is uniformly bounded and

$$\operatorname*{ess\,sup}_u \|b_u - b_u'\|_{\mathrm{HS}} \le \sqrt{K}\, \|B\|_\infty \max_\ell \operatorname*{ess\,sup}_u \|\mathsf{d}_\ell(u,\cdot) - \mathsf{d}_\ell'(u,\cdot)\|_\infty.$$

Expanding the difference of the integrands shows continuity of $\mathcal{H}_\vartheta$ in trace norm. For any bounded self-adjoint D, the Schatten Hölder inequality with exponents $(4,\infty,4)$ gives

$$\|\mathcal{K}^{1/2} D \mathcal{K}^{1/2}\|_{\mathrm{HS}} \le \|\mathcal{K}^{1/2}\|_{\mathfrak{S}_4}^2 \|D\|_{\mathrm{op}} = \|\mathcal{K}\|_{\mathrm{HS}} \|D\|_{\mathrm{op}}. \tag{H.33}$$

Here $\|T\|_{\mathfrak{S}_4} = (\sum_j s_j(T)^4)^{1/4}$ is the Schatten fourth norm; the estimate does not require D to be positive. It follows that $\mathcal{A}_\vartheta$ is Hilbert–Schmidt and depends continuously on the regular configuration. □

Lemma H.8 (Finite-state approximation) *For every admissible model ϑ in Definition H.6 and every $\epsilon_0 > 0$, there is a finite-state regular configuration $\vartheta_0 \in \mathfrak{M}_B$ with $d_{\mathrm{sp}}(\vartheta, \vartheta_0) < \epsilon_0$, with the same loadings B and fixed-effect marginals.*

Proof of Lemma H.8. Write $r_\ell = dp_\ell/dp$ and $v_{\ell,w} = \pi_\ell r_\ell \mathsf{d}_{\ell,w}$, where $\mathsf{d}_{\ell,w} = dp_\ell(\cdot \mid w)/dp_\ell$. The nonnegative functions satisfy $v_{\ell,w} \le \mathsf{d}_w$, so (G.2) puts them in $L^2(P_W; L^2(p))$ with uniformly bounded statewise norms. Choose increasing finite sigma-fields $\mathcal{F}_n$ on the state space such that conditional expectations of all $v_{\ell,W}$ and $\Omega(W)$ converge in their respective

L^2 spaces. The finitely many strongly measurable maps are essentially separably valued. Their joint map therefore generates a countably generated sigma-field up to null sets; take $\mathcal{F}_n$ generated by its first n generators.

For an atom C of $\mathcal{F}_n$ of probability $\gamma_C > 0$, set

$$\mathsf{d}_{\ell,C} = E[\mathsf{d}_{\ell,W} \mid C], \qquad \Omega_C = E[\Omega(W) \mid C].$$

Tonelli's theorem and the type-by-type consistency identities give $0 \le \mathsf{d}_{\ell,C} \le \gamma_C^{-1}$ and $\sum_C \gamma_C \mathsf{d}_{\ell,C} = 1$ p_ℓ-almost everywhere. Moreover $\underline{\omega} I \preceq \Omega_C \preceq \bar{\omega} I$. Let $b_w : \mathbb{R}^K \to L^2(p)$ be

$$(b_w a)(x) = \sum_\ell v_{\ell,w}(x) B(x, s_\ell)' a.$$

Then $\mathcal{H} = E[b_W \Omega(W) b_W^*]$ and $\|b_w\|_{\mathrm{HS}} \le L_B \|\mathsf{d}_w\|_2$, where $L_B = \sup_{x,s} \|B(x,s)\|$. The finite-state model has $b_C = E[b_W \mid C]$ and $\mathcal{H}_n = \sum_C \gamma_C b_C \Omega_C b_C^*$. Conditional-expectation convergence, the uniform bounds just given, and $\|uv^*\|_1 \le \|u\|_{\mathrm{HS}} \|v\|_{\mathrm{HS}}$ show that $\|\mathcal{H}_n - \mathcal{H}\|_1 \to 0$.

We next smooth the finitely many densities while preserving all marginals. For a Borel probability measure ν on $\mathbb{R}^d$, there are positive, constant-preserving contractions T_j on $L^2(\nu)$ that converge strongly to the identity and map bounded functions to bounded continuous functions. One construction uses a finite continuous partition of unity $(q_{j,l})_l$ and the Markov kernel

$$(T_j h)(x) = \sum_l q_{j,l}(x) \frac{\int q_{j,l}(y) h(y)\, \nu(dy)}{\int q_{j,l}\, d\nu}.$$

Terms with zero denominator are omitted. Take partitions with shrinking supports on expanding compact sets and one remaining tail component. Uniform continuity proves convergence on bounded continuous functions on those compact sets, the tail has vanishing

measure, and contractivity plus density of bounded continuous functions proves strong convergence on $L^2(\nu)$. The displayed kernel is symmetric, so T_j also preserves integrals.

Apply this construction separately with $\nu = p_\ell$. For $0 < \eta < 1$, put

$$\mathsf{d}_{\ell,C}^{j,\eta} = (1-\eta)T_{\ell,j}\mathsf{d}_{\ell,C} + \eta.$$

These densities are continuous, bounded below by η, integrate to one, and preserve the type marginals:

$$\sum_C \gamma_C \mathsf{d}_{\ell,C}^{j,\eta} = 1 \quad p_\ell\text{-a.e.}$$

They converge to $\mathsf{d}_{\ell,C}$ in $L^2(p_\ell)$ as $j \to \infty$ and $\eta \downarrow 0$. Since $r_\ell \le 1/\pi_\ell$,

$$\|\pi_\ell r_\ell(\mathsf{d}_{\ell,C}^{j,\eta} - \mathsf{d}_{\ell,C})\|_{L^2(p)}^2 \le \pi_\ell \|\mathsf{d}_{\ell,C}^{j,\eta} - \mathsf{d}_{\ell,C}\|_{L^2(p_\ell)}^2.$$

Thus their loading representers, and then their comovement operators, converge in trace norm to those of the finite-state model.

Represent the finitely many state probabilities by intervals of $[0,1]$, with the smoothed densities and factor moments constant on each interval. This is an element of $\mathfrak{M}_B$, whose admissibility is proved in Lemma H.7. Its comovement operator is the one just constructed. Inequality (H.33) transfers trace-norm convergence of the comovement operators to the required d_{sp} approximation. □

Lemma H.9 (Dense infinite-rank configurations) *For every $\vartheta \in \mathfrak{M}_B$ and $\epsilon_0 > 0$, there is $\vartheta' \in \mathfrak{M}_B$ with $d_{\mathrm{reg}}(\vartheta, \vartheta') < \epsilon_0$ and $\operatorname{rank} \mathcal{H}_{\vartheta'} = \infty$. The space contains such a configuration with $\widetilde{\Omega} \equiv I_K$.*

Proof of Lemma H.9. Choose s_0, a_0 from Assumption G.1(c), with $\|a_0\| = 1$. Let ℓ_0 satisfy $s_{\ell_0} = s_0$, and write $r_\ell = dp_\ell/dp$. The restriction of p_{ℓ_0} to $E = \{x : a_0' B(x, s_0) \neq 0\}$

is nonatomic and nonzero. Choose disjoint open sets, each containing two disjoint neighborhoods of positive $p_{\ell_0}|_E$ measure. In each pair, take two nonnegative continuous bump functions, subtract a multiple of one from the other to make their p_{ℓ_0}-integral zero, and rescale. The resulting bounded continuous functions have disjoint supports and satisfy

$$|h_n| \le 1, \qquad \int h_n \, dp_{\ell_0} = 0, \qquad a_0' B(\cdot, s_0) h_n \neq 0 \quad \text{in } L^2(p_{\ell_0}).$$

Realize independent Rademacher signs ξ_n as the binary digits of $U \sim \mathrm{Unif}[0,1]$. Choose $0 < \delta_n \le \delta < 1$ decreasing to zero and set

$$\mathsf{d}_{\ell_0}(U, x) = 1 + \sum_{n \ge 1} \delta_n \xi_n h_n(x), \qquad \mathsf{d}_\ell(U, x) = 1 \ (\ell \neq \ell_0), \qquad \widetilde{\Omega} = I_K. \tag{H.34}$$

Call this configuration ϑ_{wit} and put $\mathcal{H}_{\mathrm{wit}} := \mathcal{H}_{\vartheta_{\mathrm{wit}}}$. Disjoint supports give uniform convergence, including uniformly over U. The limit is strongly measurable as a C_b-valued map because its partial sums take finitely many values. Each density integrates to one and has mean one at every characteristic. Moreover,

$$1 - \delta \le \mathsf{d}_\ell(U, x) \le 1 + \delta, \qquad \operatorname*{ess\,sup}_w \|\mathsf{d}_w\|_2 \le 1 + \delta. \tag{H.35}$$

Thus the witness belongs to $\mathfrak{M}_B$ by Lemma H.7. Put

$$b = \sum_\ell \pi_\ell a_0' B(\cdot, s_\ell) r_\ell, \qquad u_n = \pi_{\ell_0} a_0' B(\cdot, s_{\ell_0}) h_n r_{\ell_0}.$$

The u_n are nonzero with disjoint supports and $\sum_n \|u_n\|_2^2 \le \|B\|_\infty^2$. Therefore

$$E\left[\left(b + \sum_n \delta_n \xi_n u_n\right) \otimes \left(b + \sum_n \delta_n \xi_n u_n\right)\right] = b \otimes b + \sum_n \delta_n^2 u_n \otimes u_n \preceq \mathcal{H}_{\mathrm{wit}}, \tag{H.36}$$

with trace-norm convergence. The u_n are nonzero and orthogonal by disjoint supports, and every $\delta_n > 0$, so the domination implies $\operatorname{rank} \mathcal{H}_{\mathrm{wit}} = \infty$.

For any $\vartheta \in \mathfrak{M}_B$, interpolate its primitive densities and factor moments with those of the witness: $\vartheta_t = (1-t)\vartheta + t\vartheta_{\mathrm{wit}}$, $0 \le t \le 1$. The normalization constraints and strict bounds are preserved, so every ϑ_t belongs to $\mathfrak{M}_B$. Its loading map $b_{u,t}$ and factor moment are affine in t; hence every matrix element of $\mathcal{H}_{\vartheta_t} = E[b_{U,t}\widetilde{\Omega}_t(U)b^*_{U,t}]$ is a polynomial of degree at most three. For each j, choose j test vectors whose compression of $\mathcal{H}_{\mathrm{wit}}$ is positive definite. The determinant of the corresponding compression of $\mathcal{H}_{\vartheta_t}$ is a polynomial, strictly positive at $t = 1$. It therefore has only finitely many zeros. Outside the countable union of these exceptional sets over j, $\mathcal{H}_{\vartheta_t}$ has infinite rank. Such t occur arbitrarily close to zero, and $d_{\mathrm{reg}}(\vartheta_t, \vartheta) = t\, d_{\mathrm{reg}}(\vartheta_{\mathrm{wit}}, \vartheta)$. This proves

$$\forall \vartheta \in \mathfrak{M}_B \ \forall \epsilon_0 > 0 \quad \exists \vartheta' \in \mathfrak{M}_B : \quad d_{\mathrm{reg}}(\vartheta, \vartheta') < \epsilon_0, \qquad \operatorname{rank} \mathcal{H}_{\vartheta'} = \infty. \tag{H.37}$$

□

Proposition H.10 (Generic infinite factor counts) *Let $\mathfrak{M}_B$ be a regular configuration space. The sets $\mathcal{G}_j = \{\vartheta : \kappa_j(\vartheta) > 0\}$ are open and dense in $\mathfrak{M}_B$, and $\mathcal{G} = \bigcap_{j\ge1} \mathcal{G}_j = \{\vartheta : \operatorname{rank} \mathcal{H}_\vartheta = \infty\}$ is a dense comeager set. Its complement is not comeager, and $\mathcal{G}$ is not open.*

Proof of Proposition H.10. Lemma H.7 and Weyl's inequality (Lemma A.7, part 1) give continuity of every κ_j in d_{reg}, since

$$|\kappa_j(\vartheta) - \kappa_j(\vartheta')| \le \|\mathcal{A}_\vartheta - \mathcal{A}_{\vartheta'}\|_{\mathrm{op}} \le d_{\mathrm{sp}}(\vartheta, \vartheta').$$

Thus each $\mathcal{G}_j$ is open. Completeness of the menu identifies its positive eigenvalue count with $\operatorname{rank} \mathcal{H}_\vartheta$, so Lemma H.9 makes every $\mathcal{G}_j$ and their intersection dense. The complement of $\mathcal{G}$ is a countable union of closed sets with empty interior, hence is meager. Since $\mathfrak{M}_B$ is a

nonempty Baire space by Lemma H.7, the whole space is not meager in itself. Consequently the complement of $\mathcal{G}$ cannot also be comeager.

Finally, truncate the witness in (H.34) after n signs, keeping $\widetilde{\Omega} = I_K$. The resulting regular configuration has at most 2^n directing states and therefore $\operatorname{rank} \mathcal{H} \leq K2^n$. Its primitive distance to the witness is at most $\sup_{k>n} \delta_k \to 0$, since the h_k have disjoint supports. An infinite-rank configuration is therefore approached by finite-rank configurations within the same space, proving that $\mathcal{G}$ is not open. □

Remark H.11 (The growth constant is an iterated rather than joint limit) For $\kappa_j > 0$, the reading $\lambda_j(\Psi_P) \sim P\,\kappa_j$ of the limits in Propositions H.4 and H.5 is shorthand for the two iterated steps: $P \to \infty$ first, then $N \to \infty$. It is not, by itself, a theorem about a joint path $(P, N) \to \infty$, because that would require quantitative, uniform-in-N control of the menu approximation. We neither assume nor prove that control in the general model. Multiplicity also means that individual eigenvectors inside a limiting group need not stabilize, although the group projection can.

I Verification and Exact Double Limits

This appendix does two things. First, it verifies the quadratic-form convergence used in Section 3 under explicit laws of large numbers: the conditional factor model of Assumption 3 delivers the convergence (D.7) of Assumption D.2. The factor structure alone supplies neither the infinite-rank requirement (D.8) nor the menu exposure condition (D.9). Once these hold, Lemma D.8 derives the uniform lower bound. Second, the appendix shows that the iterated lower limits of Theorem 1 are exact limits for the three standard menus.

Write $L_B := \sup_{x,s} \|B(x,s)\|$ and $\Omega_N := \Omega(X_t; S)$ in the N-stock economy, with $\bar{\omega}_0 := \sup_N \sup_{X,S} \|\Omega(X;S)\|_{\mathrm{op}}$, and let $\bar{\sigma}^2$ denote the residual upper bound in Assumption 3. Let $\mathcal{V}$ be the space of characteristic weightings in Assumption D.2. For $f \in \mathcal{V}$, let $Z_{N,f} =$

$N^{-1}\sum_{i\le N} f(X_{i,t})B(X_{i,t},S_i)$ be the cross-sectional loading average, and define

$$\Gamma_w f := \int f(u)\,\bar{\beta}_w(u)\,p(du \mid w) \in \mathbb{R}^K, \qquad \mathcal{Q}_w := \Gamma_w^* \Omega(w)\Gamma_w. \tag{I.1}$$

Thus $\Gamma_w^{\Omega} = \Omega(w)^{1/2}\Gamma_w$ for the exposure map in (H.13). The uniform laws of large numbers used below are

$$e_N^2 := \sup_{\|f\|_{\mathcal{V}}\le 1} E\|Z_{N,f} - \Gamma_{W_t} f\|^2 \longrightarrow 0, \qquad E\|\Omega_N - \Omega(W_t)\|_{\mathrm{op}} \longrightarrow 0. \tag{I.2}$$

For $\mathcal{V} = L^2(p)$ and finitely many positive-share types, the first convergence follows directly from conditional independence and bounded loadings: conditional variances sum to $O(N^{-1})$ uniformly on the unit ball, while the conditional mean differs from its limit by the converging type shares. The second convergence follows from boundedness and the almost-sure limit of Assumption B.3(ii).

Proposition I.1 (The common-shock limit) *Suppose Assumptions 3, B.3(ii),(iv), and G.1(a)–(b) hold, and suppose the portfolio weights are square integrable uniformly across stocks: for some $C_X < \infty$,*

$$\sup_{i\ge 1} E[f(X_{i,t})^2] \le C_X \|f\|_{\mathcal{V}}^2, \qquad f \in \mathcal{V}. \tag{I.3}$$

(i) If the fixed-f conditional cross-sectional law of large numbers

$$Z_{N,f} \xrightarrow{L^2} \Gamma_{W_t} f, \qquad f \in \mathcal{V}, \tag{I.4}$$

holds, then the convergence in (D.7) holds for each fixed $f \in \mathcal{V}$. The idiosyncratic and diagonal systematic terms are $o_{L^1}(1)$, and the off-diagonal factor term converges to $\langle f, \mathcal{Q}_{W_t} f\rangle_{\mathcal{V}}$.

(ii) If the uniform laws (I.2) hold, then the conditional second moment $\Sigma_N(X_t)$ satisfies (D.7).

(iii) If $\mathcal{V} = L^2(p)$, then

$$\mathcal{H} = E_W[\mathcal{Q}_W], \qquad \overline{\text{range}(\mathcal{H})} = \operatorname*{ess}_w \overline{\text{span}}\, \text{range}(\mathcal{Q}_w) \tag{I.5}$$

where the right side is the smallest closed subspace containing $\text{range}(\mathcal{Q}_w)$ *for almost every w. Consequently, (D.8) is equivalent to* $\text{rank}(\mathcal{H}) = \infty$.

If $\mathcal{V}$ contains the full uniformly bounded-second-moment class in Assumption 2, this proposition verifies its Condition (ii). Equation (D.8) gives its Condition (i); in the K-factor model this rank condition requires W_t to take infinitely many values. Meanwhile, (D.9) is its Condition (iii). For Theorem 1, use part (ii) of Proposition I.1 to obtain the uniform convergence (D.7), and impose the remaining requirements of Assumptions 1 and D.2. By part (iii) of the proposition, the rank requirement is $\text{rank}(\mathcal{H}) = \infty$; Proposition H.10 establishes its genericity in the regular configuration spaces of Definition H.6.

Proof of Proposition I.1. By the convention stated after Assumption B.3, $\Sigma_N(X_t) = \Sigma_N(X_t; S)$. Set $f_N = (f(X_{i,t}))_{i \le N}$. Substitution into (17) yields

$$\frac{1}{N^2} f_N' \Sigma_N(X_t) f_N = Z_{N,f}' \Omega(X_t; S) Z_{N,f} + \frac{1}{N^2} \sum_{i=1}^{N} f(X_{i,t})^2 \sigma_\varepsilon^2(X_{i,t}, S_i). \tag{I.6}$$

By (I.3) and the upper residual-variance bound, the expectation of the second term is at most $\bar{\sigma}^2 C_X \|f\|_{\mathcal{V}}^2 / N$, uniformly over the unit ball; a bounded cross-sectional average $N^{-1} \sum_i E[f(X_{i,t})^2]$ suffices. The stronger bound (I.3) is automatic with $C_X = 1$ under a common law, or $C_X = \max_\ell \pi_\ell^{-1}$ for finitely many types relative to the limiting pooled law. The same argument applies to the diagonal part of the factor term.

For part (i), the L^2 convergence (I.4) makes the squared loading averages uniformly integrable. Boundedness and almost-sure convergence of $\Omega(X_t; S)$ to $\Omega(W_t)$, together with this uniform integrability, therefore give convergence of the quadratic factor term in (I.6) in L^1. Together with the $O(N^{-1})$ residual bound, this proves the fixed-f limit.

For part (ii), write $C_{\mathsf{d}} = \operatorname{ess\,sup}_w \|\mathsf{d}_w\|_2$. For $\mathcal{V} = L^2(p)$, conditional Cauchy–Schwarz gives $\|\Gamma_w f\| \le C_\Gamma \|f\|_2$, where $C_\Gamma = L_B C_{\mathsf{d}}$. For a continuously embedded $\mathcal{V}$, include its embedding constant in C_Γ. In particular, $\|\mathcal{Q}_w\|_{\mathrm{op}} \le \bar{\omega}_0 C_\Gamma^2$ almost surely. Since $\|\Omega_N\| \le \bar{\omega}_0$,

$$\begin{aligned}
&\sup_{\|f\|_{\mathcal{V}} \le 1} E\left| Z'_{N,f} \Omega_N Z_{N,f} - (\Gamma_W f)' \Omega(W) \Gamma_W f \right| \\
&\qquad \le \bar{\omega}_0 e_N (e_N + 2C_\Gamma) + C_\Gamma^2 E\|\Omega_N - \Omega(W)\|_{\mathrm{op}} \longrightarrow 0.
\end{aligned}$$

Together with the $O(N^{-1})$ residual bound, this proves (D.7).

For part (iii), the first identity in (I.5) follows from the definition of $\mathcal{H}$ in (H.13). Hence, for $\mathcal{V} = L^2(p)$, the operator Q_∞ of Assumption D.2 coincides with $\mathcal{H}$. The sentence after that assumption identifies $\ker Q_\infty$ with the weightings that $\mathcal{Q}_W$ annihilates almost surely. Taking orthogonal complements yields the range identity and the rank equivalence. □

Lemma I.2 (Passing an exact portfolio limit through the stock limit) *Let $\mathcal{R}_N$ and $\mathcal{R}_\infty$ be bounded, positive, self-adjoint operators on a common Hilbert space, and write $\Psi_{P,N}$ for the matrix Ψ_P in the N-stock economy. Suppose that, for every fixed N and j,*

$$\lim_{P\to\infty} \frac{\lambda_j(\Psi_{P,N})}{P} = \lambda_j(\mathcal{R}_N), \tag{I.7}$$

and that $\|\mathcal{R}_N - \mathcal{R}_\infty\|_{\mathrm{op}} \to 0$. Then

$$\lim_{N\to\infty} \lim_{P\to\infty} \frac{\lambda_j(\Psi_{P,N})}{P} = \lambda_j(\mathcal{R}_\infty) \tag{I.8}$$

for every fixed j.

Proof of Lemma I.2. For bounded self-adjoint operators, the ordered Courant–Fischer values are Lipschitz in operator norm:

$$|\lambda_j(\mathcal{R}_N) - \lambda_j(\mathcal{R}_\infty)| \le \|\mathcal{R}_N - \mathcal{R}_\infty\|_{\mathrm{op}}.$$

Combine this inequality with (I.7). □

Proposition I.3 (The exact double limit for standard menus) *Let the hypotheses of Theorem 1 hold, and use one of the three constructions in Proposition F.7, with its stated finite-N exact-limit conditions. For $N_{\max} = \infty$, impose the following additional common-space and uniform-convergence conditions. Use one fixed reference space $\mathcal{V} = L^2(p)$ for every N, and suppose the payoff maps $J_N f = N^{-1} \sum_{i \le N} f(X_{i,t}) r_{i,t+1}$ satisfy $E[(J_N f)^2] \le C_{J,N} \|f\|^2_{L^2(p)}$ for some $C_{J,N} < \infty$ at each N. Require (D.7) on the unit ball of this same $L^2(p)$, and require the condition for the construction used:*

(a) for random features, hold the feature family and label law μ_θ fixed across N, reuse one independent label sequence, and require

$$\int \|f(\cdot;\theta)\|^2_{L^2(p)} \, \mu_\theta(d\theta) < \infty; \tag{I.9}$$

(b) for a fixed basis, require that p be the pooled marginal at every N, and hold $\{H_k\}$ and the bounded coefficients $\{q_k\}$ fixed across N;

(c) for partitions, run the limits in P along the refining sizes P_n, require that p be the pooled marginal at every N, and hold the generating partition sequence and its limiting density g fixed across N, with $c_- \le g_P, g \le c_+$ and $\|g_P - g\|_\infty \to 0$.

Then there are positive operators $\mathcal{R}_N$ and $\mathcal{R}_\infty$ satisfying the two hypotheses of Lemma I.2.

Consequently, for every fixed j,

$$\lim_{N\to N_{\max}} \lim_{P\to\infty} \frac{\lambda_j(\Psi_{P,N})}{P} = \lambda_j(\mathcal{R}_\infty) = \lambda_j^* > 0. \tag{I.10}$$

When $N_{\max} < \infty$, no stock limit is taken and $\mathcal{R}_\infty := \mathcal{R}_{N_{\max}}$. For random features the assertion holds almost surely over the common label sequence.

These conditions concern fixed menu geometries; they do not verify the claim for labels or features estimated from the same returns. No conditional factor model is assumed.

Proof of Proposition I.3. The fixed-N conclusion is the corresponding exact result in Appendices F.1, F.2, and F.3. For the growing cross-section, define bounded positive operators on the common $L^2(p)$ by

$$\langle f, Q_N h\rangle = E[(J_N f)(J_N h)], \qquad \langle f, Q_\infty h\rangle = E\langle f, \mathcal{Q}_W h\rangle.$$

For a bounded self-adjoint operator the norm is the supremum of the absolute quadratic form on the unit sphere. Thus (D.7) gives

$$\|Q_N - Q_\infty\|_{\mathrm{op}} \le \sup_{\|f\|_{L^2(p)}\le 1} E\left|N^{-2} f_N' \Sigma_N(X_t) f_N - \langle f, \mathcal{Q}_W f\rangle\right| \longrightarrow 0. \tag{I.11}$$

In particular $\sup_N \|Q_N\| < \infty$.

For random features, let $F : L^2(\mu_\theta) \to L^2(p)$ be $Fa = \int f(\cdot;\theta)a(\theta)\,\mu_\theta(d\theta)$. Condition (I.9) makes F bounded (indeed Hilbert–Schmidt). The continuum managed-return second-moment operator is $\mathcal{R}_N = F^* Q_N F$, whose nonzero eigenvalues are the exact finite-N limits. Set $\mathcal{R}_\infty = F^* Q_\infty F$. The finite-$N$ almost-sure assertions hold simultaneously for all integer N by a countable intersection of probability-one events, and $\|\mathcal{R}_N - \mathcal{R}_\infty\| \le \|F\|^2 \|Q_N - Q_\infty\| \to 0$.

For a fixed basis, let $U : \ell^2 \to L^2(p)$ be the unitary synthesis map $Ua = \sum_k a_k H_k$, and

let $D_q = \mathrm{diag}(q_k)$. Then $\mathcal{R}_N = D_q U^* Q_N U D_q$ and $\mathcal{R}_\infty = D_q U^* Q_\infty U D_q$. The normalized finite matrices are their leading $P \times P$ compressions, and the exact inner limit holds for their Courant–Fischer values even when the operators are noncompact. Boundedness of D_q and (I.11) give norm convergence.

For partitions, let Π_P project onto the cell-constant subspace $\mathcal{V}_P \subset L^2(p)$. In the orthonormal cell basis $\mathbf{1}_{C_k}/\sqrt{p(C_k)}$, direct calculation gives

$$P^{-1}\Psi_{P,N} = \Pi_P M_{\sqrt{g_P}} Q_N M_{\sqrt{g_P}} \Pi_P\big|_{\mathcal{V}_P}.$$

Uniform convergence of g_P replaces g_P by g with an operator-norm error tending to zero at each fixed N. Since the partitions generate, $\Pi_P \to I$ strongly. The fixed-j Courant–Fischer values of the compressions of any bounded positive operator then converge to those of the operator itself (Lemma A.6). Hence the exact inner limit is $\mathcal{R}_N = M_{\sqrt{g}} Q_N M_{\sqrt{g}}$. Set $\mathcal{R}_\infty = M_{\sqrt{g}} Q_\infty M_{\sqrt{g}}$; once again (I.11) gives norm convergence.

Lemma I.2 now proves the exact double limit in every case. Its value is the iterated liminf denoted by λ_j^* in Theorem 1, whose strict positivity follows from Lemma D.8. Under the additional conditional-factor assumptions of Proposition I.1, one may identify $Q_\infty = \mathcal{H}$; the argument above does not require that identification. □

J Risk Premia and Spanning

This appendix proves Theorem 5 with the conditional stock second moments held fixed. It then develops the conditional factor-model calculations used for the Gaussian example and for genericity within the more restricted space of factor premia.

J.1 Conditional means at fixed second moments

Definition J.1 (General premia conditions) Fix the characteristic laws, $\Sigma = (\Sigma_N)_N$, and the menus along the sequences of Theorem 1. Let $\mathcal{V}$ be a separable Hilbert space of measurable characteristic weightings containing every $f_k^{(P)}$. The menu functions are the same across stock counts. Writing $f_N(X_t) = (f(X_{i,t}))_{i \le N}$, require bounded positive operators Q_N on $\mathcal{V}$ with

$$\langle f, Q_N g \rangle_{\mathcal{V}} = E\left[N^{-2} f_N(X_t)' \Sigma_N(X_t) g_N(X_t)\right], \qquad \sup_N \|Q_N\|_{\mathrm{op}} < \infty. \tag{J.1}$$

Require operator-norm convergence of the menu geometry,

$$\mathcal{K}_P := \frac{1}{P} \sum_{k=1}^{P} f_k^{(P)} \otimes f_k^{(P)} \longrightarrow \mathcal{K} \qquad \text{in operator norm on } \mathcal{V}. \tag{J.2}$$

When $N_{\max} = \infty$, use the same $\mathcal{V}$ and common-state operators as in Assumption D.2, including (D.7). Define

$$\mathcal{A}_N = \mathcal{K}^{1/2} Q_N \mathcal{K}^{1/2}, \qquad \mathcal{A} = \begin{cases} \mathcal{A}_{N_{\max}}, & N_{\max} < \infty, \\ \mathcal{K}^{1/2} Q_\infty \mathcal{K}^{1/2}, & N_{\max} = \infty. \end{cases}$$

For $\delta \in [0, 1)$, let $\mathcal{P}_\delta(\Sigma)$ consist of measurable conditional-mean families $\mu = (\mu_N)_N$ satisfying

$$\mu_N(X_t)\mu_N(X_t)' \preceq (1 - \delta)\Sigma_N(X_t) \qquad \text{almost surely for every } N, \tag{J.3}$$

whose expected-return profiles $\varpi_N \in \mathcal{V}$ are defined by

$$\langle f, \varpi_N \rangle_{\mathcal{V}} = E\left[N^{-1} f_N(X_t)' \mu_N(X_t)\right]. \tag{J.4}$$

For $N_{\max} = \infty$, additionally require

$$\|\varpi_N - \varpi\|_{\mathcal{V}} \longrightarrow 0 \quad \text{for some } \varpi \in \mathcal{V}.$$

At finite $N_{\max}$, set $\varpi = \varpi_{N_{\max}}$ and index the family by that single stock count. Set $\mathcal{P}(\Sigma) = \mathcal{P}_0(\Sigma)$ and use the metric

$$d_\Sigma(\mu, \widetilde{\mu}) = \sup_N \left(E\left[(\mu_N - \widetilde{\mu}_N)' \Sigma_N^{-1} (\mu_N - \widetilde{\mu}_N) \right] \right)^{1/2}. \tag{J.5}$$

Genericity is relative to this metric on the specified admissible class.

Each element of $\mathcal{P}(\Sigma)$ specifies conditional means for the economies in the stock-count sequence; the mean-profile condition is the restriction linking those means. For a menu with random labels, fix a label sequence satisfying the convergence conditions, as in Assumption 1. At a fixed stock count, the characteristic-weighting space with norm $\|f\|_{\mathcal{V}}^2 = E[(R^f_{N,t+1})^2]$ is one possible choice, with $Q_N = I$. In that representation (J.2) is operator-norm convergence of the normalized menu operators in Assumption 1. It strengthens their strong convergence and makes the limit compact. Thus the additional condition controls the principal-component groups, without requiring an unconditional stock-trace moment.

For a growing stock count, (D.7) implies $\|Q_N - Q_\infty\|_{\text{op}} \to 0$: take expectations and the supremum over unit weightings in the difference of the quadratic forms. Consequently $\|\mathcal{A}_N - \mathcal{A}\|_{\text{op}} \to 0$. The compactness of $\mathcal{K}$ follows from (J.2), since every $\mathcal{K}_P$ has finite rank. These conditions concern the second moments and menu alone.

Definition J.2 (Full Sharpe recovery for managed portfolios) Let $N_{\max} = \infty$ and $\mu \in \mathcal{P}(\Sigma)$ as in Definition J.1. The Moore–Penrose inverse $\Psi^\dagger_{P,N}$ inverts the positive

eigenvalues of $\Psi_{P,N}$ and is zero on its kernel. Define

$$x_{P,N} := \mu'_{P,N}\Psi^{\dagger}_{P,N}\mu_{P,N}, \qquad x_N := \sup_P x_{P,N}. \tag{J.6}$$

These are the full squared uncentered Sharpe optima in a finite menu and over all menu sizes at stock count N. Set $a^{\star}_{P,N} := \Psi^{\dagger}_{P,N}\mu_{P,N}$, the canonical optimal weights when $\mu_{P,N} \neq 0$, and let $\|\cdot\|$ denote the Euclidean norm. The recovery conditions are that $(x_N)_N$ is nondecreasing along the specified stock-count sequence and that either

$$\lim_{\varepsilon\downarrow 0} \limsup_{N\to\infty} \limsup_{P\to\infty} \sum_{j:\, 0<\lambda_j(\Psi_{P,N})\le \varepsilon P} \frac{(u'_j\mu_{P,N})^2}{\lambda_j(\Psi_{P,N})} = 0 \tag{J.7}$$

or

$$\sup_{N,P} P\|a^{\star}_{P,N}\|^2 < \infty. \tag{J.8}$$

Proof of Theorem 5. We first identify the spectral groups, then show that every such group can be priced by a compatible conditional mean. All second-moment hypotheses remain unchanged when the means vary in $\mathcal{P}(\Sigma)$.

Step 1: spectral and mean limits. Define the sampling map $F_P : \mathcal{V} \to \mathbb{R}^P$ and the corresponding operators by

$$(F_P f)_k = \langle f_k^{(P)}, f\rangle_{\mathcal{V}}, \qquad \Psi_{P,N} = F_P Q_N F_P^*, \qquad \mu_{P,N} = F_P \varpi_N,$$

$$\mathcal{A}_{P,N} = \mathcal{K}_P^{1/2} Q_N \mathcal{K}_P^{1/2}, \qquad \beta_{P,N} = \mathcal{K}_P^{1/2}\varpi_N, \qquad \beta = \mathcal{K}^{1/2}\varpi.$$

The polar decomposition of $P^{-1/2}F_P$ identifies the positive spectral spaces of $\Psi_{P,N}/P$ and $\mathcal{A}_{P,N}$. Continuity of the positive square root in operator norm gives $\mathcal{A}_{P,N} \to \mathcal{A}_N$ and $\beta_{P,N} \to \mathcal{K}^{1/2}\varpi_N$ as $P \to \infty$. The subsequent stock limit gives $\mathcal{A}_N \to \mathcal{A}$ and $\mathcal{K}^{1/2}\varpi_N \to \beta$

when $N_{\max} = \infty$. At finite $N_{\max}$ this subsequent operation is evaluation at $N_{\max}$. Eigenvalue continuity and Theorem 1 therefore give

$$\lim_{N \to N_{\max}} \lim_{P \to \infty} \frac{\lambda_j(\Psi_{P,N})}{P} = \lambda_j(\mathcal{A}) = \kappa_j > 0 \qquad \text{for every fixed } j.$$

The operator $\mathcal{A}$ is compact and has infinite rank, so there are infinitely many distinct positive eigenvalues, each of finite multiplicity.

Fix one such eigenvalue κ_{E}, the interval I_{E} used in the main text, and its spectral projection $P_{\mathsf{E}} = \mathbf{1}_{I_{\mathsf{E}}}(\mathcal{A})$. For $s > 0$ set $g_{\mathsf{E}}(s) = \mathbf{1}_{I_{\mathsf{E}}}(s)/s$ and set $g_{\mathsf{E}}(0) = 0$. The same polar decomposition gives the exact finite-menu identity

$$\mathrm{SR}^2_{\mathsf{E},P,N} = \langle \beta_{P,N}, g_{\mathsf{E}}(\mathcal{A}_{P,N}) \beta_{P,N} \rangle_{\mathcal{V}}. \tag{J.9}$$

For clarity, if $P^{-1/2} F_P = U_P \mathcal{K}_P^{1/2}$, then $\Psi_{P,N}/P = U_P \mathcal{A}_{P,N} U_P^*$ and $\mu_{P,N}/\sqrt{P} = U_P \beta_{P,N}$. On each positive spectral space U_P is an isometry, so summing the matrix ratios in (29) gives (J.9).

The endpoints of I_{E} are outside the spectrum of $\mathcal{A}_N$ for every sufficiently large N, and outside that of $\mathcal{A}_{P,N}$ for all sufficiently large P at each such N. On a common neighborhood of these spectra, g_{E} agrees with a bounded continuous function, since the endpoints lie in spectral gaps. Continuous functional calculus in operator norm, followed by the mean-profile convergence, yields

$$\mathrm{SR}^2_{\mathsf{E},\infty} = \lim_{N \to N_{\max}} \lim_{P \to \infty} \mathrm{SR}^2_{\mathsf{E},P,N} = \frac{\| P_{\mathsf{E}} \mathcal{K}^{1/2} \varpi \|^2_{\mathcal{V}}}{\kappa_{\mathsf{E}}}. \tag{J.10}$$

No individual eigenvector within a repeated eigenspace is selected.

Step 2: the admissible means form a complete space. Fix $\delta \in [0, 1)$. With $h_N = \Sigma_N^{-1/2} \mu_N$, (J.3) is the pointwise constraint $\|h_N(X_t)\| \le \sqrt{1 - \delta}$. The metric (J.5) is the

uniform norm on the sequence of L^2 spaces for these normalized means. Their pointwise constraints define a closed convex subset of this Banach space. The profiles exist by the Riesz representation theorem, since

$$|\langle f, \varpi_N \rangle| \le \|h_N\|_{L^2} \langle f, Q_N f \rangle^{1/2}, \qquad \|\varpi_N - \widetilde{\varpi}_N\|_{\mathcal{V}} \le \sqrt{\|Q_N\|_{\mathrm{op}}}\, \|h_N - \widetilde{h}_N\|_{L^2}.$$

The uniform bound in (J.1) makes the map to the sequence of profiles continuous in the uniform norm. Convergent sequences of profiles form a closed subspace in that norm. Hence imposing profile convergence when $N_{\max} = \infty$ preserves closedness, and $\mathcal{P}_\delta(\Sigma)$ is convex and complete.

Step 3: every positive group is reachable. Choose $0 \neq \zeta$ in the κ_{E}-eigenspace of $\mathcal{A}$ and put $f = \mathcal{K}^{1/2}\zeta$ and $c_\delta = \frac{1}{2}\sqrt{1-\delta} > 0$. Define

$$v_{N,f}(X_t) = N^{-2} f_N(X_t)' \Sigma_N(X_t) f_N(X_t), \qquad \mu_N^{(f)}(X_t) = \frac{c_\delta \Sigma_N(X_t) f_N(X_t)}{N\sqrt{1 + v_{N,f}(X_t)}}.$$

This conditional mean respects the fixed second moments:

$$(\mu_N^{(f)})' \Sigma_N^{-1} \mu_N^{(f)} = c_\delta^2 \frac{v_{N,f}}{1 + v_{N,f}} \le \frac{1-\delta}{4}. \tag{J.11}$$

At finite $N_{\max}$,

$$\langle \zeta, \mathcal{K}^{1/2} \varpi_{N_{\max}}^{(f)} \rangle = c_\delta E \frac{v_{N_{\max},f}}{\sqrt{1 + v_{N_{\max},f}}} > 0,$$

because $E[v_{N_{\max},f}] = \kappa_{\mathsf{E}} \|\zeta\|^2 > 0$.

For $N_{\max} = \infty$, write $v_f(W) = \langle f, \mathcal{Q}_W f \rangle_{\mathcal{V}}$. The probe profiles converge in $\mathcal{V}$ to the profile defined by

$$\langle g, \varpi^{(f)} \rangle_{\mathcal{V}} = c_\delta E \frac{\langle g, \mathcal{Q}_W f \rangle_{\mathcal{V}}}{\sqrt{1 + v_f(W)}}. \tag{J.12}$$

Indeed, polarization of (D.7) gives

$$\sup_{\|g\|_{\mathcal{V}}\le 1} E\left|N^{-2} g_N' \Sigma_N f_N - \langle g, \mathcal{Q}_W f\rangle_{\mathcal{V}}\right| \longrightarrow 0.$$

The denominators in the probe are at least one. The function $s \mapsto (1+s)^{-1/2}$ is Lipschitz on $[0,\infty)$, and $|\langle g, \mathcal{Q}_W f\rangle| \le \operatorname{ess\,sup}_w \|\mathcal{Q}_w\|_{\mathrm{op}} \|g\|_{\mathcal{V}} \|f\|_{\mathcal{V}}$. Together with $v_{N,f} \to v_f(W)$ in L^1, these bounds give convergence of the probe functionals uniformly over unit g, which is norm convergence of the profiles. Thus the probe belongs to $\mathcal{P}_\delta(\Sigma)$. Moreover,

$$\langle \zeta, \mathcal{K}^{1/2} \varpi^{(f)}\rangle = c_\delta E \frac{v_f(W)}{\sqrt{1+v_f(W)}} > 0,$$

since $E[v_f(W)] = \kappa_{\mathsf{E}} \|\zeta\|^2 > 0$. In both cases $P_{\mathsf{E}} \mathcal{K}^{1/2} \varpi^{(f)} \neq 0$.

Step 4: genericity and Sharpe bounds. For each group the map $\mu \mapsto P_{\mathsf{E}} \mathcal{K}^{1/2} \varpi$ is continuous and linear. Its zero set in $\mathcal{P}_\delta(\Sigma)$ is closed. If μ is in that zero set, the convex perturbation $(1-\eta)\mu + \eta\mu^{(f)}$, $0 < \eta < 1$, remains admissible, converges to μ as $\eta \downarrow 0$, and has a nonzero group projection. Thus each zero set is nowhere dense. There are countably many groups, and completeness allows the Baire theorem to give a dense residual subset pricing all groups at once.

For any $\mu \in \mathcal{P}(\Sigma)$, put

$$\bar{x}(\mu) = \sup_N E[\mu_N' \Sigma_N^{-1} \mu_N] \le 1.$$

Conditional-moment Cauchy–Schwarz gives, for every square-integrable managed return,

$$\left(E[R^{\phi}_{N,t+1}]\right)^2 \le \bar{x}(\mu) E[(R^{\phi}_{N,t+1})^2].$$

Taking limits on the weightings $\phi = \mathcal{K}^{1/2} z$ gives $|\langle z, \beta\rangle|^2 \le \bar{x}(\mu) \langle z, \mathcal{A} z\rangle$. In particular, β is orthogonal to $\ker \mathcal{A}$. Expansion in an orthonormal eigenbasis of $\mathcal{A}$ shows that the maximal

squared uncentered ratio over finite positive spectral spans is

$$x_\infty = \sum_{\mathsf{E}} \frac{\|P_{\mathsf{E}}\beta\|^2}{\kappa_{\mathsf{E}}} \le \bar{x}(\mu) \le 1. \tag{J.13}$$

Each finite partial sum is attained by taking the corresponding coefficients of z proportional to those of β divided by their positive eigenvalues. For a return with squared uncentered ratio $x < 1$, its squared centered ratio is $x/(1-x)$. The bound for each managed return is uniform over menu sizes and stock counts. Hence, if $\bar{x}(\mu) < 1$,

$$\mathrm{S}^2_{\mathrm{full}} \le \frac{\bar{x}(\mu)}{1-\bar{x}(\mu)} < \infty.$$

The set $\{\mu : \bar{x}(\mu) < 1\}$ is open and dense in $\mathcal{P}(\Sigma)$: $\sqrt{\bar{x}(\mu)} = d_\Sigma(\mu, 0)$ is continuous, and multiplication of all means by $1-\eta$ preserves admissibility and profile convergence. Intersecting this open dense set with the residual pricing set proves the assertion for $\mathcal{P}(\Sigma)$. In $\mathcal{P}_\delta(\Sigma)$ with $\delta > 0$, $\bar{x}(\mu) \le 1-\delta$ for every mean, giving $\mathrm{S}^2_{\mathrm{full}} \le (1-\delta)/\delta$.

Step 5: truncation. Order the distinct positive eigenvalues of $\mathcal{A}$ by decreasing growth constant. Compactness and infinite rank give countably infinitely many groups, each of finite multiplicity, with growth constants tending to zero. By (J.13), $x_J \uparrow x_\infty \le \bar{x}(\mu)$ for every admissible mean. If $\bar{x}(\mu) < 1$, the exact identity

$$\mathrm{S}^2_{\mathrm{sys}} - \mathrm{S}^2_J = \frac{x_\infty - x_J}{(1-x_\infty)(1-x_J)} \longrightarrow 0 \tag{J.14}$$

proves $\mathrm{S}_J \uparrow \mathrm{S}_{\mathrm{sys}}$. On the residual pricing set, every omitted group contributes a positive term, so $x_J < x_\infty$ and $\mathrm{S}_J < \mathrm{S}_{\mathrm{sys}}$ for every finite J. Intersecting with $\{\bar{x} < 1\}$ gives both conclusions generically in $\mathcal{P}(\Sigma)$. For $\delta > 0$, the same proof applies throughout $\mathcal{P}_\delta(\Sigma)$ for convergence and on its residual pricing set for strictness; the denominator in (J.14) is at least δ^2. All group limits in (J.10) precede $J \to \infty$. For fixed J, choose the isolating intervals

of the retained groups pairwise disjoint. The sum of their finite-menu squared uncentered ratios is the maximum over their joint span, and its centered Sharpe ratio is bounded by $\mathrm{S}_{\mathrm{full}}$. Taking the iterated limits for these finitely many groups gives $\mathrm{S}_J \leq \mathrm{S}_{\mathrm{full}}$. Then $J \to \infty$ yields $\mathrm{S}_{\mathrm{sys}} \leq \mathrm{S}_{\mathrm{full}}$ without interchanging an infinite sum with either cross-section limit.

Step 6: full recovery at a fixed stock count. Fix $N = N_{\max} < \infty$ and let $\mathcal{D}$ be the managed-payoff space in (D.1). This space is closed in L^2. To see this without uniform ellipticity, put $a(X_t) = \min\{1, \lambda_{\min}(\Sigma_N(X_t))\} > 0$. The squared payoff norm dominates the squared weighting norm for the finite measure

$$B \longmapsto N^{-2} \sum_{i=1}^{N} E\big[a(X_t)\, \mathbf{1}_{\{X_{i,t} \in B\}}\big].$$

This measure has the same null sets as the average characteristic law. A sequence of weightings $f^{(m)}$ with Cauchy payoffs therefore converges in this weighted L^2 space to a measurable weighting f. A subsequence converges at every $X_{i,t}$ almost surely. The vectors $N^{-1}\Sigma_N^{1/2} f_N^{(m)}$ are Cauchy in L^2 and their limit is the vector associated with f, by this almost-sure convergence. Thus $R^f_{N,t+1}$ is square integrable and is the payoff limit.

Let $J_N : \mathcal{V} \to \mathcal{D}$ send f to $R^f_{N,t+1}$. It is bounded and $J_N^* J_N = Q_N$ by (J.1). On $\mathcal{D}$, the normalized menu operators satisfy

$$A_P = J_N \mathcal{K}_P J_N^* \longrightarrow T := J_N \mathcal{K} J_N^* \qquad \text{in operator norm.}$$

The positive operator T is compact by (J.2) and injective by Assumption 1. Its positive eigenvectors therefore form an orthonormal basis of $\mathcal{D}$. For each unit eigenvector ζ of $\mathcal{A}$ with eigenvalue $\kappa > 0$, the corresponding normalized payoff is $J_N \mathcal{K}^{1/2} \zeta / \sqrt{\kappa}$, whose expectation is $\langle \zeta, \beta \rangle / \sqrt{\kappa}$. The positive spectral correspondence between $J_N \mathcal{K} J_N^*$ and $\mathcal{K}^{1/2} J_N^* J_N \mathcal{K}^{1/2}$,

followed by Parseval's identity for the expectation functional on $\mathcal{D}$, gives

$$\sup_{0\neq\Phi\in\mathcal{D}} \frac{(E\Phi)^2}{E[\Phi^2]} = \sum_{\mathsf{E}} \frac{\|P_{\mathsf{E}}\beta\|^2}{\kappa_{\mathsf{E}}} = x_\infty.$$

Finally, the range of T is dense in $\mathcal{D}$, and $A_P\Phi \to T\Phi$ with $A_P\Phi$ in the P-menu span. Hence the distance from each fixed payoff in $\mathcal{D}$ to that span tends to zero, even when the menus are not nested. Every menu span is contained in $\mathcal{D}$, so its maximal squared uncentered ratio converges to x_∞ and never exceeds x_∞. Its supremum over P is therefore x_∞. Applying the centered Sharpe transformation gives

$$\mathrm{S}_{\mathrm{full}} = \mathrm{S}_{\mathrm{sys}} \qquad (N = N_{\max} < \infty). \tag{J.15}$$

This argument at a fixed stock count supplies no uniform control of omitted Sharpe contributions as N grows.

Step 7: full recovery with a growing stock count. Let $N_{\max} = \infty$ and fix any $\mu \in \mathcal{P}(\Sigma)$ with $\mathrm{S}_{\mathrm{full}} < \infty$. A zero-second-moment menu combination has zero mean, so $\mu_{P,N} \perp \ker \Psi_{P,N}$ and $\Psi_{P,N} a^\star_{P,N} = \mu_{P,N}$. Cauchy–Schwarz in the positive eigenspaces then identifies $x_{P,N}$ in (J.6) as the maximal squared uncentered menu Sharpe ratio, attained by $a^\star_{P,N}$ when $\mu_{P,N} \neq 0$. The argument for (J.15) applies at each fixed N, since Assumption 1 holds at every stock count. It gives

$$x_N = \sup_P x_{P,N} = \lim_{P\to\infty} x_{P,N}, \qquad \mathrm{S}^2_{\mathrm{full}} = \frac{\sup_N x_N}{1-\sup_N x_N}, \qquad \sup_N x_N < 1. \tag{J.16}$$

The strict inequality follows from the assumed finiteness of $\mathrm{S}_{\mathrm{full}}$ and the increasing transformation $x \mapsto x/(1-x)$.

For $\varepsilon > 0$, denote the sum in (J.7) by $T_{P,N}(\varepsilon)$. The identity $u_j'\mu_{P,N} = \lambda_j(\Psi_{P,N})u_j' a^\star_{P,N}$

gives

$$T_{P,N}(\varepsilon) = \sum_{j:\,0<\lambda_j(\Psi_{P,N})\le\varepsilon P} \lambda_j(\Psi_{P,N})(u_j' a^\star_{P,N})^2 \\ \le \varepsilon P \|a^\star_{P,N}\|^2. \tag{J.17}$$

Thus (J.8) implies (J.7).

Choose $\varepsilon > 0$ different from every positive eigenvalue of $\mathcal{A}$, and write $x_{>\varepsilon} := \sum_{\mathsf{E}:\,\kappa_{\mathsf{E}}>\varepsilon} \mathrm{SR}^2_{\mathsf{E},\infty}$. Compactness makes this a finite sum. Choose disjoint isolating intervals for these groups, all above ε. The operator-norm convergences $\mathcal{A}_{P,N} \to \mathcal{A}_N$ as $P \to \infty$ and $\mathcal{A}_N \to \mathcal{A}$ as $N \to \infty$ imply that, for all sufficiently large N and then all sufficiently large P at each such N, these intervals contain exactly the eigenvalues of $\Psi_{P,N}/P$ above ε. There are no additional eigenvalues above the threshold, by spectral stability under operator-norm convergence and the positive spectral correspondence in (J.9). Hence (J.10) gives

$$\lim_{N\to\infty} \lim_{P\to\infty} \big[x_{P,N} - T_{P,N}(\varepsilon)\big] = x_{>\varepsilon}.$$

Using (J.16) and nonnegativity of the omitted terms,

$$x_{>\varepsilon} \le \liminf_{N\to\infty} x_N \le \limsup_{N\to\infty} x_N \le x_{>\varepsilon} + \limsup_{N\to\infty} \limsup_{P\to\infty} T_{P,N}(\varepsilon).$$

Letting $\varepsilon \downarrow 0$ through such thresholds gives $x_{>\varepsilon} \uparrow x_\infty$. Under (J.7), the last term vanishes, proving

$$\lim_{N\to\infty} x_N = x_\infty. \tag{J.18}$$

If $(x_N)_N$ is also nondecreasing along the specified sequence, then $\sup_N x_N = x_\infty$. Equations (30) and (J.16) yield $\mathrm{S}_{\mathrm{sys}} = \mathrm{S}_{\mathrm{full}}$. This implication holds for every admissible mean satisfying the additional conditions; it does not require generic pricing. □

J.2 Unconditional stock premia

Definition J.3 (Unconditional stock mean conditions) Fix the unconditional second moments $\Sigma_N = E[r_{t+1}r'_{t+1}]$ and the fixed-effect array of a model satisfying the hypotheses of Theorem 3. For $\delta \in [0,1)$, let $\mathcal{P}^r_\delta(\Sigma)$ consist of families $\bar{\mu} = (\bar{\mu}_N)_N$, $\bar{\mu}_N \in \mathbb{R}^N$, satisfying

$$\bar{\mu}_N \bar{\mu}'_N \preceq (1-\delta)\Sigma_N \qquad \text{for every } N,$$

and

$$\frac{1}{N}\sum_{i \le N} g(S_i)\bar{\mu}_{i,N} \quad \text{converges for every } g \in C_b(\mathcal{T}), \tag{J.19}$$

where $C_b(\mathcal{T})$ is the space of bounded continuous real functions on $\mathcal{T}$. Set $\mathcal{P}^r(\Sigma) = \mathcal{P}^r_0(\Sigma)$ and use the metric

$$d^r_\Sigma(\bar{\mu}, \widetilde{\mu}) := \sup_N \big((\bar{\mu}_N - \widetilde{\mu}_N)'\Sigma_N^{-1}(\bar{\mu}_N - \widetilde{\mu}_N)\big)^{1/2}. \tag{J.20}$$

Genericity means a dense residual subset in this metric.

The matrix inequality is exactly the condition that the unconditional covariance $\Sigma_N - \bar{\mu}_N\bar{\mu}'_N$ be positive semidefinite, with a margin when $\delta > 0$. Every such pair of moments is realized, for each N, by a Gaussian stock-return vector. Condition (J.19) links the means as the stock count grows through expected returns on static portfolios.

Definition J.4 (Full Sharpe recovery for stocks) For $\bar{\mu} \in \mathcal{P}^r(\Sigma)$ as in Definition J.3, define the full squared uncentered Sharpe optimum and canonical optimal weights by

$$x^r_N := \bar{\mu}'_N \Sigma_N^{-1} \bar{\mu}_N, \qquad a^{r,\star}_N := \Sigma_N^{-1}\bar{\mu}_N. \tag{J.21}$$

The recovery conditions are that $(x^r_N)_N$ is nondecreasing along the specified stock-count

sequence and that either

$$\lim_{\varepsilon\downarrow 0}\limsup_{N\to\infty}\sum_{j:\,\lambda_j(\Sigma_N)\le\varepsilon N}\frac{(u'_{j,N}\bar{\mu}_N)^2}{\lambda_j(\Sigma_N)}=0 \tag{J.22}$$

or

$$\sup_N N\|a_N^{r,\star}\|^2<\infty. \tag{J.23}$$

Proof of Theorem 6. Use the profiles $\varphi_s\in\mathcal{H}_W$ and overlap operator $\mathcal{O}$ from (B.11) and (B.12). Define

$$(F_Nh)_i=N^{-1/2}\langle\varphi_{S_i},h\rangle_{\mathcal{H}_W},\quad A_N=\Sigma_N/N,$$
$$C_N=F_NF_N^*,\qquad\qquad \mathcal{O}_N=F_N^*F_N=\frac{1}{N}\sum_{i\le N}\varphi_{S_i}\otimes\varphi_{S_i}.$$

Equation (B.19) and Lemma B.8 give

$$\|A_N-C_N\|_{\mathrm{op}}\longrightarrow 0,\qquad \|\mathcal{O}_N-\mathcal{O}\|_{\mathrm{op}}\longrightarrow 0. \tag{J.24}$$

All these operators are uniformly bounded. The residual lower bound makes every Σ_N positive definite.

Step 1: convergence of mean exposures. Put

$$y_N=\bar{\mu}_N/\sqrt{N},\qquad \beta_N=F_N^*y_N=\frac{1}{N}\sum_{i\le N}\bar{\mu}_{i,N}\varphi_{S_i}.$$

Compatibility implies $\|y_N\|^2\le(1-\delta)\|A_N\|_{\mathrm{op}}$ and, for every $h\in\mathcal{H}_W$,

$$|\langle h,\beta_N\rangle|^2\le(1-\delta)\langle h,F_N^*A_NF_Nh\rangle,\qquad F_N^*A_NF_N\longrightarrow\mathcal{O}^2\quad\text{in operator norm.} \tag{J.25}$$

The last convergence follows from (J.24), since $F_N^* C_N F_N = \mathcal{O}_N^2$. For each h, the function $s \mapsto \langle \varphi_s, h \rangle$ is bounded and continuous. Condition (J.19) therefore makes $\langle h, \beta_N \rangle$ converge. Uniform boundedness gives a weak limit $\beta \in \mathcal{H}_W$.

Let Π_m project onto the first m positive eigenvectors of $\mathcal{O}$, taking its whole range if that range is finite dimensional. By (J.25),

$$\limsup_N \|(I - \Pi_m)\beta_N\|^2 \leq (1-\delta)\|(I - \Pi_m)\mathcal{O}^2(I - \Pi_m)\|_{\text{op}} \longrightarrow 0 \quad \text{as } m \to \infty.$$

On each finite-dimensional range of Π_m, weak convergence is norm convergence. Combining this fact with the uniform tail bound gives $\beta_N \to \beta$ in norm. If $\mathcal{O} = 0$, the same bound directly gives $\beta_N \to 0$.

Step 2: the group pricing limit. Let Π_{E} be the spectral projection of $\mathcal{O}$ at $\kappa_{\mathsf{E}}^r > 0$. For $s > 0$, put $g_{\mathsf{E}}(s) = \mathbf{1}_{I_{\mathsf{E}}}(s)/s$ and $h_{\mathsf{E}}(s) = \mathbf{1}_{I_{\mathsf{E}}}(s)/s^2$, with both functions zero at $s = 0$. The endpoints of I_{E} lie outside the spectrum of $\mathcal{O}$. Since the positive spectra of C_N and $\mathcal{O}_N$ coincide, (J.24) gives common spectral gaps around these endpoints for A_N, C_N, and $\mathcal{O}_N$ when N is large. Replacing g_{E} and h_{E} by bounded continuous functions agreeing with them off the gaps, continuous functional calculus gives $\|g_{\mathsf{E}}(A_N) - g_{\mathsf{E}}(C_N)\|_{\text{op}} \to 0$ and $h_{\mathsf{E}}(\mathcal{O}_N) \to h_{\mathsf{E}}(\mathcal{O})$ in operator norm. The identity $g_{\mathsf{E}}(C_N) = F_N h_{\mathsf{E}}(\mathcal{O}_N) F_N^*$ then yields

$$\begin{aligned} \mathrm{SR}_{\mathsf{E},N}^{2,r} &= \langle y_N, g_{\mathsf{E}}(A_N) y_N \rangle \\ &= \langle \beta_N, h_{\mathsf{E}}(\mathcal{O}_N)\beta_N \rangle + o(1) \longrightarrow \frac{\|\Pi_{\mathsf{E}}\beta\|_{\mathcal{H}_W}^2}{(\kappa_{\mathsf{E}}^r)^2}. \end{aligned} \tag{J.26}$$

In particular, the lower limit in the main text is a limit.

Step 3: completeness and a pricing perturbation. Under $\bar{\mu}_N \mapsto \Sigma_N^{-1/2}\bar{\mu}_N$, the compatible families form a closed convex subset of the Banach space of uniformly bounded

vector sequences. For $g \in C_b(\mathcal{T})$ and $g_N = (g(S_i))_{i \le N}$, Cauchy–Schwarz gives

$$\begin{aligned}\sup_N \left| \frac{1}{N} g_N'(\bar{\mu}_N - \widetilde{\mu}_N) \right| &\le d_\Sigma^r(\bar{\mu}, \widetilde{\mu}) \sup_N \sqrt{N^{-2} g_N' \Sigma_N g_N} \\ &\le d_\Sigma^r(\bar{\mu}, \widetilde{\mu}) \|g\|_\infty \sqrt{\sup_N \|A_N\|_{\mathrm{op}}}.\end{aligned}$$

Convergent scalar sequences form a closed subspace in the uniform norm. Requiring (J.19) for every g therefore preserves closedness and convexity. Hence $\mathcal{P}_\delta^r(\Sigma)$ is complete. The map $\bar{\mu} \mapsto \beta$ is continuous and linear, since

$$\|\beta_N(\bar{\mu}) - \beta_N(\widetilde{\mu})\| \le \|F_N\|_{\mathrm{op}} \sqrt{\|A_N\|_{\mathrm{op}}} d_\Sigma^r(\bar{\mu}, \widetilde{\mu})$$

uniformly in N.

Fix a unit ζ in a positive group, so $\mathcal{O}\zeta = \kappa_{\mathsf{E}}^r \zeta$. Let $t_N = (\langle \varphi_{S_i}, \zeta \rangle)_{i \le N}$ and define

$$v_N = \frac{t_N' \Sigma_N t_N}{N^2}, \qquad \bar{\mu}_N^{(\zeta)} = \frac{c_\delta \Sigma_N t_N}{N\sqrt{1 + v_N}}, \qquad c_\delta = \frac{\sqrt{1-\delta}}{2}. \tag{J.27}$$

Then

$$(\bar{\mu}_N^{(\zeta)})' \Sigma_N^{-1} \bar{\mu}_N^{(\zeta)} = c_\delta^2 \frac{v_N}{1 + v_N} \le \frac{1-\delta}{4}.$$

Since $v_N = \langle \zeta, F_N^* A_N F_N \zeta \rangle$, (J.25) gives $v_N \to (\kappa_{\mathsf{E}}^r)^2$ and

$$\beta_N^{(\zeta)} = \frac{c_\delta F_N^* A_N F_N \zeta}{\sqrt{1 + v_N}} \longrightarrow \frac{c_\delta (\kappa_{\mathsf{E}}^r)^2}{\sqrt{1 + (\kappa_{\mathsf{E}}^r)^2}} \zeta.$$

This limit has a nonzero projection onto the chosen group. To check (J.19) for the probe, fix $g \in C_b(\mathcal{T})$. Lemma B.8 gives

$$\frac{1}{N} \sum_{i \le N} g(S_i) \varphi_{S_i} \longrightarrow \int g(s) \varphi_s \, \nu(ds).$$

Using (J.24) in $N^{-1}g_N'\bar{\mu}_N^{(\zeta)}$ consequently gives the limit

$$\frac{c_\delta}{\sqrt{1+(\kappa_{\mathsf{E}}^r)^2}}\left\langle \int g(s)\varphi_s\,\nu(ds), \mathcal{O}\zeta \right\rangle.$$

The probe belongs to $\mathcal{P}_\delta^r(\Sigma)$.

Step 4: generic pricing and the Sharpe bound. For each positive group, $\{\bar{\mu} : \Pi_{\mathsf{E}}\beta = 0\}$ is relatively closed. At any mean in that set, the admissible perturbations $(1-\eta)\bar{\mu} + \eta\bar{\mu}^{(\zeta)}$, $0 < \eta < 1$, converge to it as $\eta \downarrow 0$ and give a nonzero group projection. Thus the zero set is nowhere dense. There are countably many positive groups. Baire's theorem and (J.26) give generic pricing of all of them simultaneously.

Put $\bar{x}_r(\bar{\mu}) = \sup_N \bar{\mu}_N'\Sigma_N^{-1}\bar{\mu}_N \leq 1$. For every static stock portfolio with nonzero weight vector a,

$$\frac{(a'\bar{\mu}_N)^2}{a'\Sigma_N a} \leq \bar{\mu}_N'\Sigma_N^{-1}\bar{\mu}_N \leq \bar{x}_r(\bar{\mu}).$$

The first bound is attained at $a = \Sigma_N^{-1}\bar{\mu}_N$ when $\bar{\mu}_N \neq 0$. Consequently, if $\bar{x}_r(\bar{\mu}) < 1$,

$$(\mathrm{S}_{\mathrm{full}}^r)^2 = \frac{\bar{x}_r(\bar{\mu})}{1-\bar{x}_r(\bar{\mu})} < \infty.$$

The set $\{\bar{\mu} : \bar{x}_r(\bar{\mu}) < 1\}$ is open in $\mathcal{P}^r(\Sigma)$, since $\sqrt{\bar{x}_r(\bar{\mu})} = d_\Sigma^r(\bar{\mu}, 0)$, and is dense by scaling every mean by $1-\eta$. Intersecting it with the residual pricing set gives both conclusions generically. For $\delta > 0$, $\bar{x}_r \leq 1-\delta$ gives $(\mathrm{S}_{\mathrm{full}}^r)^2 \leq (1-\delta)/\delta$ throughout $\mathcal{P}_\delta^r(\Sigma)$. Finally, Proposition B.6 gives infinite rank for generic response configurations when both supports are infinite. Each positive eigenvalue of $\mathcal{O}$ has finite multiplicity, so there are infinitely many positive groups in that case.

Step 5: truncation. The compact positive operator $\mathcal{O}$ has at most countably many distinct positive eigenvalues, each of finite multiplicity. Order the corresponding groups by

decreasing growth constant. For a fixed J, retain its first J groups, or all groups if fewer exist, and choose their intervals pairwise disjoint. The limit (J.26) is independent of the isolating intervals. Orthogonality of the stock eigenvectors gives

$$\bar{\mu}_N' \Sigma_N^{-1} \bar{\mu}_N = \sum_{j=1}^{N} \frac{(u_{j,N}' \bar{\mu}_N)^2}{\lambda_j(\Sigma_N)}.$$

The contributions of the retained intervals are nonnegative terms of this sum. Taking $N \to \infty$ for these finitely many groups, and only then increasing J, gives

$$x_J^r \uparrow x_{\mathrm{sys}}^r \le \liminf_{N\to\infty} \bar{\mu}_N' \Sigma_N^{-1} \bar{\mu}_N \le \bar{x}_r(\bar{\mu}) \le 1. \tag{J.28}$$

No convergence of the full inverse quadratic form is required. When $\bar{x}_r < 1$, the map $x \mapsto \sqrt{x/(1-x)}$ is continuous and increasing on $[0, \bar{x}_r]$, so $\mathrm{S}_J^r \uparrow \mathrm{S}_{\mathrm{sys}}^r \le \mathrm{S}_{\mathrm{full}}^r$. On the residual pricing set, infinitely many positive groups give $x_J^r < x_{\mathrm{sys}}^r$ and hence $\mathrm{S}_J^r < \mathrm{S}_{\mathrm{sys}}^r$ for every finite J. If there are finitely many groups, equality holds after all have been retained; if there are none, both partial sums and x_{sys}^r are zero. Intersecting with $\{\bar{x}_r < 1\}$ proves the assertions in $\mathcal{P}^r(\Sigma)$. For $\delta > 0$, convergence holds throughout $\mathcal{P}_\delta^r(\Sigma)$ and strictness holds on its residual pricing set when there are infinitely many groups.

The inequalities in (J.28) separate two possible shortfalls: the limiting groups need not exhaust the large-stock inverse quadratic form, and the supremum over all stock counts need not equal its lower limit. For the latter, take $\Sigma_N = \mathbf{1}\mathbf{1}' + I_N$ and $\bar{\mu}_N = a_N \mathbf{1}$, with $a_1 = 1$ and $a_N = 1/5$ for $N \ge 2$. This second-moment family satisfies the stock hypotheses with a constant loading and unit residual variance, and the mean family belongs to $\mathcal{P}_{1/2}^r(\Sigma)$. Every mean vector lies in the leading eigenspace, yet

$$x_{\mathrm{sys}}^r = \frac{1}{25}, \qquad \bar{x}_r = \sup_N \frac{a_N^2 N}{N+1} = \frac{1}{2}. \tag{J.29}$$

Thus a gap between $\mathrm{S}^r_{\mathrm{sys}}$ and $\mathrm{S}^r_{\mathrm{full}}$ need not be an idiosyncratic premium.

Step 6: equality with the full stock benchmark. Fix any $\bar{\mu} \in \mathcal{P}^r(\Sigma)$ with $\mathrm{S}^r_{\mathrm{full}} < \infty$. The quantities in (J.21) satisfy $\Sigma_N a_N^{r,\star} = \bar{\mu}_N$ and $\sup_N x_N^r = \bar{x}_r(\bar{\mu}) < 1$. For $\varepsilon > 0$, denote the sum in (J.22) by $T_N^r(\varepsilon)$. Expanding in the orthonormal eigenbasis of Σ_N gives

$$\begin{aligned} T_N^r(\varepsilon) &= \sum_{j:\, \lambda_j(\Sigma_N) \le \varepsilon N} \lambda_j(\Sigma_N)(u_{j,N}' a_N^{r,\star})^2 \\ &\le \varepsilon N \|a_N^{r,\star}\|^2. \end{aligned} \tag{J.30}$$

Thus (J.23) implies (J.22).

Choose $\varepsilon > 0$ outside the positive spectrum of $\mathcal{O}$, and put $x^r_{>\varepsilon} := \sum_{\mathsf{E}:\, \kappa^r_{\mathsf{E}} > \varepsilon} \mathrm{SR}^{2,r}_{\mathsf{E},\infty}$. There are finitely many such groups. By (J.24), the normalized stock eigenvalues above ε eventually lie precisely in disjoint isolating intervals of these groups: C_N and $\mathcal{O}_N$ have the same positive eigenvalues, and both operator-norm errors tend to zero. Equation (J.26) therefore yields

$$x_N^r - T_N^r(\varepsilon) \longrightarrow x^r_{>\varepsilon},$$

including the case of an empty retained sum. It follows that

$$x^r_{>\varepsilon} \le \liminf_{N\to\infty} x_N^r \le \limsup_{N\to\infty} x_N^r \le x^r_{>\varepsilon} + \limsup_{N\to\infty} T_N^r(\varepsilon).$$

Let $\varepsilon \downarrow 0$ through thresholds outside the positive spectrum. Then $x^r_{>\varepsilon} \uparrow x^r_{\mathrm{sys}}$, so (J.22) gives

$$\lim_{N\to\infty} x_N^r = x^r_{\mathrm{sys}}. \tag{J.31}$$

If $(x_N^r)_N$ is also nondecreasing, then $\bar{x}_r = \sup_N x_N^r = x^r_{\mathrm{sys}} < 1$. The centered Sharpe formulas in (31) and (32) give $\mathrm{S}^r_{\mathrm{sys}} = \mathrm{S}^r_{\mathrm{full}}$. As in the managed result, this conclusion does not require generic pricing. □

J.3 Conditional factor-model calculations

The remaining calculations concern the conditional K-factor model. They describe its limiting principal components and the smaller class of conditional means obtained by varying factor premia. Genericity in this class needs its own reachability argument; it does not follow by restricting a generic set of unrestricted stock means.

Definition J.5 (Factor-premia conditions) Fix $\vartheta \in \mathfrak{M}_B$ as in Definition H.6, with its types, complete random-feature menu, and operator $\mathcal{H}$. Fix $\delta \in (0,1)$ and define

$$\mathcal{P}_\delta = \{\mu_G \in L^2(P_W; \mathbb{R}^K) : \mu_G(W)'\Omega(W)^{-1}\mu_G(W) \leq 1-\delta \quad P_W\text{-a.s.}\}. \tag{J.32}$$

Here W is the vector of type directing laws. For $\mu_G \in \mathcal{P}_\delta$, put $\varpi = E_W[\bar{\beta}_W'\mu_G(W)\mathsf{d}_W] \in L^2(p)$ and define the limiting centered maximal squared Sharpe ratio by

$$\mathrm{S}^2_{\mathrm{sys}} = \sup_{\psi:\, \langle\psi,\mathcal{H}\psi\rangle>0} \frac{\langle\psi,\varpi\rangle^2}{\langle\psi,\mathcal{H}\psi\rangle - \langle\psi,\varpi\rangle^2}.$$

The supremum is zero if its indexing set is empty. For statements transferring this limit to finite menu matrices, require a finite-N realization satisfying the hypotheses of Proposition H.4 and Assumption G.2, the actual mean limit in (H.6), and

$$\langle\psi,\varpi_N\rangle := E\left[N^{-1}\sum_{i\leq N}\psi(X_{i,t})B(X_{i,t},S_i)'\mu_{G,N}(X_t;S)\right],$$
$$\varpi_N \in L^2(p), \qquad \|\varpi_N - \varpi\|_2 \longrightarrow 0.$$

The finite-matrix conclusion applies to every realization satisfying these additional conditions.

Model genericity is taken in the regular configuration spaces of Definition H.6. For each

fixed configuration and fixed $\delta \in (0,1)$, premium genericity is taken in $\mathcal{P}_\delta$ with its L^2 metric. These are separate quantifiers.

For a growing stock count, $\mathcal{P}(\Sigma)$ requires the expected return of each fixed managed weighting to converge, and Assumption D.2 controls its second moment. Under the conditional factor-model conditions of Definition J.5, uniformly bounded individual-stock second moments make the own-stock contribution of order $1/N$ for weightings with uniformly bounded cross-sectional second moments. That contribution vanishes, while the co-moments of distinct stocks can sustain systematic comovement and nonzero expected returns.

Let $\{\zeta_j\}$ be orthonormal eigenfunctions of the compact positive operator $\mathcal{A} = \mathcal{K}^{1/2}\mathcal{H}\mathcal{K}^{1/2}$, with eigenvalues κ_j. The ratios below concern positive eigenvalues.

A managed portfolio applied stock by stock through a weighting $\psi \in L^2(p)$, weight $\frac{1}{N}\psi(X_{i,t})$ on stock i, has return $\Pi_N = \frac{1}{N}\sum_{i=1}^N \psi(X_{i,t})\, r_{i,t+1}$. Its two moments follow from Assumption 3. The formulas for ϖ and $\mathcal{H}$ below are large-N limits rather than finite-N identities: the conditional premium in (15) is $E[G_{t+1} \mid X_t] = \mu_G(X_t)$, a functional of the empirical distribution of the date-t cross-section, and only its common-shock limit $\mu_G(X_t) \to \mu_G(W_t)$ of (H.6) is measurable with respect to W_t. Since $E[r_{i,t+1} \mid X_t, S] = B(X_{i,t}, S_i)'\mu_G(X_t; S)$, the expected return is an average over the pooled cross-section,

$$E[\Pi_N] \;=\; \tfrac{1}{N}\sum_{i=1}^{N} E\big[\psi(X_{i,t})B(X_{i,t},S_i)'\mu_G(X_t;S)\big] \;=\; E\big[\psi(X_{\iota,t})B(X_{\iota,t},S_\iota)'\mu_G(X_t;S)\big]$$

for every N, where ι is independent and uniform on $\{1,\dots,N\}$. Thus with $M > 1$ fixed effects the average weights each fixed effect by its share $N_\ell/N\,(\to \pi_\ell)$. In the case $M = 1$, exchangeability makes every stock representative. In every case, (H.6) with uniform

integrability gives the expected return in the limit,

$$E[\Pi_N] \xrightarrow[N\to\infty]{} E_W\big[\int \psi(x)\,\bar\beta_W(x)'\mu_G(W)\,p(dx \mid W)\big] \;=\; \big\langle \psi,\, \varpi\big\rangle_{L^2(p)}\,, \qquad \varpi := E_W\big[\bar\beta_W'\mu_G(W)\,\mathsf{d}_W\big]\,, \tag{J.33}$$

with the premium profile $\varpi \in L^2(p)$ and $\mathsf{d}_W = dp(\cdot \mid W)/dp$. The second moment, in turn, is the quadratic form $E[\Pi_N^2] = \frac{1}{N^2}E[\,\psi_N'\Sigma_N(X_t)\psi_N\,]$ with $\psi_N = (\psi(X_{i,t}))_i$; by the reduction behind Proposition H.4 it too converges to a finite limit,

$$E[\Pi_N^2] \xrightarrow[N\to\infty]{} \big\langle \psi, \mathcal{H}\psi\big\rangle_{L^2(p)}\,, \tag{J.34}$$

the $N-1$ off-diagonal (pairwise-comovement) terms per row contributing $N(N-1)$ terms in total, which survive the $1/N^2$ normalization while the N diagonal terms wash out. The j-th principal-component portfolio is the weighting that maximizes the limiting second moment (J.34) subject to the normalization fixed by the portfolio-weight geometry $\mathcal{K}$,

$$\max_{\psi\in L^2(p)} \langle \psi, \mathcal{H}\psi\rangle_{L^2(p)} \qquad \text{subject to} \qquad \|\mathcal{K}^{-1/2}\psi\|^2 \;=\; 1\,, \qquad \psi \in \operatorname{ran}(\mathcal{K}^{1/2}), \tag{J.35}$$

solved at rank j with $\mathcal{K}^{-1/2}\psi$ orthogonal to $\mathcal{K}^{-1/2}\psi_1, \ldots, \mathcal{K}^{-1/2}\psi_{j-1}$. This is a generalized eigenvalue problem; the change of variables $\psi = \mathcal{K}^{1/2}\zeta$ turns it into the ordinary one $\mathcal{A}\zeta_j = \kappa_j\zeta_j$. For the fixed optimal limiting weighting ψ_j, let $\Pi_{N,j} = N^{-1}\sum_{i\le N}\psi_j(X_{i,t})r_{i,t+1}$. Then

$$\psi_j \;=\; \mathcal{K}^{1/2}\zeta_j\,, \qquad E[\Pi_{N,j}^2] \;\to\; \langle \mathcal{K}^{1/2}\zeta_j, \mathcal{H}\mathcal{K}^{1/2}\zeta_j\rangle \;=\; \langle \zeta_j, \mathcal{A}\zeta_j\rangle \;=\; \kappa_j\,. \tag{J.36}$$

Its second moment thus has the same limit κ_j as the normalized eigenvalue $\kappa_{N,j}$ in Proposition H.4; the fixed weighting ψ_j need not be a finite-N principal component. Its expected

return, from (J.33), likewise converges,

$$E[\Pi_{N,j}] \xrightarrow[N\to\infty]{} \langle \mathcal{K}^{1/2}\zeta_j,\, \varpi\rangle \;=\; \langle \zeta_j,\, \beta\rangle\,, \qquad \beta := \mathcal{K}^{1/2}\varpi\,. \tag{J.37}$$

The uncentered Sharpe ratio of this fixed-weighting portfolio, written $\mathrm{SR}_{N,j}$ to display its dependence on N, is the ratio of the left-hand side of (J.37) to the square root of the left-hand side of (J.36); both have finite limits and

$$\mathrm{SR}_{N,j} \;=\; \frac{E[\Pi_{N,j}]}{\sqrt{E[\Pi_{N,j}^2]}} \xrightarrow[N\to\infty]{} \mathrm{SR}_j \;=\; \frac{\langle \zeta_j, \beta\rangle}{\sqrt{\kappa_j}}\,, \tag{J.38}$$

the limiting Sharpe ratio of the j-th principal component, the ratio of two finite limits. When $\mathcal{A}$ has a repeated eigenvalue the individual ζ_j inside that eigenspace are not determined; the basis-independent object is the group sum $\sum_{j\in\mathsf{E}} \mathrm{SR}_j^2 = \|P_{\mathsf{E}}\beta\|^2/\kappa_{\mathsf{E}}$, with κ_{E} the common value of κ_j on E. This is the group ratio of (29).

The dictionary below needs a law of large numbers for the weight kernel $\frac{1}{P}\sum_k f(\cdot;\theta_k)\otimes f(\cdot;\theta_k)$ on $L^2(p\otimes p)$, where a single characteristic lives. It differs from the law (D.6) for the return kernel $\frac{1}{P}\sum_k R_k(Z)R_k(\tilde{Z})$ on $L^2(\mathbb{P}\otimes\mathbb{P})$: under an upper moment bound alone, averaging the weights against returns can cancel weight variation. Under the two-sided ellipticity of (B.8), however, (F.2) makes the two convergence requirements equivalent at fixed N. For the random-feature menu, with labels drawn i.i.d. from μ_θ, the summands $f(\cdot;\theta_k)\otimes f(\cdot;\theta_k)$ are i.i.d. elements of the Hilbert space of Hilbert–Schmidt operators with mean $\mathcal{K}$. Their norm is $\|f(\cdot;\theta_k)\|^2_{L^2(p)}$, so Mourier's strong law requires the diagonal moment

$$\int_\Theta \|f(\cdot;\theta)\|^2_{L^2(p)}\,\mu_\theta(d\theta) < \infty. \tag{J.39}$$

This is a trace condition, not a consequence of (H.10). In every application here it follows

from (H.18) and the idiosyncratic lower bound: fix an N large enough that every type appears. The residual contribution bounds from below a positive weighted sum of the finitely many integrals $\int \|f(\cdot;\theta)\|^2_{L^2(p_\ell)}\,\mu_\theta(d\theta)$. Each is finite, and their mixture is (J.39). Mourier's strong law (Mourier, 1953) then gives the weight-kernel convergence almost surely, the same argument that Step 2 of the proof of Lemma F.1 runs one level up.

To connect the finite menu matrices to the factor-model limits κ_j, ζ_j, β, write $\bar{\Psi}_P$ and $\bar{r}_P$ for the large-N menu second moment and mean: $(\bar{\Psi}_P)_{kl} = \langle f(\cdot;\theta_k), \mathcal{H} f(\cdot;\theta_l)\rangle$ of (J.34) and $(\bar{r}_P)_k = \langle f(\cdot;\theta_k), \varpi\rangle$ of (J.33).

Lemma J.6 (Matrix–operator dictionary) *Let $F_P : L^2(p) \to \mathbb{R}^P$ be the sampling operator $(F_P\psi)_k = \langle f(\cdot;\theta_k), \psi\rangle_{L^2(p)}$, so that $\bar{\Psi}_P = F_P\,\mathcal{H}\,F_P^*$ and $\bar{r}_P = F_P\varpi$. Suppose the menu's **weight** kernel settles,*

$$\begin{aligned} &\frac{1}{P}\sum_{k\le P} f(z;\theta_k) f(u;\theta_k) \longrightarrow k_0(z,u) \quad \text{in } L^2(p\otimes p), \\ &\frac{1}{P} F_P^* F_P \longrightarrow \mathcal{K} \quad \text{in Hilbert–Schmidt norm.} \end{aligned} \tag{J.40}$$

Then for every fixed j

$$\frac{\lambda_j(\bar{\Psi}_P)}{P} \xrightarrow[P\to\infty]{} \kappa_j = \lambda_j(\mathcal{A}), \tag{J.41}$$

and for each distinct positive value κ_{E} of κ_j, with P_{E} the projection onto its $\mathcal{A}$-eigenspace and $\beta = \mathcal{K}^{1/2}\varpi$, choose an interval $I_{\mathsf{E}} \subset (0,\infty)$ isolating κ_{E} with endpoints outside the spectrum of $\mathcal{A}$, and put $\mathsf{E}_P = \{j : \lambda_j(\bar{\Psi}_P)/P \in I_{\mathsf{E}}\}$. Then

$$\mathrm{SR}^2_{\mathsf{E}} \;=\; \sum_{j\in\mathsf{E}_P} \frac{(u_j'\bar{r}_P)^2}{\lambda_j(\bar{\Psi}_P)} \;\xrightarrow[P\to\infty]{}\; \frac{\|P_{\mathsf{E}}\beta\|^2}{\kappa_{\mathsf{E}}}\,. \tag{J.42}$$

For $C_P = F_P\mathcal{H}^{1/2}$ and a unit eigenvector v_j of $P^{-1}C_P^*C_P$ with positive eigenvalue, $u_j =$

$C_P v_j/\|C_P v_j\|$ *satisfies*

$$u_{j,k} \;=\; \frac{\langle f(\cdot;\theta_k),\, \mathcal{H}^{1/2} v_j\rangle}{\sqrt{\lambda_j(\bar{\Psi}_P)}}\,. \tag{J.43}$$

The two convergences in (J.40) are equivalent. The exact identity (J.43) and convergence of the spectral projections suffice for the arguments below; within a group only the projection has an invariant limit.

Proof of Lemma J.6. Put $C_P = F_P \mathcal{H}^{1/2}$, so $\bar{\Psi}_P = C_P C_P^*$ and $C_P^* C_P = \mathcal{H}^{1/2} F_P^* F_P \mathcal{H}^{1/2}$. The operator $\mathcal{H}^{1/2}$ is bounded and self-adjoint, so sandwiching the settled convergence (J.40) between its two copies preserves the norm mode: $P^{-1} C_P^* C_P \to \mathcal{H}^{1/2}\mathcal{K}\mathcal{H}^{1/2}$ in operator norm, with a compact limit ($\mathcal{K}$ compact, $\mathcal{H}^{1/2}$ bounded). By Lemma A.8 the limit's nonzero eigenvalues are those of $\mathcal{A} = \mathcal{K}^{1/2}\mathcal{H}\mathcal{K}^{1/2}$, the κ_j; since the convergence is in operator norm and the limit is compact, Weyl's inequality gives $\lambda_j(P^{-1}C_P^*C_P) \to \kappa_j$ for each fixed j. Then $C_P C_P^*$ shares the nonzero spectrum of $C_P^* C_P$, so $\lambda_j(\bar{\Psi}_P)/P \to \kappa_j$, which is (J.41). For the group calculation put

$$B_P := P^{-1} C_P^* C_P, \qquad B := \mathcal{H}^{1/2}\mathcal{K}\mathcal{H}^{1/2}, \qquad a_P := \mathcal{H}^{1/2}(P^{-1}F_P^* F_P)\varpi, \qquad a := \mathcal{H}^{1/2}\mathcal{K}\varpi.$$

Then $B_P \to B$ in operator norm and $a_P \to a$ in norm. Fix an isolated nonzero eigenvalue κ_{E} of B (equivalently of $\mathcal{A}$), let $P^B_{\mathsf{E},P}$ and P^B_{E} be the Riesz projections of B_P and B for this group. These differ from the $\mathcal{A}$-eigenprojection P_{E}. Choose an orthonormal eigenbasis $v_{j,P}$ of $\operatorname{ran} P^B_{\mathsf{E},P}$, with $B_P v_{j,P} = \kappa_{P,j} v_{j,P}$. Standard compact self-adjoint perturbation theory gives $P^B_{\mathsf{E},P} \to P^B_{\mathsf{E}}$ in operator norm, equality of the group dimensions for all large P, and $\max_j |\kappa_{P,j} - \kappa_{\mathsf{E}}| \to 0$. The matching matrix eigenvectors are $u_{j,P} = C_P v_{j,P}/\|C_P v_{j,P}\|$, and $\|C_P v_{j,P}\|^2 = P\kappa_{P,j}$; reading this equality in coordinate k proves the exact identity (J.43).

Moreover,

$$u'_{j,P}\bar{r}_P = \sqrt{P}\,\frac{\langle v_{j,P}, a_P\rangle}{\sqrt{\kappa_{P,j}}}, \qquad \frac{(u'_{j,P}\bar{r}_P)^2}{\lambda_j(\bar{\Psi}_P)} = \frac{|\langle v_{j,P}, a_P\rangle|^2}{\kappa^2_{P,j}}.$$

Consequently the sum over the group converges, without selecting limits for its individual basis vectors, to $\|P^B_{\mathsf{E}}a\|^2/\kappa^2_{\mathsf{E}}$. The map $Uv = \mathcal{K}^{1/2}\mathcal{H}^{1/2}v/\sqrt{\kappa_{\mathsf{E}}}$ is an isometry from $\operatorname{ran} P^B_{\mathsf{E}}$ onto the κ_{E}-eigenspace of $\mathcal{A} = \mathcal{K}^{1/2}\mathcal{H}\mathcal{K}^{1/2}$, and

$$\langle Uv, \beta\rangle = \frac{\langle v, \mathcal{H}^{1/2}\mathcal{K}\varpi\rangle}{\sqrt{\kappa_{\mathsf{E}}}} = \frac{\langle v, a\rangle}{\sqrt{\kappa_{\mathsf{E}}}}.$$

Parseval on the group gives

$$\frac{\|P^B_{\mathsf{E}}a\|^2}{\kappa^2_{\mathsf{E}}} = \frac{\|P_{\mathsf{E}}\beta\|^2}{\kappa_{\mathsf{E}}},$$

which proves (J.42). The portfolio-count factor P cancels between numerator and denominator; no interchange of the $N \to \infty$ and $P \to \infty$ limits is used, since $\bar{\Psi}_P$ is already the $N \to \infty$ object. For a simple eigenvalue, choose signs so $v_{j,P} \to v_j$ and put $\zeta_j = \mathcal{K}^{1/2}\mathcal{H}^{1/2}v_j/\sqrt{\kappa_j}$. Then (J.43) gives

$$P^{-1/2}F^*_P u_{j,P} \longrightarrow \frac{\mathcal{K}\mathcal{H}^{1/2}v_j}{\sqrt{\kappa_j}} = \mathcal{K}^{1/2}\zeta_j = \psi_j.$$

This links the normalized finite-menu principal component to the limiting weighting in (J.36). For repeated eigenvalues only the corresponding spectral subspace is invariant. The same lemma applies at each fixed N with $\mathcal{H}$ replaced by Q_N and the premium profile ϖ by ϖ_N, giving the matrix–operator dictionary in that order of limits. □

The set $\mathcal{P}_\delta$ in Definition J.5 imposes a uniform second-moment margin. Every $\mu_G \in \mathcal{P}_\delta$

satisfies

$$\operatorname{Var}(G \mid W) = \Omega(W) - \mu_G(W)\mu_G(W)' \succeq \delta\Omega(W) \succ 0 \qquad P_W\text{-almost surely.} \tag{J.44}$$

Indeed, for $a = \Omega(W)^{-1/2}\mu_G(W)$, the smallest eigenvalue of $I - aa'$ is $1 - \|a\|^2 \geq \delta$. This proves (J.44) and excludes conditionally riskless nonzero-mean factor combinations. It does not by itself exclude an ordinary nonnegative-payoff arbitrage; such a conclusion requires information about the conditional distribution. For example, $G = 2$ with probability $1/4$ and $G = 0$ otherwise has $\Omega = 1$, $\mu_G = 1/2$, positive variance, and a nonnegative nonzero payoff. The Gaussian realization in the next proposition excludes such arbitrage. The moment condition is sufficient for the Sharpe bounds proved below.

Proposition J.7 (A Gaussian realization) *Fix the regular configuration ϑ of Definition J.5 and $\mu_G \in \mathcal{P}_\delta$. Conditional on W, draw G independently of the characteristics with law*

$$G \mid W \sim \mathcal{N}\big(\mu_G(W), \Omega(W) - \mu_G(W)\mu_G(W)'\big),$$

and put $\varepsilon_i = \sigma_\varepsilon(X_i, S_i)Z_i$, where the Z_i are independent standard normals independent of (W, X, G). Every nonzero finite or limiting managed return has positive probability of a negative payoff. If, in addition, the type-specific conditional densities relative to p_ℓ are continuous in characteristics and bounded uniformly over types and states between two positive constants, the finite-N conditional factor moments are continuous empirical-law functions, have the limits (H.6), and satisfy $\|\varpi_N - \varpi\|_2 \to 0$.

Proof of Proposition J.7. Write $V(w) = \Omega(w) - \mu_G(w)\mu_G(w)' \succeq \delta\Omega(w)$. A limiting managed return is $(\Gamma_W\psi)'G$; conditional on W its variance is $(\Gamma_W\psi)'V(W)\Gamma_W\psi$, which is strictly positive wherever $\Gamma_W\psi \neq 0$. A nonzero limiting return has such a set of positive

probability, and a nondegenerate normal variable has positive probability of a negative value. At finite N, conditional on (X, W), the independent residuals contribute variance

$$\frac{1}{N^2}\sum_{i\le N}\psi(X_i)^2\sigma_\varepsilon^2(X_i,S_i).$$

It is positive whenever the stock weights are not all zero, proving the same assertion for every nonzero finite managed return.

Under the stated density bounds, write $\mathsf{d}_{\ell,w} = dp_\ell(\cdot \mid w)/dp_\ell$. The posterior factor moments are ratios with denominator $E_W\prod_{i\le N}\mathsf{d}_{\ell(i),W}(X_i) > 0$ and numerators obtained by inserting $\mu_G(W)$ or $\Omega(W)$. These are continuous functions of the empirical laws at fixed N: the log densities are bounded continuous functions and dominated convergence applies to their likelihood products. The typewise strong laws identify the canonical directing-law state from the infinite characteristic sample. The sigma-fields $\sigma(X_1,\dots,X_N)$ increase because the samples are initial segments of one infinite array. Bounded martingale convergence with this filtration therefore gives (H.6).

For the premium profiles, the tower property gives

$$E[\psi(X_i)B(X_i,S_i)'\mu_{G,N}(X)] = E[\psi(X_i)B(X_i,S_i)'\mu_G(W)].$$

Put $r_\ell = dp_\ell/dp$ and $h_\ell(x) = B(x,s_\ell)'E[\mu_G(W) \mid X_\ell = x]$, where X_ℓ denotes a draw from type ℓ. Then

$$\varpi_N = \sum_\ell \pi_{N,\ell}r_\ell h_\ell, \qquad \varpi = \sum_\ell \pi_\ell r_\ell h_\ell.$$

The bounds $\|h_\ell\|_\infty \le L_B\sqrt{\bar\omega}$ and $\|r_\ell\|_2 \le \pi_\ell^{-1/2}$ show that

$$\|\varpi_N - \varpi\|_2 \le L_B\sqrt{\bar\omega}\sum_\ell \frac{|\pi_{N,\ell}-\pi_\ell|}{\sqrt{\pi_\ell}} \longrightarrow 0.$$

□

The positive continuous-density models constructed in Lemma H.8 and the proof of Lemma H.9 satisfy this proposition's density hypotheses. Thus their entire limiting $\mathcal{P}_\delta$ admits the stated primitive realization. For general characteristic laws, the finite-N regularity in Definition J.5 must be checked directly.

Lemma J.8 (The limiting Sharpe bound) *For any linear family of square-integrable returns with maximal squared uncentered ratio $x < 1$, the maximal squared centered Sharpe ratio is $x/(1-x)$. For a limiting configuration as in Definition J.5 and a compatible premium μ_G, with $\mu_G(W)\mu_G(W)' \preceq \Omega(W)$, define*

$$\bar{x} = E[\mu_G(W)'\Omega(W)^{-1}\mu_G(W)], \qquad x_\infty = \sup_{\psi:\langle\psi,\mathcal{H}\psi\rangle>0} \frac{\langle\psi,\varpi\rangle^2}{\langle\psi,\mathcal{H}\psi\rangle},$$

with an empty supremum equal to zero. For the distinct positive eigenspaces E of $\mathcal{A}$, with eigenvalues κ_E and projections P_E,

$$x_\infty = \sum_{\mathsf{E}} \frac{\|P_\mathsf{E}\beta\|^2}{\kappa_\mathsf{E}} \leq \bar{x} \leq 1.$$

If $\bar{x} < 1$, then $\mathrm{S}^2_{\mathrm{sys}} \leq \bar{x}/(1-\bar{x}) < \infty$; if $\mu_G \in \mathcal{P}_\delta$, then $\bar{x} \leq 1-\delta$ and $\mathrm{S}^2_{\mathrm{sys}} \leq (1-\delta)/\delta$.

Proof of Lemma J.8. For any nonzero return Y, put $z_Y = (E[Y])^2/E[Y^2]$. The identity

$$\mathrm{Var}(Y) = E[Y^2] - (E[Y])^2$$

gives the squared centered ratio $z_Y/(1-z_Y)$ whenever $z_Y < 1$. Taking suprema and using continuity and monotonicity of $z \mapsto z/(1-z)$ on $[0,1)$ proves the first assertion.

For the limiting experiment, write $a(W) = \Omega(W)^{-1/2}\mu_G(W)$. Then $\varpi = E[(\Gamma^\Omega_W)^* a(W)]$,

and Cauchy–Schwarz gives

$$\frac{\langle \psi, \varpi \rangle^2}{\langle \psi, \mathcal{H}\psi \rangle} = \frac{\big(E\langle \Gamma_W^\Omega \psi, a(W)\rangle\big)^2}{E\|\Gamma_W^\Omega \psi\|^2} \le E\|a(W)\|^2 = \bar{x}. \tag{J.45}$$

Compatibility gives $\|a(W)\|^2 \le 1$, while membership in $\mathcal{P}_\delta$ gives $\|a(W)\|^2 \le 1-\delta$.

Since $\mathcal{K}$ is injective, $\operatorname{ran}(\mathcal{K}^{1/2})$ is dense in $L^2(p)$. Both the second-moment form and the premium are continuous on $L^2(p)$, so their ratio at any weighting with positive denominator can be approximated using $\psi = \mathcal{K}^{1/2} z$. On these weightings it equals $\langle z, \beta\rangle^2/\langle z, \mathcal{A}z\rangle$. If $z \in \ker \mathcal{A}$, then $E\|\Gamma_W^\Omega \mathcal{K}^{1/2} z\|^2 = 0$, so $\langle z, \beta\rangle = E\langle \Gamma_W^\Omega \mathcal{K}^{1/2} z, a(W)\rangle = 0$. Thus $\beta \perp \ker \mathcal{A}$. Expanding in an orthonormal eigenbasis of $\mathcal{A}$, Cauchy–Schwarz bounds this ratio by $\sum_{\mathsf{E}} \|P_{\mathsf{E}}\beta\|^2/\kappa_{\mathsf{E}}$; each finite partial sum is attained on the corresponding finite spectral span by coefficients proportional to the coordinates of β divided by the positive eigenvalues. Taking the increasing union of these spans proves the sum identity. The sum is finite because every finite-span ratio is bounded by (J.45).

For the menu weights, $\mathcal{K}_P = P^{-1}F_P^* F_P \to \mathcal{K}$ in operator norm by (J.40). Every $\mathcal{K}_P z$ lies in the span of the first P menu weights and converges to $\mathcal{K}z$. Injectivity of $\mathcal{K}$ makes its range dense, so these spans approximate every fixed weighting in $L^2(p)$. Continuity of both moments at weightings with positive second moment, together with the bound by x_∞, gives $x_\infty = \lim_{P\to\infty} \bar{r}_P' \bar{\Psi}_P^\dagger \bar{r}_P$ for the already-diversified experiment of Lemma J.6. Zero-denominator weightings have zero premium and contribute nothing. This equality concerns the barred moments; it does not interchange the stock limit with an inverse or an infinite sum over finite-stock components.

For positive conditional factor covariance, the rank-one inverse identity gives

$$\mu_G(W)'\Omega(W)^{-1}\mu_G(W) = \frac{\mathrm{SR}_G^2(W)}{1+\mathrm{SR}_G^2(W)}, \qquad \mathrm{SR}_G^2(W) := \mu_G(W)'\operatorname{Var}(G \mid W)^{-1}\mu_G(W). \tag{J.46}$$

Thus $\bar{x} < 1$ whenever $\mathrm{SR}_G^2(W) < \infty$ almost surely. In particular, (J.44) supplies this property. Whenever $\bar{x} < 1$, the first assertion of the lemma and (J.45) give

$$\mathrm{S}_{\mathrm{sys}}^2 = \frac{x_\infty}{1 - x_\infty} \leq \frac{\bar{x}}{1 - \bar{x}} < \infty. \tag{J.47}$$

For $\mu_G \in \mathcal{P}_\delta$, substitute $\bar{x} \leq 1 - \delta$ to obtain the stated quantitative bound. □

The maximal ratio here belongs to the limiting managed-return experiment. Convergence of second moments and premium profiles alone does not imply convergence of the maximal ratios of entire finite-stock menus, which can put increasing weight on vanishing eigenvalues.

Proposition J.9 (Generic factor premia) *Let $N_{\max} = \infty$, fix $\delta \in (0, 1)$, and impose Definition J.5. For each fixed admissible loading function, menu, type shares, and type marginals, for a generic regular configuration in Definition H.6 and, conditional on that configuration, a generic $\mu_G \in \mathcal{P}_\delta$,*

$$\mathrm{SR}_{\mathsf{E},\infty}^2 > 0 \quad \textit{for every group } \mathsf{E} \textit{ with } \kappa_{\mathsf{E}} > 0, \qquad \mathrm{S}_{\mathrm{sys}}^2 \leq \frac{1 - \delta}{\delta} < \infty.$$

There are infinitely many such groups, and these conclusions hold almost surely over the label draws.

Proof of Proposition J.9. Write $\mu_{P,N}$ and $\Psi_{P,N}$ for the moments of Section 5.2 at stock count N, with principal components as in (27)–(28). All group statements below concern the uncentered ratios $(u_j' \mu_{P,N})^2 / \lambda_j(\Psi_{P,N})$. Lemma J.8 supplies the centered maximal Sharpe ratio and its bound.

For the group ratios we take $P \to \infty$ at fixed N, followed by $N \to \infty$. The Sharpe bound of Lemma J.8 concerns the limiting experiment directly. Let Q_N be the finite-N payoff operator of (H.20), put

$$\mathcal{A}_N := \mathcal{K}^{1/2} Q_N \mathcal{K}^{1/2}, \qquad \beta_N := \mathcal{K}^{1/2} \varpi_N,$$

and recall from (H.27) that

$$\|\mathcal{A}_N - \mathcal{A}\|_{\mathrm{op}} \longrightarrow 0, \qquad \mathcal{A} = \mathcal{K}^{1/2}\mathcal{H}\mathcal{K}^{1/2}. \tag{J.48}$$

The premium-profile condition in Definition J.5 gives $\|\beta_N - \beta\|_2 \to 0$, where $\beta = \mathcal{K}^{1/2}\varpi$.

Fix a distinct positive eigenvalue κ_{E} of $\mathcal{A}$. Choose a closed interval $I_{\mathsf{E}} \subset (0,\infty)$ whose interior contains κ_{E}, whose endpoints lie in the resolvent of $\mathcal{A}$, and which contains no other point of $\sigma(\mathcal{A})$. Let $P_{\mathsf{E}} = \mathbf{1}_{I_{\mathsf{E}}}(\mathcal{A})$ and, for large N, $P_{\mathsf{E},N} = \mathbf{1}_{I_{\mathsf{E}}}(\mathcal{A}_N)$. Operator-norm convergence and the Riesz-projection calculus give

$$\|P_{\mathsf{E},N} - P_{\mathsf{E}}\|_{\mathrm{op}} \longrightarrow 0. \tag{J.49}$$

This formulation permits the eigenvalue κ_{E} to have any finite multiplicity and permits the corresponding finite-N eigenvalues to split.

Choose N_{E} so that the endpoints of I_{E} lie in the resolvent of $\mathcal{A}_N$ and $\operatorname{rank} P_{\mathsf{E},N} = \operatorname{rank} P_{\mathsf{E}}$ for every $N \geq N_{\mathsf{E}}$. At each fixed $N \geq N_{\mathsf{E}}$, the trace condition and the weight-kernel law of large numbers make the matrix–operator dictionary applicable with $\mathcal{H}$ replaced by Q_N and ϖ by ϖ_N. For a random-feature menu, this holds almost surely over the label draws. Consequently, if $\mathsf{E}_{P,N}$ denotes the eigenvalues $\lambda_j(\Psi_{P,N})/P$ lying in I_{E}, then

$$\lim_{P\to\infty} \sum_{j\in\mathsf{E}_{P,N}} \frac{(u_j'\mu_{P,N})^2}{\lambda_j(\Psi_{P,N})} = \left\langle \beta_N,\, P_{\mathsf{E},N}\big(\mathcal{A}_N|_{\operatorname{ran} P_{\mathsf{E},N}}\big)^{-1} P_{\mathsf{E},N}\beta_N \right\rangle. \tag{J.50}$$

This is the group version of Lemma J.6; no individual eigenvector is selected.

The scalar function equal to x^{-1} on I_{E} and zero on a neighborhood of $\sigma(\mathcal{A}) \setminus I_{\mathsf{E}}$ is continuous on a neighborhood of the spectra in (J.48). Continuous functional calculus,

together with (J.49) and $\beta_N \to \beta$, therefore sends the right side of (J.50) to

$$\lim_{N\to\infty}\lim_{P\to\infty} \mathrm{SR}^2_{\mathsf{E},P,N} = \left\langle \beta, \frac{1}{\kappa_{\mathsf{E}}} P_{\mathsf{E}}\beta \right\rangle = \frac{\|P_{\mathsf{E}}\beta\|^2}{\kappa_{\mathsf{E}}}. \tag{J.51}$$

Thus the group defined by the paper's P-first constants is priced exactly when $P_{\mathsf{E}}\beta \neq 0$. No limit is interchanged, and no nesting of finite-P eigenspace partitions is asserted. It remains to show that, for a generic premium, $P_{\mathsf{E}}\beta \neq 0$ for all positive groups at once.

Every group is reachable. The premium enters the pricing of a group only through the linear map

$$\ell_{\mathsf{E}} : \ \mu_G(\cdot) \ \longmapsto \ P_{\mathsf{E}}\beta, \qquad \beta = \mathcal{K}^{1/2}\varpi, \quad \varpi = E_W[\bar{\beta}'_W \mu_G(W) \mathsf{d}_W].$$

The covariance side, hence the group E and its eigenvectors, stays fixed. We need only that ℓ_{E} is not the zero map, that is, that some premium projects onto E. To certify $\ell_{\mathsf{E}} \not\equiv 0$, evaluate it at the single probe premium $\mu_{G,0}(W) = \Omega(W)^{1/2}\,\Gamma^{\Omega}_W\,\mathcal{K}^{1/2}\zeta$ (a well-defined $\mathbb{R}^K$-valued function of W, for any fixed $\zeta \in \mathsf{E}$, $\zeta \neq 0$). Its premium profile is $\varpi_0 = E_W[(\Gamma^{\Omega}_W)^*\,\Gamma^{\Omega}_W\,\mathcal{K}^{1/2}\zeta] = \mathcal{H}\mathcal{K}^{1/2}\zeta$, so

$$\ell_{\mathsf{E}}(\mu_{G,0}) \ = \ P_{\mathsf{E}}\,\mathcal{K}^{1/2}\varpi_0 \ = \ P_{\mathsf{E}}\,\mathcal{K}^{1/2}\mathcal{H}\mathcal{K}^{1/2}\zeta \ = \ P_{\mathsf{E}}\,\mathcal{A}\zeta \ = \ \kappa_{\mathsf{E}}\,\zeta \ \neq \ 0.$$

Hence $\ell_{\mathsf{E}} \not\equiv 0$ for every group E. The probe relies on the state dependence of the premium. The premium $\mu_G(\cdot)$ ranges over an infinite-dimensional space, and the displayed probe reaches every positive eigenspace of $\mathcal{A}$, since $\kappa_{\mathsf{E}}\zeta = \mathcal{K}^{1/2}(\mathcal{H}\mathcal{K}^{1/2}\zeta)$. A constant premium $\mu_G(\cdot) \equiv \bar{\mu}_G$ has only K degrees of freedom, so the probe is unavailable and reachability of every group would need a separate check.

A generic premium prices every group. Each ℓ_{E} is a bounded linear map on the space of square-integrable premia $L^2(P_W;\mathbb{R}^K)$, and boundedness makes $\ker \ell_{\mathsf{E}}$ closed. It

costs only the bounded loadings of Assumption B.3(iv) and the square-integrable densities (G.2); the factor moment Ω does not enter this bound. Put $L_B := \sup_{x,s} \|B(x,s)\|$. For $\mu_G \in L^2(P_W; \mathbb{R}^K)$,

$$\|\ell_{\mathsf{E}}(\mu_G)\| \;\le\; \|\mathcal{K}^{1/2}\|_{\mathrm{op}}\, \|\varpi\|_{L^2(p)}\,, \qquad \varpi \;=\; E_W\big[\bar{\beta}'_W \mu_G(W) \mathsf{d}_W\big]\,,$$

$$\|\varpi\|_{L^2(p)} \;\le\; L_B\, E_W\big[\|\mu_G(W)\|\, \|\mathsf{d}_W\|_{L^2(p)}\big] \;\le\; L_B \left(E_W \|\mu_G(W)\|^2\right)^{1/2} \left(E_W \|\mathsf{d}_W\|^2_{L^2(p)}\right)^{1/2},$$

by the triangle inequality and then Cauchy–Schwarz in W, finite because B is bounded (Assumption B.3(iv)), $\mu_G \in L^2(P_W; \mathbb{R}^K)$, and (G.2) holds.

Fix $\delta \in (0,1)$ and use the admissible premium space $\mathcal{P}_\delta$ in (J.32). Lemma J.8 already supplies its quantitative Sharpe bound; it remains to establish generic pricing inside this space. When P_W is nonatomic, $\mathcal{P}_\delta$ has empty interior in $L^2(P_W; \mathbb{R}^K)$ (an L^2-small spike on a set of small P_W-mass leaves L^2-distance small but violates the pointwise bound), so a property comeager in the ambient L^2 can miss $\mathcal{P}_\delta$ entirely; the argument must therefore run inside $\mathcal{P}_\delta$. Now $\mathcal{P}_\delta$ is convex (the map $\mu_G \mapsto \|\mu_G(W)\|_{\Omega(W)^{-1}}$ is a seminorm pointwise, and the constraint is its sublevel set) and closed in $L^2(P_W; \mathbb{R}^K)$: every convergent sequence has an almost-surely convergent subsequence, and the pointwise constraint passes to its limit. Consequently, a closed subset of a Hilbert space is complete, so $\mathcal{P}_\delta$ is a complete metric space and hence a Baire space. Within $\mathcal{P}_\delta$, fix a group E. The set $\mathcal{N}_{\mathsf{E}} := \{\mu_G \in \mathcal{P}_\delta : \ell_{\mathsf{E}}(\mu_G) = 0\} = \ker \ell_{\mathsf{E}} \cap \mathcal{P}_\delta$ is closed and convex, and it is nowhere dense in $\mathcal{P}_\delta$. The probe is uniformly bounded in its conditional second-moment norm. Indeed, with $C_{\mathsf{d}} = \operatorname{ess\,sup}_w \|\mathsf{d}_w\|_2$,

$$\|\Gamma^{\Omega}_w \mathcal{K}^{1/2} \zeta\| \le \sqrt{\bar{\omega}}\, L_B C_{\mathsf{d}} \|\mathcal{K}^{1/2}\zeta\|_2 =: C_\zeta < \infty.$$

Since $\kappa_{\mathsf{E}} > 0$, $C_\zeta > 0$. For $\mu_{G,0} = \Omega^{1/2} \Gamma^{\Omega}_W \mathcal{K}^{1/2} \zeta$, set $\hat{\mu}_{G,0} = \sqrt{1-\delta}\, \mu_{G,0}/C_\zeta$. Then $\hat{\mu}_{G,0} \in \mathcal{P}_\delta$ and $\ell_{\mathsf{E}}(\hat{\mu}_{G,0}) = \sqrt{1-\delta}\, \kappa_{\mathsf{E}} \zeta / C_\zeta \neq 0$. All functions in this construction are measurable with respect to the canonical directing-law state.

To see that $\mathcal{N}_{\mathsf{E}}$ has empty interior in $\mathcal{P}_\delta$, take any $\mu_G \in \mathcal{N}_{\mathsf{E}}$ (so $\ell_{\mathsf{E}}(\mu_G) = 0$) and any $\varepsilon > 0$. The convex combination $\mu_{G,\eta} = (1-\eta)\mu_G + \eta\,\hat{\mu}_{G,0}$, $\eta \in (0,1)$, lies in $\mathcal{P}_\delta$ (convexity) and satisfies $\ell_{\mathsf{E}}(\mu_{G,\eta}) = (1-\eta)\ell_{\mathsf{E}}(\mu_G) + \eta\,\ell_{\mathsf{E}}(\hat{\mu}_{G,0}) = \eta\,\ell_{\mathsf{E}}(\hat{\mu}_{G,0}) \neq 0$ for $\eta > 0$, with $\|\mu_{G,\eta} - \mu_G\| \to 0$ as $\eta \to 0$, so every $\mathcal{P}_\delta$-ball around μ_G meets the complement of $\mathcal{N}_{\mathsf{E}}$. Thus $\mathcal{N}_{\mathsf{E}}$ contains no $\mathcal{P}_\delta$-ball; being also closed, it is nowhere dense in $\mathcal{P}_\delta$. The groups are countable, so $\bigcup_{\mathsf{E}} \mathcal{N}_{\mathsf{E}}$ is meager in $\mathcal{P}_\delta$; off it, for a premium generic in $\mathcal{P}_\delta$, $P_{\mathsf{E}}\beta \neq 0$ simultaneously for every group, i.e. $\mathrm{SR}^2_{\mathsf{E},\infty} > 0$ for all E. Genericity is thus stated in $\mathcal{P}_\delta$, a closed convex subset of a Hilbert space, hence complete and a Baire space, whose elements satisfy the uniform second-moment margin. The sets $\mathcal{P}_\delta$ increase as $\delta \downarrow 0$, and their union $\bigcup_{\delta>0} \mathcal{P}_\delta = \{\mu_G : \operatorname{ess\,sup}_W \mu_G' \Omega^{-1} \mu_G < 1\}$ sits inside the premia with positive conditional factor covariance $\{\mu_G' \Omega^{-1} \mu_G < 1\}$ without exhausting them: a premium whose pointwise bound approaches one without attaining it belongs to the latter and to no $\mathcal{P}_\delta$. The statement is made at a fixed δ because the proof uses the completeness of that particular closed set; we make no category claim for the larger union. Finally, for a model generic in the sense of Proposition H.10, infinitely many κ_j are positive, and a compact operator has finite multiplicities at its nonzero eigenvalues, so there are infinitely many distinct positive κ_{E}, infinitely many groups, every one priced. Hence infinitely many principal-component groups have nonzero premium projections, so no finite collection of those groups spans the limiting premium profile.

The category conclusion concerns the limiting premium space. For each such premium and each realization satisfying Definition J.5, (J.51) transfers the group projections to the iterated finite-menu limits. □

K Proofs for Section 5.1: The Hermite-Degree Spectrum

We make Theorem 4 precise and prove it, in the model (19)–(21) with the state-dependent factor moments (23). We recall its ingredients. Characteristics are $X_i = \sqrt{\rho}\,W + \sqrt{1-\rho}\,\eta_i$

with $W, \eta_i \sim \mathcal{N}(0, I_d)$ independent. Weights are $f(x;\theta) = \sqrt{2}\cos(\omega' x + b)$, with labels $\theta = (\omega, b)$, $\omega \sim \mathcal{N}(0, \gamma^2 I_d)$ and $b \sim \mathrm{U}[0, 2\pi]$. Returns are $r_i = G + \varepsilon_i$ with $E[G^2 \mid W] = v(W) = e^{-a\|W-s\|^2}$. Conditional on W, the factor G is independent of the characteristics, and the residuals satisfy Assumption 3. The menu return $R_k = \frac{1}{N}\sum_i f(X_i;\theta_k)\, r_i$ has limiting systematic part $\bar{f}(\theta_k; W)\, G$, where $\bar{f}(\theta; W) = E[f(X;\theta) \mid W]$. The main text calls the eigenvalues of degree m the m-th step of the staircase. The proofs below are organized in numbered steps, so this appendix calls those eigenvalues rung m instead.

We also specify the reward; it does not enter Ψ_P. Conditional Cauchy–Schwarz forces $E[G \mid W]^2 \le E[G^2 \mid W]$, so the factor's conditional premium is its conditional root second moment times a conditional uncentered Sharpe ratio,

$$E[G \mid W] \;=\; h(W)\,\sqrt{E[G^2 \mid W]}\,, \qquad |h(W)| \le 1\,, \tag{K.1}$$

with h the factor's own conditional uncentered Sharpe ratio. In the constant-second-moment case $a = 0$ of (23), a constant $h \equiv \mu_G$ returns the textbook specification: $E[G] = \mu_G$, $E[G^2] = 1$ and $\sigma_G^2 = 1 - \mu_G^2$.

Lemma K.1 (The factor moment conditional on the panel)

Let $X = (X_1, \ldots, X_N)$ follow (19), and let G be independent of X conditional on W, with second moment (23). Define

$$t_N^2 = \frac{1-\rho}{1+(N-1)\rho}, \qquad \widehat{W}_N = \frac{\sqrt{\rho}}{1+(N-1)\rho}\sum_{i=1}^{N} X_i.$$

Then $W \mid X \sim \mathcal{N}(\widehat{W}_N, t_N^2 I_d)$ and

$$\Omega_N(X) := E[G^2 \mid X] = E[v(W) \mid X] = (1 + 2at_N^2)^{-d/2} \exp\left(-\frac{a\|\widehat{W}_N - s\|^2}{1 + 2at_N^2}\right). \tag{K.2}$$

Moreover, $0 < \Omega_N(X) \le 1$ and $\Omega_N(X) \to v(W)$ almost surely and in L^1.

Proof of Lemma K.1. The conditional density of W given X is proportional to

$$\exp\left(-\frac{\|w\|^2}{2}-\frac{1}{2(1-\rho)}\sum_{i=1}^{N}\|X_i-\sqrt{\rho}\,w\|^2\right).$$

Completing the square gives the normal density with mean $\widehat{W}_N$ and covariance $t_N^2 I_d$. Conditional independence gives $E[G^2 \mid W, X] = v(W)$; the tower property therefore yields $\Omega_N(X) = E[v(W) \mid X]$. Integrating $e^{-a\|w-s\|^2}$ against this normal density, and completing the square once more in each coordinate, gives (K.2). The sample average $N^{-1}\sum_i X_i$ converges almost surely to $\sqrt{\rho}\,W$ by the strong law for the independent η_i. Hence $\widehat{W}_N \to W$ and $t_N^2 \to 0$, which proves the almost-sure limit. Since $0 < v \le 1$, dominated convergence also gives convergence in L^1. □

Lemma K.2 *$\bar{f}(\theta; W) = c(\omega)\cos(\sqrt{\rho}\,\omega' W + b)$, with $c(\omega) = \sqrt{2}\,e^{-(1-\rho)\|\omega\|^2/2}$.*

Proof of Lemma K.2. Conditioning on W, only the idiosyncratic term is random. Since $\omega' X = \sqrt{\rho}\,\omega' W + \sqrt{1-\rho}\,\omega'\eta$, write $\psi = \sqrt{\rho}\,\omega' W + b$ (a constant given W) and $Z = \omega'\eta$, which given W is $\mathcal{N}(0, \|\omega\|^2)$, so that

$$\bar{f}(\theta; W) = E\big[\sqrt{2}\cos(\omega' X + b) \mid W\big] = \sqrt{2}\,E\big[\cos(\psi + \sqrt{1-\rho}\,Z)\big].$$

Expand with the angle-addition identity $\cos(\psi + u) = \cos\psi\,\cos u - \sin\psi\,\sin u$, taking $u = \sqrt{1-\rho}\,Z$:

$$E\big[\cos(\psi + \sqrt{1-\rho}\,Z)\big] = \cos\psi\;E\big[\cos(\sqrt{1-\rho}\,Z)\big] - \sin\psi\;E\big[\sin(\sqrt{1-\rho}\,Z)\big].$$

Because Z is centered Gaussian, its law is symmetric, so $E[\sin(\sqrt{1-\rho}\,Z)] = 0$ (odd integrand) and the second term vanishes. For the first term, the Gaussian characteristic function

$E[e^{itZ}] = e^{-t^2\|\omega\|^2/2}$ at $t = \sqrt{1-\rho}$ gives, on taking real parts,

$$E\big[\cos(\sqrt{1-\rho}\,Z)\big] = \operatorname{Re} E\big[e^{i\sqrt{1-\rho}\,Z}\big] = e^{-(1-\rho)\|\omega\|^2/2}.$$

Substituting back,

$$\bar{f}(\theta; W) = \sqrt{2}\,\cos(\psi)\,e^{-(1-\rho)\|\omega\|^2/2} = \underbrace{\sqrt{2}\,e^{-(1-\rho)\|\omega\|^2/2}}_{c(\omega)}\,\cos(\sqrt{\rho}\,\omega' W + b),$$

which is the claimed identity. □

Lemma K.3 (Finite-stock moments) *Let $\sigma_\varepsilon^2(x)$ be the conditional residual variance. Define the label-side kernels*

$$\tilde{\mathfrak{k}}_W(\theta,\theta') := E_W[\bar{f}(\theta;W)\bar{f}(\theta';W)v(W)], \qquad \mathfrak{k}_X^v(\theta,\theta') := E[f(X;\theta)f(X;\theta')(v(W)+\sigma_\varepsilon^2(X))].$$

At every finite N, the portfolio second-moment matrix satisfies

$$\Psi_{P,kl} \;=\; \underbrace{\frac{N-1}{N}\,\tilde{\mathfrak{k}}_W(\theta_k,\theta_l)}_{\Psi_{P,kl}^{\mathrm{sys}}} \;+\; \underbrace{\frac{1}{N}\,\mathfrak{k}_X^v(\theta_k,\theta_l)}_{D_{kl}}. \tag{K.3}$$

The corresponding label-space operator is

$$L_N = \frac{N-1}{N} T_{\tilde{\mathfrak{k}}_W} + \frac{1}{N} T_{\mathfrak{k}_X^v}.$$

Both operators on the right are positive and trace-class, with

$$\operatorname{tr} T_{\tilde{\mathfrak{k}}_W} \le 2, \qquad \operatorname{tr} T_{\mathfrak{k}_X^v} \le 2(1+\bar{\sigma}^2).$$

Consequently,

$$\|L_N - T_{\tilde{\mathfrak{k}}_W}\|_{\mathrm{op}} \leq \frac{2 + 2(1+\bar{\sigma}^2)}{N} \longrightarrow 0.$$

Proof of Lemma K.3. Throughout we use the $1/N$ convention of (21) and (E.2), $R_k = \frac{1}{N}\sum_{i=1}^{N} f(X_i;\theta_k)\, r_i$, with $r_i = G + \varepsilon_i$. Write $f_i^k = f(X_i;\theta_k)$ and $\Omega_N(X) = E[G^2 \mid X]$. Conditioning first on the characteristic cross-section, the residual conditions in Assumption 3 give

$$E[R_k R_l] = \frac{1}{N^2}\sum_{i,j=1}^{N} E[f_i^k f_j^l \Omega_N(X)] + \frac{1}{N^2}\sum_{i=1}^{N} E[f_i^k f_i^l \sigma_\varepsilon^2(X_i)].$$

The tower property and conditional independence of G and the characteristics given W imply

$$E[f_i^k f_j^l \Omega_N(X)] = E[f_i^k f_j^l G^2] = E[f_i^k f_j^l v(W)].$$

For $i \neq j$, conditional independence of X_i and X_j given W now gives

$$E[f_i^k f_j^l v(W)] = E_W[v(W)\bar{f}(\theta_k;W)\bar{f}(\theta_l;W)] = \tilde{\mathfrak{k}}_W(\theta_k,\theta_l).$$

This kernel reduces to $\mathfrak{k}_W(\theta,\theta') = E_W[\bar{f}(\theta;W)\bar{f}(\theta';W)]$ at $a = 0$. For $i = j$, the systematic and residual contributions add to $E[f(X;\theta_k)f(X;\theta_l)(v(W)+\sigma_\varepsilon^2(X))] = \mathfrak{k}_X^v(\theta_k,\theta_l)$. There are $N(N-1)$ off-diagonal pairs and N diagonal terms, proving (K.3) without requiring residual independence from characteristics.

Both kernels are Gram kernels. Their diagonal integrals are bounded because $|f| \leq \sqrt{2}$, $|\bar{f}| \leq \sqrt{2}$, $0 < v \leq 1$, and $\sigma_\varepsilon^2(X) \leq \bar{\sigma}^2$. For either label-space Gram operator, Parseval and Tonelli give $\operatorname{tr} T = \int k(\theta,\theta)\,\mu_\theta(d\theta)$, proving the stated bounds. Subtracting $T_{\tilde{\mathfrak{k}}_W}$ from L_N and bounding the norm of each positive operator by its trace gives the convergence claim. □

Remark K.4 (The order of limits) The limits in Theorem 4 and in the propositions below are iterated, in the order of (11) with $N_{\max} = \infty$: $P \to \infty$ first at a fixed cross-section, then $N \to \infty$. At each fixed N, the inner menu limit is governed, by (K.3), by the operator L_N of Lemma K.3, and that lemma gives $L_N \to T_{\tilde{\mathfrak{k}}_W}$ in operator norm. The menu mean vectors computed below do not depend on N. For an isolated group of positive eigenvalues, the Riesz projections of L_N therefore converge in operator norm to those of $T_{\tilde{\mathfrak{k}}_W}$; because the group is bounded away from zero, the inverses restricted to the group converge as well. Applying the empirical Gram-operator limit first at fixed N and then letting $N \to \infty$ thus justifies computing each group-pricing functional from the limiting systematic operator. At finite N, the inner limit still contains the own-name contribution in (K.3); the geometric staircase is the limiting systematic spectrum.

Lemma K.5 *Let* $\mathfrak{k}(W, W') := E_\theta[\bar{f}(\theta; W)\bar{f}(\theta; W')]$. *Then*

$$\mathfrak{k}(W, W') = c_0 \exp\Big(- q_\star \, \|W - W'\|^2 \Big),$$
$$q_\star := \frac{\rho\,\gamma^2}{2 + 4(1-\rho)\gamma^2}, \qquad c_0 = \big(1 + 2(1-\rho)\gamma^2\big)^{-d/2}.$$

Proof of Lemma K.5. By definition the kernel is the population second moment of the conditional mean feature, $\mathfrak{k}(W, W') = E_\theta\big[\bar{f}(\theta; W)\, \bar{f}(\theta; W')\big]$ with $\theta = (\omega, b)$. By Lemma K.2, $\bar{f}(\theta; W) = c(\omega)\cos(\psi + b)$ and $\bar{f}(\theta; W') = c(\omega)\cos(\psi' + b)$, where $\psi = \sqrt{\rho}\,\omega' W$ and $\psi' = \sqrt{\rho}\,\omega' W'$. Since b and ω are independent we may average over b first.

Step 1: averaging over the phase b. The product-to-sum identity gives

$$\cos(\psi + b)\cos(\psi' + b) = \tfrac{1}{2}\cos(\psi - \psi') + \tfrac{1}{2}\cos(\psi + \psi' + 2b).$$

Averaging over $b \sim \mathrm{U}[0, 2\pi]$, the second term integrates to zero (a full period of cos), so

$$E_b\big[\cos(\psi + b)\cos(\psi' + b)\big] = \tfrac{1}{2}\cos(\psi - \psi').$$

Writing $u = W - W'$, we have $\psi - \psi' = \sqrt{\rho}\,\omega' u$, hence

$$\mathfrak{k}(W, W') = \tfrac{1}{2}\, E_\omega\big[c(\omega)^2 \cos(\sqrt{\rho}\,\omega' u)\big] = E_\omega\big[e^{-(1-\rho)\|\omega\|^2} \cos(\sqrt{\rho}\,\omega' u)\big],$$

using $c(\omega)^2 = 2e^{-(1-\rho)\|\omega\|^2}$ from Lemma K.2.

Step 2: factoring over coordinates. With $\omega \sim \mathcal{N}(0, \gamma^2 I_d)$ the density factors as

$$\prod_{i=1}^{d} (2\pi\gamma^2)^{-1/2} e^{-\omega_i^2/(2\gamma^2)},$$

and likewise $e^{-(1-\rho)\|\omega\|^2} = \prod_i e^{-(1-\rho)\omega_i^2}$. To split the cosine, write it as a real part of a product,

$$\cos(\sqrt{\rho}\,\omega' u) = \mathrm{Re}\; e^{i\sqrt{\rho}\,\omega' u} = \mathrm{Re} \prod_{i=1}^{d} e^{i\sqrt{\rho}\,u_i \omega_i}.$$

Each coordinate integral below is real (the imaginary part of its integrand is odd in ω_i), so the real part passes through and $\mathfrak{k}$ becomes a product of one-dimensional integrals,

$$\mathfrak{k}(W, W') = \prod_{i=1}^{d} \underbrace{\int_{\mathbb{R}} (2\pi\gamma^2)^{-1/2} e^{-\omega_i^2/(2\gamma^2)}\, e^{-(1-\rho)\omega_i^2} \cos(\sqrt{\rho}\,u_i\,\omega_i)\, d\omega_i}_{=:I_i}.$$

Step 3: the one-dimensional integral. Collect the two Gaussian factors: the quadratic coefficient is

$$A := \tfrac{1}{2\gamma^2} + (1 - \rho), \qquad \text{so} \qquad e^{-\omega_i^2/(2\gamma^2)}\, e^{-(1-\rho)\omega_i^2} = e^{-A\omega_i^2}.$$

Thus $I_i = (2\pi\gamma^2)^{-1/2} \int_{\mathbb{R}} e^{-A\omega_i^2} \cos(\sqrt{\rho}\,u_i\,\omega_i)\, d\omega_i$. For the standard Gaussian–cosine integral,

complete the square in the complex exponential with $t = \sqrt{\rho}\, u_i$:

$$-A\omega_i^2 + it\omega_i = -A\Big(\omega_i - \tfrac{it}{2A}\Big)^2 - \tfrac{t^2}{4A},$$

so that, shifting the contour and using $\int_{\mathbb{R}} e^{-Au^2}\, du = \sqrt{\pi/A}$,

$$\int_{\mathbb{R}} e^{-A\omega_i^2} \cos(t\omega_i)\, d\omega_i = \mathrm{Re} \int_{\mathbb{R}} e^{-A\omega_i^2} e^{it\omega_i}\, d\omega_i = \sqrt{\tfrac{\pi}{A}}\; e^{-t^2/(4A)}.$$

With $t^2 = \rho u_i^2$ this yields

$$I_i = (2\pi\gamma^2)^{-1/2} \sqrt{\tfrac{\pi}{A}}\; e^{-\rho u_i^2/(4A)} = (2A\gamma^2)^{-1/2}\, e^{-\rho u_i^2/(4A)},$$

where the prefactor simplifies via $(2\pi\gamma^2)^{-1/2}\sqrt{\pi/A} = \big(2A\gamma^2\big)^{-1/2}$.

Step 4: taking the product. Multiplying the d identical-form factors,

$$\mathfrak{k}(W, W') = \prod_{i=1}^{d} I_i = (2A\gamma^2)^{-d/2}\, \exp\Big(-\tfrac{\rho}{4A} \textstyle\sum_{i=1}^{d} u_i^2\Big) = (2A\gamma^2)^{-d/2}\, \exp\Big(-\tfrac{\rho}{4A}\, \|u\|^2\Big).$$

Finally evaluate the two constants:

$$2A\gamma^2 = 2\gamma^2\Big(\tfrac{1}{2\gamma^2} + (1-\rho)\Big) = 1 + 2(1-\rho)\gamma^2, \qquad \frac{\rho}{4A} = \frac{\rho}{\frac{2}{\gamma^2} + 4(1-\rho)} = \frac{\rho\gamma^2}{2 + 4(1-\rho)\gamma^2} = q_\star.$$

Hence $\mathfrak{k}(W, W') = \big(1 + 2(1-\rho)\gamma^2\big)^{-d/2} \exp\big(-q_\star \|W - W'\|^2\big) = c_0 \exp\big(-q_\star \|W - W'\|^2\big)$ with c_0 and $q_\star$ as stated, which is a Gaussian (RBF) kernel. □

Proposition K.6 (Operator spectrum) *Let $\rho \in (0,1)$ and $\gamma > 0$. Set $a_H = \frac{1}{4}$ and $c_\star = \sqrt{a_H^2 + 2a_H q_\star}$. The eigenvalues of $T_{\mathfrak{k}}$ on $L^2(\mathcal{N}(0, I_d))$ are $\nu_m = \nu_0 Q_{\mathrm{feat},0}^m$ $(m \geq 0)$ with multiplicity $\mathfrak{g}_d(m) = \binom{m+d-1}{d-1}$, where $\nu_0 = c_0\, (2a_H/(a_H + q_\star + c_\star))^{d/2}$ and $Q_{\mathrm{feat},0} = q_\star/(a_H + q_\star + c_\star)$. The eigenfunctions are Hermite polynomials in a rescaled argument times a Gaussian factor. Moreover, $Q_{\mathrm{feat},0}$ is strictly increasing in γ, with $Q_{\mathrm{feat},0} \sim \rho\gamma^2 \to 0$ as*

$\gamma \to 0$. *As* $\gamma \to \infty$, $Q_{\text{feat},0} \uparrow Q_{\text{feat},*}(\rho)$, *the value of* $Q_{\text{feat},0}$ *at* $q_\star = \rho/[4(1-\rho)]$, *given in closed form by*

$$Q_{\text{feat},*}(\rho) \;=\; \frac{\rho}{1+\sqrt{1-\rho^2}} \;<\; \rho\,. \tag{K.4}$$

Proof of Proposition K.6. Step 1: tensorization. Both factors of the integral operator split over coordinates: by Lemma K.5, $\mathfrak{k}(W,W') = c_0 \prod_{i=1}^d e^{-q_\star (w_i - w_i')^2}$, and the measure factors as $d\mathcal{N}(0,I_d) = \prod_{i=1}^d d\mathcal{N}(0,1)(w_i)$. Hence $T_{\mathfrak{k}} = c_0 \bigotimes_{i=1}^d T^{(1)}$, where $T^{(1)}$ is the one-dimensional operator on $L^2(\mathcal{N}(0,1))$ with kernel $k_1(w,w') = e^{-q_\star (w-w')^2}$.

Step 2: symmetrization to a Mehler kernel. $T^{(1)}$ is self-adjoint on $L^2(\mathcal{N}(0,1))$, and conjugating by $\sqrt{p}$, with $p(w) = \sqrt{2a_H/\pi}\, e^{-2a_H w^2}$ the $\mathcal{N}(0,1)$ density (with $a_H = \frac{1}{4}$), gives an operator on $L^2(\mathbb{R},dx)$ with the same spectrum and symmetric kernel

$$\tilde{k}(x,y) = \sqrt{p(x)}\, k_1(x,y) \sqrt{p(y)} = \sqrt{\tfrac{2a_H}{\pi}}\, \exp\big(-a_H x^2 - q_\star (x-y)^2 - a_H y^2\big).$$

Expanding $-q_\star(x-y)^2 = -q_\star x^2 + 2q_\star xy - q_\star y^2$ collects this into the Mehler form

$$\tilde{k}(x,y) = \sqrt{\tfrac{2a_H}{\pi}}\, K(x,y), \qquad K(x,y) = \exp\big(-\alpha(x^2+y^2) + 2q_\star xy\big), \quad \alpha = a_H + q_\star.$$

Note the discriminant-type quantity $\alpha^2 - q_\star^2 = (a_H + q_\star)^2 - q_\star^2 = a_H^2 + 2a_H q_\star = c_\star^2$, that is, $c_\star = \sqrt{\alpha^2 - q_\star^2}$.

Step 3: ground state and the constant $\nu_0^{(1)}$**.** Try the Gaussian $\phi(x) = e^{-b_H x^2}$. Using $\int_{\mathbb{R}} e^{-Ay^2 + By}\, dy = \sqrt{\pi/A}\, e^{B^2/(4A)}$ with $A = \alpha + b_H$, $B = 2q_\star x$,

$$(K\phi)(x) = e^{-\alpha x^2} \int_{\mathbb{R}} e^{-(\alpha + b_H) y^2 + 2q_\star xy}\, dy = \sqrt{\tfrac{\pi}{\alpha + b_H}}\, \exp\Big(\big(\tfrac{q_\star^2}{\alpha + b_H} - \alpha\big) x^2\Big).$$

This equals $\mu_0\, e^{-b_H x^2}$ if and only if $\frac{q_\star^2}{\alpha + b_H} - \alpha = -b_H$, i.e., $q_\star^2 = \alpha^2 - b_H^2$, so $b_H = \sqrt{\alpha^2 - q_\star^2} = c_\star$

(the integrable root) and

$$\mu_0 = \sqrt{\tfrac{\pi}{\alpha+c_\star}}.$$

Carrying the prefactor $\sqrt{2a_H/\pi}$ from Step 2, the leading eigenvalue of $T^{(1)}$ is

$$\nu_0^{(1)} = \sqrt{\tfrac{2a_H}{\pi}}\sqrt{\tfrac{\pi}{\alpha+c_\star}} = \sqrt{\frac{2a_H}{a_H+q_\star+c_\star}}.$$

The conjugated operator has ground state $\propto e^{-c_\star x^2}$, which is the ground state of the rescaled Hermite system of scale $\ell_H = \sqrt{2c_\star}$ (rather than the standard one, whose ground state is $e^{-x^2/2}$; the two coincide only at $c_\star = \frac{1}{2}$).

Step 4: Mehler ratio. Mehler's formula (Szegő, 1975, Problems and Exercises, Problem 23) expands a kernel of the form $\exp(-\alpha(x^2+y^2)+2q_\star xy)$ in the $L^2(dx)$-orthonormal Hermite functions of scale $\ell_H = \sqrt{2c_\star}$, $\{\psi_k^{(\ell_H)}\}$, as $K = \mu_0 \sum_{k\geq 0} u^k\, \psi_k \otimes \psi_k$, a geometric sequence whose ratio $u \in (0,1)$ is fixed by matching the quadratic forms:

$$\frac{2u}{1+u^2} = \frac{q_\star}{\alpha}.$$

Clearing denominators, $q_\star u^2 - 2\alpha u + q_\star = 0$, whose roots are $u = \dfrac{\alpha \pm \sqrt{\alpha^2 - q_\star^2}}{q_\star} = \dfrac{\alpha \pm c_\star}{q_\star}$. Only the $(-)$ root lies in $(0,1)$, and rationalizing it,

$$Q_{\text{feat},0} = \frac{\alpha - c_\star}{q_\star} = \frac{\alpha - c_\star}{q_\star}\cdot\frac{\alpha + c_\star}{\alpha + c_\star} = \frac{\alpha^2 - c_\star^2}{q_\star(\alpha+c_\star)} = \frac{q_\star}{\alpha+c_\star} = \frac{q_\star}{a_H+q_\star+c_\star},$$

using $\alpha^2 - c_\star^2 = q_\star^2$. Thus the eigenvalues of $T^{(1)}$ are $\nu_k^{(1)} = \nu_0^{(1)} Q_{\text{feat},0}^k$, $k \geq 0$, with Hermite eigenfunctions.

Step 5: tensorizing back. Eigenvalues of a tensor product multiply and eigenfunctions

are products of the ψ_{k_i}. So at multi-index $k \in \mathbb{Z}^d_{\geq 0}$,

$$\nu_{(k)} = c_0 \prod_{i=1}^{d} \nu_{k_i}^{(1)} = c_0 \Big(\tfrac{2a_H}{a_H+q_\star+c_\star}\Big)^{d/2} Q_{\text{feat},0}^{\sum_i k_i} = \nu_0\, Q_{\text{feat},0}^{|k|}, \qquad \nu_0 = c_0 \Big(\tfrac{2a_H}{a_H+q_\star+c_\star}\Big)^{d/2}.$$

Collecting the multi-indices with $|k| = m$, the distinct eigenvalue is $\nu_m = \nu_0 Q_{\text{feat},0}^m$ with multiplicity $\#\{k \in \mathbb{Z}^d_{\geq 0} : |k| = m\} = \binom{m+d-1}{d-1} = \mathfrak{g}_d(m)$ (stars and bars).

Step 6: monotonicity and limits. Write $t = \gamma^2$. From Lemma K.5, $q_\star(t) = \dfrac{\rho t}{2 + 4(1-\rho)t}$, and

$$q_\star'(t) = \frac{\rho\big(2 + 4(1-\rho)t\big) - \rho t \cdot 4(1-\rho)}{\big(2 + 4(1-\rho)t\big)^2} = \frac{2\rho}{\big(2 + 4(1-\rho)t\big)^2} > 0,$$

so $q_\star$ is strictly increasing in γ^2, with $q_\star \sim \rho\gamma^2/2$ as $\gamma \to 0$ and $q_\star \uparrow \rho/[4(1-\rho)]$ as $\gamma \to \infty$.

For $Q_{\text{feat},0}$ as a function of $q_\star$, the relation of Step 4 reads $\Upsilon(Q_{\text{feat},0}) = \frac{q_\star}{a_H+q_\star}$, where $\Upsilon(u) = \frac{2u}{1+u^2}$. Now $q_\star \mapsto q_\star/(a_H + q_\star)$ is increasing (derivative $a_H/(a_H+q_\star)^2 > 0$), and Υ is a strictly increasing bijection of $[0,1)$ onto $[0,1)$ since $\Upsilon'(u) = \frac{2(1-u^2)}{(1+u^2)^2} > 0$; hence $Q_{\text{feat},0} = \Upsilon^{-1}\big(q_\star/(a_H+q_\star)\big)$ is strictly increasing in $q_\star$, and therefore in γ. As $q_\star \to 0$, $\Upsilon(u) \approx 2u$ and $q_\star/(a_H+q_\star) \approx q_\star/a_H$, so $Q_{\text{feat},0} \sim q_\star/(2a_H) = 2q_\star$ (using $a_H = \frac{1}{4}$); combined with $q_\star \sim \rho\gamma^2/2$ this gives $Q_{\text{feat},0} \sim 2q_\star \sim \rho\gamma^2 \to 0$ as $\gamma \to 0$. As $\gamma \to \infty$, $q_\star \uparrow \rho/[4(1-\rho)]$ and, by monotonicity and continuity, $Q_{\text{feat},0} \uparrow Q_{\text{feat},*}(\rho) := Q_{\text{feat},0}\big(q_\star = \rho/[4(1-\rho)]\big)$. This ceiling is available in closed form. At $q_\star = \rho/[4(1-\rho)]$ one has $\alpha = a_H + q_\star = 1/[4(1-\rho)]$ and $c_\star = \sqrt{a_H^2 + 2a_H q_\star} = \frac{1}{4}\sqrt{(1+\rho)/(1-\rho)}$, so

$$Q_{\text{feat},*}(\rho) = \frac{q_\star}{\alpha + c_\star} = \frac{\rho/(1-\rho)}{1/(1-\rho) + \sqrt{(1+\rho)/(1-\rho)}} = \frac{\rho}{1+\sqrt{1-\rho^2}},$$

which is (K.4). It lies strictly below ρ: equivalently, at this $q_\star$ the relation of Step 4 reads $\Upsilon(Q_{\text{feat},*}) = q_\star/(a_H+q_\star) = \rho$, and $\Upsilon(u) = 2u/(1+u^2) > u$ for $u \in (0,1)$, so $Q_{\text{feat},*} < \Upsilon(Q_{\text{feat},*}) = \rho < 1$.

The hypothesis $\rho < 1$ is what makes the ceiling a ceiling. At $\rho = 1$ the features are noiseless, $q_\star = \gamma^2/2$ is unbounded, and $Q_{\text{feat},0} = q_\star/(a_H + q_\star + c_\star) \to 1$ as $\gamma \to \infty$: the staircase flattens completely and no $Q_{\text{feat},*}(\rho) < 1$ exists. □

Proposition K.7 (Premia of the principal components) *Assume $a = 0$, $E[G \mid W] = \mu_G$, $|\mu_G| \le 1$, $\rho \in (0,1)$, and $\gamma > 0$, and write $Q = Q_{\text{feat},0}$. The normalization (23) gives $E[G^2] = 1$. Let E_m be the degree-m eigenspace of $T_{\mathfrak{k}}$, with orthogonal projection P_{E_m}, and set* $\text{group}(m) = \{j : J_{m-1} < j \le J_m\}$. *In the iterated limit $P \to \infty$ followed by $N \to \infty$,*

$$\text{SR}^2_{(m)} := \sum_{j \in \text{group}(m)} \text{SR}_j^2 \;\longrightarrow\; \frac{\mu_G^2}{E[G^2]} \|P_{E_m} 1\|^2 = \frac{\mu_G^2}{E[G^2]} (1-Q^2)^{d/2} Q^m \sum_{|n|=m/2} \prod_{i=1}^{d} \frac{(2n_i)!}{4^{n_i}(n_i!)^2} \tag{K.5}$$

for even m, and the limit is zero for odd m; the limiting ratios satisfy $\sum_{j \ge 1} \text{SR}_j^2 = \mu_G^2/E[G^2]$. Let $\varphi(w) = (2\pi)^{-1/2} e^{-w^2/2}$ be the standard normal density and let $e_k^W(w) = \prod_{i=1}^d \frac{\psi_{k_i}^{(\ell_H)}(w_i)}{\sqrt{\varphi(w_i)}}$, where $\ell_H = \sqrt{2c_\star}$ and $\psi_n^{(\ell_H)}$ are the normalized Hermite functions in the proof of Proposition K.6. Then

$$\langle e_k^W, 1 \rangle = \begin{cases} (1-Q^2)^{d/4} Q^{m/2} \displaystyle\prod_{i=1}^{d} \frac{\sqrt{(2n_i)!}}{2^{n_i} n_i!}, & k = 2n, \quad m = 2|n|, \\ 0, & \text{some } k_i \text{ is odd.} \end{cases} \tag{K.6}$$

The ratios SR_j are uncentered, as defined after (28). The limit of $\text{SR}^2_{(m)}$ is the group quantity $\text{SR}^2_{\mathsf{E},\infty}$ of Section 5.2, with $\mathsf{E} = \text{group}(m)$. Within a repeated-eigenvalue group, the Hermite basis represents the limiting ratio vector by $\mu_G(\langle e_k^W, 1\rangle)_{|k|=m}/\sqrt{E[G^2]}$. Only its norm is invariant under rotations of the eigenspace; individual coordinates require a choice of basis. In the displayed Hermite basis, (K.6) vanishes whenever any coordinate index is odd, including some components on even rungs. For example, $k = (1,1)$ has degree two and

zero overlap. Thus the rung total, rather than a particular component coordinate, is the intrinsic pricing quantity.

Proof of Proposition K.7. Step 1: premium of one portfolio. In this constant-premium case $E[G \mid W] = \mu_G$. Conditioning first on X removes the residual contribution because $E[\varepsilon_i \mid X] = 0$. Conditional independence of G and the characteristics given W then gives, at every N,

$$E[R_a] = E[\bar{f}(\theta_a; W)G] = \mu_G g_0(\theta_a), \qquad g_0(\theta) := E_W[\bar{f}(\theta; W)].$$

Step 2: singular value decomposition of the exposure map. Write

$$\bar{f}(\theta; W) = \sum_k \sqrt{\nu_{(k)}}\, e_k^{\Theta}(\theta)\, e_k^{W}(W),$$

with $\{e_k^{\Theta}\}$ orthonormal in $L^2(\mu_\theta)$ and $\{e_k^{W}\}$ orthonormal in $L^2(\mathcal{N}(0, I_d))$: the two Gram operators of this map are the θ-side kernel $\mathfrak{k}_W(\theta, \theta') = E_W[\bar{f}(\theta; W)\bar{f}(\theta'; W)]$ and the state-side $T_{\mathfrak{k}}$, which share the nonzero spectrum $\{\nu_{(k)}\}$ of Proposition K.6. Taking E_W and using $E_W[e_k^W(W)] = \langle e_k^W, 1\rangle$,

$$g_0(\theta) \;=\; \sum_k \sqrt{\nu_{(k)}}\, \langle e_k^W, 1\rangle\, e_k^{\Theta}(\theta)\,. \tag{K.7}$$

Step 3: the two moments, summed over a rung. At $a = 0$ the limiting label-side operator is $E[G^2]\, T_{\mathfrak{k}_W}$, with $T_{\mathfrak{k}_W}$ the label-side systematic operator, and the mean vector is exactly $\mu_P = \mu_G g_{0,P}$ for every N. By Remark K.4, the pricing functional below has the asserted P-first, N-second limit.

We may therefore calculate it from the limiting systematic block, which is $E[G^2]$ times the Gram matrix $K_P = [\mathfrak{k}_W(\theta_a, \theta_b)]_{a,b \le P}$. Fix a degree m and let $U_{(m)} \in \mathbb{R}^{P \times \mathfrak{g}_d(m)}$ collect the unit eigenvectors of K_P belonging to the eigenvalues in group(m). The dictionary identifies the

invariant limit through the group's spectral projection, without selecting limits for individual columns: the limiting level ν_m has multiplicity $\mathfrak{g}_d(m) > 1$ as soon as $m \geq 1$ and $d \geq 2$, so its limiting eigenspace admits any orthonormal basis. The finite-menu eigenvalues within the group may split; their individual eigenvectors need not converge. We therefore work with the rung as a whole. The degree-m eigenspace of $T_{\mathfrak{k}_W}$ is $E_m^{\Theta} = \operatorname{span}\{e_k^{\Theta} : |k| = m\}$ and, by (K.7),

$$P_{E_m^{\Theta}} g_0 = \sqrt{\nu_m} \sum_{|k|=m} \langle e_k^W, 1\rangle e_k^{\Theta}.$$

Apply Lemma J.6 on state space to the sampled functions $\bar{f}(\theta;\cdot)$, with $\mathcal{H} = I$, $\varpi = 1$, $\bar{\Psi}_P = K_P$, and $\bar{r}_P = g_{0,P}$. Its weight operator is $T_{\mathfrak{k}}$. The singular-value identity in Step 2 translates (J.42) to the label projection below; multiplying by the limiting eigenvalue ν_m gives

$$\frac{\lambda_j(K_P)}{P} \longrightarrow \nu_m, \qquad \frac{1}{P}\left\|U'_{(m)} g_{0,P}\right\|^2 \longrightarrow \left\|P_{E_m^{\Theta}} g_0\right\|^2, \qquad g_{0,P} = (g_0(\theta_a))_{a \leq P}.$$

Since $\mu_P = \mu_G g_{0,P}$ and the eigenvalues of the systematic second-moment block are $E[G^2]$ times those of K_P, functional calculus on the finite-dimensional eigenspace of the group now yields

$$\sum_{j \in \text{group}(m)} \frac{(u_j' \mu_P)^2}{\lambda_j(\Psi_P)} \longrightarrow \frac{\mu_G^2}{E[G^2]\,\nu_m} \left\|P_{E_m^{\Theta}} g_0\right\|^2 = \frac{\mu_G^2}{E[G^2]} \sum_{|k|=m} \langle e_k^W, 1\rangle^2.$$

Thus the rung sum is the squared norm of a state-side projection,

$$\mathrm{SR}_{(m)}^2 = \frac{\mu_G^2}{E[G^2]} \sum_{|k|=m} \langle e_k^W, 1\rangle^2 = \frac{\mu_G^2}{E[G^2]} \left\|P_{E_m} 1\right\|^2,$$

and the right-hand side depends only on P_{E_m}.

Step 4: the overlaps in closed form. The eigenfunctions tensorize, so it suffices to take $d = 1$. Undoing the $\sqrt{p}$ conjugation of Step 2 in the proof of Proposition K.6, the eigenfunctions of $T^{(1)}$ orthonormal in $L^2(\mathcal{N}(0,1))$ are

$$e_k^W(w) = \frac{(c_\star/a_H)^{1/4}}{\sqrt{2^k\, k!}}\, e^{-(c_\star - a_H)w^2}\, H_k\big(\sqrt{2c_\star}\, w\big).$$

With $p(w) = \sqrt{2a_H/\pi}\, e^{-2a_H w^2}$, $\langle e_k^W, 1\rangle = \int e_k^W\, p = \frac{(c_\star/a_H)^{1/4}}{\sqrt{2^k k!}} \sqrt{\frac{2a_H}{\pi}} \int H_k(\ell_H w) e^{-b_I w^2} dw$ with $\ell_H = \sqrt{2c_\star}$ and $b_I = (c_\star - a_H) + 2a_H = c_\star + a_H$. The generating function $\sum_k H_k(x) t^k/k! = e^{2xt - t^2}$ integrates to $\sqrt{\pi/b_I}\, e^{t^2(\ell_H^2/b_I - 1)}$, so the integral vanishes for odd k and equals

$$\sqrt{\pi/b_I}\, \frac{(2n)!}{n!} \Big(\frac{\ell_H^2 - b_I}{b_I}\Big)^n \qquad \text{for } k = 2n.$$

Now

$$\frac{\ell_H^2 - b_I}{b_I} = \frac{c_\star - a_H}{c_\star + a_H} = Q,$$

the last step because $c_\star^2 = a_H^2 + 2a_H q_\star$ gives $a_H + q_\star + c_\star = (a_H + c_\star)^2/(2a_H)$, whence $Q = q_\star/(a_H + q_\star + c_\star) = 2a_H q_\star/(a_H + c_\star)^2 = (c_\star - a_H)/(c_\star + a_H)$. Collecting constants and using $(c_\star/a_H)^{1/2}\, 2a_H/(a_H + c_\star) = 2\sqrt{a_H c_\star}/(a_H + c_\star) = \sqrt{1 - Q^2}$ gives (K.6) for $d = 1$; the general case is the product over coordinates.

Step 5: odd rungs, by coordinatewise symmetry. Substituting (K.6) into Step 3 and squaring gives, for even m, the closed form in (K.5), the sum running over the $n \in \mathbb{Z}_{\geq 0}^d$ with $|n| = m/2$. For odd m the rung earns nothing. The clean way to see this is a symmetry that does not depend on the basis. The law $\mathcal{N}(0, I_d)$ of W and the kernel $\mathfrak{k}$, which depends on W, W' only through $\|W - W'\|^2$, are both invariant under the reflection $W_i \mapsto -W_i$ of any single coordinate i. So $T_{\mathfrak{k}}$ commutes with each of the d reflections, each eigenspace E_m decomposes into their joint parity classes, and the constant function 1 lies in the all-even

class. Hence $P_{E_m}1$ picks up only the part of E_m that is even in every coordinate separately. In the Hermite basis this is exactly the vanishing of $\langle e_k^W, 1\rangle$ whenever some k_i is odd, a stronger statement than vanishing when $|k|$ is odd, and the reason the reflection must be applied coordinatewise rather than globally as $W \mapsto -W$. When m is odd, every k with $|k| = m$ has some k_i odd, so $P_{E_m}1 = 0$ and $\mathrm{SR}_{(m)} = 0$.

Step 6: the total. Every $\nu_m = \nu_0 Q^m$ is strictly positive, and the Hermite eigenfunctions form a complete orthonormal basis of $L^2(\mathcal{N}(0, I_d))$, so the eigenspaces E_m exhaust the space and every eigenfunction corresponds to a limiting principal component. Parseval applied to the constant function is basis free, $\sum_{m\geq 0} \|P_{E_m}1\|^2 = \|1\|_{L^2}^2 = 1$, so summing (K.5) over m yields $\sum_j \mathrm{SR}_j^2 = \mu_G^2/E[G^2]$. □

Proposition K.8 (State-dependent factor second moments) *Let $\rho \in (0,1)$, $\gamma > 0$, $v(W) = e^{-a\|W-s\|^2}$, $a \geq 0$, and $\tilde{\mathfrak{k}}(w,w') = \sqrt{v(w)}\mathfrak{k}(w,w')\sqrt{v(w')}$. Then*

$$E[G^2] = (1+2a)^{-d/2} e^{-a\|s\|^2/(1+2a)}. \tag{K.8}$$

Write $a_H^{\mathrm{eff}} = a_H + a/2$, $\alpha^{\mathrm{eff}} = a_H^{\mathrm{eff}} + q_\star$, and $c_\star^{\mathrm{eff}} = \sqrt{(a_H^{\mathrm{eff}})^2 + 2a_H^{\mathrm{eff}} q_\star}$. The eigenvalues of $T_{\tilde{\mathfrak{k}}}$ are $\nu_m = \nu_0(a,s) Q_{\mathrm{feat},a}^m$, with multiplicity $\binom{m+d-1}{d-1}$, where

$$Q_{\mathrm{feat},a} = \frac{q_\star}{a_H^{\mathrm{eff}} + q_\star + c_\star^{\mathrm{eff}}}, \qquad \nu_0(a,s) = c_0 \left(\frac{2a_H}{a_H^{\mathrm{eff}} + q_\star + c_\star^{\mathrm{eff}}}\right)^{d/2} e^{-a\|s\|^2/(1+2a)}. \tag{K.9}$$

The eigenfunctions are products of the normalized one-dimensional functions

$$e_k^W(w) = \left(\frac{c_\star^{\mathrm{eff}}}{a_H}\right)^{1/4} \frac{e^{a_H w^2 - c_\star^{\mathrm{eff}}(w-\tau)^2}}{\sqrt{2^k k!}} H_k\big(\sqrt{2c_\star^{\mathrm{eff}}}(w-\tau)\big), \qquad \tau = \frac{as}{2a_H^{\mathrm{eff}}}. \tag{K.10}$$

The ratio $Q_{\mathrm{feat},a}$ strictly decreases in a, with $Q_{\mathrm{feat},0} = Q$ and $Q_{\mathrm{feat},a} \to 0$ as $a \to \infty$. Let E_m be the degree-m eigenspace of $T_{\tilde{\mathfrak{k}}}$. For $E[G \mid W] = h(W)\sqrt{v(W)}$, $|h| \leq 1$, the uncentered

ratios in the iterated limit $P \to \infty$ followed by $N \to \infty$ satisfy

$$\mathrm{SR}^2_{(m)} = \|P_{E_m} h\|^2, \qquad \sum_{j \geq 1} \mathrm{SR}^2_j = E[h(W)^2] \leq 1.$$

The scale factor $\nu_0(a, s)$ in (K.9) retains the original $2a_H = 1/2$, inherited from the unchanged state density. The location s multiplies all eigenvalues by the same factor but moves the eigenspaces, so it can change the projections of a fixed premium h. The Gaussian factor $e^{a_H w^2}$ in (K.10) remains centered at the origin.

For a Gaussian premium $h(W) = h_0 e^{-\varkappa \|W - \varsigma\|^2/2}$, with $\varkappa \geq 0$ and $|h_0| \leq 1$, the overlaps follow coordinatewise from

$$\begin{aligned} \int_{\mathbb{R}} H_k(\ell_H u) e^{-b_I u^2 + \eta u}\, du &= \sqrt{\frac{\pi}{b_I}} e^{\eta^2/(4b_I)} k! [t^k] \exp\left(c t^2 + \frac{\ell_H \eta}{b_I} t \right), \\ c = \frac{\ell_H^2 - b_I}{b_I}, &\qquad \ell_H = \sqrt{2 c_\star^{\mathrm{eff}}}. \end{aligned} \tag{K.11}$$

where $[t^k]$ extracts the coefficient of t^k and

$$b_I = a_H + c_\star^{\mathrm{eff}} + \frac{\varkappa}{2}, \qquad \eta = -2a_H \tau - \varkappa(\tau - \varsigma). \tag{K.12}$$

The coefficient η in this integral differs from the kernel's linear coefficient *as*. The offset-tuning condition is $\eta = 0$, equivalently $\varkappa\varsigma = (2a_H + \varkappa)\tau$, coordinatewise. When $s = \varsigma = 0$, odd coefficients vanish and the even coefficients have quadratic rate

$$c_{a,\varkappa} = \frac{c_\star^{\mathrm{eff}} - a_H - \varkappa/2}{c_\star^{\mathrm{eff}} + a_H + \varkappa/2}.$$

In particular, for $s = 0$ and constant h, the overlaps with the constant function are

$$\langle e_k^W, 1 \rangle = (1 - \tilde{Q}^2_{\mathrm{feat},a})^{d/4} \tilde{Q}^{m/2}_{\mathrm{feat},a} \prod_{i=1}^{d} \frac{\sqrt{(2n_i)!}}{2^{n_i} n_i!}, \quad k = 2n, \quad m = 2|n|, \qquad \tilde{Q}_{\mathrm{feat},a} = \frac{c_\star^{\mathrm{eff}} - a_H}{c_\star^{\mathrm{eff}} + a_H}.$$

(K.13)

Here $\tilde{Q}_{\mathrm{feat},a} > Q_{\mathrm{feat},a}$ for $a > 0$, with equality at $a = 0$: the geometric part of the premium coefficients decays more slowly than the eigenvalue ratio. If $\eta \neq 0$, every coefficient in (K.11) is nonzero for $c \geq 0$. For $c < 0$, the exceptional zero sets are given by

$$k![t^k]e^{ct^2+(\ell_H\eta/b_I)t} = (-c)^{k/2}H_k\left(\frac{\ell_H\eta}{2b_I\sqrt{-c}}\right) = 0.$$

Thus nonzero Gaussian premia generically price every rung, subject to the offset-tuning and Hermite-zero exceptions.

Proof of Proposition K.8. Write $v(W) = E[G^2 \mid W] = e^{-a\|W-s\|^2}$. By Lemma K.3, the limiting systematic return is $\bar{f}(\theta_a; W)\, G$. Conditioning on W gives

$$\begin{aligned}\lim_{N\to\infty} \Psi_{P,ab} &= E\big[\bar{f}(\theta_a; W)\bar{f}(\theta_b; W)\, G^2\big] = E_W\big[\bar{f}(\theta_a; W)\, \bar{f}(\theta_b; W)\, E[G^2 \mid W]\big] \\ &= E_W\big[\bar{f}(\theta_a; W)\, \bar{f}(\theta_b; W)\, v(W)\big],\end{aligned}$$

which uses only $E[G^2 \mid W] = v(W)$ and nothing about $E[G \mid W]$, so the premium may itself be state dependent. The state-side operator is that of Lemma K.5 conjugated by $\sqrt{v}$, with kernel $\tilde{\mathfrak{k}}(w, w') = \sqrt{v(w)}\,\mathfrak{k}(w, w')\,\sqrt{v(w')}$ and $\mathfrak{k}(w, w') = c_0 e^{-q_\star\|w-w'\|^2}$. Exactly as in Step 1 of the proof of Proposition K.6, both the kernel and the measure factor over coordinates, so $T_{\tilde{\mathfrak{k}}} = c_0 \bigotimes_{i=1}^{d} \tilde{T}^{(1)}$ with $\tilde{T}^{(1)}$ the one-dimensional operator with kernel $\sqrt{v_1(w)}\, e^{-q_\star(w-w')^2}\sqrt{v_1(w')}$, $v_1(w) = e^{-a(w-s)^2}$. We pull the single constant c_0 out here and take $d = 1$ until the tensorizing step, where it is reattached once. Completing the square in W gives (K.8): $E[G^2] = E_W[e^{-a\|W-s\|^2}] = (1+2a)^{-d/2}e^{-a\|s\|^2/(1+2a)}$, which is 1 at $a = 0$.

Step 1: one quadratic form. Conjugating by $\sqrt{p}$ as in Step 2 of the proof of Proposition K.6, with $p(w) = \sqrt{2a_H/\pi}\, e^{-2a_H w^2}$ still the $\mathcal{N}(0, 1)$ density and still carrying the original

$a_H = \frac{1}{4}$, the symmetrized kernel of $\tilde{T}^{(1)}$ on $L^2(\mathbb{R}, dx)$ is

$$\begin{aligned}\tilde{k}(x,y) &= \sqrt{\tfrac{2a_H}{\pi}}\,\exp\Big(-a_H x^2 - \tfrac{a}{2}(x-s)^2 - q_\star(x-y)^2 - \tfrac{a}{2}(y-s)^2 - a_H y^2\Big)\\ &= \; C\,\exp\Big(-\alpha^{\mathrm{eff}}(x^2+y^2) + 2q_\star xy + \eta_{\mathfrak{k}}\,(x+y)\Big),\end{aligned}$$

with $\alpha^{\mathrm{eff}} = a_H + \frac{a}{2} + q_\star = a_H^{\mathrm{eff}} + q_\star$, $\eta_{\mathfrak{k}} = as$, and $C = \sqrt{2a_H/\pi}\; e^{-as^2}$. The tilt by $\sqrt{v}$ adds $\frac{a}{2}$ to the harmonic coefficient and creates a linear term. The prefactor $\sqrt{2a_H/\pi}$ comes from the untilted state density p, so the substitution $a_H \mapsto a_H^{\mathrm{eff}}$ propagates to $\alpha, c_\star, Q$ but not to ν_0.

Step 2: removing the linear term, and the constant it leaves behind. Translate $x = u + \tau,\ y = z + \tau$. The coefficient of $(u+z)$ becomes $-2(\alpha^{\mathrm{eff}} - q_\star)\tau + \eta_{\mathfrak{k}}$, which vanishes at

$$\tau = \frac{\eta_{\mathfrak{k}}}{2(\alpha^{\mathrm{eff}} - q_\star)} = \frac{as}{2a_H^{\mathrm{eff}}}\,,$$

using $\alpha^{\mathrm{eff}} - q_\star = a_H^{\mathrm{eff}}$. The translation leaves the constant term

$$-2(\alpha^{\mathrm{eff}} - q_\star)\tau^2 + 2\eta_{\mathfrak{k}}\tau = -2a_H^{\mathrm{eff}}\tau^2 + 2as\,\tau = \frac{a^2 s^2}{2a_H^{\mathrm{eff}}}\,,$$

so the surviving kernel is $C'\exp\big(-\alpha^{\mathrm{eff}}(u^2+z^2) + 2q_\star uz\big)$, the pure Mehler kernel of Step 2 of Proposition K.6 with α^{eff} in place of α, and with

$$C' = \sqrt{\tfrac{2a_H}{\pi}}\;\exp\Big(-as^2 + \frac{a^2 s^2}{2a_H^{\mathrm{eff}}}\Big) = \sqrt{\tfrac{2a_H}{\pi}}\;\exp\Big(-\frac{a\,a_H\,s^2}{a_H^{\mathrm{eff}}}\Big) = \sqrt{\tfrac{2a_H}{\pi}}\;\exp\Big(-\frac{a\,s^2}{1+2a}\Big),$$

where the middle equality uses $2a_H^{\mathrm{eff}} = \frac{1}{2} + a$ (as $a_H = \frac{1}{4}$) and the last uses $a_H/a_H^{\mathrm{eff}} = 1/(1+2a)$. Lebesgue measure is translation invariant, so the map $u \mapsto u + \tau$ is unitary on $L^2(\mathbb{R}, dx)$ and leaves the spectrum unchanged. The constant C' multiplies the spectrum, and the center s enters the eigenvalues through C'.

Step 3: the spectrum. Steps 3–5 of the proof of Proposition K.6 now apply to the Mehler kernel of Step 2 with α^{eff} in place of α: the discriminant is $c_\star^{\mathrm{eff}} = \sqrt{(\alpha^{\mathrm{eff}})^2 - q_\star^2} = \sqrt{(a_H^{\mathrm{eff}})^2 + 2a_H^{\mathrm{eff}} q_\star}$, the Mehler ratio solves $2Q_{\mathrm{feat},a}/(1+Q_{\mathrm{feat},a}^2) = q_\star/\alpha^{\mathrm{eff}}$ and equals (K.9), and the leading eigenvalue of the one-dimensional operator is $C'\sqrt{\pi/(\alpha^{\mathrm{eff}} + c_\star^{\mathrm{eff}})}$, that is

$$\nu_0^{(1)}(a,s) = \sqrt{\frac{2a_H}{a_H^{\mathrm{eff}} + q_\star + c_\star^{\mathrm{eff}}}}\, e^{-as^2/(1+2a)}.$$

Tensorizing back as in Step 5 of the proof of Proposition K.6, the eigenvalue at a multi-index k is $\nu_{(k)} = c_0 \prod_{i=1}^d \nu_{k_i}^{(1)}(a, s_i)$, so that

$$\nu_{(k)} = c_0 \Big(\frac{2a_H}{a_H^{\mathrm{eff}} + q_\star + c_\star^{\mathrm{eff}}}\Big)^{d/2} e^{-a\|s\|^2/(1+2a)}\, Q_{\mathrm{feat},a}^{|k|} = \nu_0(a,s)\, Q_{\mathrm{feat},a}^{|k|},$$

with $\nu_0(a,s)$ as in (K.9), the d exponential factors having combined into $e^{-a\|s\|^2/(1+2a)}$. Collecting the multi-indices with $|k| = m$ gives $\nu_m = \nu_0(a,s) Q_{\mathrm{feat},a}^m$ with multiplicity $\binom{m+d-1}{d-1}$, by the same stars-and-bars count. The center s enters only through the common factor $e^{-a\|s\|^2/(1+2a)}$, hence not through $Q_{\mathrm{feat},a}$. Undoing the $\sqrt{p}$ conjugation as in Step 4 of the proof of Proposition K.7, and noting that the leftover Gaussian $e^{a_H w^2}$ stays at the origin with the original a_H while the Mehler–Hermite factor is centered at τ, gives the eigenfunctions (K.10).

Step 4: a trace check on ν_0. The operator $\tilde{T}$ with kernel $\tilde{\mathfrak{k}}$ has

$$\operatorname{tr}\tilde{T} = \int v(w)\, \mathfrak{k}(w,w)\, d\mathcal{N}(0, I_d)(w) = c_0\, E_W\big[e^{-a\|W-s\|^2}\big] = c_0\,(1+2a)^{-d/2}\, e^{-a\|s\|^2/(1+2a)}.$$

On the other side, sum the eigenvalues with their multiplicities, using $\sum_{m\geq 0} \mathfrak{g}_d(m) Q_{\mathrm{feat},a}^m = (1 - Q_{\mathrm{feat},a})^{-d}$ together with the identity

$$\Big(\tfrac{2a_H^{\mathrm{eff}}}{a_H^{\mathrm{eff}} + q_\star + c_\star^{\mathrm{eff}}}\Big)^{1/2} \;=\; 1 - Q_{\mathrm{feat},a},$$

immediate from $Q_{\text{feat},a} = q_\star/(\alpha^{\text{eff}} + c_\star^{\text{eff}})$ and $(\alpha^{\text{eff}})^2 - (c_\star^{\text{eff}})^2 = q_\star^2$. This gives

$$\nu_0(a,s)\sum_{m\geq 0}\mathfrak{g}_d(m)Q_{\text{feat},a}^m = c_0\Big(\frac{a_H}{a_H^{\text{eff}}}\Big)^{d/2} e^{-a\|s\|^2/(1+2a)} = c_0\,(1+2a)^{-d/2}\,e^{-a\|s\|^2/(1+2a)},$$

and the two sides agree.

Step 5: monotonicity and the shrunken state. Monotonicity is immediate: $Q_{\text{feat},a} = q_\star/(A + q_\star + \sqrt{A^2 + 2Aq_\star})$ with $A = a_H^{\text{eff}}$ is strictly decreasing in A, hence in a, with $Q_{\text{feat},0} = Q$. As $a \to \infty$, $A \sim a/2$ and the denominator is $A + q_\star + \sqrt{A^2 + 2Aq_\star} \sim 2A \sim a$, so

$$Q_{\text{feat},a} \;\sim\; \frac{q_\star}{a} \;\longrightarrow\; 0.$$

Since the standard normal density has $a_H = 1/4$, replacing a_H by $a_H^{\text{eff}} = a_H + a/2$ inside $\alpha, c_\star, Q$ corresponds to state variance $\sigma_{\text{eff}}^2 = 1/(4a_H^{\text{eff}}) = 1/(1+2a)$: a factor whose risk concentrates is a factor facing a less variable state, which is what makes the ratio $Q_{\text{feat},a}$ fall in a.

Step 6: premia. Write $\mu_G(W) = E[G \mid W]$. Conditional Cauchy–Schwarz gives $\mu_G(W)^2 \leq v(W)$, so $\mu_G = h\sqrt{v}$ with $|h| \leq 1$ as in (K.1). The constraint bites when $a > 0$: since $v(W) = e^{-a\|W-s\|^2} \to 0$ in distant states, a premium that stays at a nonzero constant there would violate $\mu_G^2 \leq v$, so the only admissible constant premium is $\mu_G \equiv 0$. It is h rather than μ_G that may be held constant. A portfolio's premium is $E[R_a] = E_W[\bar{f}(\theta_a; W)\,\mu_G(W)]$.

The next calculation is written for the limiting systematic operator. The menu mean vector derived below does not depend on N, so by Remark K.4 the calculation has the required P-first, N-second meaning.

Write $L^2(W)$ for $L^2(\mathcal{N}(0, I_d))$ and $L^2(\Theta)$ for $L^2(\mu_\theta)$, and define the exposure map

$$\mathsf{e}(\theta)(w) \;:=\; \sqrt{v(w)}\,\bar{f}(\theta; w)\,. \tag{K.14}$$

It turns the limiting systematic second moment into a Gram matrix:

$$\Psi_{ab} \;=\; E\big[\bar f(\theta_a;W)\,\bar f(\theta_b;W)\,G^2\big] \;=\; E_W\big[v\,\bar f(\theta_a)\,\bar f(\theta_b)\big] \;=\; \big\langle \mathsf{e}(\theta_a),\,\mathsf{e}(\theta_b)\big\rangle_{L^2(W)}\,. \tag{K.15}$$

Read e as the integral operator $\mathcal{E}: L^2(W)\to L^2(\Theta)$, $(\mathcal{E}\psi)(\theta)=\int \mathsf{e}(\theta)(w)\,\psi(w)\,\mathcal{N}(dw)$, with adjoint $(\mathcal{E}^*g)(w)=\int_\Theta \mathsf{e}(\theta)(w)\,g(\theta)\,\mu_\theta(d\theta)$. Composing the two,

$$\big(\mathcal{E}^*\mathcal{E}\big)(w,w') \;=\; \int_\Theta \mathsf{e}(\theta)(w)\,\mathsf{e}(\theta)(w')\,\mu_\theta(d\theta) \;=\; \sqrt{v(w)}\;\mathfrak{k}(w,w')\;\sqrt{v(w')} \;=\; \tilde{\mathfrak{k}}(w,w')\,, \tag{K.16}$$

so $\mathcal{E}^*\mathcal{E}=\tilde T$ is exactly the operator diagonalized in Steps 1–3 above, with $\tilde T e_k^W=\nu_{(k)}e_k^W$ and the e_k^W of (K.10) orthonormal in $L^2(W)$. Set

$$e_k^\Theta \;:=\; \nu_{(k)}^{-1/2}\,\mathcal{E}e_k^W \;\in\; L^2(\Theta) \qquad (\nu_{(k)}>0)\,. \tag{K.17}$$

These are orthonormal, because $\langle e_k^\Theta,e_l^\Theta\rangle=(\nu_{(k)}\nu_{(l)})^{-1/2}\langle e_k^W,\mathcal{E}^*\mathcal{E}e_l^W\rangle=(\nu_{(k)}\nu_{(l)})^{-1/2}\nu_{(l)}\,\delta_{kl}=\delta_{kl}$. Now expand $\mathsf{e}(\theta)$, an element of $L^2(W)$ for μ_θ-almost every θ, in the basis $\{e_k^W\}$: its coefficients are $\langle \mathsf{e}(\theta),e_k^W\rangle_{L^2(W)}=(\mathcal{E}e_k^W)(\theta)=\sqrt{\nu_{(k)}}\,e_k^\Theta(\theta)$ by (K.17), whence

$$\mathsf{e}(\theta)(w) \;=\; \sqrt{v(w)}\,\bar f(\theta;w) \;=\; \sum_k \sqrt{\nu_{(k)}}\;e_k^\Theta(\theta)\,e_k^W(w)\,, \tag{K.18}$$

the singular-value decomposition of the exposure map. The expansion loses nothing: if $\psi\in\ker\tilde T$ then $\|\mathcal{E}\psi\|^2=\langle\psi,\mathcal{E}^*\mathcal{E}\psi\rangle=0$, so $\langle\mathsf{e}(\theta),\psi\rangle=0$ for almost every θ, and $\mathsf{e}(\theta)$ has no component outside $\overline{\mathrm{ran}}(\tilde T)=\overline{\mathrm{span}}\{e_k^W\}$.

The premia now read off (K.18). Since $\bar f\,\mu_G=\mathsf{e}\cdot h$ pointwise,

$$E[R_a] \;=\; E_W\big[\bar f(\theta_a;W)\,\mu_G(W)\big] \;=\; \big\langle \mathsf{e}(\theta_a),\,h\big\rangle \;=\; \sum_k \sqrt{\nu_{(k)}}\,\big\langle e_k^W,h\big\rangle\;e_k^\Theta(\theta_a)\;: \tag{K.19}$$

the constant function of Proposition K.7 is replaced by the factor's conditional Sharpe ratio

h. No infinite series needs to be interchanged in the empirical argument. Put

$$g_\mu(\theta) \;:=\; \big\langle \mathsf{e}(\theta),\, h\big\rangle_{L^2(W)}\,, \qquad \text{so that} \qquad E[R_a] \;=\; g_\mu(\theta_a)$$

directly by (K.19). Then $g_\mu \in L^2(\Theta)$, since $\|g_\mu\|^2 \le \|h\|^2\, E_\theta\|\mathsf{e}(\theta)\|^2_{L^2(W)} = \|h\|^2\, \operatorname{tr}\tilde{T} < \infty$. Using the identity $\mathcal{E}^* e_k^\Theta = \nu_{(k)}^{-1/2}\mathcal{E}^*\mathcal{E} e_k^W = \sqrt{\nu_{(k)}}\, e_k^W$ and $g_\mu = \mathcal{E}h$,

$$P_{E_m^\Theta} g_\mu = \sqrt{\nu_m} \sum_{|k|=m} \langle e_k^W, h\rangle\, e_k^\Theta, \qquad E_m^\Theta := \operatorname{span}\{e_k^\Theta : |k| = m\}.$$

By (K.15), Ψ_P is the Gram matrix of the $\mathsf{e}(\theta_a)$. Apply the isolated-group empirical projection argument of Lemma J.6 to this sampling map. It gives, directly and without a coordinatewise eigenvector approximation,

$$\sum_{j\in\text{group}(m)} \frac{(u_j'\mu_P)^2}{\lambda_j(\Psi_P)} \;\longrightarrow\; \frac{\|P_{E_m^\Theta} g_\mu\|^2}{\nu_m} \;=\; \sum_{|k|=m} |\langle e_k^W, h\rangle|^2 \;=\; \|P_{E_m} h\|^2. \tag{K.20}$$

As after Proposition K.7, only the norm in (K.20) is invariant under rotations of the eigenspace; the coordinates $\langle e_k^W, h\rangle$ require a choice of basis. Summing over rungs and using Parseval over the complete basis $\{e_k^W\}$ gives

$$\sum_j \mathrm{SR}_j^2 \;=\; \big\|h\big\|^2_{L^2(\mathcal{N}(0,I_d))} \;=\; E_W\big[h(W)^2\big] \;\le\; 1,$$

where the bound follows from $|h| \le 1$. At $a = 0$ one has $E[G^2] = 1$ by (K.8), so a constant $h \equiv \mu_G$ is admissible exactly when $|\mu_G| \le 1$, and it returns $\sum_j \mathrm{SR}_j^2 = \mu_G^2 = \mu_G^2/E[G^2]$, as in Proposition K.7.

Step 7: the Gaussian premium, and the loss of the parity dichotomy. Let $h(W) = h_0 e^{-\frac{\varkappa}{2}\|W-\varsigma\|^2}$, with $|h_0| \le 1$. Then $\langle e_k^W, h\rangle$ is, coordinate by coordinate, an integral of the form $\int H_k(\ell_H u)\, e^{-b_I u^2 + \eta u}\, du$ after the shift $u = w - \tau$, with Hermite scale $\ell_H = \sqrt{2c_\star^{\text{eff}}}$

and with η collecting the linear terms produced by the two offsets τ (from the volatility bump, $\tau \propto s$) and ς (from the premium). The generating function $\sum_k H_k(x)t^k/k! = e^{2xt-t^2}$ evaluates it: multiplying by $t^k/k!$ and summing,

$$\sum_{k\geq 0} \frac{t^k}{k!} \int H_k(\ell_H u) e^{-b_I u^2+\eta u} du = \int e^{2\ell_H ut - t^2 - b_I u^2 + \eta u} du = \sqrt{\tfrac{\pi}{b_I}}\, e^{\eta^2/(4b_I)} \exp\Big(\tfrac{\ell_H^2 - b_I}{b_I} t^2 + \tfrac{\ell_H \eta}{b_I} t\Big),$$

by completing the square in u with $\int e^{-b_I u^2 + Bu} du = \sqrt{\pi/b_I}\, e^{B^2/(4b_I)}$ at $B = 2\ell_H t + \eta$. Reading off the coefficient of $t^k/k!$ gives (K.11). Each SR_k is therefore explicit.

The parity structure is where the general case parts company with Proposition K.7. When $\eta = 0$, the generating function is even in t and its odd coefficients vanish, so the odd-degree components are unpriced. By (K.12), $\eta = 0$ exactly when the two offsets satisfy $\varkappa\varsigma = (2a_H + \varkappa)\tau$, in particular when both vanish, $s = 0$ and $\varsigma = 0$. For $\varkappa = 0$, the condition is $as = 0$, and ς is immaterial. With $s = 0$ and constant h, (K.11) reduces to (K.13), which has the shape of (K.6) with Q replaced by the premium rate $\tilde{Q}_{\mathrm{feat},a}$. The premium rate exceeds $Q_{\mathrm{feat},a}$ when $a > 0$. Away from the offset-tuning condition, the eigenfunctions (K.10) need no longer be even or odd about the origin and the linear term $\ell_H \eta t / b_I$ survives. For $h_0 \neq 0$, every degree then earns a premium except on the additional Hermite-zero parameter sets on which $c < 0$ places $\ell_H \eta/(2b_I\sqrt{-c})$ at a zero of H_k. Thus $s \neq 0$ is sufficient only away from those offset-tuning and Hermite-zero sets. The eigenvalues, by Step 3, are indifferent to all of this. □

Proposition K.9 (The same model, sorted) *Take the model (19) with $\rho \in (0,1)$ and $r_i = G + \varepsilon_i$, the factor's conditional second moment (23) with $a \geq 0$ and $s \in \mathbb{R}^d$, let the idiosyncratic variances obey the paper's standing bounds $\bar{\sigma}^2 \geq \sigma_\varepsilon^2(x) \geq \underline{\sigma}^2 > 0$, and take the partition menu (C.7) formed by bisecting every cell at its conditional median, cycling through the coordinates. Along $P = 2^n$, every cell has mass $1/P$. Then:*

(i) *the partition conditions of Appendix C.2.3 and the regularity (H.29) hold, and the*

constants of Proposition H.5 are

$$\kappa_j \;\propto\; \big(Q_{\mathrm{sort},a}\big)^m \qquad \textit{for} \quad J_{m-1} < j \le J_m, \qquad J_m = \binom{m+d}{d}, \tag{K.21}$$

up to a common scale, the level carrying multiplicity $\mathfrak{g}_d(m) = \binom{m+d-1}{d-1}$*, with ratio*

$$\begin{aligned} Q_{\mathrm{sort},a} \;&=\; \frac{q_\bullet}{\alpha_\bullet + a_H + \sqrt{(\alpha_\bullet + a_H)^2 - q_\bullet^2}}\,, \\ A_\bullet = a + \frac{\rho}{1-\rho} + \frac{1}{2}, \qquad q_\bullet &= \frac{\rho}{4A_\bullet(1-\rho)^2}, \qquad \alpha_\bullet = \frac{\rho}{2(1-\rho)} - q_\bullet, \end{aligned} \tag{K.22}$$

and $a_H = \frac{1}{4}$*. At* $a = 0$ *this is exactly* $Q_{\mathrm{sort},0} = \rho$ *and the scale is* $E[G^2]$*, so* $\kappa_j = E[G^2]\rho^m$*. The ratio falls strictly as the concentration* a *rises, and does not depend on* s*. The eigenfunctions are shifted and rescaled Gaussian–Hermite functions when* $a > 0$*. At* $a = 0$ *specifically, they are the products* $\chi_n(x) = \prod_{i\le d} \mathrm{he}_{n_i}(x_i)$ *of the orthonormal Hermite polynomials* $\mathrm{he}_k = \mathrm{He}_k/\sqrt{k!}$*, and the eigenvalue at the multi-index* n *is* $E[G^2]\rho^{|n|}$*.*

(i′) *For every* $\rho \in (0,1)$*, every* $a \ge 0$ *and every bandwidth* $\gamma > 0$*,*

$$Q_{\mathrm{feat},a}(\gamma,\rho) \;<\; Q_{\mathrm{sort},a}\,,$$

with $Q_{\mathrm{feat},a}$ *defined in* (K.9)*.*

Proof of Proposition K.9. Step 1: the model satisfies the hypotheses. Here $K = 1$ and $B \equiv 1$. Lemma K.1 gives the finite-panel moment $\Omega_N(X) = E[v(W) \mid X] \in (0,1]$ and its limit $v(W)$. Hence the conditional second moment of the returns is $\Sigma_N(X) = \Omega_N(X)\,\mathbf{1}\mathbf{1}' + \mathrm{diag}(\sigma_\varepsilon^2(X_i)) \succeq \underline{\sigma}^2 I_N$, which is the ellipticity floor (C.10), and a stock's own conditional second moment is $\varsigma^2_{(N)}(z) = E[v(W) \mid X_i = z] + \sigma_\varepsilon^2(z) \le 1 + \bar{\sigma}^2$, hence is bounded uniformly in N. The characteristics are continuously distributed, so nested cells of equal

mass $1/P$ exist along the dyadic subsequence $P = 2^n$. In particular, the coordinate-cycling construction in Section 5.1 becomes the usual dyadic rectangular partition of $(0,1)^d$ after applying the standard normal distribution function to each coordinate. Every coordinate is split infinitely often, so the rectangle diameters tend to zero; their generated sigma-field is the Borel sigma-field, verifying (C.6). Equal cell masses give (C.8) and (C.9) with $g_P = g = 1$. The comovement between two distinct stocks is $c_\infty(x, \tilde{x}) = E[v(W) \mid X_1 = x, X_2 = \tilde{x}]$, bounded by one, and the average marginal $p = \mathcal{N}(0, I_d)$ and the pair law P_2 are free of N too, so the second half of (H.29) holds trivially and the first half is the uniform bound just recorded. Finiteness of $\iint (c\varrho)^2 dp\, dp$ is verified in Step 2.

Step 2: the dependence between two stocks. By (19), X_1 and X_2 are jointly Gaussian, each marginally $\mathcal{N}(0, I_d)$, with $E[X_{1,i} X_{2,l}] = \rho\, \delta_{il}$: the coordinates are independent across i, and within a coordinate the pair $(X_{1,i}, X_{2,i})$ is bivariate standard normal with correlation ρ. Mehler's formula (Szegő, 1975, Problems and Exercises, Problem 23) expands the density of a bivariate standard normal against the product of its marginals as

$$\frac{\varphi_\rho(x, \tilde{x})}{\varphi(x)\varphi(\tilde{x})} \;=\; \sum_{k \geq 0} \rho^{\,k}\, \mathrm{he}_k(x)\, \mathrm{he}_k(\tilde{x}), \qquad \mathrm{he}_k = \mathrm{He}_k / \sqrt{k!},$$

the series converging in $L^2(\varphi \otimes \varphi)$ because $\sum_k \rho^{2k} = (1 - \rho^2)^{-1} < \infty$ and the he_k are orthonormal in $L^2(\varphi)$. Since both the joint density and the product of the marginals factor across coordinates, so does the ratio, and multiplying out the d copies gives

$$\varrho(x, \tilde{x}) \;=\; \frac{dP_2}{d(p \otimes p)}(x, \tilde{x}) \;=\; \sum_{n \in \mathbb{N}_0^d} \rho^{\,|n|}\, \chi_n(x)\, \chi_n(\tilde{x}), \qquad \chi_n(x) = \prod_{i=1}^{d} \mathrm{he}_{n_i}(x_i), \tag{K.23}$$

with $\{\chi_n\}$ an orthonormal basis of $L^2(p)$. In particular $\iint \varrho^2\, dp\, dp = \sum_n \rho^{2|n|} = (1 - \rho^2)^{-d} < \infty$. The comovement $c_\infty(x, \tilde{x}) = E[v(W) \mid X_1 = x, X_2 = \tilde{x}]$ is constant only at $a = 0$; what

the bound needs is merely $0 < c_\infty \le 1$, which holds for every $a \ge 0$ because $v \le 1$. Hence $\iint (c_\infty \varrho)^2 dp\, dp \le (1-\rho^2)^{-d} < \infty$, completing Step 1.

Step 3: the comovement kernel is Gaussian. By Proposition H.5 the constants are the eigenvalues of $\mathcal{H}$, whose kernel against $p \otimes p$ is $c_\infty \varrho_\infty$. Disintegration of the pair law P_2 gives

$$\big(c_\infty \varrho_\infty\big)(x,\tilde{x}) \;=\; \frac{1}{p(x)p(\tilde{x})} \int v(w)\, p(x \mid w)\, p(\tilde{x} \mid w)\, P_W(dw)\,,$$

which is a Gaussian integral: $p(\cdot \mid w) = \mathcal{N}(\sqrt{\rho}\, w, (1-\rho) I_d)$, $P_W = \mathcal{N}(0, I_d)$ and $v(w) = e^{-a\|w-s\|^2}$. It factors across coordinates, so take $d = 1$. Collecting the exponent in w gives the quadratic coefficient $A_\bullet = a + \rho/(1-\rho) + \frac{1}{2}$, and completing the square,

$$\big(c_\infty \varrho_\infty\big)(x,\tilde{x}) \;=\; C\, \exp\!\big(-\alpha_\bullet (x^2+\tilde{x}^2) + 2q_\bullet\, x\tilde{x} + \eta_\bullet (x+\tilde{x})\big), \qquad \eta_\bullet = \frac{as\sqrt{\rho}}{A_\bullet (1-\rho)}, \tag{K.24}$$

with $q_\bullet$ and $\alpha_\bullet$ as in (K.22) and C a constant.

Step 4: diagonalizing it. Equation (K.24) is exactly the form treated in Proposition K.8: a Gaussian kernel on $L^2(\mathcal{N}(0,1))$ with a quadratic and a linear term. Conjugating by $\sqrt{p}$ adds $a_H = \frac{1}{4}$ to the quadratic coefficient. Translating the Hermite system removes the linear term $\eta_\bullet$; the translation leaves the eigenvalue ratios unchanged, and the constant it leaves in front scales all eigenvalues by one common factor. Mehler's formula then gives a geometric spectrum with ratio (K.22). This proves (K.21), with multiplicities $\mathfrak{g}_d(m)$ from tensorizing across the d coordinates.

At $a = 0$ we have $v \equiv 1$, so $c_\infty \equiv E[G^2]$ and $c_\infty \varrho_\infty = E[G^2]\varrho_\infty$ is $E[G^2]$ times the Gaussian copula density, whose Mehler expansion $\varrho_\infty = \sum_n \rho^{|n|} \chi_n \otimes \chi_n$ exhibits the eigenvalues directly: $Q_{\mathrm{sort},0} = \rho$ and the scale is $E[G^2]$. Monotonicity in a and independence of s follow from (K.22), in which s appears only through $\eta_\bullet$.

Step 5: part (i′), the sort's staircase is the shallower one. Both ratios have the form $q/(\Lambda + \sqrt{\Lambda^2 - q^2}) = F_{\mathrm{rat}}(q/\Lambda)$, where $F_{\mathrm{rat}}(u) := \Upsilon^{-1}(u) = u/(1+\sqrt{1-u^2})$ is strictly

increasing on $[0,1)$ with inverse $F_{\mathrm{rat}}^{-1}(Q) = 2Q/(1+Q^2)$; thus $F_{\mathrm{rat}}^{-1}(Q_{\mathrm{sort},a}) = q_\bullet/(\alpha_\bullet + a_H)$ for the sort, and $F_{\mathrm{rat}}^{-1}(Q_{\mathrm{feat},a}) = q_\star/(a_H^{\mathrm{eff}} + q_\star)$ for the random-feature menu of Proposition K.8, with $a_H^{\mathrm{eff}} = \frac{1}{4} + \frac{a}{2}$. Since $Q_{\mathrm{feat},a}$ increases in γ through $q_\star$, which rises to its ceiling $q_{\star,\max} := \rho/[4(1-\rho)]$, it suffices to compare at $\gamma = \infty$. Substituting $q_\star = q_{\star,\max}$ and the constants (K.22) and simplifying,

$$F_{\mathrm{rat}}^{-1}\big(\sup_\gamma Q_{\mathrm{feat},a}\big) = \frac{\rho}{1+2a(1-\rho)}\,, \qquad F_{\mathrm{rat}}^{-1}(Q_{\mathrm{sort},a}) = \frac{2\rho}{1+\rho^2+2a(1-\rho^2)}\,, \tag{K.25}$$

so that

$$F_{\mathrm{rat}}^{-1}(Q_{\mathrm{sort},a}) - F_{\mathrm{rat}}^{-1}\big(\sup_\gamma Q_{\mathrm{feat},a}\big) = \frac{\rho\big[(1-\rho^2)+2a(1-\rho)^2\big]}{\big(1+2a(1-\rho)\big)\big(1+\rho^2+2a(1-\rho^2)\big)} \;>\; 0$$

for every $\rho \in (0,1)$ and every $a \geq 0$. Since F_{rat}^{-1} is strictly increasing, $\sup_\gamma Q_{\mathrm{feat},a} < Q_{\mathrm{sort},a}$, hence $Q_{\mathrm{feat},a}(\gamma,\rho) < Q_{\mathrm{sort},a}$ at every γ, which is part (i′). At $a = 0$, Proposition K.6 and (K.4) give the sharper bound $Q_{\mathrm{feat},0}(\gamma,\rho) < Q_{\mathrm{feat},*}(\rho) < \rho$. □

Decay ratios collected. For reference, we restate side by side the two decay ratios defined in (K.9) and (K.22):

$$Q_{\mathrm{feat},a} = \frac{q_\star}{a_H^{\mathrm{eff}} + q_\star + \sqrt{(a_H^{\mathrm{eff}})^2 + 2a_H^{\mathrm{eff}} q_\star}}, \qquad Q_{\mathrm{sort},a} = \frac{q_\bullet}{\alpha_\bullet + a_H + \sqrt{(\alpha_\bullet + a_H)^2 - q_\bullet^2}}. \tag{K.26}$$

Proof of Theorem 4. Step 1: the exact decomposition at finite N. Lemma K.3 gives the exact identity (K.3), using the $1/N$ convention of (21). Its systematic part has label-side kernel $\tilde{\mathfrak{k}}_W$, and the within-name remainder has kernel $N^{-1}\mathfrak{k}_X^v$.

Step 2: Gram and integral operators share the nonzero spectrum. Regard the weighted exposure $\sqrt{v(W)}\,\bar{f}(\theta;W)$ as the kernel of a Hilbert–Schmidt operator $\mathcal{E}$ from

$L^2(\mathcal{N}(0, I_d))$ to $L^2(\mu_\theta)$,

$$(\mathcal{E}g)(\theta) = E_W\big[\sqrt{v(W)}\,\bar{f}(\theta; W)\,g(W)\big], \qquad (\mathcal{E}^*u)(W) = E_\theta\big[\sqrt{v(W)}\,\bar{f}(\theta; W)\,u(\theta)\big].$$

Composing the two ways reproduces the two kernels:

$$\mathcal{E}\mathcal{E}^* \text{ has kernel } E_W[\bar{f}(\theta; W)\bar{f}(\theta'; W)v(W)] = \tilde{\mathfrak{k}}_W(\theta, \theta'),$$
$$\mathcal{E}^*\mathcal{E} \text{ has kernel } \sqrt{v(W)v(W')}\;E_\theta[\bar{f}(\theta; W)\bar{f}(\theta; W')] = \tilde{\mathfrak{k}}(W, W').$$

So $T_{\tilde{\mathfrak{k}}_W} = \mathcal{E}\mathcal{E}^*$ and $T_{\tilde{\mathfrak{k}}} = \mathcal{E}^*\mathcal{E}$. As in the finite-dimensional SVD, $\mathcal{E}\mathcal{E}^*$ and $\mathcal{E}^*\mathcal{E}$ have identical nonzero eigenvalues with equal multiplicities (see Lemma A.8). Hence the nonzero spectrum of $T_{\tilde{\mathfrak{k}}_W}$ equals that of $T_{\tilde{\mathfrak{k}}}$, which by Proposition K.8 is $\{\nu_m = \nu_0(a, s)\,Q^m_{\text{feat},a}\}$ with multiplicities $\mathfrak{g}_d(m)$; at $a = 0$ this is $\{\nu_0 Q^m_{\text{feat},0}\}$ of Proposition K.6. By (K.9) and (K.8), $\nu_0(a, s) = E[G^2] \cdot C(\gamma, \rho, a, d)$ with C free of s: the center of the volatility bump enters the spectrum only through the common scale $E[G^2]$, as described in Section 5.1. Moreover,

$$\Upsilon(Q_{\text{feat},a}) = \frac{q_\star}{a_H^{\text{eff}} + q_\star}, \qquad \frac{d}{dq_\star}\frac{q_\star}{a_H^{\text{eff}} + q_\star} = \frac{a_H^{\text{eff}}}{(a_H^{\text{eff}} + q_\star)^2} > 0. \tag{K.27}$$

Since Υ is increasing and Lemma K.5 makes $q_\star$ strictly increasing in γ, the ratio increases strictly in γ and tends to zero as $\gamma \to 0$.

Step 3: the menu first, then the cross-section. We take the limits in the order of Remark K.4. By (K.3), Ψ_P is at every N the Gram matrix of the kernel

$$M^{(N)}(\theta, \theta') \;=\; \tfrac{N-1}{N}\,\tilde{\mathfrak{k}}_W(\theta, \theta') \;+\; \tfrac{1}{N}\,\mathfrak{k}^v_X(\theta, \theta')$$

on $L^2(\mu_\theta)$. Since $|f| \le \sqrt{2}$ and $v \le 1$, the trace condition (C.5) holds at every fixed N. The

exact-limit argument in Appendix F.1 therefore gives (F.1): letting $P \to \infty$ at fixed N,

$$\kappa_{N,j} \;=\; \lambda_j\Big(\tfrac{N-1}{N}\,T_{\tilde{\mathfrak{k}}_W} \;+\; \tfrac{1}{N}\,T_{\mathfrak{k}^v_X}\Big).$$

Now let $N \to \infty$. The perturbation is trace-class, with

$$\operatorname{tr} T_{\mathfrak{k}^v_X} = \int E[f(X;\theta)^2(v(W)+\sigma_\varepsilon^2(X))]\,d\mu_\theta \le 2(1+\bar{\sigma}^2) < \infty.$$

Thus $\|\frac{1}{N}T_{\mathfrak{k}^v_X}\|_{\mathrm{op}} \le \frac{1}{N}\operatorname{tr} T_{\mathfrak{k}^v_X} \to 0$, and Weyl's inequality (Lemma A.7) gives $\kappa_{N,j} \to \lambda_j(T_{\tilde{\mathfrak{k}}_W}) = \kappa_j$. Therefore

$$\lim_{N\to\infty}\lim_{P\to\infty}\frac{\lambda_j(\Psi_P)}{P} \;=\; \kappa_j \;=\; E[G^2]\,C(\gamma,\rho,a,d)\,Q_{\mathrm{feat},a}^m, \qquad J_{m-1} < j \le J_m,$$

which is the claim, in the iterated order of the theorem. This direct kernel argument identifies $\kappa_j = \nu_m$ for $J_{m-1} < j \le J_m$ without invoking Proposition H.4, whose uniform density and ellipticity assumptions the Gaussian model need not satisfy. The remark after this proof shows that, for random features, the fixed-rank limits also hold jointly in P and N.

Step 4: cumulative counts via the hockey-stick identity. The distinct level $\nu_0 Q^m$ has multiplicity $\mathfrak{g}_d(m) = \binom{m+d-1}{d-1}$, so the number of eigenvalues at or above that level is $J_m = \sum_{l=0}^m \binom{l+d-1}{d-1}$. By Pascal's rule $\binom{l+d-1}{d-1} = \binom{l+d}{d} - \binom{l+d-1}{d}$, the sum telescopes:

$$J_m = \sum_{l=0}^{m}\left[\binom{l+d}{d} - \binom{l+d-1}{d}\right] = \binom{m+d}{d} - \binom{d-1}{d} = \binom{m+d}{d},$$

since $\binom{d-1}{d} = 0$. This is the hockey-stick identity. Consequently the iterated-limit constant $\kappa_j \propto Q_{\mathrm{feat},a}^m$ precisely for $J_{m-1} < j \le J_m$, and because consecutive levels differ by the constant factor $Q_{\mathrm{feat},a}$, every rung-to-rung ratio equals $Q_{\mathrm{feat},a}$.

Step 5: the leading rungs and the cliff. At the top, $J_0 = \binom{d}{d} = 1$ (the ground mode,

$\mathfrak{g}_d(0) = 1$) and $J_1 = \binom{d+1}{d} = d+1$, including the d first-degree modes. At $a = 0$ and as $\gamma \to 0$, these modes approach the constant and linear characteristic exposures; at positive bandwidth they include Gaussian factors. The ratio across the boundary at rank $d+1$ is

$$\frac{\kappa_{d+2}}{\kappa_{d+1}} = \frac{\nu_0 Q_{\text{feat},a}^2}{\nu_0 Q_{\text{feat},a}} = Q_{\text{feat},a},$$

and likewise every adjacent-rung ratio is $Q_{\text{feat},a}$. The drop after rank one has the same ratio; the boundary at rank $d+1$ stands out because it separates the constant and linear exposures, which a menu linear in characteristics would span, from the higher-degree modes. As $\gamma \to 0$,

$$Q_{\text{feat},a} \sim \frac{\rho\gamma^2}{1+2a} \longrightarrow 0;$$

the formula reduces to $Q \sim \rho\gamma^2$ at $a = 0$ (Proposition K.6). Thus each such drop sharpens without bound: the leading modes dominate and the remainder collapses, i.e., an arbitrarily sharp cliff.

Step 6: the shallow staircase, in log–log chord slopes. Plot the spectrum as the points (j, κ_j); its corners sit at the right ends of the rungs, $(J_m, \nu_0 Q^m)$. In log–log coordinates a corner is $(\log J_m,\ \log \nu_0 + m \log Q)$, and the chord joining consecutive corners has slope

$$\frac{(\log \nu_0 + (m+1)\log Q) - (\log \nu_0 + m \log Q)}{\log J_{m+1} - \log J_m} = \frac{\log Q}{\log(J_{m+1}/J_m)},$$

whose magnitude, using $Q \in (0,1)$ so $\log(1/Q) > 0$, is $\zeta_m = \log(1/Q)/\log(J_{m+1}/J_m)$. The count ratio simplifies by cancellation,

$$\frac{J_{m+1}}{J_m} = \frac{\binom{m+1+d}{d}}{\binom{m+d}{d}} = \frac{(m+1+d)!/\big(d!\,(m+1)!\big)}{(m+d)!/\big(d!\,m!\big)} = \frac{m+1+d}{m+1} = \frac{d+m+1}{m+1},$$

so that

$$\zeta_m = \frac{\log(1/Q)}{\log\big((d+m+1)/(m+1)\big)} = \frac{\log(1/Q)}{\log\big(1+\frac{d}{m+1}\big)}.$$

The denominator $\log(1+\frac{d}{m+1})$ strictly decreases in m, hence ζ_m strictly increases in m: the signed slope $-\zeta_m$ decreases, so the log–log envelope is concave. For the bandwidth dependence, the numerator $\log(1/Q_{\mathrm{feat},a})$ strictly decreases as $Q_{\mathrm{feat},a}$ increases, and $Q_{\mathrm{feat},a}$ increases in γ by (K.27), so each ζ_m strictly decreases in γ. Thus a wide bandwidth ($Q_{\mathrm{feat},a}$ bounded away from 0, up to $Q_{\mathrm{feat},*}(\rho) < 1$) holds each ζ_m finite, a shallow staircase, while $\gamma \to 0$ sends $Q_{\mathrm{feat},a} \to 0$ and every $\zeta_m \to \infty$, recovering the cliff. (For instance $Q_{\mathrm{feat},*}(0.5) = 0.5/(1+\sqrt{0.75}) \approx 0.27$.)

Step 7: the sort, and the comparison. Part (b) is Proposition K.9(i): the constants have ratio $Q_{\mathrm{sort},a}$ on the degree-m rung, with the multiplicity $\mathfrak{g}_d(m)$ of Step 4 and the same cumulative count J_m; only at $a = 0$ are they exactly $E[G^2]\rho^m$. Thus the two menus produce the same staircase and differ only in its ratio. The comparison (24) is Proposition K.9(i′). Both menus use the same iterated order, first $P \to \infty$ at fixed N and then $N \to \infty$, as established above. The partition constants at finite N obey $\lambda_j^*(N) \geq \underline{\sigma}^2/N$ by (F.16) for these equal-mass cells, and (K.21) describes their limits as $N \to \infty$. □

Remark K.10 (Joint convergence in the Gaussian model) In this model, the kernels in (K.3) do not depend on N. Let $K_P = [\tilde{\mathfrak{k}}_W(\theta_k, \theta_l)]$ and $K_{X,P} = [\mathfrak{k}_X^v(\theta_k, \theta_l)]$. The diagonal bounds of Lemma K.3 imply, uniformly in P,

$$\left\| \frac{\Psi_{P,N}}{P} - \frac{K_P}{P} \right\|_{\mathrm{op}} \leq \frac{\|K_P\|_{\mathrm{op}} + \|K_{X,P}\|_{\mathrm{op}}}{NP} \leq \frac{2 + 2(1+\bar{\sigma}^2)}{N}.$$

For every fixed j, fixed-kernel Gram convergence gives $\lambda_j(K_P)/P \to \kappa_j$ almost surely. Weyl's

inequality therefore yields

$$\frac{\lambda_j(\Psi_{P,N})}{P} \longrightarrow \kappa_j \qquad \text{along every path } P, N \to \infty$$

on the same probability-one event. No additional uniform empirical-kernel assumption is needed for this special model's fixed-rank joint limit.

Growing ranks require more information. The displayed bound guarantees a negligible relative finite-stock perturbation at degree $m = m_N$ if $N\nu_{m_N} \to \infty$, equivalently $NQ^{m_N}_{\mathrm{feat},a} \to \infty$. This is a sufficient condition supplied by the norm bound, not a necessary resolution threshold. A growing-rank sample-kernel approximation must also be controlled relative to ν_{m_N}.

Centered features. In the scalar model $B \equiv 1$ with $a = 0$, centering the features replaces their state-side operator by $\Pi T_{\mathfrak{k}} \Pi$, where $\Pi = I - 1 \otimes 1$, because $E_W[\bar{f}(\theta; W)] = E_X[f(X; \theta)]$. This compression has the same nonzero spectrum as the rank-one downdate

$$T_{\mathfrak{k}}^{1/2} \Pi T_{\mathfrak{k}}^{1/2} = T_{\mathfrak{k}} - (T_{\mathfrak{k}}^{1/2} 1) \otimes (T_{\mathfrak{k}}^{1/2} 1).$$

The difference $\Pi T_{\mathfrak{k}} \Pi - T_{\mathfrak{k}}$ itself need not have rank one. Since 1 is not an eigenfunction of $T_{\mathfrak{k}}$, centering does not simply delete an eigenfunction. The next lemma gives the retained Hermite values and the interlacing secular roots; the ordered spectrum need not have adjacent ratio Q. For nonconstant loadings or $a > 0$, feature centering does not give this scalar compression.

Lemma K.11 (The centered spectrum) *Let $T_{\mathfrak{k}} = \sum_{m \geq 0} \nu_0 Q^m P_{E_m}$ with $Q \in (0, 1)$ and $\dim E_m = \mathfrak{g}_d(m)$, let $\mathbf{1}$ be a unit vector with $w_m := \|P_{E_m} \mathbf{1}\|^2$ positive for even m and zero for odd m, and let $\Pi = I - \mathbf{1} \otimes \mathbf{1}$. Then the spectrum of $\Pi T_{\mathfrak{k}} \Pi$ on $\mathbf{1}^\perp$ consists of the accumulation point 0 and the multiset union of the following nonzero values:*

(i) $\mathfrak{g}_d(m)$ *copies of* $\nu_0 Q^m$ *for odd* m*, and* $\mathfrak{g}_d(m) - 1$ *copies for even* m*; and*

(ii) *exactly one further eigenvalue in each interval* $(\nu_0 Q^{2j+2}, \nu_0 Q^{2j})$*,* $j \geq 0$*, the roots of the secular equation* $\sum_{i \geq 0} w_{2i}(\nu_0 Q^{2i} - \lambda)^{-1} = 0$*.*

Multiplicities add when a root in (ii) *equals an odd-degree value in* (i)*;* 0 *is not an eigenvalue on* $\mathbf{1}^\perp$*.*

Proof of Lemma K.11. On $\mathbf{1}^\perp$ the operator $\Pi T_{\mathfrak{k}} \Pi$ is the compression of $T_{\mathfrak{k}}$, so we describe that. For odd m, $P_{E_m}\mathbf{1} = 0$, hence $E_m \subseteq \mathbf{1}^\perp$, and $T_{\mathfrak{k}}$ acts there as $\nu_0 Q^m$, contributing $\mathfrak{g}_d(m)$ copies. For even m, put $u_m := P_{E_m}\mathbf{1} \neq 0$; then $E_m \ominus \operatorname{span}(u_m) \subseteq \mathbf{1}^\perp$ is $T_{\mathfrak{k}}$-invariant with eigenvalue $\nu_0 Q^m$, contributing $\mathfrak{g}_d(m) - 1$. This is (i), and it exhausts $\mathbf{1}^\perp$ except for the closed span $\mathcal{U}$ of $\{u_{2j}\}_{j \geq 0}$ intersected with $\mathbf{1}^\perp$, which has codimension one in $\mathcal{U}$ because $\mathbf{1} \in \mathcal{U}$.

On $\mathcal{U}$ the operator $T_{\mathfrak{k}}$ is diagonal in the orthonormal basis $\hat{u}_{2j} = u_{2j}/\sqrt{w_{2j}}$, with eigenvalues $\nu_0 Q^{2j}$, and $\mathbf{1}$ has coordinates $\sqrt{w_{2j}}$. If $x \in \mathcal{U} \cap \mathbf{1}^\perp$ satisfies $\Pi T_{\mathfrak{k}} x = \lambda x$, then $T_{\mathfrak{k}} x = \lambda x + \mu \mathbf{1}$ with $\mu = \langle \mathbf{1}, T_{\mathfrak{k}} x \rangle$, so coordinatewise $x_j = \mu \sqrt{w_{2j}} \big(\nu_0 Q^{2j} - \lambda\big)^{-1}$. Imposing $\langle \mathbf{1}, x \rangle = 0$ gives $\mu \sum_j w_{2j}(\nu_0 Q^{2j} - \lambda)^{-1} = 0$, and $\mu = 0$ would force $x = 0$; hence λ solves the secular equation. Its left side is strictly increasing in λ on each interval between consecutive $\nu_0 Q^{2j}$, running from $-\infty$ to $+\infty$ there, so it has exactly one root in each, which is (ii).

Two consequences follow, and one does not. The top level disappears: $\mathfrak{g}_d(0) = 1$, so (i) contributes no copy of ν_0. The retained Hermite values remain $\nu_0 Q^m$ at their original degree indices, and the secular roots lie between consecutive even levels. If a root equals an odd level, its copy is added to that level's multiplicity. Adjacent eigenvalues of the fully ordered spectrum generally do not have ratio Q. Positivity of $T_{\mathfrak{k}}$ gives $\langle x, \Pi T_{\mathfrak{k}} \Pi x \rangle = \langle x, T_{\mathfrak{k}} x \rangle > 0$ for every nonzero $x \in \mathbf{1}^\perp$, so zero is only a spectral accumulation point. Codimension-one interlacing, $\lambda_{j+1}(T_{\mathfrak{k}}) \leq \lambda_j(\Pi T_{\mathfrak{k}} \Pi) \leq \lambda_j(T_{\mathfrak{k}})$, is a coarser consequence of (i)–(ii); it also follows directly from the variational formula in Lemma A.5.

One thing that (i)–(ii) do not by themselves give is that the degree-one level tops the centered spectrum. The single root μ_1 of (ii) in $(\nu_0 Q^2, \nu_0)$ may sit above $\nu_0 Q$, and where it

sits depends on the weights: since the left side of the secular equation increases through the gap, $\mu_1 < \nu_0 Q$ holds exactly when

$$\sum_{i\geq 0} \frac{w_{2i}}{Q^{2i} - Q} \;>\; 0 \qquad \Longleftrightarrow \qquad w_0 \;>\; \frac{1-Q}{Q} \sum_{i\geq 1} \frac{w_{2i}}{1 - Q^{2i-1}}\,, \tag{K.28}$$

the two forms being related by $Q^{2i} - Q = -Q\big(1 - Q^{2i-1}\big)$ and division by $1 - Q > 0$. The criterion can fail if w_0 is small enough relative to w_2: at $Q = \frac{1}{2}$ with $(w_0, w_2) = (\frac{1}{2}, \frac{1}{2})$ and no higher weights, the sum on the left of (K.28) is $1 - 2 = -1$. Adding sufficiently small positive weights at every higher even level and renormalizing preserves this strict sign. For the Hermite weights (K.6), $w_{2n} = (1-Q^2)^{d/2} Q^{2n} (d/2)_n / n!$, where $(z)_n$ is the rising factorial. At small Q and fixed d, the first two contributions to the left side of (K.28) are approximately 1 and $-Qd/2$. At the rounded parameters of Figure 1, $d = 50$, the criterion is about 0.76 for $Q = 0.010$ and -3.08 for $Q = 0.186$. Thus the degree-one level tops the centered spectrum at the narrow bandwidth, whereas the largest secular root tops it at the wide bandwidth. □

K.1 Premia, Sharpe ratios, and spanning

Two pieces of the premium calculation appear earlier, because they use the eigenbases constructed there: Proposition K.7 computes the rung premia in the constant-second-moment case, and Step 6 of Proposition K.8 carries them to a state-dependent factor. What follows collects the consequences, which Section 5.2 uses.

Theorem 4 reports only eigenvalues. The same model also delivers the rewards attached to those eigenvalues in closed form, and with them the two statements the paper needs: that no finite set of principal components spans expected returns exactly, and that the limiting maximal Sharpe ratio remains finite under the strict moment condition. We collect them here, in the random-feature case (21) and in the order of limits of Theorem 4(a).

Because the degree-m level is shared by many components, the individual components within a rung are not separately identified, and the premia are stated for the rung as a

whole.[23] Write $\mathrm{SR}_{(m)}$ for the rung-level uncentered ratio of degree m, S_J for the maximal centered Sharpe ratio from the first J limiting principal components, restricted to rung boundaries $J = J_m$, and $\mathrm{S}_{\mathrm{sys}} := \sup_J \mathrm{S}_J$ for the supremum over their finite linear combinations.

Proposition K.12 (Rung premia and spanning) *Work in Theorem 4(a) with (23) and (K.1), and assume $E[h(W)^2] < 1$. All component ratios below refer to the limiting systematic spectrum.*

(i) *The limiting maximal centered Sharpe ratio satisfies*

$$\sum_{j\geq 1} \mathrm{SR}_j^2 = E[h(W)^2], \qquad \mathrm{S}_{\mathrm{sys}}^2 = \frac{E[h(W)^2]}{1 - E[h(W)^2]}. \tag{K.29}$$

If $s = 0$ and h is even in each coordinate, every odd-degree rung has zero premium.

(ii) *If $a = 0$ and $h \equiv \mu_G$, write $Q = Q_{\mathrm{feat},0}$. Then, for $m = 2n$,*

$$\mathrm{SR}_{(2n)}^2 = \frac{\mu_G^2}{E[G^2]}(1 - Q^2)^{d/2} Q^{2n} \frac{(d/2)_n}{n!}, \tag{K.30}$$

where $(b)_n = b(b+1)\cdots(b+n-1)$ and $(b)_0 = 1$; odd rungs have zero premium. Moreover, with $\sigma_G^2 = E[G^2] - \mu_G^2$,

$$\sum_{j\geq 1} \mathrm{SR}_j^2 = \frac{\mu_G^2}{E[G^2]}, \qquad \mathrm{S}_{\mathrm{sys}} = \frac{|\mu_G|}{\sigma_G}. \tag{K.31}$$

(iii) *If $P_{E_m} h \neq 0$ for infinitely many m, then $\mathrm{S}_J < \mathrm{S}_{\mathrm{sys}}$ for every finite J, and $\mathrm{S}_J \uparrow \mathrm{S}_{\mathrm{sys}} < \infty$. If h is an admissible finite Hermite combination, finitely many rungs span its mean–variance frontier.*

[23] Rotating the components inside a rung leaves the eigenvalues alone but reshuffles the individual premia, so only rung-level quantities are well defined. This is the same caveat as in Section 5.2.

The aggregate rung weight includes the multiplicity factor $(d/2)_n/n!$. Consequently, when $\mu_G \neq 0$, $\mathrm{SR}_{(2n)} \asymp Q^n n^{d/4-1/2}$ as $n \to \infty$; the exact proportionality $\mathrm{SR}_{(2n)} \propto Q^n$ holds only for $d = 2$. Every even rung is priced in the constant nonzero-premium case, although odd rungs remain unpriced. For constant h with $a > 0$ and $s = 0$, (K.13) instead supplies the rate $\tilde{Q}_{\mathrm{feat},a} > Q_{\mathrm{feat},a}$. A general h need not have a geometric coefficient rate. For Gaussian h, (K.11) gives explicit coefficients and identifies the offset-tuning and Hermite-zero exceptions to generic pricing of all rungs.

Proof of Proposition K.12. Step 1: the premia. Part (i) is Step 6 of Proposition K.8: rung by rung, $\mathrm{SR}^2_{(m)} = \|P_{E_m} h\|^2$, and, summing over rungs by Parseval, $\sum_{j\geq 1} \mathrm{SR}_j^2 = E_W[h(W)^2]$, which is (K.29). In the constant-second-moment case $a = 0$ with $h \equiv \mu_G$ this is Proposition K.7: the rung-level squared Sharpe ratio is (K.5), which gives (K.30) by the generating-function identity

$$\sum_{|\ell|=n} \prod_{i=1}^{d} \frac{(2\ell_i)!}{4^{\ell_i}(\ell_i!)^2} = [z^n](1-z)^{-d/2} = \frac{(d/2)_n}{n!}.$$

The total is $\sum_j \mathrm{SR}_j^2 = \mu_G^2/E[G^2]$ of (K.31), i.e. $\mathrm{S}_{\mathrm{sys}} = |\mu_G|/\sigma_G$. That is part (ii). Step 7 of Proposition K.8 gives the closed form for a Gaussian h and shows that the odd rungs are unpriced when $s = 0$ and h is even in each coordinate. For the Gaussian family they are generically priced off that symmetry locus, subject to the explicit Hermite-zero exceptions.

Step 2: spanning and the maximal Sharpe ratio. Write $x_J = \sum_{j\leq J} \mathrm{SR}_j^2$ and $x_\infty = E_W[h^2] < 1$. The conversion from uncentered to centered ratios in Lemma J.8 gives $\mathrm{S}_J^2 = x_J/(1-x_J)$ and $\mathrm{S}_{\mathrm{sys}}^2 = x_\infty/(1-x_\infty)$, proving the exact equality in part (i). If infinitely many rungs have $P_{E_m} h \neq 0$, then $x_J < x_\infty$ for every finite J, while Parseval gives $x_J \uparrow x_\infty$. The strict monotonicity and continuity of $x/(1-x)$ on $[0,1)$ prove $\mathrm{S}_J < \mathrm{S}_{\mathrm{sys}}$ and $\mathrm{S}_J \uparrow \mathrm{S}_{\mathrm{sys}}$. If h has only finitely many nonzero Hermite coefficients, a finite collection of rungs attains x_∞ and hence $\mathrm{S}_{\mathrm{sys}}$. This proves part (iii). □

For nonnegative-payoff arbitrage, see the discussion after (J.44).

L Finite Samples Spread the Spectrum

The spectrum of Section 5.1 is a population object; in practice one sees the sample second-moment matrix $\hat{\Psi}_P = \frac{1}{T}\sum_{t=1}^{T} R_t R_t'$ from a panel of length T, and a natural worry is that estimation noise might carve a cliff back into a shallow population staircase. Two rigorous but distinct benchmarks are useful. First, in expectation, convex spectral functionals of the sample matrix exceed their population counterparts, with the top and bottom eigenvalues moving outward in expectation. Second, in a centered independent-component model with $P/T \to y < 1$, the whitened sample covariance is bounded above and away from zero with high probability; Loewner order then prevents a bounded population eigenvalue ratio from diverging in sample. Neither statement rules out a striking gap in one particular realization, and the second does not cover the paper's nonzero-premium, dependent managed returns.

We adopt the standard model of the sampling noise: the returns are drawn as $R_t = \mu_P + V_P^{1/2} z_t$, where $\mu_P = E[R_t]$ and $V_P = \mathrm{Var}(R_t)$ are the managed-return mean and covariance. The vector z_t has mean zero and identity covariance, so $E[R_t R_t'] = \Psi_P$. Part (b) sets $\mu_P = 0$ and additionally requires i.i.d. entries from a fixed distribution with finite fourth moment. Managed portfolios do not literally satisfy this model. Their entries are smooth nonlinear functions of a common state and a common factor, hence strongly dependent. We treat the model as a stylized benchmark rather than a description.

Theorem L.1 (Spectral dispersion and a centered no-split benchmark) *Consider a panel of independent and identically distributed returns $R_1, \ldots, R_T$ with*

$$E[R_t R_t'] = \Psi_P, \qquad \hat{\Psi}_P = \frac{1}{T}\sum_{t=1}^{T} R_t R_t',$$

and write $\Psi_P = V_P + \mu_P \mu_P'$ with $\mu_P = E[R_t]$ and $V_P = \mathrm{Var}(R_t)$.

(a) ***Dispersion.*** *For every convex φ, $E\,\operatorname{tr}\varphi(\hat{\Psi}_P) \geq \operatorname{tr}\varphi(\Psi_P)$, with strict inequality for $\varphi(x) = x^2$ unless $\hat{\Psi}_P = \Psi_P$ almost surely. Also, $E\,\operatorname{tr}\hat{\Psi}_P = \operatorname{tr}\Psi_P$, $E\,\lambda_1(\hat{\Psi}_P) \geq \lambda_1(\Psi_P)$ and $E\,\lambda_P(\hat{\Psi}_P) \leq \lambda_P(\Psi_P)$. For Gaussian returns $R_t \sim \mathcal{N}(\mu_P, V_P)$ the second spectral moment is inflated by exactly*

$$E\,\operatorname{tr}\hat{\Psi}_P^2 - \operatorname{tr}\Psi_P^2 = \frac{1}{T}\Big[(\operatorname{tr}\Psi_P)^2 - \operatorname{tr}\Psi_P^2 + 2\operatorname{tr}V_P^2 + 4\,\mu_P' V_P\,\mu_P\Big]. \tag{L.1}$$

(b) ***No manufactured cliff in the centered independent-component benchmark.*** *Suppose $\mu_P = 0$ and $R_t = V_P^{1/2} z_t$, where the entries of the $P \times T$ matrix $Z = [z_1, \ldots, z_T]$ are i.i.d. from a fixed distribution with mean zero, variance one, and finite fourth moment. Let $P, T \to \infty$ with $P/T \to y \in (0,1)$. If $j = j_P < P$ satisfies $\lambda_{j+1}(V_P) > 0$ and $\lambda_j(V_P)/\lambda_{j+1}(V_P) \leq C_0$ for all sufficiently large P, then $\lambda_j(\hat{\Psi}_P)/\lambda_{j+1}(\hat{\Psi}_P) = O_{\mathbb{P}}(1)$. More precisely, for every $0 < c < (1-\sqrt{y})^2$ and $C > (1+\sqrt{y})^2$, with probability tending to one, simultaneously at every such rank,*

$$\frac{\lambda_j(\hat{\Psi}_P)}{\lambda_{j+1}(\hat{\Psi}_P)} \leq \frac{C}{c}\,\frac{\lambda_j(V_P)}{\lambda_{j+1}(V_P)} \leq \frac{C}{c} C_0. \tag{L.2}$$

Part (b) is deliberately narrower than part (a). Its point is conditional: when the whitened sample covariance is uniformly bounded above and away from zero, congruence by an arbitrary population covariance cannot turn a bounded population eigenvalue ratio into a divergent sample ratio. The standard extreme-eigenvalue law for an i.i.d. matrix with finite fourth moment supplies exactly those two bounds when $y < 1$ (Yin, Bai and Krishnaiah, 1988; Bai and Yin, 1993; Bai and Silverstein, 2010). No spiked-population hypothesis, limiting bulk law, or outlier classification is needed.

Neither part proves that a particular realized sample cannot display a conspicuous local gap. Part (a) is an expectation/convex-order statement and part (b) applies only to the centered independent-component benchmark. Managed returns have a nonzero mean and

dependent whitened coordinates, so extrapolating (L.2) to them would require additional assumptions.

Proof of Theorem L.1. (a) The map $A \mapsto \operatorname{tr}\varphi(A)$ is convex on symmetric matrices whenever φ is convex on $\mathbb{R}$, by Ky Fan's maximum principle and Schur convexity of $\Lambda \mapsto \sum_i \varphi(\Lambda_i)$ (see, e.g., Bhatia, 1997; Marshall and Olkin, 1979). Since $E\hat{\Psi}_P = \Psi_P$, Jensen's inequality gives $E \operatorname{tr}\varphi(\hat{\Psi}_P) \geq \operatorname{tr}\varphi(E\hat{\Psi}_P) = \operatorname{tr}\varphi(\Psi_P)$; equality for all such φ would force $\hat{\Psi}_P = \Psi_P$ almost surely. For $\varphi(x) = x^2$ alone the strictness is the identity $E \operatorname{tr}\hat{\Psi}_P^2 - \operatorname{tr}\Psi_P^2 = E\|\hat{\Psi}_P - \Psi_P\|_F^2$, which is positive unless $\hat{\Psi}_P = \Psi_P$ almost surely. Taking $\varphi(x) = \pm x$ gives the mean identity. The same Jensen step applied to $A \mapsto \lambda_1(A)$, which is convex, and to $A \mapsto \lambda_P(A)$, which is concave, gives $E\,\lambda_1(\hat{\Psi}_P) \geq \lambda_1(\Psi_P)$ and $E\,\lambda_P(\hat{\Psi}_P) \leq \lambda_P(\Psi_P)$: the extremes of the spectrum move outward.

For the Gaussian identity (L.1), write $\operatorname{tr}\hat{\Psi}_P^2 = T^{-2}\sum_{s,t}(R_s'R_t)^2$. For $s \neq t$ the two draws are independent, so $E[(R_s'R_t)^2] = E[R_s'\,E[R_tR_t']\,R_s] = E[R_s'\Psi_P R_s] = \operatorname{tr}(\Psi_P\Psi_P) = \operatorname{tr}\Psi_P^2$, using $E[R_sR_s'] = \Psi_P$; note that this step is exact and needs no Gaussianity. For $s = t$ we need $E[(R'R)^2]$ with $R \sim \mathcal{N}(\mu_P, V_P)$. The quadratic-form variance formula $\operatorname{Var}(R'AR) = 2\operatorname{tr}\big((AV_P)^2\big) + 4\,\mu_P'AV_PA\,\mu_P$ at $A = I$ gives $\operatorname{Var}(R'R) = 2\operatorname{tr}V_P^2 + 4\,\mu_P'V_P\mu_P$, while $E[R'R] = \operatorname{tr}V_P + \mu_P'\mu_P = \operatorname{tr}\Psi_P$, so $E[(R'R)^2] = (\operatorname{tr}\Psi_P)^2 + 2\operatorname{tr}V_P^2 + 4\,\mu_P'V_P\mu_P$. Collecting the $T(T-1)$ off-diagonal and T diagonal terms,

$$E \operatorname{tr}\hat{\Psi}_P^2 = \frac{T-1}{T}\operatorname{tr}\Psi_P^2 + \frac{1}{T}\Big[(\operatorname{tr}\Psi_P)^2 + 2\operatorname{tr}V_P^2 + 4\,\mu_P'V_P\mu_P\Big],$$

which is (L.1). At $\mu_P = 0$ one has $V_P = \Psi_P$ and the right-hand side collapses to $\operatorname{tr}\Psi_P^2 + \frac{1}{T}\big[(\operatorname{tr}\Psi_P)^2 + \operatorname{tr}\Psi_P^2\big]$, the familiar central-Wishart formula. The noncentral terms must be retained when $\mu_P \neq 0$. Writing the inflation coefficient in terms of V_P gives

$$(\operatorname{tr}V_P)^2 + \operatorname{tr}V_P^2 + 2\|\mu_P\|^2\operatorname{tr}V_P + 2\mu_P'V_P\mu_P.$$

Thus $\operatorname{tr}\Psi_P = O(P)$ implies an $O(P^2/T)$ absolute upper bound. If also $\operatorname{tr} V_P \asymp P$, the inflation is $\Theta(P^2/T)$, growing at order yP when $P/T \to y > 0$. The trace bound on Ψ_P alone does not supply this lower bound. In relative terms, $\operatorname{tr} V_P^2 \le (\operatorname{tr} V_P)^2$ and $\mu_P' V_P \mu_P \le \|\mu_P\|^2 \operatorname{tr} V_P$ give the sharper bound

$$\frac{E\operatorname{tr}\hat{\Psi}_P^2 - \operatorname{tr}\Psi_P^2}{\operatorname{tr}\Psi_P^2} \le \frac{2(\operatorname{tr}\Psi_P)^2}{T\operatorname{tr}\Psi_P^2}.$$

Whenever $\operatorname{tr}\Psi_P \asymp P$ and $\operatorname{tr}\Psi_P^2 \asymp P^2$, as under convergence of the trace and Hilbert–Schmidt norm to a nonzero positive trace-class limit, this relative inflation is $O(1/T)$ regardless of P/T. Thus absolute inflation can grow while relative inflation vanishes. At fixed P the absolute inflation also vanishes as $T \to \infty$.

(b) Since $\mu_P = 0$,

$$\hat{\Psi}_P = V_P^{1/2}\Big(\frac{ZZ'}{T}\Big)V_P^{1/2}.$$

The extreme-eigenvalue law for an i.i.d. matrix with finite fourth moment gives, in probability,

$$\lambda_{\min}(ZZ'/T) \to (1-\sqrt{y})^2, \qquad \lambda_{\max}(ZZ'/T) \to (1+\sqrt{y})^2$$

(Yin et al., 1988; Bai and Yin, 1993; Bai and Silverstein, 2010). Hence, for the fixed c, C in the statement, with probability tending to one,

$$cI_P \preceq \frac{ZZ'}{T} \preceq CI_P.$$

Congruence by $V_P^{1/2}$ preserves Loewner order, so on that event

$$cV_P \preceq \hat{\Psi}_P \preceq CV_P.$$

Eigenvalue monotonicity yields $c\lambda_j(V_P) \leq \lambda_j(\hat{\Psi}_P) \leq C\lambda_j(V_P)$ for every j. Taking the upper bound at j and the lower bound at $j+1$ proves (L.2), and therefore the asserted $O_{\mathbb{P}}(1)$ conclusion. □

Remark L.2 (Why $P > T$ is different) The restriction $y < 1$ in part (b) is essential. If $P > T$, then $\hat{\Psi}_P$ is a sum of T rank-one matrices and therefore has rank at most T: it has at least $P - T$ zero eigenvalues. If the T observed return vectors are linearly independent, as they are almost surely under an absolutely continuous joint law, its rank is exactly T and the eigenvalue ratio at rank T is infinite. This hard cliff is created by the sample size alone. It is a reminder that in the high-dimensional regime (the one of Section 5.4), the sample spectrum beyond rank T carries no information about the population spectrum, and Theorem 4 is a statement about the population.

A short panel adds substantial spectral noise. The missing cliff of Theorem 4 remains a population statement: part (a) says that spectral dispersion is inflated in expectation, while part (b) says that a bounded population ratio cannot diverge in the centered independent-component benchmark with $y < 1$. When $P > T$, the sample cliff at rank T of Remark L.2 is a fact about the panel rather than the economy. In both regimes, the covariance problem of Section 5.4 has a sample part and a population part, and the population part does not go away as T grows.